

Appendix B

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C3
		Application Number	
Title of Invention	Memory Subsystem and Method of Operation		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number		NETL.046C3	
		Application Number			
Title of Invention		Memory Subsystem and Method of Operation			
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Application Information:

Title of the Invention	Memory Subsystem and Method of Operation				
Attorney Docket Number	NETL.046C3	Small Entity Status Claimed ☒			
Application Type	Nonprovisional				
Subject Matter	Utility				
Suggested Class (if any)		Sub Class (if any)			
Suggested Technology Center (if any)					
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)	1		

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C3
		Application Number	
Title of Invention	Memory Subsystem and Method of Operation		
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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	13745790	2013-01-19
Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
13745790	Continuation of	13183253	2011-07-14
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13183253	Continuation of	12422925	2009-04-13
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12422925	non provisional of	61044801	2008-04-14
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Application Number	Country i	Filing Date (YYYY-MM-DD)	Priority Claimed
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		Application Number	
Title of Invention	Memory Subsystem and Method of Operation		

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		Application Number	
Title of Invention	Memory Subsystem and Method of Operation		

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Memory Subsystem and Method of Operation

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. Patent Application No. 13/745,790, filed January 19, 2013, which is a continuation of U.S. Patent Application No. 13/183,253, filed July 14, 2011, now U.S. Patent No. 8,395,501, which is a continuation of U.S. Patent Application No. 12/422,925, filed April 13, 2009, now U.S. Patent No. 8,001,434, which claims the benefit of priority from U.S. Provisional Application No. 61/044,801, filed April 14, 2008, U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and U.S. Provisional Application No. 61/044,839, filed April 14, 2008. Each of the foregoing applications is incorporated in their entirety by reference herein. This application is related to US. Patent Application No. 12/422,912, filed on April 13, 2009 and entitled "Self-Adjusting Damper", and to US. Application No. 12/422,853, filed on April 13, 2009 and entitled "Circuit Providing Load Isolation and Noise Reduction", both of which are incorporated in their entirety by reference herein.

BACKGROUND

Field

[0002] The present invention relates to self-testing electronic modules and, more particularly, to self-testing electronic memory modules.

Description of the Related Art

[0003] The failure of memory components in an electronic system may result in the loss of valid data. Therefore, it is important to ensure proper memory operation in an electronic system. Memory integrated circuits ("memory chips") often go through a series of tests at various stages of system manufacture. Once memory chips are deployed in a system, they also generally go through a system level memory test each time the system is booted. In addition, memory chips may undergo a parity checking process during normal system operation.

[0004] There are typically at least three test phases which memories undergo during system manufacture. Each phase generally tests for memory defects and for the correct operation of the input/output interface. The first test phase is typically conducted

by the memory chip manufacturer and generally involves checking for bit failures, correct memory access speed, etc. The second test phase is typically done by memory module manufacturers and generally involves testing the signal quality, the noise susceptibility, and the operational speed of the memory module as a single unit. The second test phase may also include checking for bit failures in individual memory chips. The third phase is usually carried out by the system manufacturer. During the third phase, the interaction of the memory subsystem with other components in the system is tested. During the third phase, the individual memory module operation is also tested again and the memory array is checked for defects. Because of the significant amount of testing that memories undergo during the manufacturing process, there is generally substantial test cost and test time associated with ensuring the proper memory operation. This test cost and test time translate into an increase in system cost and a decrease in system performance.

[0005] There are a number of memory test methodologies that employ either external test hardware, embedded self-test logic ("MBIST"), or both. However, the usefulness of these test methodologies is limited due to the high cost and other limitations associated with them. For instance, external test hardware such as automatic test equipment ("ATE") is very expensive. Moreover, the development time and cost associated with implementing MBIST is relatively high. These costs and limitations are especially significant when testing dynamic random access memory ("DRAM"). For example, technological developments, such as increases in DRAM speed, may require manufacturers to upgrade ATE machines relatively frequently. In addition, MBIST in DRAM chips generally cannot be fully utilized for system level testing of memory boards.

[0006] Because of the increasing cost, complexity, and time involved with fully testing DRAM chips, DRAM manufacturers often provide "effectively tested" ("ETT") DRAM chips to memory module manufacturers at a lower price rather than providing fully tested DRAM chips. Memory module manufacturers often prefer the ETT DRAM chips mainly due to their greater availability. Memory module manufacturers who receive ETT DRAM chips then have to assume a part of the responsibility of validating the DRAM chips, adding to the complexity of the memory module test process.

SUMMARY

[0007] A memory subsystem is operable with a system memory controller. In certain embodiments, the memory subsystem comprises memory devices mounted on a circuit board, a data module mounted on the circuit board, and a control module mounted on the circuit board to provide address and control signals to the memory devices. The memory subsystem is operable in any of a plurality of modes including a normal mode and a test mode. During the normal mode, the control module provides the address and control signals based on address and control signals from the system memory controller, and the data module enables data paths between the memory devices and the system memory controller. During the test mode, the control module generates the address and control signals, and the data module isolates the memory devices from the system memory controller.

[0008] In certain embodiments, the memory subsystem is operable to communicate data with a system memory controller via a N-bit wide system memory bus. The memory devices are operable to receive or output data in response to the address and control signals from the control module, the data including a plurality of data bits. The data module includes a plurality of data handlers, each respective data handler corresponding to a respective n-bit section of the N-bit wide system memory bus, n being a fraction of N. The respective data handler includes a plurality of data paths and is configured to enable selected data paths among the plurality of data paths to route a respective portion of the plurality of data bits based on a mode in which the memory subsystem is operating.

[0009] In certain embodiments, a method of operating a memory subsystem comprises selecting a mode of operation for the memory subsystem. The method comprises providing address and control signals to the memory devices based on address and control signals from the system memory controller and enabling data paths between the memory devices and the system memory controller if the mode of operation is the normal mode. The method further comprises generating address and control signals for providing to the memory devices and isolating the memory devices from the system memory controller if the mode of operation is the test mode.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein.

[0011] Figure 2 is a block diagram of an example self-testing memory module including eight memory devices and eight data handlers in accordance with certain embodiments described herein.

[0012] Figure 3 is a block diagram of an example data handler module and an example control module in accordance with certain embodiments described herein.

[0013] Figure 4 is a flowchart of an example method of self-testing a memory module in accordance with certain embodiments described herein.

[0014] Figure 5 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

[0015] Figure 6 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0016] Certain embodiments described herein present a self-testing electronic system, such as for example, a self-testing electronic memory module. Some embodiments described herein present a self-testing memory module that is populated with ETT DRAM chips. Certain embodiments described herein present a self-testing registered dual in-line memory module (“RDIMM”). Some embodiments described herein present a self-testing RDIMM that does not require any additional pins other than the pins on the standard JEDEC RDIMM connector. For example, the memory module may utilize the address and control signals (e.g., address and control signals generated by the memory module to test the memory module) along with a test signal to enable and execute a self testing function.

[0017] Certain embodiments described herein present a self-testing memory module that includes a control module and a data module which can generate memory addresses and data according to the JEDEC standard memory protocol.

[0018] Some embodiments described herein presents a self-testing memory module that can be configured through an I²C interface and that allows test results to be read out through the I²C interface. Certain embodiments described herein present a self-testing memory

module that allows a test function to be configured, controlled, and/or executed without substantial system memory controller involvement. Some embodiments present a self-testing memory module that can be tested without any external test equipment. For example, certain embodiments present a self-testing memory module that can be tested without any system driven test procedure. Various embodiments described herein present a self-testing memory module that can be self-tested at a target system speed. Certain embodiments described herein present a self-testing memory module that can generate data (DQ) and data-strobe (DQS) signals with wave characteristics that resemble the wave characteristics of DQ and DQS signals from a system memory controller.

[0019] MBIST is commonly used to test memories embedded in application specific integrated circuits (“ASICs”) or system on chip integrated circuits (“SoCs”) such as, for example, advanced memory buffers (“AMBs”). The MBIST implementation usually includes three distinct functional blocks: the address/control generator, the data generator/checker (sometimes referred to as a signature analyzer), and the test interface controller/register (“JTAG”). The test instructions and test patterns are generally loaded to the address/control generator and the data generator/checker through the JTAG interface. The test results are generally read out through the JTAG interface.

[0020] In memory module applications, there are obstacles associated with using MBIST. One of these obstacles is the large number of signals used to interface the three MBIST functional blocks. The inter-block timing constraints can present another obstacle. In addition, the ability to control the test is limited. For example, synchronizing the address/or control signals with the data signal can be difficult. The relatively large amount of information to be gathered and stored while testing the memory presents yet another obstacle.

[0021] Unlike on an ASIC, in which the three MBIST functional blocks are in a single chip, the MBIST functional blocks on a memory module generally would be segregated into multiple chips on the memory module due to physical and electrical limitations and requirements. This makes implementing MBIST on a memory module difficult because, while there is virtually no limit on the number of available interface signals among the three MBIST functional blocks in an ASIC, the memory module can support only a limited amount of interface signals between memory chips. In addition, because the MBIST functional blocks are

spread out to multiple chips on a memory module, the inter-block signal delay is generally much longer on a memory module than on an ASIC. It is also generally not feasible to implement cross-checking logic that operates the three MBIST functional blocks in lock-operational block. This is due to the limitation on the number of interface signals and to the relatively long inter-block signal delay time on a memory module versus on an ASIC.

[0022] In some cases such as where a memory module includes an AMB, the self-test logic (MBIST) implemented in the AMB includes command and address generation logic in addition to a data generator and checker. Each of these functional blocks may be implemented on a single physical AMB device (e.g., a single integrated circuit package). However, because all of the self-test command, address and data signals are combined in one physical area of the memory module, a memory module (e.g., DIMM) level routing problem can occur, making it difficult to route the self-test signals on the memory module and resulting in performance degradation and/or implementation difficulty. In addition, the data width of self-test logic of the memory module will be limited to the data width of the AMB (e.g., to the data width of the self-test logic implemented on the AMB, to the number of available ports on the AMB, etc.). As such, implementing memory module test logic on an AMB is not flexible (e.g., to changes in the data width of the memory module) and generally supports only memory modules having certain predetermined, fixed data widths.

[0023] Finally, in most cases, since an ASIC cannot be repaired, the ASIC MBIST is generally only capable of detecting and reporting the pass/fail status of memory tests. For memory module test results, on the other hand, it is generally advantageous to include both addresses of the memory locations where failures occur and the data patterns that were read back from the failed memory locations. This type of reporting can help to facilitate the repair of the memory module by, for example, allowing for the identification and replacement of failed components.

[0024] Figure 1 is a block diagram of an example self-testing memory module 10 in accordance with certain embodiments described herein. The memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board (PCB) 12, each memory device 20 of the plurality of memory

devices 18 comprising data, address, and control ports. The memory module 10 comprises a control module 22 configured to generate address and control signals 24 for testing the memory devices 18. The memory module 10 also includes a data module 28 comprising a plurality of data handlers 30. Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports of one or more of the plurality of memory devices 18. For example, each of the data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports. Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. The memory module 10 may further include an I²C interface 15 in certain embodiments.

[0025] As described more fully below, in certain embodiments the data module 28 generates test data patterns to write to the plurality of memory devices 18 of the memory module 12 and checks the data patterns read or received back from the plurality of memory devices 18 for agreement with corresponding data patterns that are expected to be read back from the plurality of memory devices 18. For example, in one embodiment, the data module 28 generates cyclic patterns to write to the plurality of memory devices 18. In some embodiments, the data module 18 also isolates the data path from the system board of the computer system 16 to the plurality of memory devices 18 while the memory module 10 is not accessed by the computer system 16. For example, the data module 28 may isolate the data path from the system board to the plurality of memory devices 18 when the memory module 10 is in a self-testing mode. The control module 22 may include, for example, a dual input register (e.g., the memory device controller 32 described more fully below) for registering address and control signals coming from either self-testing logic (e.g., from the test controller 36 described more fully below) or from the memory controller 14 on the system board. In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20. For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation. The data module 28 may then receive a read command to read back the data from

those locations and check the read data for agreement with the expected data. If there is a mismatch between the read data and the expected data, the data module 28 may, for example, store the failure information (e.g., the failed data word) and inform the control module 22 about the failure. The control module may save the address of the memory location where the failure occurred.

[0026] In certain embodiments, the memory module 10 is configured to be operated in a test mode in which the control module 22 selectively inputs the address and control signals to the address and control ports of the plurality of memory devices 18. Moreover, in the test mode, each of the data handlers 30 write the data generated by the data handler 30 to the corresponding plurality of data ports by selectively inputting data signals to the data ports of the plurality of memory devices 18. The data module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20. For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according the operating specification of the memory devices 20. In some embodiments, the control module 22 and the data module 28 produce memory addresses, control and/or data signals according to the JEDEC standard memory protocol. In some embodiments, for example, the control module 22 and the data module 28 generate the memory interface signals with proper edge relationships based on the JEDEC standard. In certain embodiments, the test speed, for example, may be defined by the speed of the clock (e.g., the system clock). The address sequences and/or the data patterns of certain embodiments may be programmable either through the I²C interface or they may be defaulted to pre-defined values.

[0027] In certain embodiments, data module 28 is configured to generate data signals with programmable slew rates and/or with variable peak values. In one embodiment, for example, the data module 28 is also able to generate data (DQ) and data-strobe (DQS) signals with programmable slew rates and programmable peak values so that the characteristics of the signals generated by the data module 28 generally correspond to the characteristics DQ and DQS signals generated by the system memory controller 14.

[0028] In some embodiments, the data module 28 and/or the control module 22 are configured to test the plurality of memory devices 18 under non-normal conditions. For example, the data module 28 and/or control module 22 may be configured to provide signals having frequencies which are higher or lower than the normal operating frequencies of the memory devices 20.

[0029] In certain embodiments, the memory module 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 12 has an industry-standard form factor. For example, the PCB 12 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 12 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 12 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0030] In certain embodiments, the plurality of memory devices 18 of the memory module 10 may be arranged as ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 20 of the plurality of memory devices 18 that can be accessed by the memory controller 14 to access the full bit-width of the memory bus of the memory module 10. For example, a memory module 10 in which each rank of the memory module is 64 bits wide is described as having an “x 64” organization. Similarly, a memory module 10 having 72-bit-wide ranks is described as having an “x 72” organization. The number of memory devices 20 and corresponding memory capacity of a memory module 10 can be increased by increasing the number of memory

devices 20 per rank or by increasing the number of ranks. For example, a memory module with four ranks with each rank having N 512-MB memory devices 20 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 20 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices 20. During operation, the ranks of a memory module 10 may be selected or activated by control signals that are received from a component of the system (e.g., a system memory controller 14 or a local memory controller of the memory module 10). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 10 comprises only one rank of memory devices 20.

[0031] As discussed, the PCB 12 may include at least one connector (not shown) configured to operatively couple the memory module 10 to the memory controller 14 of the computer system 16. The computer system 16 may include a host computer system. For example, the memory module is electrically coupled, logically coupled, or both, with the memory controller 14. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 14 may comprise a disk controller of the computer system 16, for example. The memory controller 14 may be mounted on a system board of the host computer 16. The connector can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system 16. The connector of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 10 and the host system 16. For example, the connector can comprise a standard DDR2, DDR3, and other future generation edge connectors. Additionally, in certain embodiments, more than one memory module 10 is coupled to the host system 16.

[0032] The plurality of memory devices 18 on the PCB 12 may include one or more volatile memory components. For example, the plurality of memory devices 18 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 20. Types of DRAM devices 20 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM

(SDRAM). The memory devices 18 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 20 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 20 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP). The plurality of memory devices 18 may further include one or more non-volatile memory devices 20, such as, for example, flash memories. The plurality of memory devices 18 of certain embodiments may include both volatile and non-volatile memory devices 20. For example, the plurality of memory devices 18 may include one or more of DRAM, SRAM, and/or flash memory devices in some embodiments.

[0033] Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28. For example, each data handler 30 is configured to write to and/or read from the corresponding plurality of data ports of one or more of the memory devices 20 without being in communication any of the other data handlers 30 or other data ports of the memory devices 20. As such, each data handler 30 can be used to generally independently test a portion of the memory space of the memory module 10. For example, each data handler 30 may be used to independently test one memory device 20 of the memory module 10. In such a configuration, the corresponding plurality of data ports of each data handler 30 may comprise each data port of the corresponding memory device 20. In other embodiments, each data handler 30 may be used to test a segment of one memory device 20, more than one memory device 20, segments more than one memory device 20, or any combination or sub-combination thereof. Because each of the data handlers 30 is operable independently of each of the other data handlers 30, the data handlers 30 are generally modular. As such, modifications in the configuration of the memory module 10 (e.g., changes in the bit-width of the memory bus, changes in the number of memory devices 20, etc.) may be less complicated to accommodate than in other types of self-testing memory modules 10. For example, where a new memory device 20 or set of memory devices 20 is added to the memory module 10, the change may be generally accommodated by adding a corresponding data handler 30. The change may be accommodated without having to

implement a major reorganization of the memory module 10 or the self-testing logic of the memory module 10, for example.

[0034] Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. Figure 2 is a block diagram of an example self-testing memory module 10 including eight memory devices 20 (e.g., memory devices 40a-40h) and a data module 28 comprising eight data handlers 30 (e.g., data handlers 30a-30h) in accordance with certain embodiments described herein. Each of the memory devices 20 includes an eight bit output data word and eight corresponding data ports. In addition, the system memory bus 50 between the memory controller 14 and the example memory module 10 is 64 bits wide and each of the data handlers 30 receives an eight bit segment of the system memory bus 50. Each of the data handlers 30 is operatively coupled to a corresponding plurality of data ports 21 of a corresponding one of the memory devices 20. As such, the data handlers 30 may be operatively coupled (e.g., electrically and/or logically coupled or connected) to the eight data ports 21 of one of the corresponding memory devices 20. For example, the data handler 30a may be operatively coupled to the eight data ports 21 of the memory device 40a of Figure 2.

[0035] The configuration shown in Figure 2 is for the purposes of illustration and is not intended to be limiting. For example, while the example memory module 10 of Figure 2 includes an equal number of memory devices 20 and data handlers 30, other configurations are possible. In some configurations there are more memory devices 20 than data handlers 30 or vice versa. Moreover, the one or more data handlers 30 may be operatively coupled to a subset of the data ports 21 of one the memory devices 20 instead of all of the data ports 21 of one of the memory devices 20. In other embodiments, one or more data handlers 30 may be operatively coupled to a subset or all of the data ports 21 of more than one of the memory devices 20. For example, in one embodiment, each of the data handlers 30 are operatively coupled to all of the data ports of two memory devices 20.

[0036] In certain embodiments, the plurality of data handlers 28 comprises at least two physically separate components mounted on the PCB 12. For example, the plurality of data handlers 28 may include at least two physically separate integrated circuit packages. The physically separate integrated circuit packages are mounted on different portions of the

PCB 12 in some embodiments. For example, each of the eight data handlers 30a-30h shown in Figure 2 may include physically separate integrated circuit packages mounted on different portions of the PCB 12. While eight data handlers 30 are shown in Figure 2, other numbers of data handlers 30 are possible including fewer or more than eight.

[0037] In certain embodiments, each of the plurality of data handlers 30 is positioned on the PCB 12 proximate to the corresponding plurality of data ports. For example, each data handler 30 of certain embodiments is positioned closer to the corresponding plurality of data ports 21 than the data handler 30 is to the other data ports 21 of the plurality of memory devices 18. For example, the data handler 30a is positioned closer to the corresponding plurality of data ports 21 of the memory device 40a than to the other data ports 21 of the other memory devices 40b-40h.

[0038] Figure 3 is a block diagram an example data module 28 and an example control module 22 in accordance with certain embodiments described herein. The control module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18. In some embodiments, the control module 22 includes a control mixer element 32. The control mixer element 32 may include a memory device controller 34 (e.g., a DRAM controller) and a test controller 36. In certain embodiments, the control mixer element 32 generally controls the address and the control signals for the self-testing function.

[0039] In certain embodiments, the memory device controller 34 generally pre-processes address and control information before it sends the information to a register 40. In one embodiment, the memory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36. The control module 22 of certain embodiments is configured to selectively input to the address and control ports of the plurality memory devices 18 either the address and control signals 38 from the system memory controller 14 or the address and control signals 42 from the control module 22 (e.g., from the test controller 36). For example, the memory device controller 34 may send either the signals 38 from the system memory controller 14 or, alternatively, the signals 42 from the test controller 36, to the register 40 depending on whether the memory module 10 is in normal (non-test) mode or in a

test mode, respectively. In one embodiment, the memory device controller 34 generates the address and control signals for memory device (e.g., DRAM device) operations. The test controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28. The control module 22 may be implemented in the control register of the memory module 10 in certain embodiments. In various embodiments, the control module 22 includes discrete logic, one or more application-specific integrated circuit (ASICs), one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic device (CPLDs).

[0040] The data module 28 and the subcomponents thereof (e.g., the data handlers 30) may be in communication with one or more of the memory devices 20, the control module 22, and the memory controller 14. In certain embodiments, the data module 28 comprises a plurality of data handlers 30. In other embodiments the data module 28 includes at least one data handler 30. Each of the data handlers 30 of certain embodiments comprises a switch 44. For example, the switch 44 may include a data multiplexer/demultiplexer (“data mux/demux”). The switch 44 may provide a bi-directional data multiplexer function. In certain embodiments, the switch 44 is configured to selectively input to the corresponding plurality of data ports either data signals 48 from the system memory controller 14 or data signals 50 from the data handler logic element 46. The switch 44 of certain embodiments may further be configured to receive data signals 52 (e.g., during a read operation) from the plurality of memory devices 18 and to propagate the data signals 52 to the data handler logic element 46 and/or the memory controller 14. In some embodiments, for example, the switch 44 selectively inputs the data signals 48 to be written to the plurality of memory devices 18 from the system memory controller 14 when the memory module 10 is a normal (non-test mode) mode and, alternatively, inputs the data signals 50 from the data handler logic element 46 during a test mode. While the switch 44 is shown as being included in the data handler 30 in the example of Figure 3, other configurations are possible. For example, in other embodiments the switch 44 may be logically and/or physically separated from the data handler module 28 and/or the data handlers 30.

[0041] Each of the data handlers 30 of certain embodiments further includes a data handler logic element 46. The data handler logic element 46 of certain embodiments comprises a data generation element 54 and a verification element 56. The data generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports, for example. The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information) the data handler logic element 46 receives from the control module 22, for example. The data may be cyclic data in some embodiments or non-cyclic data in other embodiments. For example, the cyclic data may comprise at least one predetermined pattern of data which repeats or is cycled two or more times. In various embodiments, the data comprises one or more incrementing patterns or decrementing patterns, for example. In other embodiments, the data comprises a pattern which alternates each bit on successive memory writes. For example, a memory write comprising one or more hexadecimal “A” characters (each corresponding to a four-bit binary word of “1010”) may be followed by a memory write comprising one or more hexadecimal “5” characters (each corresponding to a four-bit binary word of “0101”). The data may be generated in a variety of ways. In one embodiment, the data is generated based on a current write address value. For example, in one example configuration, on a first write cycle, hexadecimal “A’s” are generated and written to even address locations and hexadecimal “5’s” are generated and written to odd address locations, and on a second write cycle, “5’s” are written to even addresses and “A’s” are written to odd addresses, and this pattern repeats in subsequent cycles. The data may be generated based previously written data (e.g., inverting each of the bits of a previously written data word) in some embodiments. In general, any manner of generating a cyclic or otherwise deterministic data pattern may be compatible with embodiments described herein. In other embodiments, random or pseudorandom data may be generated and written to the corresponding plurality of data ports. For example, a linear feedback shift register (LFSR) may be used in some embodiments. In addition, the data patterns may be programmable. For example, the data patterns may be programmable based on information received by the data generation element 54 from the memory controller 14 (e.g., through the control module 22), or from the control module 22.

[0042] The plurality of data handlers 30 are further configured to read data from the corresponding plurality of data ports. For example, the verification element 56 may be configured to receive data from the corresponding plurality of data ports (e.g., through the switch 44 during a test mode). The verification element 56 may further be configured to check for failures in the operation of the plurality memory devices 18 by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler 30 and written to the corresponding plurality of data ports.

[0043] In certain embodiments, the verification element 56 is configured to perform the verification without storing a copy of the data written to the corresponding plurality of data ports or accessing a stored copy beyond the data read from the plurality of memory devices 18. For example, the verification element 56 does not store or access a copy of the data that is written to the corresponding plurality of data ports except for the data stored and read back from the plurality of memory devices 18. As such, the memory module 10 of certain embodiments advantageously does not require separate memory for storing duplicate copies of test data that is written to the plurality of memory devices 18 for later comparison. For example, the verification element 56 may calculate comparison data and may compares the comparison data to the data read from the corresponding plurality of data ports. In certain embodiments, the comparison data comprises data which expresses the data or values expected to be received from the plurality of memory devices 18 if the write, store, and read processes of the data using the data module 28 and the plurality of memory devices 18 are performed correctly or as expected. The calculation may be performed simultaneously or substantially simultaneously with receiving the data read from the corresponding plurality of data ports in certain embodiments. In other embodiments, the calculation is performed either before or after receiving the data. Other configurations are possible. For example, in one embodiment, the verification element 56 does store a separate copy of the data written to the corresponding plurality of data ports upon writing the data and compares the separate copy to the read data received from the plurality of memory devices 18.

[0044] The verification element 56 of certain embodiments calculates the comparison data based on the cyclic data. For example, in one embodiment, the verification element 56 calculates the comparison data in substantially the same manner that the data

generation element 54 generates the data as described above (e.g., based on a current write address, using an LFSR, etc.). As such, the comparison data of certain embodiments is substantially a repeat of the data written. In one example embodiment, the data handler 30 is configured (e.g., is programmed by the control module 22) to write an alternating series of “A’s” and “5’s” to the data ports of the corresponding plurality of data ports as described herein. For example, the data handler 30 may be configured to write one or more “A’s” to the first address location of an N-word memory device 20 including the corresponding plurality of data ports. The data handler 30 may then write one or more “5’s” to the second address location, one or more “A’s” to the third address location and so on until the data handler 30 has written to all N memory locations. The verification element 56 of the example embodiment then calculates the comparison data based on the cyclic data written to the corresponding plurality of data ports. For example, the verification element 56 calculates a comparison word including one or more “A’s”, “5’s,” and “A’s,” respectively, to be compared to the data read from the first, second, and third address locations based on the cyclic data (e.g., based on the known cycle of the data). In some embodiments, the comparison data is calculated based on a current read address.

[0045] In certain embodiments, data associated with failures in the operation of the plurality of memory devices 18 are stored in the data module 28. For example, data read from the corresponding plurality of data ports which do not correspond to (e.g., match) the comparison data calculated by the verification element 56 may be stored in the data module. Moreover, in some embodiments, memory addresses associated with the failures in the operation of the plurality of memory devices 18 are stored in the control module 22. For example, the data handler 30 may communicate data failures (e.g., when data read from the corresponding plurality of data ports does not correspond to calculated comparison data) to the control module 22 which may then store the addresses corresponding to the data failure. In certain embodiments, the memory module 12 is configured to report failures (e.g., the failed data, the address corresponding to the memory location of the failed data, and/or expected data) via the I²C interface 15 to the memory controller 14. In addition, in certain embodiments the test controller 36 and/or the data handlers 30 may be updated through the

I²C interface 15 with new data patterns and/or with alternative memory access sequences to conduct AC tests (e.g., tests of the power, current, I/O speed, etc.).

[0046] In various embodiments, the components of the data module 28 (e.g., the switch 44, the data handlers 30, the data handler logic element 46, the data generation element 54, and/or verification element 56) may include discrete logic, one or more application-specific integrated circuits (ASICs) one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic devices (CPLDs). Additionally, one or more of the various functional blocks (e.g., the switch 44) of the data module 28 of Figure 3 may not be included. In some embodiments, additional functional blocks may be included. Moreover, some of the functional blocks are described as separate functional blocks for illustration purposes and may comprise one physical component. For example, in one embodiment, each of the data handlers 30 and the corresponding switch 44, data generation element 54, and verification element 56 comprise one physical component (e.g., are included in one integrated circuit package). In another embodiment, the data module 28 comprises one physical component.

[0047] Referring again to Figure 1, a self-testing memory module 10 of certain embodiments comprises a printed circuit board (PCB) 12 and is configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board 12 where each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. The memory module 10 further comprises a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. In certain embodiments, the memory module 10 further comprises a data module 28 comprising at least one data handler 30 and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices 20. The data handler 30 is configured to generate cyclic data for writing to the corresponding plurality of data ports. In certain embodiments, there may be one data handler 30, for example. In other embodiments, there may be more than one data handler 30. The elements of the memory module 10 may be compatible with any of the embodiments described herein.

[0048] Figure 4 is a flowchart of an example method 70 of self-testing a memory module 10 in accordance with certain embodiments described herein. While the method 70 is described herein by reference to the memory module 10, other memory modules, electronic systems or subsystems, and/or circuits are also compatible with the embodiments described herein. The method 70 of certain embodiments comprises providing a self-testing memory module 10 at operational block 72. The memory module 10 may comprise a printed circuit board (PCB) 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 may further comprise a plurality of memory devices 18 on the printed circuit board 12. Each memory device 20 of the plurality of memory devices 18 may comprise data, address, and control ports. The memory module 10 may comprise a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. The memory module 10 may further comprise a data module 28 comprising a plurality of data handlers 30. In certain embodiments, the data module 28 comprises at least one data handler 30. Each data handler 30 of certain embodiments is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports. At operational block 74, the method 70 further comprises generating, by each of the data handlers 30, data for writing to the corresponding plurality of data ports.

[0049] Figure 5 is a flow diagram 100 illustrating self-testing operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. At operational block 110 the memory module 10 enters an idle state after power up or, in some embodiments, when the memory board of the computer system 16 is connected to a mother board or to a test board. In one embodiment, at operational block 110, the contents of the register 40 are undefined and the controllers (e.g., the test controller 36 and the memory device controller 34) are in unknown state. After power up, the memory module 10 is reset (e.g., a reset command is executed) at operational block 111. At operational block 111, the logic of the data handler 30 (e.g., data generation element 54 and/or verification element 56) and the logic of the control mixer 32 are set to default values and/or states. For example, the memory module 10 may default to a non-test mode (e.g., normal operational mode) and the switches 44 of the data handlers 30 and the memory device controller 34 are not configured in

a test mode. At operational block 112, the test mode is configured. For example, the computer system 16 (e.g., through the memory controller 14) may configure the test mode. The configuration may be through the I²C interface 15, for example, and may include configuring the test controller 36 for test mode. For example, configuring the test mode may include switching (e.g., by configuring the test controller 36) the mode of the memory module 10 from a normal operation mode to the test mode.

[0050] At operational block 113, the test mode is initiated. Initiating the test mode may comprise configuring a particular test case (e.g., particular test data or data patterns, particular write and/or read address sequences, etc.). The initiation may be achieved by activating the test controller 36 to initiate the test mode. For example, at operational block 113, the data handlers 30 and the control mixer 32 are readied (e.g., via a tester) beginning the self-testing of the memory module 10. For example, the tester may comprise an ATM tester, a server, a specialized tester. The tester may ready the data handlers 30 and the control mixer through the I²C interface 15 or through the control signals 38, for example. For example, the data handlers 30 are configured to input data generated by the data handlers 30 to the corresponding plurality of data ports of the plurality of memory devices 18 and the control mixer 32 is configured to input address and control signals from the test controller 36 to the plurality of memory devices 18. At operational block 113 the test controller 36 updates each of the data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.).

[0051] In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal (“Par-in”) is un-asserted (e.g., set to a “low” value), it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. The memory module 10 executes the test mode (e.g., writes and reads test data patterns) at operational block 114. The memory module 10 continues executing the test mode at operational block 114 until the self-test is complete or until the memory module 10 detects a certain number of memory failures such that a failure count exceeds a preset number.

[0052] If the failure count exceeds the preset number, the memory module 10 enters operational block 115 and “errors out” of the test mode. If the self-test is completed at

operational block 114 and the failure count does not exceed the preset number, the memory module 10 enters operational block 115. At operational block 115, the memory module 10 sends out a test completion indication signal (e.g., through the test controller 36 to the memory controller 14). At operational block 116, the fail status of the test may be read out through, for example, the I²C interface 15. For example, one or more address values associated with failed memory locations, and/or the data read from those locations, and/or expected data may be read. In other cases, where the test fails, the memory module 10 may report the failure to the memory controller 14 through any available signal (e.g., bidirectional data or data strobe signal) between the memory module 10 and the memory controller 14.

[0053] In some embodiments, the memory module 10 is generally interruptible. For example, the memory module 10 may exit the self-test and return to operational block 110 if the signal going into the parity-in pin is, for example, de-asserted or removed at operational block 155. In other embodiments, the interrupt signal or condition may be different and may not be a parity-in signal. For example, in one embodiment, the memory module 10 may exit the self-test and return to operational block 110 if a timer having a pre-determined count expires. When the self-testing is interrupted, the memory module 10 will return to operational block 110 and the bit failure information in the control mixer 32 and the data handler 30 becomes invalid.

[0054] Figure 6 is a flow diagram 200 illustrating the operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. For example, one or more of the operational blocks of the flow diagram 200 of Figure 6 may correspond to one or more of the operational blocks of Figure 5. At operational block 210, the memory module 10 may be generally inactive. For example, the testing logic (e.g., the control module 22 and/or the data module 28) may be generally inactive and the memory module 10 may default to a functional (e.g., non-test) mode. In certain embodiments, the memory module 10 executes one or more tasks which configure the self-test mode of the self-testing memory module using an I²C interface 15. For example, the memory module 200 may configure the test controller 36 through the I²C interface 15 (e.g., as described with respect to Figure 5). In other embodiments, the test mode is configured using the control signals 38. For

example, the test mode may be updated at operational block 211 in certain embodiments. In certain embodiments, updating the test mode comprises programming the test controller 36 with test configuration information such as test data patterns, test data sequences, test failure conditions, etc. After updating the test mode at operational block 211, the test controller 36 may then be activated through the I²C interface 15 (e.g., as described with respect to Figure 5) in certain embodiments. Alternatively, as shown, the test controller 36 may be activated through the I²C interface 15 without first updating the test mode.

[0055] At operational block 212, the memory module test mode is ready for execution. In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal is low, it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. At operational block 113, a set of status registers (e.g., one or more registers of the control module 22 or the data module 28) are cleared.

[0056] The memory module 200 generates address and data signals for testing the plurality of memory devices 18 at operational block 214. For example, the data generation elements 54 of the data handlers 30 may generate the data for writing to the plurality of memory devices 18 as described herein. In addition, the control module 22 may generate address and control signals for testing the plurality of memory devices 18 as described herein. At operational block 215, the memory module 10 performs a burst write function. For example, in certain embodiments, the memory module 10 writes multiple memory locations in the memory module 10 that are to be tested at operational block 215 in a relatively short amount of time (or in a burst). In certain embodiments, the memory module 10 can perform multiple write bursts at operational block 215. Next, at operational block 216, the memory module 10 performs a read & compare function. For example, in certain embodiments, the memory module 10 reads back certain memory locations at operational block 216 that have been written at operational block 215 and compares the values with certain expected data. For example, a verification element 56 of each of the data handlers 30 may calculate the expected data and/or compare the values as described herein. In some embodiments, the memory module 200 reads from multiple memory locations in the memory module 10 in a relatively

short amount of time (or in a burst). In certain embodiments, the memory module 10 can execute multiple read bursts at operational block 216 as it compares the read data with expected data. In some embodiments, the memory module 200 (e.g., the data handlers 30 of the memory module 200) stores any failed read data and the associated expected data at operational block 217. In some embodiments, the memory module sends one or more failure indications to the memory controller 14 over one or more available signals between memory module 10 and the memory controller 14. For example, the I²C interface 15 or one or more bi-directional data pins (e.g., one or more of the data pins 48) may be used.

[0057] The memory module 200 stores the addresses of the memory locations associated with the failed read data using the test controller 32 at operational block 217. In other embodiments the failed read data, expected data, and the addresses of the memory locations associated with the failed read data may be stored using the test controller 32. For example, in one embodiment, the test controller 32 stores the failed read data, expected data, and the addresses of the memory locations associated with the failed read data at operational block 217. In some embodiments, the test controller 32 is generally inaccessible through the I²C interface 15 during self-test operation. For example, the test controller is inaccessible through the I²C interface 15 until a test failure occurs at operational block 217 or until the test completes at operational block 219. In some embodiments, when the test is complete, the test results can be read out of the memory module 10 (e.g., through the I²C interface 15). In some cases, the test can be interrupted using the I²C interface 15 or using the parity-in signal (e.g., by de-asserting the parity-in signal).

[0058] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects

or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory subsystem operable with a system memory controller, comprising:
memory devices mounted on a circuit board;
a data module mounted on the circuit board; and
a control module mounted on the circuit board to provide address and control signals to the memory devices; and
wherein the memory subsystem is operable in any of a plurality of modes including a normal mode and a test mode;
wherein, during the normal mode, the control module provides the address and control signals based on address and control signals from the system memory controller, and the data module enable data paths between the memory devices and the system memory controller; and
wherein, during the test mode, the control module generates the address and control signals, and the data module isolates the memory devices from the system memory controller.
2. The memory subsystem of claim 1, wherein the memory devices are arranged in at least one rank, each rank having a width of N bits, and wherein the data module includes a plurality of data handlers, each respective data handler being configured to handle a respective n-bit section of the N bits, n being a fraction of N.
3. The memory subsystem of claim 2, wherein each memory device is n-bit wide.
4. The memory subsystem of claim 2, wherein the data module is further configured to receive a N-bit wide read data signal from the memory devices and wherein the respective data handler is further configured to check, during the test mode, a data pattern in a respective n-bit section of the N-bit wide read data signal with respect to an expected data pattern.
5. The memory subsystem of claim 4, wherein the expected data pattern is generated in the respective data handler.

6. The memory subsystem of claim 4, wherein the respective data handler is further configured to receive the respective n-bit section of the N-bit wide read data signal from a respective group of at least one memory device among the memory devices.
7. The memory subsystem of claim 4, wherein the data module is further configured to store failure information when a test failure is indicated by the data pattern not in agreement with the expected data pattern.
8. The memory subsystem of claim 4, wherein the data module is further configured to report the failure information to the control module.
9. The memory subsystem of claim 2, wherein the data module is further configured to transmit a N-bit wide write data signal to the memory devices, and wherein the N-bit write data signal has at least one of a programmable slew rate and a programmable peak value that can be programmed to correspond to characteristics of data signals from the system memory controller.
10. The memory subsystem of claim 8, wherein the respective data handler is further configured to send a respective n-bit section of the N-bit wide write data signal to a respective group of at least one memory device among the memory devices.
11. The memory subsystem of claim 2, wherein the plurality of data handlers operate independently of each other.
12. The memory subsystem of claim 2, wherein the plurality of data handlers include at least two physically separate integrated circuit packages.
13. A memory subsystem operable to communicate data with a system memory controller via a N-bit wide system memory bus, comprising:
 - a control module mounted on a circuit board;
 - memory devices mounted on the circuit board and operable to receive or output data in

response to address and control signals from the control module, the data including a plurality of data bits; and

a data module including a plurality of data handlers, each respective data handler corresponding to a respective n-bit section of the N-bit wide system memory bus, n being a fraction of N, wherein the memory subsystem is operable in any one of a plurality of modes, and wherein the respective data handler includes a plurality of data paths and is configured to enable selected data paths among the plurality of data paths to route a respective portion of the plurality of data bits based on a mode in which the memory subsystem is operating.

14. The memory subsystem of claim 13, wherein the plurality of modes includes a normal mode and a test mode, and wherein:

during the normal mode, the control module provides the address and control signals based on address and control signals received from the system memory controller, and the respective data handler enables data paths between one or more of the memory devices and the system memory controller; and

wherein, during the test mode, the control module generates the address and control signals, and the respective data handler enables data paths between one or more of the memory devices and self-test logic in the memory subsystem.

15. The memory subsystem of claim 14, wherein the data module isolates the memory devices from the N-bit wide data bus when the memory subsystem is operating in the test mode.

16. The memory subsystem of claim 13, wherein the plurality of data paths in the respective data handler include a first data path coupled to the respective n-bit section of the N-bit wide data bus and a second data path coupled to a logic element in the memory subsystem.

17. The memory subsystem of claim 16, wherein the logic element is included in the respective data handler.

18. The memory subsystem of claim 16, wherein the logic element includes a data generation element, the data generation element generating data bits that are written to the memory devices.
19. The memory subsystem of claim 14, wherein the logic element includes a verification element, the verification element checking for data patterns represented by data bits read from the memory devices.
20. A method of operating a memory subsystem coupled to a system memory controller, the memory subsystem including memory devices, the method comprising:
- selecting a mode of operation for the memory subsystem, the mode of operation being any one of a normal mode and a test mode;
 - if the mode of operation is the normal mode:
 - providing address and control signals to the memory devices based on address and control signals from the system memory controller; and
 - enabling data paths between the memory devices and the system memory controller;
 - if the mode of operation is the test mode:
 - generating address and control signals for providing to the memory devices; and
 - isolating the memory devices from the system memory controller.

Memory Subsystem and Method of Operation

ABSTRACT OF THE DISCLOSURE

A memory subsystem is operable with a system memory controller. The memory subsystem comprises memory devices mounted on a circuit board, a data module mounted on the circuit board; and a control module mounted on the circuit board to provide address and control signals to the memory devices. The memory subsystem is operable in any of a plurality of modes including a normal mode and a test mode. During the normal mode, the control module provides the address and control signals based on address and control signals from the system memory controller, and the data module enables data paths between the memory devices and the system memory controller. During the test mode, the control module generates the address and control signals, and the data module isolates the memory devices from the system memory controller.

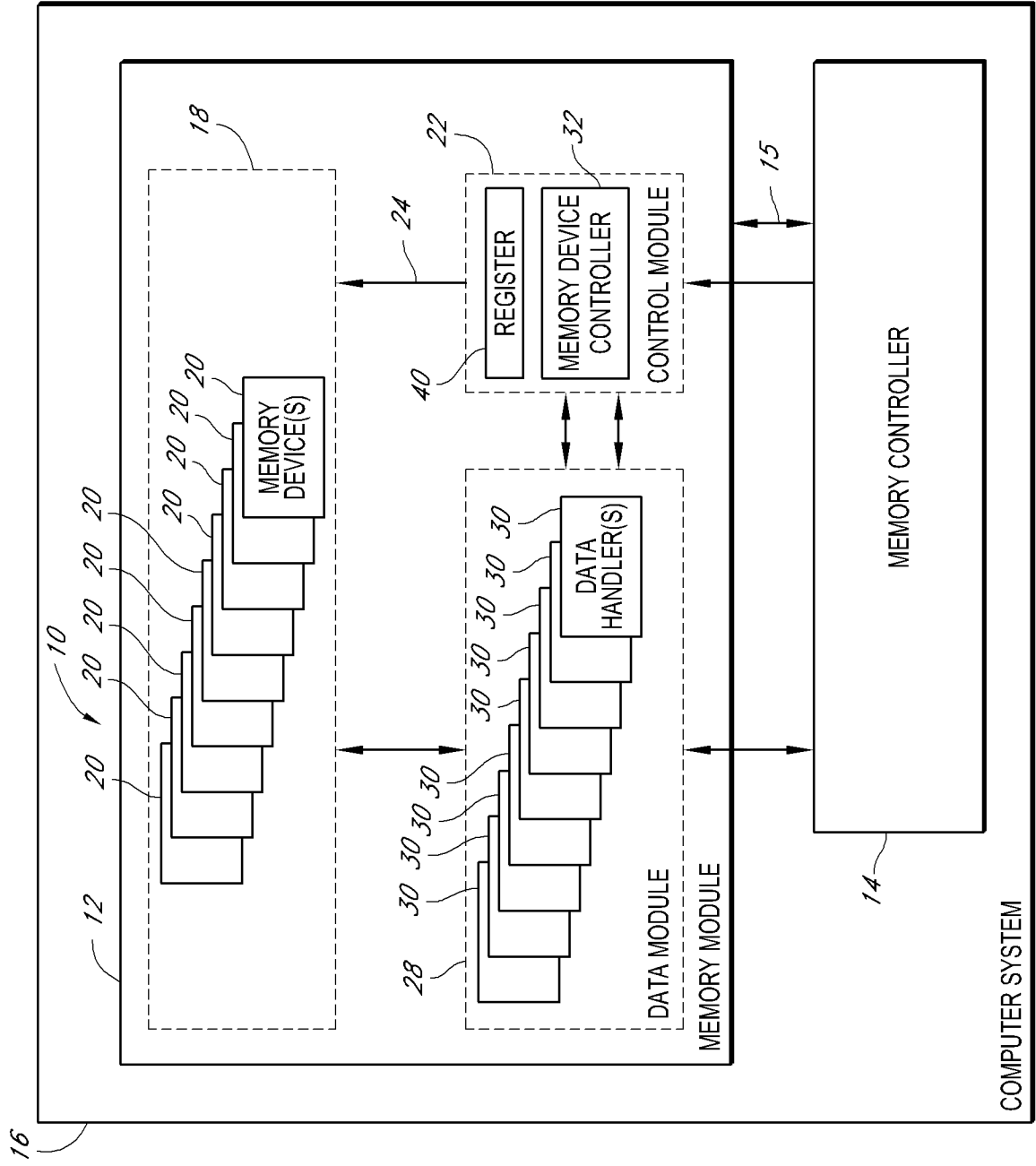
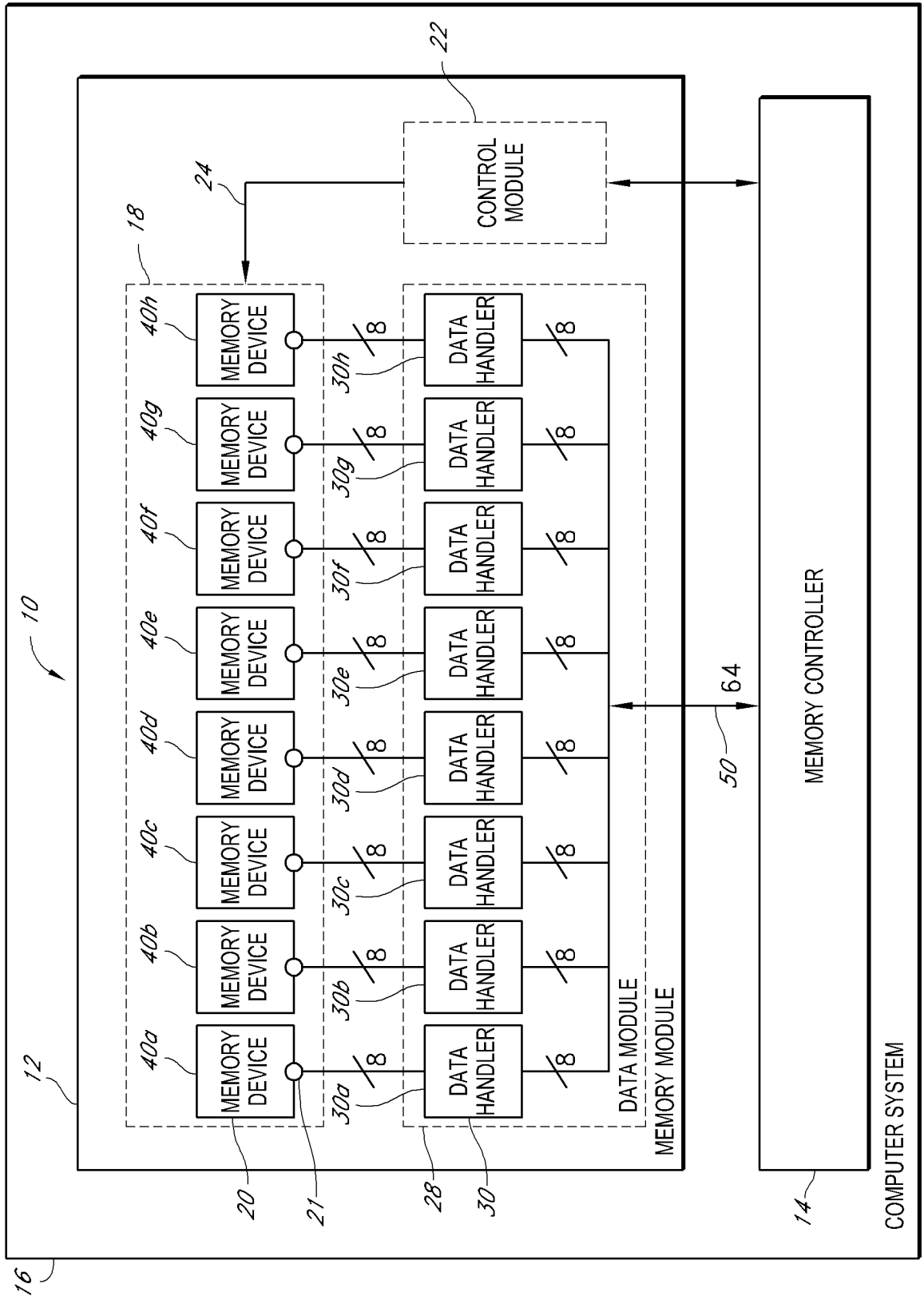


FIG. 1



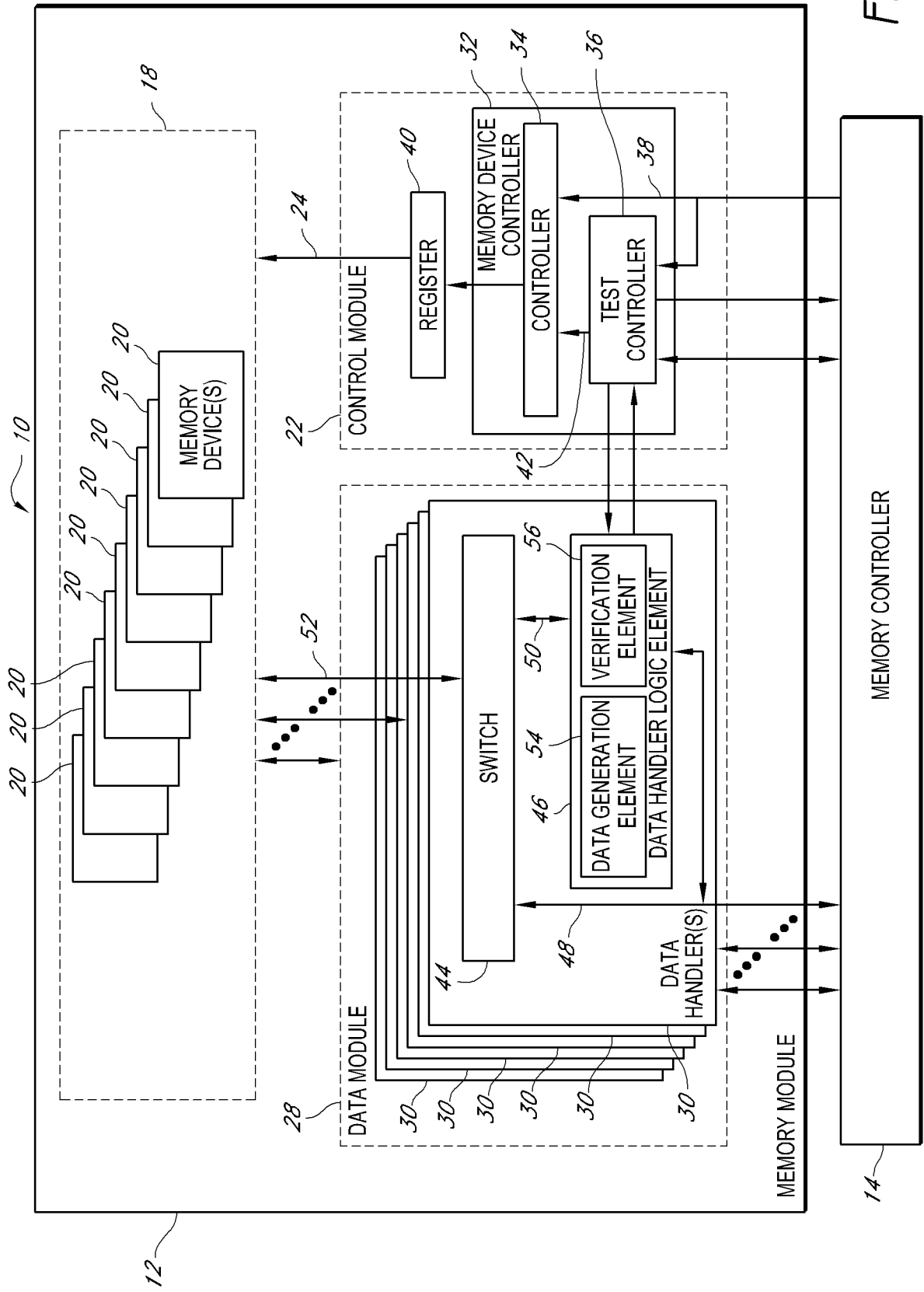


FIG. 3

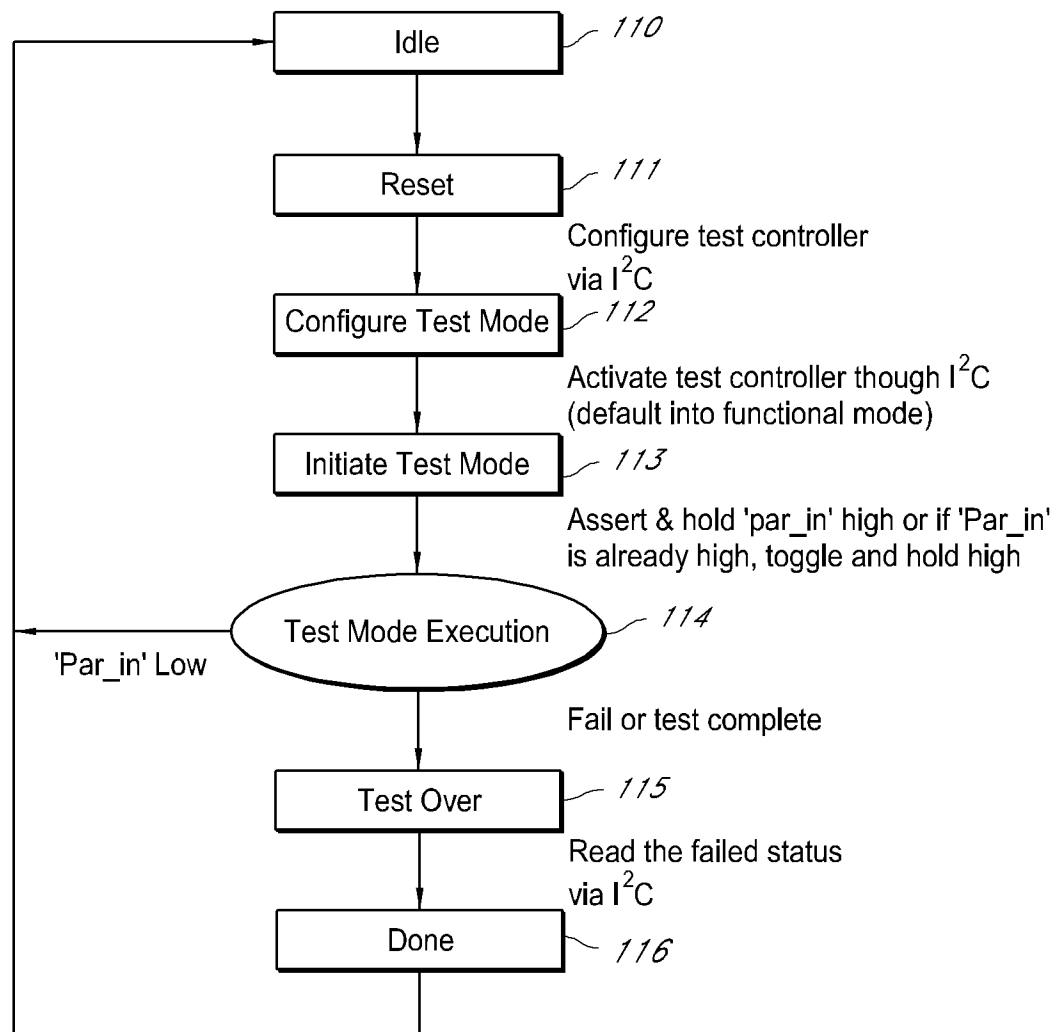
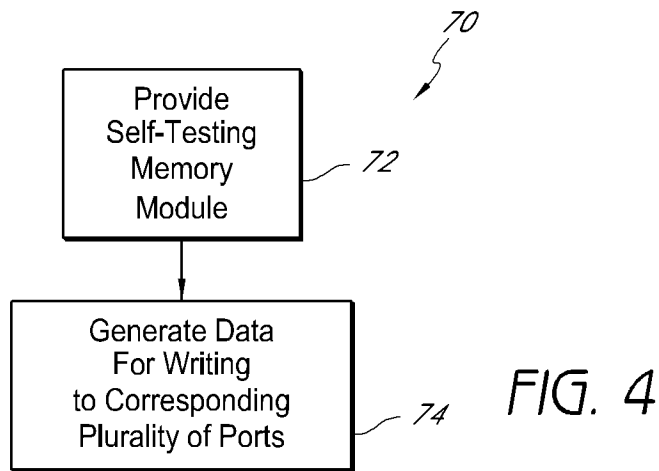
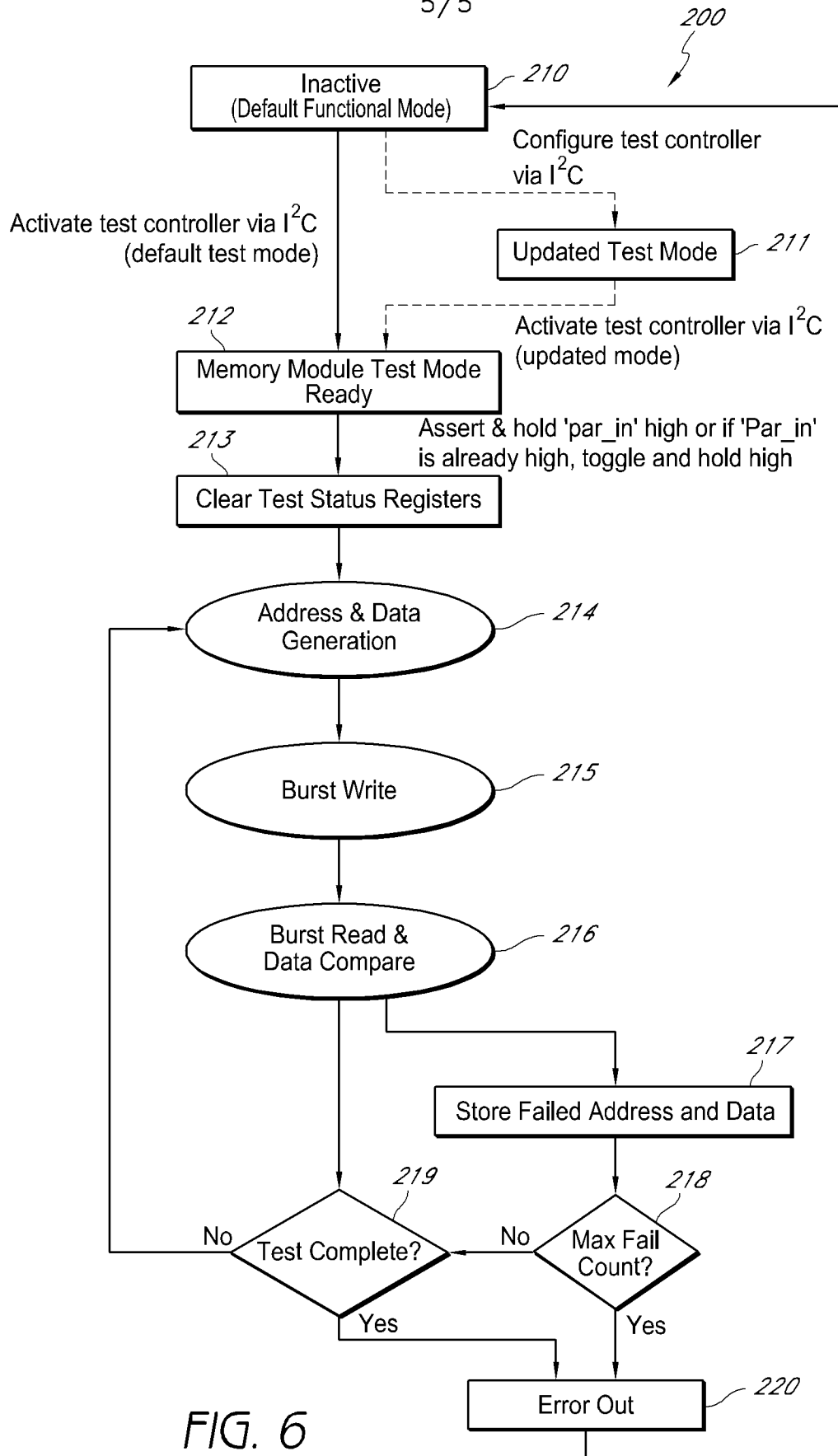


FIG. 5



Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

Memory Subsystem and Method of Operation

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Jamie Jie Zheng

Attorney Docket Number:

NETL.046C3

Filed as Small Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				730

Electronic Acknowledgement Receipt

EFS ID:	18619741
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	29-MAR-2014
Filing Date:	
Time Stamp:	02:19:20
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	2014-03-28_ApplicationDataSheet_NETL046C3.pdf	1437656 2944bfdd2de86ed275b6791e35e5b12a0a71ba15	no	7

Warnings:

Information:

2		2014-03-28_Specification_NETL046C3.pdf	14200815 ef2339a132fc446d15b7f3c807a9e9fad35aa535	yes	29
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	24	
	Claims		25	28	
	Abstract		29	29	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	2014-03-28_Drawings-NETL046C3.pdf	102122 3a434e20f5e761f341b218106e090c7f7691e057	no	5
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	32400 027659054bfb510b1fb9ad4c2b5223140dc60661	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			15772993		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/229,844	03/29/2014	Hyun Lee	NETL.046C3

CONFIRMATION NO. 1018

FORMALITIES LETTER



OC000000067925179

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 04/22/2014

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **70** basic filing fee.
- \$ **70** surcharge.
- \$ **300** search fee.
- \$ **360** examination fee.
- \$(**0.00**) previous unapplied payment amount.
- \$ **800** TOTAL FEE BALANCE DUE.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's

oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

Hyun Lee
Jayesh R. Bhakta
Soonju Choi

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
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Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/229,844

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

OR

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/229,844	03/29/2014	2189	0.00	NETL.046C3	20	3

CONFIRMATION NO. 1018

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

FILING RECEIPT



Date Mailed: 04/22/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/745,790 01/19/2013 PAT 8689064
which is a CON of 13/183,253 07/14/2011 PAT 8359501
which is a CON of 12/422,925 04/13/2009 PAT 8001434
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 04/21/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/229,844**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Memory Subsystem and Method of Operation

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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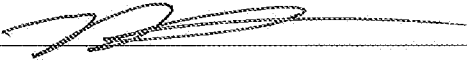
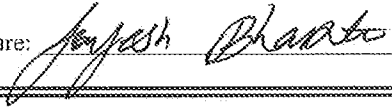
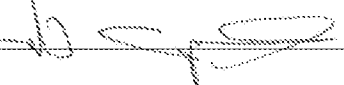
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DECLARATION FOR UTILITY OR DESIGN APPLICATION	
Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
As a below named inventor, I hereby declare that: This declaration is directed to: ☐ The attached application, or ☒ United States application or PCT international application number <u>13/745,790</u> filed on <u>January 19, 2013</u>. The above-identified application was made or authorized to be made by me. I believe that I am the original inventor or an original joint inventor of a claimed invention in the application. I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both. Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."	
First Inventor (Legal Name): <u>Hyun Lee</u>	
Signature: <u></u>	Date (Optional): <u>4/17/2013</u>
Second Inventor (Legal Name): <u>Jayesh R. Bhakta</u>	
Signature: <u></u>	Date (Optional): <u>4/17/2013</u>
Third Inventor (Legal Name): <u>Soonju Choi</u>	
Signature: <u></u>	Date (Optional): <u>4/17/2013</u>

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C3			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Filing Fee for Oath or Declaration	2051	1	70	70
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				800

Electronic Acknowledgement Receipt

EFS ID:	19241990
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	09-JUN-2014
Filing Date:	29-MAR-2014
Time Stamp:	01:09:27
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$800
RAM confirmation Number	7243
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
-----------------	----------------------	-----------	-------------------------------------	------------------	------------------

1	Oath or Declaration filed	2013-04-17_Declaration_Signed_NETL046C2.pdf	381396 c44cf7f9988fae05e4b03b32f230809360c177b8	no	1
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	36655 d988ce29a8ed34ab974dc400bb0a699beb78a9f1a	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			418051		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/229,844

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

OR

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/229,844	03/29/2014	2189	800	NETL.046C3	20	3

CONFIRMATION NO. 1018

UPDATED FILING RECEIPT

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306



Date Mailed: 06/13/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/745,790 01/19/2013 PAT 8689064
which is a CON of 13/183,253 07/14/2011 PAT 8359501
which is a CON of 12/422,925 04/13/2009 PAT 8001434
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 04/21/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/229,844**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Memory Subsystem and Method of Operation

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	2	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6011710	01-04-2000	Wiggers	
	2.	20010008006	07-12-2001	Klein	
	3.	20050257109	11-17-2005	Averbuj	
	4.	20070070669	03-29-2007	Tsern	

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	5.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,516,185, IPR Case Number IPR2014-01029, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	6.	Petition for Inter Partes Review filed on 6/22/2014 for U.S. Patent No. 7,881,150, IPR Case Number IPR2014-00882 and IPR Case Number IPR2014-01011, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	7.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,081,536, IPR Case Number IPR2014-00883, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	8.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,001,434, IPR Case Number IPR2014-00970, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	9.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,359,501, IPR Case Number IPR2014-00971, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	10.	Altera, ACEX iK, Programmable Logic Device Family, Data Sheet, May 2003, Ver 3.4	
	11.	Horowitz, "The Art of Electronics," Cambridge Univ. Press, 1989, selected pages	
	12.	Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379	
	13.	Jacob, Bruce L.; "Synchronous DRAM Architectures, Organizations, and Alternative Technologies". University of Maryland, Dec. 10, 2002.	
	14.	JEDEC Standard No. 21-C Section 4.5.7, 168 Pin Registered SDRAM DIMM Family, Release 7	
	15.	JEDEC 21-C, Section 4.6.1, 278 Pin Buffered SDRAM DIMM Family	
	16.	JEDEC Standard No. 21-C Section 4.1.2.5, Appendix E, "Specific PD's for Synchronous DRAM (SDRAM)," pp. 1-25,	
	17.	JEDEC Standard, "Fully Buffered DIMM (FBDIMM): DfX Design for Validation and Test," JESD82-28, February 2008	

EXAMINER**DATE CONSIDERED**

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	2	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	18.	Reese, "Introduction to Logic Synthesis using Verilog HDL," Morgan&Claypool Publishers, 2006, pp. 1-28	

EXAMINER**DATE CONSIDERED**

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Electronic Acknowledgement Receipt

EFS ID:	19804932
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	07-AUG-2014
Filing Date:	29-MAR-2014
Time Stamp:	15:17:34
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2014-08-07_IDS_Cover_NETL046C3.pdf	58349 23e25bcbdd3669340b7850fd7f2a2a4d4c1c57d6d	no	2

Warnings:

Information:

2	Information Disclosure Statement (IDS) Form (SB08)	2014-08-07_IDS_NETL046C3.pdf	111501 7ddf1bf9ebe2d7ce0ea20d79c03c54ce98a11754	no	2
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	2003-05_Altera_ACEX_1K_Data sheet.pdf	23673114 d3153cf33657081c125ece8a06e13d43ffad67cd	no	86
Warnings:					
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Information:					
4	Non Patent Literature	1989_Horowitz_Art_of_Electronics.pdf	804824 4fe58ae9db8168f345983f24ffe9b752877cd2c4	no	15
Warnings:					
Information:					
5	Non Patent Literature	2001_Huang_BIST_Method_VI_Sl_Design.pdf	3496109 6bc20bd6ff74761ea00ffdc2e5032ebb5bdf1321	no	17
Warnings:					
Information:					
6	Non Patent Literature	2002-12-10_Jacob_SDRAM_Architectures.pdf	5974132 f24bc23ec4f9e3bb8affeb17f74c9bc8fe428859	no	22
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
7	Non Patent Literature	0000_JEDEC_21-C_4-5-7_168_Pin_RSDRAM.pdf	801227 346b9c985eafcea6da69a3ad7b3fca4d79e5f17b	no	8
Warnings:					
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8	Non Patent Literature	0000_JEDEC_21C_4-6-1-278_Pinn.pdf	4025672 757e358ba523ebc1c7f068d3e3cf05349542c252	no	24
Warnings:					
Information:					
9	Non Patent Literature	0000_JEDEC_21-C_4-1-2-5_Appendix_E.pdf	3114809 cff2ebcdb681dc41dd4ad0494d49109d2e25e34d	no	25

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10	Non Patent Literature	2008-02_JEDEC_FBDIMM_DFX_Design_JESD82-28.pdf	21675078 5d5deecd1996940b5e14b00bf2c9e06f0cd158	no	102
Warnings:					
Information:					
11	Non Patent Literature	2006_Reese_Logic_Synthesis.pdf	1719895 25fabbb91adf16030c84b4a7f5395d11df5c71d0c	no	35
Warnings:					
Information:					
Total Files Size (in bytes):			65454710		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14229844
Filed	: August 20, 2013
For	: LOAD-REDUCING CIRCUIT FOR MEMORY MODULE
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 18 references. Copies of foreign documents and/or their translation and non-patent literature are submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits

Application No.: 13/971,231
Filing Date: August 20, 2013

was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: August 7, 2014

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 273-6008

51167
(Reg. No.)

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	1	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	1.	PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification JEDEC, Standard No. 21-C, Revision 1.3, Jan. 2002, pages 4.20.4-1	

EXAMINER**DATE CONSIDERED**

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Electronic Acknowledgement Receipt

EFS ID:	19823168
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	10-AUG-2014
Filing Date:	29-MAR-2014
Time Stamp:	01:59:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2014-08-09_IDS2_Cover_NETLO 46C3.pdf	58235 85ad5040546212214d20753a732fed1cf97 4075a	no	2

Warnings:

Information:

2	Information Disclosure Statement (IDS) Form (SB08)	2014-08-08_IDS2_NETL046C3. pdf	85602 cff8d781dccec1c6e5848d74f770f5878179 946	no	1
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	2002-01- JEDEC_PC2100_PC1600_DDR_ SDRAM_Reg_Design.pdf	3571816 abf1a0c4e0c639a658bcd212408e8d33498 4b844	no	82
Warnings:					
Information:					
Total Files Size (in bytes):			3715653		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14229844
Filed	: March 29, 2014
For	: LOAD-REDUCING CIRCUIT FOR MEMORY MODULE
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 1 non-patent literature, a copy of which is submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

Application No.: 13/971,231
Filing Date: August 20, 2013

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: August 9, 2014

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 273-6008

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(Reg. No.)

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NETL.046C3
Sheet	1	of	5		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	5655113	08-05-1997	Leung	
	2.	5878051	03-02-1999	Sharma	
	3.	5936900	08-10-1999	Hii	
	4.	6321320	11-20-2001	Fleischman	
	5.	6747887	06-08-2004	Halbert et al.	
	6.	6832303	12-14-2004	Tanaka	
	7.	6934205	08-23-2005	Pandey et al.	
	8.	7093066	08-15-2006	Klein	
	9.	7168005	01-23-2007	Adams	
	10.	7203873	04-10-2007	Adams	
	11.	7379316	05-27-2008	Rajan	
	12.	7392442	06-24-2008	Averbuj	
	13.	7464225	12-09-2008	Tsern	
	14.	7865674	01-04-2011	Gower et al.	
	15.	8089795	01-03-2012	Rajan	
	16.	8189328	05-29-2012	Kanapathippillai	
	17.	8233303	07-31-2012	Best	
	18.	20040006729	01-08-2004	Pendurkar	
	19.	20080104352	05-01-2008	Talbot	

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.		T ²
	20.	Inter Partes Review Case No. IPR2014-00970, Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on June 18, 2014		
	21.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1007 to Petition for Inter Partes Review, “Declaration of Donald Albert, Ph.D.,” filed on June 18, 2014		
	22.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2002 to Petition for Inter Partes Review, “Excerpt from Merriam-Webster’s Collegiate Dictionary,” filed on June 18, 2014		
	23.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2003 to Petition for Inter Partes Review, “Excerpt from Webster Handy College Dictionary,” filed on June 18, 2014		
	24.	Inter Partes Review Case No. IPR2014-00970, Patent Owner’s Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on September 24, 2014		

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				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	5	Attorney Docket Number	NETL.046C3

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	25.	Inter Partes Review Case No. IPR2014-00970, Decision – Institution of Inter Partes Review, issued on December 16, 2014		
	26.	Inter Partes Review Case No. IPR2014-00971, Petition for Inter Partes Review for U.S. Patent No. 8,359,501, filed on June 18, 2014		
	27.	Inter Partes Review Case No. IPR2014-00971, Exhibit 1007 to Petition for Inter Partes Review, “Declaration of Donald Albert, Ph.D,” filed on June 18, 2014		
	28.	Inter Partes Review Case No. IPR2014-00971, Patent Owner’s Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on September 24, 2014		
	29.	Inter Partes Review Case No. IPR2014-00971, Decision – Institution of Inter Partes Review, issued on December 16, 2014		
	30.	Inter Partes Review Case No. IPR2014-01373, Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on August 25, 2014.		
	31.	Inter Partes Review Case No. IPR2014-01373, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68,” filed on August 23, 2014.		
	32.	“IEEE Standard Test Access Port and Boundary-Scan Architecture” (“JTAG 2001”, IEEE-SA Standards Board, IEEE Std1149.1-2001, June 14, 2001.		
	33.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2001 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Webster’s II New College Dictionary,” filed on August 23, 2014		
	34.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “IEEE Standard Dictionary of Electrical and Electronics Terms,” filed on August 22, 2014.		
	35.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2003 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “JEDEC Standard, Fully Buffered DIMM (FBDIMM) DfX Design for Validation and Test,” filed on August 22, 2014.		
	36.	Inter Partes Review Case No. IPR2014-01372, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.		
	37.	Inter Partes Review Case No. IPR2014-01372, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68,” filed on August 23, 2014.		
	38.	Inter Partes Review Case No. IPR2014-01374, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.		
	39.	Inter Partes Review Case No. IPR2014-01374, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68,” filed on August 23, 2014.		
	40.	Inter Partes Review Case No. IPR2014-01375, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.		

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	5	Attorney Docket Number	NETL.046C3

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	41.	Inter Partes Review Case No. IPR2014-01375, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, "Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68," filed on August 23, 2014.		
	42.	Inter Partes Review Case No. IPR2014-01029, Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on June 24, 2014.		
	43.	Inter Partes Review Case No. IPR2014-01029, Exhibit 1008 to Petition for Inter Partes Review, "Declaration of Charles J. Neuhauser, Ph.D. under 37 C.F.R. § 1.68," filed on June 24, 2014.		
	44.	Inter Partes Review Case No. IPR2014-01029, Supplemental Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on July 23, 2014.		
	45.	Inter Partes Review Case No. IPR2014-01029, Patent Owner's Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on October 17, 2014.		
	46.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014.		
	47.	Inter Partes Review Case No. IPR2014-01369, Corrected Petition for Inter Partes Review of Claims 1-19 of U.S. Patent No. 8,516,185, filed on September 22, 2014.		
	48.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1008 to Corrected Petition for Inter Partes Review, "Declaration of Dr. Nader Bagherzadeh under 37 C.F.R. § 1.68," filed on September 22, 2014.		
	49.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1013 to Corrected Petition for Inter Partes Review, "Webster's II New College Dictionary," filed on September 22, 2014.		
	50.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1014 to Corrected Petition for Inter Partes Review, "Standard Dictionary of Electrical and Electronics Terms," IEEE 1988, filed on September 22, 2014.		
	51.	Inter Partes Review Case No. IPR2014-00882, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.		
	52.	Inter Partes Review Case No. IPR2014-00882, Exhibit 1007 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 22, 2014.		
	53.	Inter Partes Review Case No. IPR2014-00883, Corrected Petition for Inter Partes Review of U.S. Patent No. 8,081,536, filed on July 8, 2014.		
	54.	Inter Partes Review Case No. IPR2014-00883, Exhibit 1011 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 21, 2014.		
	55.	Inter Partes Review Case No. IPR2014-01011, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.		
	56.	Inter Partes Review Case No. IPR2014-01011, Exhibit 1007 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 22, 2014.		
	57.	McCluskey, Edward J., <i>Logic Design Principles with Emphasis on Testable Semicustom Circuits</i> , Prentice Hall, 1986, pp. 104-107 and 119-120.		

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	4	of	5	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	58.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.'s Invalidity Contentions, dated June 6, 2014	
	59.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits A.1-A.10 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	60.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits B.1-B.8 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	61.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	62.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F.1-F.5 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	63.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G.1-G.6 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	64.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	65.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Diablo Technologies, Inc.'s Invalidity Contentions, dated June 6, 2014.	
	66.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits A-1 to A-10 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	67.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits C-1 to C-7 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	68.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits D-1 to D-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	69.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F-1 to F-5 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	

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				First Named Inventor	Hyun Lee
				Art Unit	2189
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Sheet	5	of	5	Attorney Docket Number	NETL.046C3

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	70.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G-1 to G-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
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INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/229,844
Filed	: March 29, 2014
For	: LOAD-REDUCING CIRCUIT FOR MEMORY MODULE
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 71 patent documents and non-patent literature, a copy of which is submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits

Application No.: 14/229,844
Filing Date: March 29, 2014

was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: December 24, 2014

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 273-6008

51167
(Reg. No.)

Electronic Acknowledgement Receipt

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Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
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Application Type:	Utility under 35 USC 111(a)

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	2014-06-18_petition_IPR2014-00970.pdf	754299 97fb3eb7a1319d18eb792099fa973941252d5a3a	no	66

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2	Non Patent Literature	2014-06-18_Albert_Declaration_IPR2014-00970.pdf	1583795	no	145
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3	Non Patent Literature	IPR970_Exhibit-2002.pdf	641415	no	3
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4	Non Patent Literature	IPR970_Exhibit-2003.pdf	393989	no	3
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5	Non Patent Literature	2014-09-24_preliminary_response_IPR970.pdf	2426015	no	67
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6	Non Patent Literature	2014-12-16_institution_decision_IPR970.pdf	430709	no	24
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7	Non Patent Literature	2014-06-18_petition_IPR2014-00971.pdf	730600	no	60
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12	Non Patent Literature	Exhibit-1002_Declaration_Bagherzadeh.pdf	3843743	no	252
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25	Non Patent Literature	2014-07-23_Supp_Petition_IPR2014-01029.pdf	1058145	no	65
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26	Non Patent Literature	2014-10-17_Prelim_Rsps_IPR2014-01029.pdf	3272498	no	56
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38	Non Patent Literature	20140606_Smart_Storage_Invalidity_Contentions.pdf	13104462	no	85
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51	Non Patent Literature	20140606_DIABLO_Invalidity_Contentions.pdf	1703900	no	41
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52	Non Patent Literature	Diablo_EXHIBIT_A1-A10.pdf	12661621	no	493
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53	Non Patent Literature	Diablo_EXHIBIT_C1-C7.pdf	7965781	no	216
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54	Non Patent Literature	Diablo_EXHIBIT_D1-D6.pdf	4503878	no	145
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56	Non Patent Literature	Exhibit_G1_G6.pdf	814795	no	79
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57	Non Patent Literature	Exhibit_H_Diablo.pdf	45048	no	3
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58	Information Disclosure Statement (IDS) Form (SB08)	2014-12-24_IDS_NETL046C3.pdf	186114	no	5
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59	Transmittal Letter	2014-12-24_IDS_Cover_NETL046C3.pdf	58515	no	2
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant	: Netlist, Inc.
App. No.	: 14/229,844
Filed	: March 29, 2014
For	: Memory Subsystem and Method of Operation
Examiner	: Unknown
Art Unit	: 2189
Conf. No.	: 1018

PRELIMINARY AMENDMENT**Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

Applicant submits the following amendments and remarks in the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 10 of this paper.

In the Specification:

Please change the title to the following:

“A Self-Test Memory Module and Method of Operation”

In the Claims:

1. (Currently Amended) A memory ~~subsystem-module~~ operable in a computer system with a system memory controller, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports, and operating according to a standard memory protocol;

a data module mounted on the circuit board and coupled to the data ports of the memory devices; and

a control module mounted on the circuit board and coupled to the address and control ports of the memory devices, ~~to provide address and control signals to the memory devices;~~ and

wherein the memory ~~subsystem-module~~ is operable in any of a plurality of modes including a normal mode and a test mode;

wherein, during the normal mode, the control module provides ~~the~~ memory address and control signals to the memory devices according to the standard memory protocol, the memory address and control signals being based on system address and control signals received by the control module from the system memory controller, and the data module propagates data signals ~~enable data paths~~ between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals; and

wherein, during the test mode, the control module generates ~~the self-test~~ address and control signals according to the standard memory protocol without using any system driven test procedure, the control module provides the self-test address and control signals to the address and control ports of the memory devices, and the data module generates self-test data patterns, isolates the memory devices from the system memory controller, and transmits the self-test data patterns to the data ports of the memory devices in accordance with the standard memory protocol, wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals received from the control module.

2. (Currently Amended) The memory ~~subsystem-module~~ of claim 1, wherein the memory devices are arranged in at least one rank, each rank having a width of N bits, ~~and~~

wherein the data module includes a plurality of data handlers, each respective data handler ~~being configured to handle~~ handles a respective n-bit section of the N bits ~~by propagating a respective n-bit section of each N-bit wide data signal between the memory devices and the system memory controller during the normal mode, and by generating respective n-bit wide self-test data patterns and transmitting the respective n-bit wide self-test data patterns to a respective n-bit section of the at least one rank during the test mode,~~ n being a fraction of N.

3. (Currently Amended) The memory ~~subsystem~~ module of claim 2, wherein each memory device is n-bit wide.

4. (Currently Amended) The memory ~~subsystem~~ module of claim 2, wherein the control module generates additional self-test address and control signals and provides the additional self-test address and control signals to the address and control ports of the memory devices during the test mode, the additional self-test address and control signals causing the memory devices to output a N-bit wide read data signal, wherein the data module is further configured to receive a N-bit wide read data signal from the memory devices and wherein the each respective data handler is further configured to receives a respective n-bit section of the N-bit wide read data signal and check, during the test mode, checks a data pattern in the a ~~respective n-bit section of the N-bit wide read data signal with respect to an expected data pattern.~~

5. (Currently Amended) The memory ~~subsystem~~ module of claim 4, wherein the expected data pattern is generated in the respective data handler after the control module generates the additional self-test address and control signals.

6. (Currently Amended) The memory ~~subsystem~~ module of claim 1, wherein the standard memory protocol is a JEDEC standard memory protocol 4, wherein the respective data handler is further configured to receive the respective n-bit section of the N-bit wide read data signal from a respective group of at least one memory device among the memory devices.

7. (Currently Amended) The memory ~~subsystem-module~~ of claim 4, wherein the data module is ~~further configured to store~~ stores failure information when a test failure is indicated by the data pattern not in agreement with the expected data pattern.
8. (Currently Amended) The memory ~~subsystem-module~~ of claim 4, wherein the data module is further configured to report the failure information to the control module.
9. (Currently Amended) The memory ~~subsystem-module~~ of claim ~~1-2~~, wherein the ~~data module is further configured to transmit a N-bit wide write data signal to the memory devices, and wherein the N-bit write data signal has~~ self-test data patterns transmitted to the memory devices have at least one of a programmable slew rate and a programmable peak value that can be programmed to correspond to characteristics of data signals from the system memory controller.
10. (Currently Amended) The memory ~~subsystem-module~~ of claim ~~1-8~~, wherein the data module generates and transmits the self-test data patterns to the memory devices in response to a write command from the control module, and wherein the data module read back test data patterns from the memory devices in response to a read command from the control module ~~respective data handler is further configured to send a respective n-bit section of the N-bit wide write data signal to a respective group of at least one memory device among the memory devices.~~
11. (Currently Amended) The memory ~~subsystem-module~~ of claim 2, wherein the plurality of data handlers operate independently of each other without being in communication with each other, and wherein, during the test mode, the each respective data handler generates n-bit self-test data patterns to test a corresponding portion of a memory space of the memory module independently of the other data handlers.
12. (Currently Amended) The memory ~~subsystem-module~~ of claim 2, wherein the plurality of data handlers include ~~at least two~~ physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective n-bit section of the at least one rank.

13. (Currently Amended) A memory ~~subsystem-module~~ operable to communicate data with a system memory controller via a N-bit wide system memory bus, comprising:

a control module mounted on a circuit board;

memory devices mounted on the circuit board and operable to receive or output data signals in response to address and control signals from the control module, wherein the memory devices are arranged in at least one N-bit wide rank ~~the data including a plurality of data bits~~; and

a data module including a plurality of data handlers, wherein each respective data handler ~~corresponding to~~ is coupled between a respective n-bit section of the N-bit wide system memory bus and a respective n-bit section of the at least one N-bit wide rank, n being a fraction of N, wherein the memory subsystem is operable in any one of a plurality of modes including a test mode, and wherein, during the self test mode: ~~the respective data handler includes a plurality of data paths and is configured to enable selected data paths among the plurality of data paths to route a respective portion of the plurality of data bits based on a mode in which the memory subsystem is operating;~~

the control module generates self-test address and control signals and provide the self-test address and control signals to the memory devices; and

the each respective data handler produces respective n-bit wide test data patterns and transmits the respective n-bit wide test data patterns to the respective n-bit section of the at least one N-bit wide rank so that one or more memory devices in the respective n-bit section of the at least one N-bit wide rank receive the respective n-bit wide test data patterns in accordance with the self-test address and control signals.

14. (Currently Amended) The memory ~~subsystem-module~~ of claim 13, wherein the plurality of modes further includes a normal mode ~~and a test mode~~, and wherein:

during the normal mode, the control module provides ~~the memory~~ address and control signals to the memory devices based on system address and control signals received from the system memory controller, and the each respective data handler propagates data ~~enables data paths~~ between one or more of the memory devices in the respective n-bit section of the at least one N-bit wide rank and the system memory controller; ~~and~~

~~wherein, during the test mode, the control module generates the address and~~

~~control signals, and the respective data handler enables data paths between one or more of the memory devices and self test logic in the memory subsystem.~~

15. (Currently Amended) The memory ~~subsystem module~~ of claim 14, wherein the data module isolates the memory devices from the N-bit wide data bus when the memory ~~subsystem module~~ is operating in the test mode.

16. (Currently Amended) The memory ~~subsystem module~~ of claim 13, wherein the memory devices operate according to a standard memory protocol, wherein the control module generates the self-test address and control signals according to the standard memory protocol without using any system driven test procedure, and wherein the each respective data handler transmits the respective n-bit wide data patterns to the respective n-bit section of the at least one N-bit wide rank according to the standard memory protocol~~the plurality of data paths in the respective data handler include a first data path coupled to the respective n-bit section of the N-bit wide data bus and a second data path coupled to a logic element in the memory subsystem.~~

17. (Currently Amended) The memory ~~subsystem module~~ of claim 16, wherein the standard memory protocol is a JEDEC standard memory protocol~~the logic element is included in the respective data handler.~~

18. (Currently Amended) The memory ~~subsystem module~~ of claim 16, wherein the ~~logic element includes a~~ each respective data handler receives control information from the control module and includes a respective data generation element that produces the n-bit wide data patterns in response to the control information from the control module,~~the data generation element generating data bits that are written to the memory devices.~~

19. (Currently Amended) The memory ~~subsystem module~~ of claim 14, wherein the each respective data handler further includes ~~logic element includes a~~ respective verification element, the respective verification element checking for data patterns represented by data bits read from the respective portion of the memory devices for errors.

20. (Currently Amended) A method of operating a memory ~~subsystem-module~~ coupled to a system memory controller, the memory ~~subsystem-module~~ including memory devices, the memory devices having address and control ports and data ports and operating according to a standard memory protocol, the method comprising:

selecting a mode of operation for the memory ~~module-subsystem~~, the mode of operation being ~~any~~ one of a plurality of modes including a normal mode and a test mode;

~~if the mode of operation is in~~ the normal mode:

receiving system address and control signals from the system memory controller;

providing memory address and control signals to the memory devices according to the standard memory protocol, the memory address and control signals being based on the system address and control signals from the system memory controller; and

~~enabling data paths propagating data signals between the memory devices and the system memory controller~~;

~~if the mode of operation is in~~ the test mode:

generating self-test address and control signals on the memory module according to the standard memory protocol;

~~for~~ providing the self-test address and control signals to the memory devices; ~~and~~

generating self-test data patterns on the memory module;

isolating the memory devices from the system memory controller; and

transmitting the self-test data patterns to the memory devices according to the standard memory protocol, wherein the self-test data patterns are received by

at least a portion of the memory devices in response to the self-test address and control signals.

21. (New) The method of claim 20, wherein the memory devices are arranged in at least one rank, each rank having a width of N bits, wherein the self-test data patterns include a set of independently generated n-bit wide data patterns, and wherein transmitting the self-test data patterns comprises transmit each respective n-bit wide data pattern in the set of independently generated n-bit wide data patterns to a respective n-bit section of the at least one rank, n being a fraction of N.
22. (New) The method of claim 21, wherein each memory device is n-bit wide.
23. (New) The method of claim 21, further comprising, during the test mode:
 - generating additional self-test address and control signals;
 - providing the additional self-test address and control signals to the address and control ports of the memory devices, the additional self-test address and control signals causing the memory devices to output a N-bit wide read data signal;
 - receiving each respective n-bit section of the N-bit wide read data signal; and
 - checking a respective data pattern in the each respective n-bit section of the N-bit wide read data signal against a respective comparison data pattern.
24. (New) The method of claim 23, further comprising storing failure information on the memory module when a test failure is indicated by the respective data pattern not in agreement with the respective comparison data pattern.
25. (New) The method of claim 20, wherein the standard memory protocol is a JEDEC standard memory protocol.

REMARKS

Claims 1-20 are amended. New claims 21-25 are presented. No new matter is added. After entry of this amendment, the pending claims are 1-25.

The Examiner is invited to call the undersigned attorney at (650) 273-6008, if a telephone call could help resolve any issues.

Respectfully submitted,

Date: April 17, 2015

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C3			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Claims in excess of 20	2202	5	40	200
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				200

Electronic Acknowledgement Receipt

EFS ID:	22101619
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	18-APR-2015
Filing Date:	29-MAR-2014
Time Stamp:	01:35:50
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$200
RAM confirmation Number	6317
Deposit Account	
Authorized User	

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Preliminary Amendment	2015-04-17_Prelim_NETL046C3.pdf	116112	no	10
			3327ae937b60432f4f21d9cfb67c0b533e5ed0a1		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30128	no	2
			942e11fdcae2631fc50346bc5e90383c83219a1d		

Warnings:

Information:

Total Files Size (in bytes):			146240
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/229,844	Filing Date 03/29/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO**APPLICATION AS FILED – PART I**

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
	TOTAL ADD'L FEE					

(Column 1)

(Column 2)

(Column 3)

	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	04/18/2015					
	Total (37 CFR 1.16(i))	*	25	Minus	** 20	= 5
	Independent (37 CFR 1.16(h))	*	3	Minus	*** 3	= 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE					200	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
/Chantae Dessau/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NETL.046C3
Sheet	1	of	5		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	20010048342	12-06-2001	Yoshikazu Yoshida	
	2.	20020000847	01-03-2002	Masao Taguchi	
	3.	20020131535	09-19-2002	Huber Brian W.	
	4.	20020140523	10-03-2002	Edwin Park	
	5.	20030098742	05-29-2003	Nakagawa Osamu Samuel	
	6.	20030126346	07-03-2003	Kuo Sung H.	
	7.	20030197797	10-23-2003	Josep Segura	
	8.	20030218491	11-27-2003	Makoto Nagasue	
	9.	20040155702	08-12-2004	Siemens Elema Ab	
	10.	20040199843	10-07-2004	Hansquine David W.	
	11.	20050093620	05-05-2005	International Business Machines Corporation	
	12.	20050127989	06-16-2005	Hiroshi Miyagi	
	13.	20050289423	12-29-2005	Kabushiki Kaisha Toshiba	
	14.	20060082383	04-20-2006	Choi Jung-Hwan	
	15.	20060107156	05-18-2006	Kee-Hoon Lee	
	16.	20060140015	06-29-2006	Rambus Inc.	
	17.	20060144015	07-06-2006	Cash John W Iii	
	18.	20060147217	07-06-2006	Hahin Jayne C	
	19.	20060192653	08-31-2006	Paul Atkinson	
	20.	20060242458	10-26-2006	Daniel Feldman	
	21.	20060262586	11-23-2006	Solomon Jeffrey C	
	22.	20060271748	11-30-2006	Intel Corporation	
	23.	20070030814	02-08-2007	Samsung Electronics Co., Ltd.	
	24.	20070058471	03-15-2007	Rajan Suresh N	
	25.	20070079199	04-05-2007	Asset Intertech, Inc.	
	26.	20070109707	05-17-2007	International Rectifier Corporation	
	27.	20070152743	07-05-2007	Brent Keeth	
	28.	20070204075	08-30-2007	Rajan Suresh N	
	29.	20070223296	09-27-2007	Christopher Miller	

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	30.	3660675	05-02-1972	Honeywell Inc	
	31.	3757235	09-04-1973	Shell Oil Co	
	32.	4305091	12-08-1981	Cooper J Carl	
	33.	4586168	04-29-1986	Motorola, Inc.	
	34.	4701845	10-20-1987	Unisys Corporation	
	35.	4752741	06-21-1988	Honeywell Inc.	
	36.	4782487	11-01-1988	Digital Equipment Corporation	
	37.	4837743	06-06-1989	Texas Instruments Incorporated	
	38.	4885799	12-05-1989	Motorola, Inc.	
	39.	4903266	02-20-1990	International Business Machines Corporation	
	40.	4910597	03-20-1990	Canon Kabushiki Kaisha	
	41.	4942556	07-17-1990	Hitachi, Ltd.	
	42.	4958346	09-18-1990	Advantest Corporation	
	43.	4987321	01-22-1991	Eastman Kodak Company	
	44.	5033048	07-16-1991	Digital Equipment Corporation	
	45.	5051997	09-24-1991	Mitsubishi Denki Kabushiki Kaisha	
	46.	5138619	08-11-1992	National Semiconductor Corporation	
	47.	5173906	12-22-1992	Dreibelbis Jeffrey H	
	48.	5222066	06-22-1993	Motorola, Inc.	
	49.	5241503	08-31-1993	Motorola, Inc.	
	50.	5304856	04-19-1994	At&T Bell Laboratories	
	51.	5337254	08-09-1994	Hewlett-Packard Company	
	52.	5359235	10-25-1994	Digital Equipment Corporation	

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	53.	5394037	02-28-1995	Lattice Semiconductor Corporation	
	54.	5430335	07-04-1995	Oki Electric Industry Co., Ltd.	
	55.	5525917	06-11-1996	Altera Corporation	
	56.	5732241	03-24-1998	Mos Electronics, Corp.	
	57.	5841296	11-24-1998	Xilinx, Inc.	
	58.	5914543	06-22-1999	Dallas Semiconductor Corporation	
	59.	6000048	12-07-1999	Cirrus Logic, Inc.	
	60.	6044481	03-28-2000	Artisan Components, Inc.	
	61.	6070217	05-30-2000	International Business Machines Corporation	
	62.	6169696	01-02-2001	Micron Technology, Inc.	
	63.	6194959	02-27-2001	Kabushiki Kaisha Toshiba	
	64.	6216240	04-10-2001	Samsung Electronics Co., Ltd.	
	65.	6467056	10-15-2002	Hitachi, Ltd.	
	66.	6560740	05-06-2003	Advanced Micro Devices, Inc.	
	67.	6614702	09-02-2003	Samsung Electronics Co., Ltd	
	68.	6681358	01-20-2004	Lsi Logic Corporation	
	69.	6721150	04-13-2004	Advanced Micro Devices, Inc.	
	70.	6812869	11-02-2004	Lattice Semiconductor Corporation	
	71.	6829728	12-07-2004	Wu-Tung Cheng	
	72.	6910162	06-21-2005	Kingston Technology Corp.	

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	4	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	73.	6918072	07-12-2005	Micron Technology, Inc.	
	74.	6928024	08-09-2005	Infineon Technologies Ag	
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	78.	7036064	04-25-2006	Omar Kebichi	
	79.	7053470	05-30-2006	Azul Systems, Inc.	
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	85.	7284166	10-16-2007	Stmicroelectronics S.R.L.	
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NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	90.	Bayard "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, vol. 47, Issue 5, pp. 1288-1296.	

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NETL.046C3
Sheet	5	of	5		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	91.	Der-Chang et al. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, Apr. 2002, vol. 21, Issue 4, pp. 449-465.	
	92.	Mutoh et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, Nov. 2-6, 2004, vol. 1, pp. 963-968.	
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EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Electronic Acknowledgement Receipt

EFS ID:	24915764
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	16-FEB-2016
Filing Date:	29-MAR-2014
Time Stamp:	01:43:45
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-02-15_IDS1_Cover_NETL046C3.pdf	58533 675facfd8d531d219cd54c07a59d45e3cd669429	no	2

Warnings:

Information:

2	Information Disclosure Statement (IDS) Form (SB08)	2016-02-15_IDS1_NETL046C3. pdf	184207 4f966d1c0b8ba4eac78f4e8bb1a22204cd5 e09a2	no	5
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Information:					
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INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/229,844
Filed	: March 29, 2014
For	: Memory Subsystem and Method of Operation
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 93 patent documents. These patent documents are included in the information disclosure statements filed in the parent application US Pat. Appl. No. 13/183,253, filed July 14, 2011, now US Pat. No. 8/359,501. Copies of foreign documents and/or their translation and non-patent literature in the listing can be found in the file-wrapper of the above-referenced parent application and are available on PAIR.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

Application No.: 14/229,844
Filing Date: March 29, 2014

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: February 15, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167
(Reg. No.)

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NETL.046C3
Sheet	1	of	4		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	1.	JEDEC Global Standards for the Microelectronics Industry, Standards http://www.jedec.org/standards-documents/about-JEDEC-standards (2014)	
	2.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Response pursuant to 37 C.F.R. § 42.120, filed March 17, 2015	
	3.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Reply to Patent Owner Response, June 8, 2015	
	4.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Request for Oral Argument, filed July 2, 2015	
	5.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Motion for Observation re Cross-Examination of Dr. Donald Alpert, filed July 2, 2015	
	6.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Second Amended Exhibit List, filed July 2, 2015	
	7.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Request for Oral Argument, filed July 2, 2015	
	8.	Inter Partes Review Case No. IPR2014-00970, Order: Request for Oral Argument, issued July 7, 2015	
	9.	Inter Partes Review Case No. IPR2014-00970, Petitioner Response to Netlist's Motion for Observations on Cross-Examination, filed July 13, 2015	
	10.	Inter Partes Review Case No. IPR2014-00970, Record of Oral Hearing, issued August 24, 2015	
	11.	Inter Partes Review Case No. IPR2014-00970, Final Written Decision - 35 USC 318 and 37 CFR 42.73, issued December 14, 2015	
	12.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Request for Rehearing, filed January 13, 2016	
	13.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1009 to Petitioner's Reply to Patent Owner Response, "Declaration of Donald Alpert PhD," filed June 8, 2015	
	14.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1010 to Petitioner's Reply to Patent Owner Response, "Deposition Transcript of Dr. Carl Sechen," filed June 8, 2015	
	15.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1011 to Petitioner's Reply to Patent Owner Response, "American Heritage Dictionary," filed June 8, 2015	
	16.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1012 to Petitioner's Reply to Patent Owner Response, "Webster's Encyclopedic Unabridged Dictionary," filed June 8, 2015	
	17.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1013 to Petitioner's Reply to Patent Owner Response, "IEEE Standard 1450," filed June 8, 2015	
	18.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2004 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Defendants' Proposed Preliminary Claim Constructions and Extrinsic Evidence, Case No. 13-cv-05889-YGR July 28, 2014," filed March 17, 2015	

EXAMINER

DATE CONSIDERED

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				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	4	Attorney Docket Number	NETL.046C3

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	19.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2005 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Enlarged Figure 3 from Exhibit 1004," filed March 17, 2015	
	20.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2006 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Enlarged Figure 3 from Exhibit 1004 to Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on June 18, 2014, 'Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379," filed March 17, 2015	
	21.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2007 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Color copy of page 35 to Patent Owner's Preliminary Response filed September 24, 2014," filed March 17, 2015	
	22.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2008 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Color copy of page 38 to Patent Owner's Preliminary Response filed September 24, 2014," filed March 17, 2015	
	23.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2009 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from American Heritage Dictionary," filed March 17, 2015	
	24.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2010 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Webster's Third New International Dictionary," filed March 17, 2015	
	25.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2011 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from New Riverside University Dictionary," filed March 17, 2015	
	26.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2012 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Declaration of Carl Sechen, Ph.D.," filed March 17, 2015	
	27.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2013 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Cross-Examination transcript of Donald Alpert, Ph.D., taken February 12, 2015," filed March 17, 2015	
	28.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2014 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Design of High-Performance Microprocessor Circuits," filed March 17, 2015	
	29.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2015 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Logic and Computer Design Fundamentals," filed March 17, 2015	
	30.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2016 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Digital Design, Principles and Practices," filed March 17, 2015	

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				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	4	Attorney Docket Number	NETL.046C3

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	31.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2017 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Digital Logic with Verilog Design," filed March 17, 2015	
	32.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2018 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Decision on Institution of IPR for 8,001,434 (IPR2014-01372)," filed March 17, 2015	
	33.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2019 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Recent Advances in BIST," filed March 17, 2015	
	34.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2020 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Decision on Institution of IPR for 8,359,501 (IPR2014-01375)," filed March 17, 2015	
	35.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2021 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Cross-Examination transcript of Dr. Alpert, June 24, 2015," filed March 17, 2015	
	36.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Response, filed March,17,2015	
	37.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's First Amended Exhibit List, filed March,18,2015	
	38.	Inter Partes Review Case No. IPR2014-00971, Motion to File Corrected Patent Owner Response, filed April,15,2015	
	39.	Inter Partes Review Case No. IPR2014-00971, Corrected Patent Owner Response, filed April,15,2015	
	40.	Inter Partes Review Case No. IPR2014-00971, Petitioner's Reply to Patent Owner Response, filed June,8,2015	
	41.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Request for Oral Argument, filed July,2,2015	
	42.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Motion for Observation re Cross-Examination of Dr. Donald Alpert, filed July,2,2015	
	43.	Inter Partes Review Case No. IPR2014-00971, Petitioner Request for Oral Argument, filed July,2,2015	
	44.	Inter Partes Review Case No. IPR2014-00971, Order Request for Oral Argument, filed July,7,2015	
	45.	Inter Partes Review Case No. IPR2014-00971, Petitioner Response to Netlist's Motion for Observations on Cross-Examination, filed July,13,2015	
	46.	Inter Partes Review Case No. IPR2014-00971, Record of Oral Hearing, filed August,24,2015	
	47.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December,14,2015	
	48.	Inter Partes Review Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January,13,2016	

EXAMINER

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	4	of	4	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	49.	Inter Partes Review Case No. IPR2014-00971, Exhibit 1009, 'Declaration of Donald Alpert PhD,' filed June,8,2015	

EXAMINER**DATE CONSIDERED**

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INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/229,844
Filed	: March 29, 2014
For	: Memory Subsystem and Method of Operation
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 49 non-patent literature documents, copies of which are submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

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Application No.: 14/229,844
Filing Date: March 29, 2014

was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: February 17, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167

(Reg. No.)

Electronic Acknowledgement Receipt

EFS ID:	24944321
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	17-FEB-2016
Filing Date:	29-MAR-2014
Time Stamp:	22:40:51
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	2015-03-17_Pat_Owner_Rsps. pdf	1544749 f09425369c388e97de1c64cff9fe3d9a475f6790	no	65

Warnings:

Information:

2	Non Patent Literature	2015-06-08_petitioner_Reply_Pat_Owner_Response.pdf	733989	no	31
			03e8a573a3e42f64ef21fbfce899c2b4a156e647		
Warnings:					
Information:					
3	Non Patent Literature	2015-07-02_Pat_Owner_Oral_Argument.pdf	95093	no	4
			7ea0c5b94e4eeec3a93398e24f91af03a41e7911		
Warnings:					
Information:					
4	Non Patent Literature	2015-07-02_Pat_Owner_Motion.pdf	150587	no	18
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Warnings:					
Information:					
5	Non Patent Literature	2015-07-02_Pat_Owner_Exhibit_2.pdf	100587	no	4
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Warnings:					
Information:					
6	Non Patent Literature	Items_7_thru_12.pdf	5003447	no	205
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7	Non Patent Literature	Items_13-14.pdf	2829492	no	165
			60891067f3d78db1e36b93abcc01ae913ab0b0ad		
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Information:					
8	Non Patent Literature	Items_15-22.pdf	17490500	no	144
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9	Non Patent Literature	items_36-49.pdf	13808640	no	395
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10	Information Disclosure Statement (IDS) Form (SB08)	2016-02-17_IDS1_NETL046C3.pdf	143707	no	4
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Warnings:					
Information:					

This is not an USPTO supplied IDS fillable form					
11	Transmittal Letter	2016-02-17_IDS1_Cover_NETL046C3.pdf	57859 96d1df215ba6cfb3cb05f615f077ef4a7f4e5263	no	2
Warnings:					
Information:					
12	Non Patent Literature	items_25-28.pdf	24545046 818f76ccc29dbb3093b13126f6725db7f8432793	no	238
Warnings:					
Information:					
13	Non Patent Literature	Items_23-24.pdf	23712175 459ad89f327b6efc05179bdfc9efc39450040b	no	16
Warnings:					
Information:					
14	Non Patent Literature	items_29-30.pdf	20304204 bce7da539e89a7bbf956f87ec93f01af27b78506	no	9
Warnings:					
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Information:					
15	Non Patent Literature	items_31-35.pdf	8746319 f7fe0db8768c4eaf8be197897d838be6719a13ff	no	274
Warnings:					
Information:					
Total Files Size (in bytes):			119266394		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

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				Application Number	14/299,844
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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NETL.046C3
Sheet	1	of	4		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	1.	Inter Partes Review Case No. IPR2014-01372, Related Matters, filed September 12, 2014	
	2.	Inter Partes Review Case No. IPR2014-01372, Corrected Petition For Inter Partes Review, filed September 22, 2014	
	3.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Preliminary Response, filed December 18, 2014	
	4.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Exhibit List, filed December 18, 2014	
	5.	Inter Partes Review Case No. IPR2014-01372, Decision - Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 10, 2015	
	6.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Response, filed June 15, 2015	
	7.	Inter Partes Review Case No. IPR2014-01372, Petitioners REPLY To POs RESPONSE, filed September 10, 2015	
	8.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Objections to Evidence Filed With Petitioner Reply, filed September 17, 2015	
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	10.	Inter Partes Review Case No. IPR2014-01372, Order -- Trial Hearing - 37 CFR 42.70, filed October 8, 2015	
	11.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Request for Oral Argument, filed October 16, 2015	
	12.	Inter Partes Review Case No. IPR2014-01372, Patent Owner Motion to Exclude, filed October 16, 2015	
	13.	Inter Partes Review Case No. IPR2014-01372, Motion to Exclude Portions of Petitioner's Reply to Patent Owner's Response, filed October 19, 2015	
	14.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Opposition to Petitioner's Motion to Exclude, filed October 26, 2015	
	15.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Fourth Amended Exhibit List, filed October 26, 2015	
	16.	Inter Partes Review Case No. IPR2014-01372, IPR2014-01372 SMART Response - PO Motion To Exclude Exhibits, filed October 26, 2015	
	17.	Inter Partes Review Case No. IPR2014-01372, IPR2014-01372 SMART Response - PO Motion To Exclude Reply Portions, filed October 26, 2015	
	18.	Inter Partes Review Case No. IPR2014-01372, Reply In Support of Patent Owner Motion to Exclude, filed November 2, 2015	
	19.	Inter Partes Review Case No. IPR2014-01372, SMART Reply in Support of Motion to Exclude, filed November 2, 2015	

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Sheet	2	of	4	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	20.	Inter Partes Review Case No. IPR2014-01372, Record of Oral Hearing, filed December 18, 2015	
	21.	Inter Partes Review Case No. IPR2014-01372, Ex 1026 - Websters New Collegiate, filed September 10, 2015	
	22.	Inter Partes Review Case No. IPR2014-01372, Ex 1027 - May 2015 Dolin Article, filed September 10, 2015	
	23.	Inter Partes Review Case No. IPR2014-01372, Ex 1028 - Portola Packaging Reexam Guidance, filed September 10, 2015	
	24.	Inter Partes Review Case No. IPR2014-01372, Ex 1029 - Staunstrup Textbook Re Self Timed Signalling, filed September 10, 2015	
	25.	Inter Partes Review Case No. IPR2014-01372, Ex 1030 - DIC Rabaey Textbook, filed September 10, 2015	
	26.	Inter Partes Review Case No. IPR2014-01372, Ex 1031 - Dp Tr Dr Carl Sechen, filed September 10, 2015	
	27.	Inter Partes Review Case No. IPR2014-01372, Ex 1032 - Websters II Dict, filed September 10, 2015	
	28.	Inter Partes Review Case No. IPR2014-01372, Ex 1033 - McGraw Hill Electronics Dict, filed September 10, 2015	
	29.	Inter Partes Review Case No. IPR2014-01372, Ex 1034 - IEEE Dict, filed September 10, 2015	
	30.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2009, 'Excerpts from Merriam Webster's Collegiate Dictionary, Tenth Edition (2001),' filed December 18, 2014	
	31.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2010, 'Excerpts from the American Webster Hand College Dictionary (1981),' filed December 18, 2014	
	32.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2011, 'Excerpts from Merriam Webster's Collegiate Dictionary, Eleventh Edition (2003),' filed December 18, 2014	
	33.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2012, 'Cross-Examination transcript of Nader Bagherzadeh, Ph.D. , 5/21/2015,' filed June 15, 2015	
	34.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2013, 'Report to the Association for Computing,' filed June 15, 2015	
	35.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2014, 'MorphoSys - Case Study,' filed June 15, 2015	
	36.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2016, 'Declaration of Nadar Baghezadeh, Ph.D. ISO Motion to Stay,' filed June 15, 2015	
	37.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2017, 'Design and Implementation,' filed June 15, 2015	
	38.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2018, 'Deposition of Doanld Alpert, Ph.D., 2/12/15,' filed June 15, 2015	
	39.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2019, 'Declaration of Carl Sechen, Ph.D.,' filed June 15, 2015	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	3	of	4	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	40.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2020, 'Excerpts from Logic and Computer Design,' filed June 15, 2015	
	41.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2021, 'Excerpts from Digital Design Principles,' filed June 15, 2015	
	42.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2022, 'CMOS LVSI,' filed June 15, 2015	
	43.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2025, 'References reviewed by Examiner,' filed June 15, 2015	
	44.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2026, 'Excerpts from Digital Integrated Circuits,' filed June 15, 2015	
	45.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2027, 'Transcript of the September 30, 2015 Conference Call,' filed October 2, 2015	
	46.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2028, 'Attachment to Oct. 19, 2015 Patent Owner Motion to Exclude,' filed October 19, 2015	
	47.	Inter Partes Review Case No. IPR2014-01374, Corrected Petition For Inter Partes Review, filed September 22, 2014	
	48.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Preliminary Response, filed December 18, 2014	
	49.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Exhibit List, filed December 18, 2014	
	50.	Inter Partes Review Case No. IPR2014-01374, Decision - Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 10, 2015	
	51.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Response, filed June 15, 2015	
	52.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Corrected Response, filed June 16, 2015	
	53.	Inter Partes Review Case No. IPR2014-01374, Petitioner's Motion To Exclude Evidence, filed July 20, 2015	
	54.	Inter Partes Review Case No. IPR2014-01374, Petitioners REPLY To POs RESPONSE, filed September 10, 2015	
	55.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Objections to Evidence Filed With Petitioner Reply, filed September 17, 2015	
	56.	Inter Partes Review Case No. IPR2014-01374, Petitioner SMART Request For Oral Argument, filed September 24, 2015	
	57.	Inter Partes Review Case No. IPR2014-01374, Order - Trial Hearing - 37 CFR 42.70, filed October 8, 2015	
	58.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Request for Oral Argument, filed October 16, 2015	
	59.	Inter Partes Review Case No. IPR2014-01374, Patent Owner Motion to Exclude, filed October 16, 2015	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	4	of	4	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	60.	Inter Partes Review Case No. IPR2014-01374, Motion to Exclude Portions of Petitioner's Reply to Patent Owner's Response, filed October 19, 2015	
	61.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Opposition to Petitioner's Motion to Exclude, filed October 26, 2015	
	62.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Fourth Amended Exhibit List, filed October 26, 2015	
	63.	Inter Partes Review Case No. IPR2014-01374, IPR2014-01374 SMART Response - PO Motion To Exclude Exhibits, filed October 26, 2015	
	64.	Inter Partes Review Case No. IPR2014-01374, IPR2014-01374 SMART Response - PO Motion To Exclude Reply Portions, filed October 26, 2015	
	65.	Inter Partes Review Case No. IPR2014-01374, Reply In Support of Patent Owner Motion to Exclude, filed November 2, 2015	
	66.	Inter Partes Review Case No. IPR2014-01374, SMART Reply in Support of Motion to Exclude, filed November 2, 2015	
	67.	Inter Partes Review Case No. IPR2014-01374, Record of Oral Hearing, filed December 18, 2015	
	68.	Inter Partes Review Case No. IPR2014-01373, Corrected Petition For Inter Partes Review, filed September 22, 2014	
	69.	Inter Partes Review Case No. IPR2014-01373, Patent Owner's Preliminary Response, filed December 18, 2014	
	70.	Inter Partes Review Case No. IPR2014-01373, Patent Owner's Exhibit List, filed December 18, 2014	
	71.	Inter Partes Review Case No. IPR2014-01373, Decision - Denying Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 13, 2015	
	72.	Inter Partes Review Case No. IPR2014-01375, Corrected Petition For Inter Partes Review, filed September 22, 2014	
	73.	Inter Partes Review Case No. IPR2014-01375, Patent Owner's Preliminary Response, filed December 18, 2014	
	74.	Inter Partes Review Case No. IPR2014-01375, Patent Owner's Exhibit List, filed December 18, 2014	
	75.	Inter Partes Review Case No. IPR2014-01375, Decision - Denying Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 13, 2015	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Electronic Acknowledgement Receipt

EFS ID:	24944508
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	18-FEB-2016
Filing Date:	29-MAR-2014
Time Stamp:	00:46:27
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-02-17_IDS2_Cover_NETL046C3.pdf	57853 1b738bda4fa7ae520fddf3b4b7ea0d327a708cec	no	2

Warnings:

Information:

2	Information Disclosure Statement (IDS) Form (SB08)	2016-02-17_IDS2_NETL046C3.pdf	156143	no	4
			bd8d053e9525c117544fe7418c08bede012eadd7		
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	items_1-20.pdf	9517679	no	417
			4bb53b99b352d19008c6f56b967d260a68d96109		
Warnings:					
Information:					
4	Non Patent Literature	items_21-29.pdf	12933544	no	120
			9e9e95ad22b797176ff1ca5d5f56c3f73a5f0428		
Warnings:					
Information:					
5	Non Patent Literature	items_30-41.pdf	18832067	no	435
			925e802241ba2ccad10f1a54a10d0f2c28f3daea		
Warnings:					
Information:					
6	Non Patent Literature	items_42-46.pdf	8629301	no	75
			ba93b1e1d53a467e187e6f51f7ec291846be8609		
Warnings:					
Information:					
7	Non Patent Literature	items_47-67.pdf	9703676	no	474
			d3a13fd5c3e93edf1b93aa234acbdd483657ab67		
Warnings:					
Information:					
8	Non Patent Literature	items_68-71.pdf	3586959	no	143
			5d63e76019c3ad93e4dc88a71bfa9467166ec57e		
Warnings:					
Information:					
9	Non Patent Literature	items_72-75.pdf	3924077	no	135
			8221b8dd7d322e8d1eac794bba7cba797a3eff53		
Warnings:					
Information:					
Total Files Size (in bytes):			67341299		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/229,844
Filed	: March 29, 2014
For	: Memory Subsystem and Method of Operation
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 75 non-patent literature documents, copies of which are submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits

Application No.: 14/229,844
Filing Date: March 29, 2014

was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: February 17, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	NETL.046C3	1018

79141 7590 03/09/2016
Jamie J. Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

CHUNG, PHUNG M

ART UNIT	PAPER NUMBER
----------	--------------

2117

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

03/09/2016

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
eofficeaction@appcoll.com
infohub.j@gmail.com

Office Action Summary	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 4/18/15.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-25 is/are pending in the application.
5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 1-25 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☒ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 3/29/14 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 2/18/16, 2/17/17, 2/16/16, 12/24/14, 8/10/14, 8/7/14.
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____.
- 4) ☐ Other: ____.

DETAILED ACTION

The present application is being examined under the pre-AIA first to invent provisions.

Specification

The disclosure is objected to because of the following informalities:

Paragraph (0001) the U.S. Patent Applications No., for example “No. 13/745,790”; “No.12/422,912” and “No. 12/422,853” are not up-to-date. Appropriate correction is required.

Information disclosure statement (IDSs)

The Applicant has submitted a large number of IDS's and references for consideration by the Examiner, IDS filed on 2/18/16, 2/17/16, 2/16/18, 12/24/14, 8/10/14 and 8/7/14 contain more than 112+ references including 161+ non patent literatures.

In view of the large number of references submitted by the Applicant for consideration in this application, the Applicant is requested to identify any references submitted which have particular significance in the prosecution of this application for further consideration by the Examiner. In indicating reference(s) for further consideration, applicant should also indicate the specific features, corresponding passages, and figures of such reference(s)

which are believed to be relevant to the invention as presented in the claims of this application.

Due to the large number of references cited in the I.D.S. filed on 2/18/16, 2/17/16, 2/16/18, 12/24/14, 8/10/14 and 8/7/14, applicant is requested to highlight those documents which have been specifically brought to applicant's attention and/or are known to be of most significance. See *Penn Yan Boats, Inc. v. Sea Lark Boats, Inc.*, 359 F. Supp. 948, 175 USPQ 260 (S.D. Fla. 1972), *aff'd*, 479 F.2d 1338, 178 USPQ (5th Cir. 1973), *cert. denied*, 414 U.S. 874 (1974). But cf. *Molins PLC v. Textron Inc.*, 48 F.3d 1172, 33 USPQ2d 1823 (Fed. Cir. 1995). See MPEP § 2004.

Note: The I.D.S. filed on 2/18/16, 2/17/16, 2/16/18, 12/24/14, 8/10/14 and 8/7/14 are not considered by Examiner because they are too many, too long and no statement of relevant. Examiner unable to examine them within a limited amount of time per Application. Also some of the non-patent literatures (NPLs) submitted without dates, and/or un-match dates, un-match Title, and blurred. Applicant is responsible to the rest of the IDSs and is requested to resubmit them with readable and with dates and title that match the dates (MM/DD/YR) and title on the IDSs.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

Art Unit: 2117

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-25 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

As per claim 1, lines 1 and 10, "operable" a positive term should be used; and line 4, "a standard memory protocol" is unclear. What it meant by memory protocol standard?

Lines 19-27, "during the test mode..., wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals received from the control module" is not clear how the memory module doing the test and/or what is being test? Are these test patterns used to test the memory devices? or just for storing them in the memory devices? Appropriate correction and/or clarification are required.

As per claim 2, line 2, "at least one rank, each rank having a width of N bits" it should be changed to - - at least one range, the at least one rank having a width of N bits - -. Appropriate correction is required.

As per claim 6, line 2, "JEDEC standard memory protocol" is unclear. What is meant by the JEDEC memory protocol standard? In addition, "JEDEC" should be spelled out when first usage in the claim. Appropriate correction is required.

As per claim 8, line 2, "the failure information" does not have a clear antecedent basis. Appropriate correction is required.

As per claims 13-14, lines 1 and 11, "operable" a positive term should be used;
Lines 8-11, "a data module...is coupled between a respective n-bit section of the N-bit wide system memory bus and a respective n-bit section of the at least one N-bit wide rank, n being a fraction of N" is not clear as to what it meant. Is applicant meant - - a respective n-bit section of the at least one N-bit wide rank of the memory devices- -?

Line 11, "the memory subsystem" does not have a clear antecedent basis;

Lines 11-12, "the memory subsystem is operable in any one of a plurality of modes" is vague and indefinite, (beside test mode) what are they? and how it operates in any one of the plurality of modes? And the interrelationship and/or interconnection between the plurality of modes are unclear;

Lines 12-15, "a test mode", and wherein, during the self-test mode:" the inconsistent used of "test mode" and "self-test mode" are confusing. Are they the same or different? And the ":" follow by ";" is not clear as to what it meant. For a suggestion the ":" should be deleted; and

Lines 12-22, “during the self-test mode: the control module generates self-test address and control signals and provide the self-test address and control signals to the memory devices; and

The each respective data handler produces respective n-bit wired test data patterns and transmits the respective n-bit wide test data patterns to the respective n-bit section of the at least on N-bit wide rank so that one or more memory devices in the respective n-bit section of the at least one N-bit wide rank receive the respective n-bit wide test data patterns in accordance with the self-test address and control signals" is not clear how the memory module doing the test and/or what being test? Are these test data patterns used to test the memory devices? What is the result of the test? Are these test data patterns just for transmitting to the memory devices?

Lines 18-22, “the each respective data handler...and transmits the respective n-bit wide test data patterns to the respective n-bit section of the at least one N-bit wide rank so that one or more memory devices in the respective n-bit section of the at least one N-bit wide rank receive the respective n-bit wide test data patterns in accordance with the self-test address and control signals” is not clear as to what it meant.

Appropriate correction and/or clarification are required.

As per claim 16-17, line 2, “a standard memory protocol” and “a JEDEC standard memory protocol” is unclear. What is meant by the memory protocol standard or the JEDEC memory protocol standard? In addition, the “JEDEC” first usage should be spelled out. Appropriate correction and/or clarification are required.

As per claim 19, lines 2-5, “each respective data handler further includes a respective verification element, the respective verification element checking data patterns represented by data bits read from the respective portion of the memory devices for errors” is not clear how it check the data patterns represented by data bits read from the respective portion of the memory devices for errors? Appropriate correction and/or clarification are required.

As per claim 20, line 4, “a standard memory protocol” is unclear. What it meant by memory protocol standard?

Line 8,, “in the normal mode:” should be changed to

- - in the normal mode, the memory module:- -;

Line 19, “in the test mode:” should be changed to

- - in the test mode, the memory module:- -;

Lines 19-33, “in the test mode:...; and transmitting the self-test data patterns to the memory devices according to the standard memory protocol, wherein the self-test data patterns are received by at least a portion of the memory device in response to the self-test address and control signals” is not clear how the memory module doing the test and/or what being test, what is the result of the test? Are these self-test data patterns just for transmitted to the at least a portion of the memory device? and

Lines 20 and 25, “generating self-test address and control signals on the memory module according to the standard memory protocol” is applicant meant - -

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generating...by the memory module...- -? Appropriate correction and/or clarification are required.

As per claim 21, line 2, "at least one rank, each rank having a width of N bits" it should be changed to - - at least one range, the at least one rank having a width of N bits - -; and

Lines 4-6, "transmitting...to a respective n-bit section of the at least one rank" is applicant meant - - transmitting...to a respective n-bit section of the at least one rank of the memory devices- -? Appropriate correction and/or clarification are required.

As per claim 23, line 1, "during the test mode:" should be changed to - -during the test mode, the memory module:- -. Appropriate correction is required.

As per claim 25, "the standard memory protocol is a JEDEC standard memory protocol" the "JEDEC" first usage should be spelled out, and what is meant by "standard memory protocol" or "JEDEC standard memory protocol"? Appropriate correction and/or clarification are required.

As per claims 3-5, 7-12, 15, 18, 22 and 24, these claims are also rejected because they dependent upon the rejected base claim.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

This application currently names joint inventors. In considering patentability of the claims under pre-AIA 35 U.S.C. 103(a), the examiner presumes that the subject matter of the various claims was commonly owned at the time any inventions covered therein were made absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and invention dates of each claim that was not commonly owned at the time a later invention was made in order for the examiner to consider the applicability of pre-AIA 35 U.S.C. 103(c) and potential pre-AIA 35 U.S.C. 102(e), (f) or (g) prior art under pre-AIA 35 U.S.C. 103(a).

Claims 1-25, as best understood, are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Zuraski, Jr. et al (6,560,740) (hereafter referred to as Zuraski)

in view of Jacob (Synchronous DRAM architectures, organizations and alternative technologies, 12/10/2002).

As per claims 1-12, Zuraski discloses a memory module (microprocessor (10)) operable with a system memory controller (test control unit (26)), comprising:

memory devices having address and control ports, (Fig. 1, embedded memory unit (16 a-b), address and control ports A and C);

a data module coupled to the address and control ports of the memory devices (Fig.1, col. 3, lines 25-45, BIST logic unit (20a, 20b), each BIST logic unit provides the data input signals to the memory unit coupled thereto, receives the data outputs signals from the memory unit, and compares the data output signals to the data input signals to from BIST results); and

a control module to provide address and control signals to the memory devices (Fig. 1, Master test unit (18)); and

wherein the memory module is operable in any of a plurality of modes including a normal mode and a test mode;

wherein, during the normal mode, the control module provides the address and control signals to the memory devices according to the memory address and control signals being based on system address and control signals received by the control module from the system memory controller (Fig. 1, test control unit (26))

based on address and control signals from the system memory controller, and the data module enable data paths between the memory devices and the system memory controller; and

wherein, during the test mode, the control module generates the address and control signals, and the data module isolates the memory devices from the system memory controller. (See Fig. 1, col. 3, lines 25-60, col. 5, lines 40-67 to col. 6, lines 1-65).

Zuraski does not specifically disclose that the memory devices, the data module and the control module are mounted on a circuit board. However Jacob disclose a memory module, which is a small computer board ("printed circuit board") with a handful of DRAM chips (DRAMs are arranged in at least on rank, pg. 2, col. 2, second and third paragraphs) and associated circuitry attached to it. A computer's central processing unit (CPU) can access an individual storage cell inside a DRAM chip so as to read or write the data held there. This is accomplished by sending both a row address and a column address to the DRAM (section 1.1, first paragraph).

Every DRAM chip is equipped with pins (i.e., very short wires), each one of which connects the DRAM to one of many possible busses. Each bus is a group of wires that carry electrical signals; busses connect the CPU, memory controller, and DRAM chips. The busses in a JEDEC-style organization are classified by their function and organization into data, address, control, and chip-select busses. There is a relatively wide data bus that transmit data to and from the DRAMs (pg. 2, col. 1-2, fifth paragraph).

Jacob, pg. 4, col. 2, third paragraph, further discloses "the printed circuit board that contains the CPU and memory bus". (See section 1.1: Basis Organization of a Conventional DRAM, pgs. 1-5). Therefore, it would have been obvious to a person of

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ordinary skill in the art, at the time the invention was made, to implement the teaching of Jacob with the one taught by Zuraski in order to mount the memory devices, the data module and the control module on the circuit board in the teaching of Jacob so that the computer (CPU or Microprocessor) can access an individual storage cell inside a DRAM chip so as to read or write the data held there during functional mode.

As per claims 13-8 and 20-25, these claims are rejected under similar rationale as set forth in claims 1-12.

2.

References Cited by Examiner

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. (See the attached form 892).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571) 272-

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3818. The examiner can normally be reached on Monday to Friday from 8:00am-4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, April Y. Blair can be reached on 571-270-1014. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

Notice of References Cited	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	Page 1 of 1

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-5,138,619 A	08-1992	Fasang; Patrick P.	G11C29/20	714/718
*	B	US-6,175,529 B1	01-2001	Otsuka; Hidefumi	G11C29/14	365/200
*	C	US-6,560,740 B1	05-2003	Zuraski, Jr.; Gerald D.	G01R31/3187	714/727
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	02/21/2016								
	1	v								
	2	v								
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	24	v								
	25	v								

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	195048	((memory devices) or (memory units) or (memory module)) and (((application specific integrated circuits) or ("ASICs")) or ((system on chip integrated circuits) or ("SoCs")) or (embedded memories) or ((printed circuit board) or (PCB)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 11:58
L2	210352	(data module) or (BIST logic units) or (data handlers) or (data controllers) or (data managements)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:02
L3	607535	(control module) or (memory controller) or (master test unit)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:14
L4	3514	I1 and I2 and I3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:14
L5	104815	(controller or gate or switch or multiplexer or MUX or selector) and ((test mode or self-test mode or testing mode) or (normal mode or functional mode or instruction mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:15
L6	248	I4 and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:16
L7	15905	("BIST") or ("logic BIST") or (in-circuit test) or ("MBIST")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:16
L8	21	I6 and I7	US-PGPUB;	ADJ	ON	2016/02/28

			USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			12:17
L9	311881	(generat\$3 or produc\$3) and ((address and control signals) or (address signal and control signal))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:20
L10	15	I8 and I9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:21
L11	42	(standard memory protocol) or ("JEDEC standard memory protocol")	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:23
L12	3	I10 and I11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 12:23
L14	5763	(compar\$3 or detect\$3 or determin\$3 or identify\$3) and (test failure or fault or fail or error or defect or bad or abnormal or interrupted or test result) and ((data pattern or test data or test pattern or test vector) and (expected or predetermined or known or reference or predefined) adj (data pattern or test data or test vector))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:05
L16	2	I14 and I8	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:07
L17	6	I14 and I6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:13
L18	384	I14 and I7	US-PGPUB; USPAT; USOCR; FPRS;	ADJ	ON	2016/02/28 13:13

			EPO; JPO; DERWENT; IBM_TDB			
L19	208817	((memory devices) or (memory units) or (memory module)) and (((application specific integrated circuits) or ("ASICs")) or ((system on chip integrated circuits) or ("SoCs")) or (embedded memories) or ((printed circuit board) or (PCB)) or (circuit board))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:32
L20	3850	I19 and 2 and I3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:33
L21	301	I5 and I20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:33
L22	21	I21 and I7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:33
L23	3	I22 and I9 and I11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:33
L25	3850	I19 and I2 and I3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:34
L26	21	I25 and I6 and I7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:35
L28	15	26 AND ((G06F11/1068 OR G06F11/27 OR G06F3/0683 OR G11C5/04 OR G11C29/12 OR G11C2207/104 OR G01R31/3187).CPC. OR (714/718 OR 714/733 OR 714/719 OR 714/734 OR 714/E11.056 OR 714/E11.169 OR 714/42 OR 365/201 OR 716/136).CCLS. OR (G11C29/04 OR G11C29/12 OR	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:42

		G01R31/28 OR G01R31/3187 OR G06F11/27).IPCR.)				
L29	3109	((memory self-test address\$2 and control signal\$1 and test data patterns or test data or test vectors) and (memory device\$1) and (test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:42
L30	4	I26 and I29	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 13:43
L31	3183	((memory devices) or (memory units) or (memory module)) and ((printed circuit board) or (PCB)) or (circuit board)) and ((data module) or (BIST logic units) or (data handlers) or (data controllers) or (data managements)) and ((control module) or (memory controller) or (master test unit)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 14:11
L32	68	I31 and (I7 or I11 or I14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 14:12
L33	52	I32 and I9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 14:12
L34	28	33 AND ((G06F11/1068 OR G06F11/27 OR G06F3/0683 OR G11C5/04 OR G11C29/12 OR G11C2207/104 OR G01R31/3187).CPC. OR (714/718 OR 714/733 OR 714/719 OR 714/734 OR 714/E11.056 OR 714/E11.169 OR 714/42 OR 365/201 OR 716/136).CCLS. OR (G11C29/04 OR G11C29/12 OR G01R31/28 OR G01R31/3187 OR G06F11/27).IPCR.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 14:13
L35	26	I34 and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/02/28 14:14

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	1	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	5655113	08-05-1997	Leung	
	2.	5878051	03-02-1999	Sharma	
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	16.	8189328	05-29-2012	Kanapathippillai	
	17.	8238303	07-31-2012	Best	
	18.	2004006729	01-08-2004	Pendurkar	
	19.	20080104352	05-01-2008	Talbot	

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	20.	Inter Partes Review Case No. IPR2014-00970, Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on June 18, 2014		
	21.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1007 to Petition for Inter Partes Review, “Declaration of Donald Albert, Ph.D.,” filed on June 18, 2014		
	22.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2002 to Petition for Inter Partes Review, “Excerpt from Merriam-Webster’s Collegiate Dictionary,” filed on June 18, 2014		
	23.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2003 to Petition for Inter Partes Review, “Excerpt from Webster Handy College Dictionary,” filed on June 18, 2014		
	24.	Inter Partes Review Case No. IPR2014-00970, Patent Owner’s Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on September 24, 2014		

EXAMINER

/Phung Chung/

DATE CONSIDERED

03/02/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	2	of	5		

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Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.		T ²
	25.	Inter Partes Review Case No. IPR2014-00970, Decision – Institution of Inter Partes Review, issued on December 16, 2014		
	26.	Inter Partes Review Case No. IPR2014-00971, Petition for Inter Partes Review for U.S. Patent No. 8,359,501, filed on June 18, 2014		
	27.	Inter Partes Review Case No. IPR2014-00971, Exhibit 1007 to Petition for Inter Partes Review, “Declaration of Donald Albert, Ph.D,” filed on June 18, 2014		
	28.	Inter Partes Review Case No. IPR2014-00971, Patent Owner’s Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on September 24, 2014		
	29.	Inter Partes Review Case No. IPR2014-00971, Decision – Institution of Inter Partes Review, issued on December 16, 2014		
	30.	Inter Partes Review Case No. IPR2014-01373, Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on August 25, 2014.		
	31.	Inter Partes Review Case No. IPR2014-01373, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68,” filed on August 23, 2014.		
	32.	“IEEE Standard Test Access Port and Boundary-Scan Architecture” (“JTAG 2001”, IEEE-SA Standards Board, IEEE Std1149.1-2001, June 14, 2001.		
	33.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2001 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Webster’s New College Dictionary,” filed on August 23, 2014		
	34.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “IEEE Standard Dictionary of Electrical and Electronics Terms,” filed on August 22, 2014.		
	35.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2003 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “JEDEC Standard, Fully Buffered DIMM (FBDIMM) DfX Design for Validation and Test,” filed on August 22, 2014.		
	36.	Inter Partes Review Case No. IPR2014-01372, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.		
	37.	Inter Partes Review Case No. IPR2014-01372, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68,” filed on August 23, 2014.		
	38.	Inter Partes Review Case No. IPR2014-01374, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.		
	39.	Inter Partes Review Case No. IPR2014-01374, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68,” filed on August 23, 2014.		
	40.	Inter Partes Review Case No. IPR2014-01375, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.		

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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	3	of	5		

NON PATENT LITERATURE DOCUMENTS				
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	41.	Inter Partes Review Case No. IPR2014-01375, Exhibit 1002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, "Declaration of Dr. Nader Bagherzadeh, Ph.D. under 37 C.F.R. § 1.68," filed on August 23, 2014.		
	42.	Inter Partes Review Case No. IPR2014-01029, Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on June 24, 2014.		
	43.	Inter Partes Review Case No. IPR2014-01029, Exhibit 1008 to Petition for Inter Partes Review, "Declaration of Charles J. Neuhauser, Ph.D. under 37 C.F.R. § 1.68," filed on June 24, 2014.		
	44.	Inter Partes Review Case No. IPR2014-01029, Supplemental Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on July 23, 2014.		
	45.	Inter Partes Review Case No. IPR2014-01029, Patent Owner's Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on October 17, 2014.		
	46.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014.		
	47.	Inter Partes Review Case No. IPR2014-01369, Corrected Petition for Inter Partes Review of Claims 1-19 of U.S. Patent No. 8,516,185, filed on September 22, 2014.		
	48.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1008 to Corrected Petition for Inter Partes Review, "Declaration of Dr. Nader Bagherzadeh under 37 C.F.R. § 1.68," filed on September 22, 2014.		
	49.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1013 to Corrected Petition for Inter Partes Review, "Webster's II New College Dictionary," filed on September 22, 2014.		
	50.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1014 to Corrected Petition for Inter Partes Review, "Standard Dictionary of Electrical and Electronics Terms," IEEE 1988, filed on September 22, 2014.		
	51.	Inter Partes Review Case No. IPR2014-00882, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.		
	52.	Inter Partes Review Case No. IPR2014-00882, Exhibit 1007 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 22, 2014.		
	53.	Inter Partes Review Case No. IPR2014-00883, Corrected Petition for Inter Partes Review of U.S. Patent No. 8,081,536, filed on July 8, 2014.		
	54.	Inter Partes Review Case No. IPR2014-00883, Exhibit 1011 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 21, 2014.		
	55.	Inter Partes Review Case No. IPR2014-01011, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.		
	56.	Inter Partes Review Case No. IPR2014-01011, Exhibit 1007 to Petition for Inter Partes Review, "Declaration of Dr. Srinivasan Jagannathan," filed on June 22, 2014.		
	57.	McCluskey, Edward J., <i>Logic Design Principles with Emphasis on Testable Semicustom Circuits</i> , Prentice Hall, 1986, pp. 104-107 and 119-120.		

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				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	4	of	5	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
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	58.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.'s Invalidity Contentions, dated June 6, 2014	
	59.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits A.1-A.10 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	60.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits B.1-B.8 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	61.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	62.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F.1-F.5 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	63.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G.1-G.6 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	64.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	65.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Diablo Technologies, Inc.'s Invalidity Contentions, dated June 6, 2014.	
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	67.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits C-1 to C-7 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	68.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits D-1 to D-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	69.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F-1 to F-5 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	

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				First Named Inventor	Hyun Lee
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				Examiner Name	Unknown /Phung Chung/
Sheet	5	of	5	Attorney Docket Number	NETL.046C3

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	70.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G-1 to G-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	
	71.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.	

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BIB DATA SHEET

CONFIRMATION NO. 1018

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
14/229,844	03/29/2014	714	2117	NETL.046C3		
APPLICANTS Netlist, Inc., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Soonju Choi, Irvine, CA; ** CONTINUING DATA ***** This application is a CON of 13/745,790 01/19/2013 PAT 8689064 which is a CON of 13/183,253 07/14/2011 PAT 8359501 which is a CON of 12/422,925 04/13/2009 PAT 8001434 which claims benefit of 61/044,801 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,839 04/14/2008 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 04/21/2014						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /PHUNG M CHUNG/ Acknowledged Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 5	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS Jamie J. Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES						
TITLE Memory Subsystem and Method of Operation						
FILING FEE RECEIVED 1000	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

Search Notes 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner
G06F11/1068, G06F11/27, G06F3/0683, G11C5/04, G11C29/12, G11C2207/104, G01R31/3187	2/28/16	pc

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
714	718, 724, 733, 734, 736, E11.056, E11.169, 42	2/28/16	pc
365	201	2/28/16	pc
716	136	2/28/16	pc

SEARCH NOTES		
Search Notes	Date	Examiner
EAST-CPC Searched.	2/28/16	pc

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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	1.	JEDEC Global Standards for the Microelectronics Industry, Standards http://www.jedec.org/standards-documents/about-JEDEC-standards (2014)	
	2.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Response pursuant to 37 C.F.R. § 42.120, filed March 17, 2015	
	3.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Reply to Patent Owner Response, June 8, 2015	
	4.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Request for Oral Argument, filed July 2, 2015	
	5.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Motion for Observation re Cross-Examination of Dr. Donald Alpert, filed July 2, 2015	
	6.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Second Amended Exhibit List, filed July 2, 2015	
	7.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Request for Oral Argument, filed July 2, 2015	
	8.	Inter Partes Review Case No. IPR2014-00970, Order: Request for Oral Argument, issued July 7, 2015	
	9.	Inter Partes Review Case No. IPR2014-00970, Petitioner Response to Netlist's Motion for Observations on Cross-Examination, filed July 13, 2015	
	10.	Inter Partes Review Case No. IPR2014-00970, Record of Oral Hearing, issued August 24, 2015	
	11.	Inter Partes Review Case No. IPR2014-00970, Final Written Decision - 35 USC 318 and 37 CFR 42.73, issued December 14, 2015	
	12.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Request for Rehearing, filed January 13, 2016	
	13.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1009 to Petitioner's Reply to Patent Owner Response, "Declaration of Donald Alpert PhD," filed June 8, 2015	
	14.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1010 to Petitioner's Reply to Patent Owner Response, "Deposition Transcript of Dr. Carl Sechen," filed June 8, 2015	
	15.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1011 to Petitioner's Reply to Patent Owner Response, "American Heritage Dictionary," filed June 8, 2015	
	16.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1012 to Petitioner's Reply to Patent Owner Response, "Webster's Encyclopedic Unabridged Dictionary," filed June 8, 2015	
	17.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1013 to Petitioner's Reply to Patent Owner Response, "IEEE Standard 1450," filed June 8, 2015	
	18.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2004 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Defendants' Proposed Preliminary Claim Constructions and Extrinsic Evidence, Case No. 13-cv-05889-YGR July 28, 2014," filed March 17, 2015	

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	19.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2005 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Enlarged Figure 3 from Exhibit 1004," filed March 17, 2015	
	20.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2006 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Enlarged Figure 3 from Exhibit 1004 to Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on June 18, 2014, 'Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379," filed March 17, 2015	
	21.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2007 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Color copy of page 35 to Patent Owner's Preliminary Response filed September 24, 2014," filed March 17, 2015	
	22.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2008 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Color copy of page 38 to Patent Owner's Preliminary Response filed September 24, 2014," filed March 17, 2015	
	23.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2009 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from American Heritage Dictionary," filed March 17, 2015	
	24.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2010 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Webster's Third New International Dictionary," filed March 17, 2015	
	25.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2011 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from New Riverside University Dictionary," filed March 17, 2015	
	26.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2012 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Declaration of Carl Sechen, Ph.D.," filed March 17, 2015	
	27.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2013 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Cross-Examination transcript of Donald Alpert, Ph.D., taken February 12, 2015," filed March 17, 2015	
	28.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2014 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Design of High-Performance Microprocessor Circuits," filed March 17, 2015	
	29.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2015 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Logic and Computer Design Fundamentals," filed March 17, 2015	
	30.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2016 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Digital Design, Principles and Practices," filed March 17, 2015	

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	31.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2017 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Digital Logic with Verilog Design," filed March 17, 2015		
	32.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2018 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Decision on Institution of IPR for 8,001,434 (IPR2014-01372)," filed March 17, 2015		
	33.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2019 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Recent Advances in BIST," filed March 17, 2015		
	34.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2020 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Decision on Institution of IPR for 8,359,501 (IPR2014-01375)," filed March 17, 2015		
	35.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2021 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Cross-Examination transcript of Dr. Alpert, June 24, 2015," filed March 17, 2015		
	36.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Response, filed March,17,2015		
	37.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's First Amended Exhibit List, filed March,18,2015		
	38.	Inter Partes Review Case No. IPR2014-00971, Motion to File Corrected Patent Owner Response, filed April,15,2015		
	39.	Inter Partes Review Case No. IPR2014-00971, Corrected Patent Owner Response, filed April,15,2015		
	40.	Inter Partes Review Case No. IPR2014-00971, Petitioner's Reply to Patent Owner Response, filed June,8,2015		
	41.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Request for Oral Argument, filed July,2,2015		
	42.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Motion for Observation re Cross-Examination of Dr. Donald Alpert, filed July,2,2015		
	43.	Inter Partes Review Case No. IPR2014-00971, Petitioner Request for Oral Argument, filed July,2,2015		
	44.	Inter Partes Review Case No. IPR2014-00971, Order Request for Oral Argument, filed July,7,2015		
	45.	Inter Partes Review Case No. IPR2014-00971, Petitioner Response to Netlist's Motion for Observations on Cross-Examination, filed July,13,2015		
	46.	Inter Partes Review Case No. IPR2014-00971, Record of Oral Hearing, filed August,24,2015		
	47.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December,14,2015		
	48.	Inter Partes Review Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January,13,2016		

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	4	of	4	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
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	49.	Inter Partes Review Case No. IPR2014-00971, Exhibit 1009, 'Declaration of Donald Alpert PhD,' filed June,8,2015	

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	1.	Inter Partes Review Case No. IPR2014-01372, Related Matters, filed September 12, 2014	
	2.	Inter Partes Review Case No. IPR2014-01372, Corrected Petition For Inter Partes Review, filed September 22, 2014	
	3.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Preliminary Response, filed December 18, 2014	
	4.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Exhibit List, filed December 18, 2014	
	5.	Inter Partes Review Case No. IPR2014-01372, Decision - Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 10, 2015	
	6.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Response, filed June 15, 2015	
	7.	Inter Partes Review Case No. IPR2014-01372, Petitioners REPLY To POs RESPONSE, filed September 10, 2015	
	8.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Objections to Evidence Filed With Petitioner Reply, filed September 17, 2015	
	9.	Inter Partes Review Case No. IPR2014-01372, Petitioner SMART Request For Oral Argument, filed September 24, 2015	
	10.	Inter Partes Review Case No. IPR2014-01372, Order -- Trial Hearing - 37 CFR 42.70, filed October 8, 2015	
	11.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Request for Oral Argument, filed October 16, 2015	
	12.	Inter Partes Review Case No. IPR2014-01372, Patent Owner Motion to Exclude, filed October 16, 2015	
	13.	Inter Partes Review Case No. IPR2014-01372, Motion to Exclude Portions of Petitioner's Reply to Patent Owner's Response, filed October 19, 2015	
	14.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Opposition to Petitioner's Motion to Exclude, filed October 26, 2015	
	15.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Fourth Amended Exhibit List, filed October 26, 2015	
	16.	Inter Partes Review Case No. IPR2014-01372, IPR2014-01372 SMART Response - PO Motion To Exclude Exhibits, filed October 26, 2015	
	17.	Inter Partes Review Case No. IPR2014-01372, IPR2014-01372 SMART Response - PO Motion To Exclude Reply Portions, filed October 26, 2015	
	18.	Inter Partes Review Case No. IPR2014-01372, Reply In Support of Patent Owner Motion to Exclude, filed November 2, 2015	
	19.	Inter Partes Review Case No. IPR2014-01372, SMART Reply in Support of Motion to Exclude, filed November 2, 2015	

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	20.	Inter Partes Review Case No. IPR2014-01372, Record of Oral Hearing, filed December 18, 2015		
	21.	Inter Partes Review Case No. IPR2014-01372, Ex 1026 - Websters New Collegiate, filed September 10, 2015		
	22.	Inter Partes Review Case No. IPR2014-01372, Ex 1027 - May 2015 Dolin Article, filed September 10, 2015		
	23.	Inter Partes Review Case No. IPR2014-01372, Ex 1028 - Portola Packaging Reexam Guidance, filed September 10, 2015		
	24.	Inter Partes Review Case No. IPR2014-01372, Ex 1029 - Staunstrup Textbook Re Self Timed Signalling, filed September 10, 2015		
	25.	Inter Partes Review Case No. IPR2014-01372, Ex 1030 - DIC Rabaey Textbook, filed September 10, 2015		
	26.	Inter Partes Review Case No. IPR2014-01372, Ex 1031 - Dp Tr Dr Carl Sechen, filed September 10, 2015		
	27.	Inter Partes Review Case No. IPR2014-01372, Ex 1032 - Websters II Dict, filed September 10, 2015		
	28.	Inter Partes Review Case No. IPR2014-01372, Ex 1033 - McGraw Hill Electronics Dict, filed September 10, 2015		
	29.	Inter Partes Review Case No. IPR2014-01372, Ex 1034 - IEEE Dict, filed September 10, 2015		
	30.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2009, 'Excerpts from Merriam Webster's Collegiate Dictionary, Tenth Edition (2001),' filed December 18, 2014		
	31.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2010, 'Excerpts from the American Webster Hand College Dictionary (1981),' filed December 18, 2014		
	32.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2011, 'Excerpts from Merriam Webster's Collegiate Dictionary, Eleventh Edition (2003),' filed December 18, 2014		
	33.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2012, 'Cross-Examination transcript of Nader Bagherzadeh, Ph.D. , 5/21/2015,' filed June 15, 2015		
	34.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2013, 'Report to the Association for Computing,' filed June 15, 2015		
	35.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2014, 'MorphoSys - Case Study,' filed June 15, 2015		
	36.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2016, 'Declaration of Nadar Baghezadeh, Ph.D. ISO Motion to Stay,' filed June 15, 2015		
	37.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2017, 'Design and Implementation,' filed June 15, 2015		
	38.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2018, 'Deposition of Doanld Alpert, Ph.D., 2/12/15,' filed June 15, 2015		
	39.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2019, 'Declaration of Carl Sechen, Ph.D.,' filed June 15, 2015		

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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
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	40.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2020, 'Excerpts from Logic and Computer Design,' filed June 15, 2015	
	41.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2021, 'Excerpts from Digital Design Principles,' filed June 15, 2015	
	42.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2022, 'CMOS LVSI,' filed June 15, 2015	
	43.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2025, 'References reviewed by Examiner,' filed June 15, 2015	
	44.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2026, 'Excerpts from Digital Integrated Circuits,' filed June 15, 2015	
	45.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2027, 'Transcript of the September 30, 2015 Conference Call,' filed October 2, 2015	
	46.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2028, 'Attachment to Oct. 19, 2015 Patent Owner Motion to Exclude,' filed October 19, 2015	
	47.	Inter Partes Review Case No. IPR2014-01374, Corrected Petition For Inter Partes Review, filed September 22, 2014	
	48.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Preliminary Response, filed December 18, 2014	
	49.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Exhibit List, filed December 18, 2014	
	50.	Inter Partes Review Case No. IPR2014-01374, Decision - Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 10, 2015	
	51.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Response, filed June 15, 2015	
	52.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Corrected Response, filed June 16, 2015	
	53.	Inter Partes Review Case No. IPR2014-01374, Petitioner's Motion To Exclude Evidence, filed July 20, 2015	
	54.	Inter Partes Review Case No. IPR2014-01374, Petitioners REPLY To POs RESPONSE, filed September 10, 2015	
	55.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Objections to Evidence Filed With Petitioner Reply, filed September 17, 2015	
	56.	Inter Partes Review Case No. IPR2014-01374, Petitioner SMART Request For Oral Argument, filed September 24, 2015	
	57.	Inter Partes Review Case No. IPR2014-01374, Order - Trial Hearing - 37 CFR 42.70, filed October 8, 2015	
	58.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Request for Oral Argument, filed October 16, 2015	
	59.	Inter Partes Review Case No. IPR2014-01374, Patent Owner Motion to Exclude, filed October 16, 2015	

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	60.	Inter Partes Review Case No. IPR2014-01374, Motion to Exclude Portions of Petitioner's Reply to Patent Owner's Response, filed October 19, 2015		
	61.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Opposition to Petitioner's Motion to Exclude, filed October 26, 2015		
	62.	Inter Partes Review Case No. IPR2014-01374, Patent Owner's Fourth Amended Exhibit List, filed October 26, 2015		
	63.	Inter Partes Review Case No. IPR2014-01374, IPR2014-01374 SMART Response - PO Motion To Exclude Exhibits, filed October 26, 2015		
	64.	Inter Partes Review Case No. IPR2014-01374, IPR2014-01374 SMART Response - PO Motion To Exclude Reply Portions, filed October 26, 2015		
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	67.	Inter Partes Review Case No. IPR2014-01374, Record of Oral Hearing, filed December 18, 2015		
	68.	Inter Partes Review Case No. IPR2014-01373, Corrected Petition For Inter Partes Review, filed September 22, 2014		
	69.	Inter Partes Review Case No. IPR2014-01373, Patent Owner's Preliminary Response, filed December 18, 2014		
	70.	Inter Partes Review Case No. IPR2014-01373, Patent Owner's Exhibit List, filed December 18, 2014		
	71.	Inter Partes Review Case No. IPR2014-01373, Decision - Denying Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 13, 2015		
	72.	Inter Partes Review Case No. IPR2014-01375, Corrected Petition For Inter Partes Review, filed September 22, 2014		
	73.	Inter Partes Review Case No. IPR2014-01375, Patent Owner's Preliminary Response, filed December 18, 2014		
	74.	Inter Partes Review Case No. IPR2014-01375, Patent Owner's Exhibit List, filed December 18, 2014		
	75.	Inter Partes Review Case No. IPR2014-01375, Decision - Denying Institution of Inter Partes Review 37 C.F.R. 42.108, filed March 13, 2015		

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Sheet	1	of	2	Attorney Docket Number	NETL046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	601,710	01-04-2000	Wiggers	
	2.	2001008006	07-12-2001	Klein	
	3.	20050257109	11-17-2005	Averbuj	
	4.	20070070669	03-29-2007	Tsern	

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	5.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,516,185, IPR Case Number IPR2014-01029, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	6.	Petition for Inter Partes Review filed on 6/22/2014 for U.S. Patent No. 7,881,150, IPR Case Number IPR2014-00882 and IPR Case Number IPR2014-01011, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	7.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,081,536, IPR Case Number IPR2014-00883, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	8.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,001,434, IPR Case Number IPR2014-00970, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	9.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,359,501, IPR Case Number IPR2014-00971, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	10.	Altera, ACEX iK, Programmable Logic Device Family, Data Sheet, May 2003, Ver 3.4	
	11.	Horowitz, "The Art of Electronics," Cambridge Univ. Press, 1989, selected pages	
	12.	Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379	
	13.	Jacob, Bruce L.; "Synchronous DRAM Architectures, Organizations, and Alternative Technologies". University of Maryland, Dec. 10, 2002.	
	14.	JEDEC Standard No. 21-C Section 4.5.7, 168 Pin Registered SDRAM DIMM Family, Release 7	
	15.	JEDEC 21-C, Section 4.6.1, 278 Pin Buffered SDRAM DIMM Family	
	16.	JEDEC Standard No. 21-C Section 4.1.2.5, Appendix E, "Specific PD's for Synchronous DRAM (SDRAM)," pp. 1-25, no date	
	17.	JEDEC Standard, "Fully Buffered DIMM (FBDIMM): DFX Design for Validation and Test," JESD82-28, February 2008	

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	18.	Reese, "Introduction to Logic Synthesis using Verilog HDL," Morgan&Claypool Publishers, 2006, pp. 1-28	

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	1.	PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification JEDEC, Standard No. 21-C, Revision 1.3, Jan. 2002 / pages 4.20.4-1	

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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	1	of	5		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	20010048342	12-06-2001	Yoshikazu Yoshida	
	2.	20020000847	01-03-2002	Masao Taguchi	
	3.	20020131535	09-19-2002	Huber Brian W.	
	4.	20020140523	10-03-2002	Edwin Park	
	5.	20030098742	05-29-2003	Nakagawa Osamu Samuel	
	6.	20030126346	07-03-2003	Kuo Sung H.	
	7.	20030197797	10-23-2003	Josep Segura	
	8.	20030218491	11-27-2003	Makoto Nagasue	
	9.	20040155702	08-12-2004	Siemens Elema Ab	
	10.	20040199843	10-07-2004	Hansquine David W.	
	11.	20050093620	05-05-2005	International Business Machines Corporation	
	12.	20050127989	06-16-2005	Hiroshi Miyagi	
	13.	20050289423	12-29-2005	Kabushiki Kaisha Toshiba	
	14.	20060082383	04-20-2006	Choi Jung-Hwan	
	15.	20060107156	05-18-2006	Kee-Hoon Lee	
	16.	20060140015	06-29-2006	Rambus Inc.	
	17.	20060144015	07-06-2006	Cash John W Iii	
	18.	20060147217	07-06-2006	Hahin Jayne C	
	19.	20060152653	08-31-2006	Paul Atkinson	
	20.	20060242458	10-26-2006	Daniel Feldman	
	21.	20060262586	11-23-2006	Solomon Jeffrey C	
	22.	20060271748	11-30-2006	Intel Corporation	
	23.	20070030814	02-08-2007	Samsung Electronics Co., Ltd.	
	24.	20070058471	03-15-2007	Rajan Suresh N	
	25.	20070079199	04-05-2007	Asset Intertech, Inc.	
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	27.	20070152743	07-05-2007	Brent Keeth	
	28.	20070204075	08-30-2007	Rajan Suresh N	
	29.	20070223296	09-27-2007	Christopher Miller	

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/Phung Chung/

DATE CONSIDERED

03/02/2016

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				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	2	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	30.	3660675	05-02-1972	Honeywell Inc	
	31.	3757235	09-04-1973	Shell Oil Co	
	32.	4305091	12-08-1981	Cooper J Carl	
	33.	4586168	04-29-1986	Motorola, Inc.	
	34.	4701845	10-20-1987	Unisys Corporation	
	35.	4752741	06-21-1988	Honeywell Inc.	
	36.	4782487	11-01-1988	Digital Equipment Corporation	
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	38.	4885799	12-05-1989	Motorola, Inc.	
	39.	4903266	02-20-1990	International Business Machines Corporation	
	40.	4910597	03-20-1990	Canon Kabushiki Kaisha	
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	42.	4958346	09-18-1990	Advantest Corporation	
	43.	4987321	01-22-1991	Eastman Kodak Company	
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	45.	5051997	09-24-1991	Mitsubishi Denki Kabushiki Kaisha	
	46.	5138619	08-11-1992	National Semiconductor Corporation	
	47.	5173906	12-22-1992	Dreibelbis Jeffrey H	
	48.	5222066	06-22-1993	Motorola, Inc.	
	49.	5241503	08-31-1993	Motorola, Inc.	
	50.	5304856	04-19-1994	At&T Bell Laboratories	
	51.	5337254	08-09-1994	Hewlett-Packard Company	
	52.	5359235	10-25-1994	Digital Equipment Corporation	

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				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	3	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
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	57.	5841296	11-24-1998	Xilinx, Inc.	
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DATE CONSIDERED

03/02/2016

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				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	4	of	5		

US PATENT DOCUMENTS					
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		Number-Kind Code ² (if known)			
	73.	6918072	07-12-2005	Micron Technology, Inc.	
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	75.	6928593	08-09-2005	Intel Corporation	
	76.	6930509	08-16-2005	Stmicroelectronics Limited	
	77.	6934900	08-23-2005	Global Unichip Corporation	
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	87.	7957218	06-07-2011	Freescale Semiconductor, Inc.	
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	89.	8250295	8/21/12	Amidi et al.	

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	90.	Bayard "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, vol. 47, Issue 5, pp. 1288-1296.	

EXAMINER

/Phung Chung/

DATE CONSIDERED

03/02/2016

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				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	5	of	5		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	91.	Der-Chang et al. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, Apr. 2002, vol. 21, Issue 4, pp. 449-465.	
	92.	Mutoh et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, Nov. 2-6, 2004, vol. 1, pp. 963-968.	
	93.	Sekiguchi et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, vol. 30, Issue 12, pp. 1569-1574.	

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Applicant Initiated Interview Request FormApplication No.: 14/229,844First Named Applicant: Hyun LeeExaminer: CHUNG, PHUNG MArt Unit: 2117Status of Application: Pending**Tentative Participants:**(1) Jamie Zheng

(2) _____

(3) _____

(4) _____

Proposed Date of Interview: July 1, 2016Proposed Time: 2:00 PM (AM/PM)**Type of Interview Requested:**(1) ☒ Telephonic(2) ☐ Personal(3) ☐ Video ConferenceExhibit To Be Shown or Demonstrated: ☐ YES☒ NO

If yes, provide brief description: _____

Issues To Be Discussed

Issues (Rej., Obj., etc)	Claims/ Fig. #s	Prior Art	Discussed	Agreed	Not Agreed
(1) <u>IDS</u>	_____	_____	☐	☐	☐
(2) <u>103 rejections</u>	<u>1-25</u>	<u>Zuraski in view of Jacob</u>	☐	☐	☐
(3) _____	_____	_____	☐	☐	☐
(4) _____	_____	_____	☐	☐	☐

☐ Continuation Sheet Attached☐ Proposed Amendment or Arguments Attached

Brief Description of Arguments to be Presented: Applicant hereby authorize the Examiner to communicate with the Applicant or its representative via internet email, and will provide proposed amendment or arguments to the Examiner via email prior to the requested interview.

An interview was conducted on the above-identified application on _____

NOTE: This form should be completed and filed by applicant in advance of the interview (see MPEP § 713.01). If this form is signed by a registered practitioner not of record, the Office will accept this as an indication that he or she is authorized to conduct an interview on behalf of the principal (37 CFR 1.32(a)(3)) pursuant to 37 CFR 1.34. This is not a power of attorney to any above named practitioner. See the Instruction Sheet for this form, which is incorporated by reference. By signing this form, applicant or practitioner is certifying that he or she has read the Instruction Sheet. After the interview is conducted, applicant is advised to file a statement of the substance of this interview (37 CFR 1.133(b)) as soon as possible. This application will not be delayed from issue because of applicant's failure to submit a written record of this interview.

/Jamie J. Zheng/

Applicant/Applicant's Representative Signature

Examiner/SPE Signature

Jamie J. Zheng

Typed/Printed Name of Applicant or Representative

51167

Registration Number, if applicable

This collection of information is required by 37 CFR 1.133. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 24 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Instruction Sheet for:
APPLICANT INITIATED INTERVIEW REQUEST FORM
(Not to be Submitted to the USPTO)

1. If this form is signed by a registered practitioner not of record, the authority to submit the Applicant Initiated Interview Request Form is pursuant to limited authority to act in a representative capacity under 37 CFR 1.34 and further proof of authority to act in a representative capacity may be required. See 37 CFR 1.34.

The Office will accept the signed form as an indication that the registered practitioner not of record is authorized to conduct an interview on behalf of the principal in pursuant to 37 CFR 1.34.

For more information, see the "Conducting an Interview with a Registered Practitioner Acting in a Representative Capacity" notice which is available on the USPTO Web site at: <http://www.uspto.gov/patents/law/notices/2010.jsp>.

2. This is not a power of attorney to any named practitioner. Accordingly, any registered practitioner not of record named on the form does not have authority to sign a request to change the correspondence address, a request for express abandonment, a disclaimer, a power of attorney, or other document requiring the signature of the applicant, assignee of the entire interest or an attorney of record. If appropriate, a separate power of attorney to the named practitioner should be executed and filed in the US Patent and Trademark Office.
3. Any interview concerning an unpublished application under 35 U.S.C. § 122(b) with a registered practitioner not of record, pursuant to 37 CFR 1.34, will be conducted based on the information and files supplied by the practitioner in view of the confidentiality requirements of 35 U.S.C. § 122(a).

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	26175315
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	25-JUN-2016
Filing Date:	29-MAR-2014
Time Stamp:	19:19:30
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Letter Requesting Interview with Examiner	2016-06-25_Interview_Request _NETL046C3.pdf	421201 40910c81501c4d73fb70d3e04b85b460f4e5fe5e	no	3

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/229,844
Filed : March 29, 2014
For : A Self-Test Memory Module and
Method of Operation
Examiner : CHUNG, PHUNG M
Art Unit : 2189
Conf. No. : 1018
Docket No. : NT006US

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence and
any other attachment noted on the Automated
Acknowledgement Receipt, is being
transmitted from within the Pacific Time zone
to the Commissioner for Patents via the EFS
Web Server on:

July 11, 2016

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND RESPONSE TO NON-FINAL OFFICE ACTION**DATED MARCH 9, 2016****Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir or Madam:

Applicant submits the following amendments and remarks in the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 11 of this paper.

In the Specification:

Please amend Paragraph [0001] as follows:

[0001] The present application is a continuation of U.S. Patent Application No. 13/745,790, filed January 19, 2013, now U.S. Patent No. 8,689,064, which is a continuation of U.S. Patent Application No. 13/183,253, filed July 14, 2011, now U.S. Patent No. 8,359,501-8,395,501, which is a continuation of U.S. Patent Application No. 12/422,925, filed April 13, 2009, now U.S. Patent No. 8,001,434, which claims the benefit of priority from U.S. Provisional Application No. 61/044,801, filed April 14, 2008, U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and U.S. Provisional Application No. 61/044,839, filed April 14, 2008. Each of the foregoing applications is incorporated in their entirety by reference herein. This application is related to US. Patent Application No. 12/422,912, filed on April 13, 2009 and entitled "Self-Adjusting Damper", now abandoned, and to US. Application No. 12/422,853, filed on April 13, 2009 and entitled "Circuit Providing Load Isolation and Noise Reduction", now U.S. Patent No. 8,154,901, both of which are incorporated in their entirety by reference herein.

In the Claims:

1. (Currently Amended) A memory module accessible ~~operable~~ in a computer system ~~with~~ by a system memory controller via a system memory bus, comprising:
 - memory devices mounted on a circuit board, the memory devices having address and control ports and data ports, ~~and operating according to a standard memory protocol~~;
 - a data module mounted on the circuit board and coupled ~~to~~ between the data ports of the memory devices and the system memory bus; and
 - a control module mounted on the circuit board and coupled to the address and control ports of the memory devices; andwherein the memory module is operable in any of a plurality of modes including a normal mode and a test mode;
 - wherein, during the normal mode, the control module receives system address and control signals from the system memory controller and provides memory address and control signals to the memory devices ~~according to the standard memory protocol~~, ~~the memory address and control signals being based on~~ according to the system address and control signals ~~received by the control module from the system memory controller~~, and the data module propagates data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals; and
 - wherein, during the test mode, the control module generates self-test address and control signals ~~according to the standard memory protocol~~ without using any ~~system driven~~ test procedure driven by the system memory controller, the control module provides the self-test address and control signals to the address and control ports of the memory devices, and the data module generates self-test data patterns for testing the memory devices, isolates the memory devices from the system memory controller, and transmits the self-test data patterns to the data ports of the memory devices ~~in accordance with the standard memory protocol~~, wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals received from the control module.

2. (Currently Amended) The memory module of claim 1, wherein the memory devices are arranged in at least one rank, each rank having a set of memory devices organized in a plurality groups, each group including one or more memory devices, wherein the system memory bus has a width of N bits and is organized in a plurality of n-bit sections, wherein the data module includes a plurality of data handlers, each respective data handler ~~handles being coupled between~~ a respective n-bit section of the N bits ~~by system memory bus and propagating a respective n-bit section of each N-bit wide data signal between the~~ a respective group of one or more memory devices ~~and the system memory controller during the normal mode, and by the each respective data handler generating respective n-bit wide self-test data patterns and transmitting the respective n-bit wide self-test data patterns to the respective group of one or more memory devices a respective n-bit section of the at least one rank during the test mode, n being a fraction of N.~~

3. (Previously Presented) The memory module of claim 2, wherein each memory device is n-bit wide.

4. (Previously Presented) The memory module of claim 2, wherein the control module generates additional self-test address and control signals and provides the additional self-test address and control signals to the address and control ports of the memory devices during the test mode, the additional self-test address and control signals causing the memory devices to output a N-bit wide read data signal, and wherein the each respective data handler receives a respective n-bit section of the N-bit wide read data signal and checks a data pattern in the respective n-bit section of the N-bit wide read data signal with respect to an expected data pattern.

5. (Previously Presented) The memory module of claim 4, wherein the expected data pattern is generated in the respective data handler after the control module generates the additional self-test address and control signals.

6. (Currently Amended) The memory module of claim 1, wherein the ~~standard memory protocol is~~ the control module generates the self-test addresses and control signals according to a Joint Electron Devices Engineering Council (JEDEC) standard

memory protocol, and wherein the data module transmits the self-test data patterns to the memory devices according to the JEDEC standard memory protocol.

7. (Previously Presented) The memory module of claim 4, wherein the data module stores failure information when a test failure is indicated by the data pattern not in agreement with the expected data pattern.

8. (Currently Amended) The memory module of claim 7-4, wherein the data module is further configured to report the failure information to the control module.

9. (Currently Amended) The memory module of claim 1, wherein the self-test data patterns are transmitted to the memory devices in test data signals that have at least one of a programmable slew rate and a programmable peak value that is ~~can be~~ programmed to correspond to characteristics of write data signals from the system memory controller.

10. (Previously Presented) The memory module of claim 1, wherein the data module generates and transmits the self-test data patterns to the memory devices in response to a write command from the control module, and wherein the data module read back test data patterns from the memory devices in response to a read command from the control module.

11. (Previously Presented) The memory module of claim 2, wherein the plurality of data handlers operate independently of each other without being in communication with each other, and wherein, during the test mode, the each respective data handler generates n-bit self-test data patterns to test a corresponding portion of a memory space of the memory module independently of the other data handlers.

12. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective group of one or more memory devices ~~n-bit section of the at least one rank~~.

13. (Currently Amended) A memory module ~~operable~~ to communicate data with a system memory controller via a N-bit wide system memory bus, comprising:

a control module mounted on a circuit board;

memory devices mounted on the circuit board ~~and operable~~ to receive or output data signals in response to address and control signals from the control module, wherein the memory devices are arranged in at least one N-bit wide rank; and

a data module including a plurality of data handlers, wherein each respective data handler is coupled between a respective n-bit section of the N-bit wide system memory bus and a respective n-bit section of the at least one N-bit wide rank of memory devices, n being a fraction of N, wherein the memory module ~~subsystem~~ is operable in any one of a plurality of modes including a normal mode and a self-test mode, and wherein, during the self test mode;

the control module generates first self-test address and control signals and provides ~~provide the first self-test address and control signals to the memory devices;~~ the control module further generates second self-test address and control signals and provides the second self-test address and control signals to the memory devices; and

the each respective data handler produces respective n-bit wide test data patterns for testing the respective n-bit section of at least one N-bit wide rank of memory devices and transmits the respective n-bit wide test data patterns to the respective n-bit section of the at least one N-bit wide rank of memory devices, ~~so that wherein~~ one or more memory devices in the respective n-bit section of the at least one N-bit wide rank of memory devices receive the respective n-bit wide test data patterns in accordance with the first self-test address and control signals, wherein the one or more memory devices output a respective n-bit wide read data signal in response to the second self-test address and control signals, and wherein the each respective data handler receives the n-bit wide read data signal and checks data patterns in the respective n-bit wide read data signal with respect to expected data patterns; and

wherein, during the normal mode, the control module provides memory address and control signals to the memory devices based on system address and control signals received from the system memory controller, and the each respective data handler propagates data between the respective n-bit section of the at least one N-bit wide rank of memory devices and the system memory controller.

14. (Cancelled)

15. (Previously Presented) The memory module of claim 14, wherein the data module isolates the memory devices from the N-bit wide data bus when the memory module is operating in the test mode.

16. (Currently Amended) The memory module of claim 13, ~~wherein the memory devices operate according to a standard memory protocol,~~ wherein the control module generates the self-test address and control signals ~~according to the standard memory protocol~~ without using any system driven test procedure, ~~and wherein the each respective data handler transmits the respective n-bit wide data patterns to the respective n-bit section of the at least one N-bit wide rank according to the standard memory protocol~~

17. (Currently Amended) The memory module of claim 16, wherein the control module generates the self-test address and control signals according to the standard memory protocol is a Joint Electron Devices Engineering Council (JEDEC) standard memory protocol, and wherein the each respective data handler transmits the respective n-bit wide data patterns to the respective n-bit section of the at least one N-bit wide rank according to the JEDEC standard memory protocol.

18. (Previously Presented) The memory module of claim 16, wherein the each respective data handler receives control information from the control module and includes a respective data generation element that produces the n-bit wide data patterns in response to the control information from the control module.

19. (Currently Amended) The memory module of claim 13-14, wherein the each respective data handler further includes a respective verification element, the respective verification element checking data patterns in the respective n-bit wide read data signal with respect to the expected data patterns ~~represented by data bits read from the respective portion of the memory devices for~~ and records errors if the data pattern do not match the expected data patterns.

20. (Cancelled)

21. (Cancelled)

22. (Cancelled)
23. (Cancelled)
24. (Cancelled)
25. (Cancelled)
26. (New) The memory module of claim 1, wherein the control module comprises a memory device controller and a test controller, wherein the memory device controller receives the system address and control signals from the system memory controller and the self-test address and control signals from the test controller.
27. (New) The memory module of claim 26, wherein the control module further comprises a register coupled to the address and control ports of the memory devices, and wherein the memory device controller sends address and control information from either the system address and control signals or from the self-test address and control signals to the register depending on whether the memory module is in the normal mode or test mode.
28. (New) The memory module of claim 2, wherein the each respective data handler comprises a data handler logic element that generates test data patterns and a switch coupled to the data handler logic element and to the respective n-bit section of the system memory bus, and wherein the switch is configured to selectively input to the respective group of one or more memory devices either write data signals from the system memory controller or test data patterns from the data handler logic element depending on whether the memory module is in the normal mode or test mode.
29. (New) The memory module of claim 28, wherein the switch is further configured to receive read data signals from the the respective group of one or more memory devices and sends the read data signals to the system memory controller or to the data handler logic element depending on whether the memory module is in the normal mode or the test mode.

30. (New) The memory module of claim 13, wherein the control module comprises a memory device controller and a test controller, wherein the memory device controller receives the system address and control signals from the system memory controller and the first and second self-test address and control signals from the test controller.

31. (New) The memory module of claim 30, wherein the control module further comprises a register coupled to the address and control ports of the memory devices, and wherein the memory device controller sends address and control information from either the system address and control signals or from the test controller depending on whether the memory module is in the normal mode or test mode.

32. (New) The memory module of claim 13, wherein the each respective data handler comprises a data handler logic element that generates test data patterns and a switch coupled to the data handler logic element and to the respective n-bit section of the system memory bus, and wherein the switch is configured to selectively input to the respective n-bit section of the at least one N-bit wide rank of memory devices either write data signals from the system memory controller or test data patterns from the data handler logic element.

33. (New) The memory module of claim 31, wherein the switch is further configured to receive read data signals from the the n-bit section of the at least one N-bit wide rank of memory devices and sends the read data signals to the system memory controller or to the data handler logic element depending on whether the memory module is in the normal mode or the test mode.

34. (New) The memory module of claim 13, wherein the each respective data handler transmits the respective n-bit wide test data patterns in one or more test data signals to the respective n-bit section of the at least one N-bit wide rank of memory devices, the one or more test data signals each having at least one of a programmable slew rate and a programmable peak value that is programmed to resemble that of a write data signal from the system memory controller.

35. (New) The memory module of claim 13, wherein the each respective data handler includes an integrated circuit package positioned on the circuit board proximate to the respective n-bit section of the at least one rank of memory devices.

REMARKS

This amendment is in response to the Non-Final Office Action dated March 9, 2016. In the Office Action, the Examiner:

- objected to the disclosure because of informalities in paragraph [0001];
- requested that the Applicant identify references submitted in the Information Disclosure Statements filed on 2/18/16, 2/17/16, 2/16/16, 12/24/14, 8/10/14, 8/7/14, which have particular significance in the prosecution of this Application for further consideration by the Examiner;
- rejected claims 1-25 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph; and
- rejected claims 1-25 under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Zuraski, Jr. et al (6,560,740) (hereafter referred to as Zuraski) in view of Jacob (Synchronous DRAM architectures, organizations and alternative technologies, 12/10/2002).

Objection to Disclosure

Paragraph [0001] in the Specification has been amended to address the objections thereto.

Information Disclosure Statements filed on 2/18/16, 2/17/16, 2/16/16, 12/24/14, 8/10/14, 8/7/14

The Examiner has requested identification of any references submitted in the Information Disclosure Statements (IDS) filed on 2/18/16, 2/17/16, 2/16/16, 12/24/14, 8/10/14, 8/7/14, which have particular significance in the prosecution of this Application. Applicant would like to point out that this Application is a continuation of U.S. Patent Application No. 13/745,790, filed January 19, 2013, now U.S. Patent No. 8,689,064, which is a continuation of U.S. Patent Application No. 13/183,253, filed July 14, 2011, now U.S. Patent No. 8,359,501 (*hereafter, the '501 patent*), which is a continuation of U.S. Patent Application No. 12/422,925, filed April 13, 2009, now U.S. Patent No.

8,001,434 (*hereafter, the '434 patent*). Both the '501 patent and the '434 patent are the subjects in litigation and inter partes review proceedings. This Application also has overlapping subject matter with commonly-owned U.S. Patent Nos. 8,516,185 (*hereafter, the '185 patent*), 8,081,536 (*hereafter, the '536 patent*), and 7,881,150 (*hereafter, the '150 patent*), which are also subjects in litigation and inter partes review proceedings.

Except for the IDS filed on 2/15/16, which list references submitted in the parent application, 13/183,253, filed July 14, 2011, now US Pat. No. 8,359,501, most of the references listed in the other IDS's are documents from litigation and inter partes review proceedings involving commonly-owned patents that are related to this Application either because of priority claims or overlapping subject matter or both. These proceedings are listed in the following:

- The '185, the '536, the 150, the '434 and the '501 patents were the subject of a complaint filed by Netlist on August 23, 2013, in Netlist, Inc. v. Smart Modular Technologies, Inc., Civil Action 4:13-cv-05889, pending in the Northern District of California. On April 9, 2015, the district court granted a motion to stay the case pending inter partes review. The case is stayed.
- The '185, the '536, the 150, the '434 and the '501 patents were also the subject of a declaratory judgment complaint filed by Smart Modular Technologies, Inc. on August 23, 2013, in Smart Modular Technologies, Inc. v. Netlist, Inc., Civil Action 4:13-cv-03916, previously pending in the Northern District of California. The case was dismissed on February 12, 2014.
- The '185, the '536, the 150, the '434 and the '501 patents were also the subject of a declaratory judgment complaint filed by Diablo Technologies, Inc. on August 23, 2013, in Diablo Technologies, Inc. v. Netlist, Inc., Civil Action 4:13-cv-03901, previously pending in the Northern District of California. The case was consolidated with Netlist, Inc. v. Smart Modular Technologies, Inc., Civil Action 4:13-cv-05889 on April 8, 2014.

- The '185 patent was also the subject of a petition for inter partes review filed by SanDisk Corporation on June 24, 2014, in SanDisk Corporation v. Netlist, Inc., Case IPR 2014-01029, previously pending before the USPTO Patent Trial and Appeal Board ("PTAB"). On December 16, 2014, the PTAB issued a decision denying institution of inter partes review.
- The '185 patent was also the subject of a petition for inter partes review filed by Smart Modular Technologies, Inc. on August 23, 2014, in Smart Modular Technologies, Inc. v. Netlist, Inc., Case IPR 2014-01369, previously pending before the PTAB. On March 9, 2015, the PTAB issued a decision denying institution of inter partes review.
- The '546 patent was also the subject of a petition for inter partes review filed by Diablo Technology, Inc. on June 21, 2014, in Diablo Technology, Inc. v. Netlist, Inc., Case IPR 2014-00883, previously pending before the PTAB. On December 14, 2015, the PTAB issued a final written decision finding that the claims subject to review were unpatentable.
- The '546 patent is also the subject of a petition for inter partes review filed by SanDisk Corporation on April 7, 2015, in SanDisk Corporation v. Netlist, Inc., Case IPR 2015-01021, now pending before the PTAB.
- The '150 patent was also the subject of a petition for inter partes review filed by Diablo Technology, Inc. on June 22, 2014, in Diablo Technology, Inc. v. Netlist, Inc., Case IPR 2014-00882, previously pending before the PTAB. On December 14, 2015, the PTAB issued a final written decision finding that the claims subject to review were unpatentable.
- The '150 patent was also the subject of a petition for inter partes review filed by Diablo Technology, Inc. on June 22, 2014, in Diablo Technology, Inc. v. Netlist, Inc., Case IPR 2014-01011, previously pending before the PTAB. On December 14, 2015, the PTAB issued a

final written decision finding that the claims subject to review were unpatentable.

- The '150 patent is also the subject of a petition for inter partes review filed by SanDisk Corporation on April 7, 2015, in SanDisk Corporation v. Netlist, Inc., Case IPR 2015-01020, now pending before the PTAB.
- The '434 patent was also the subject of a petition for inter partes review filed by Smart Modular Technologies, Inc. on August 23, 2014, in Smart Modular Technologies, Inc. v. Netlist, Inc., Case IPR 2014-01373, previously pending before the PTAB. On March 13, 2015, the PTAB issued a decision denying institution of inter partes review.
- The '434 patent was also the subject of another petition for inter partes review filed by Smart Modular Technologies, Inc. on August 23, 2014, in Smart Modular Technologies, Inc. v. Netlist, Inc., Case IPR 2014-01372, previously pending before the PTAB. On March 9, 2016, the PTAB issued a final written decision finding that 13 claims of the '434 patent were not unpatentable.
- The '434 patent was also the subject of another petition for inter partes review filed by SanDisk Corporation on June 18, 2014, in SanDisk Corporation v. Netlist, Inc., Case IPR 2014-00970, previously pending before the PTAB. On December 16, 2014, the PTAB issued a decision declining to institute inter partes review as to 3 claims of the '434 patent, and instituting inter partes review as to 32 claims of the '434 patent. On December 14, 2015, the PTAB issued a final written decision finding 14 claims of the '434 patent were not unpatentable and 18 claims of the '434 patent were unpatentable.
- The '501 patent was also the subject of a petition for inter partes review filed by SanDisk Corporation on June 18, 2014, in SanDisk Corporation v. Netlist, Inc., Case IPR2014-00971, previously pending before the PTAB. On April 27, 2016, the PTAB issued a final written

decision finding 1 claim of the '501 patent was not unpatentable and 19 claims of the '501 patent were unpatentable.

- The '501 patent was also the subject of another petition for inter partes review filed by Smart Modular Technologies, Inc. on August 23, 2014, in Smart Modular Technologies, Inc. v. Netlist, Inc., Case IPR 2014-01375, previously pending before the PTAB. On March 13, 2015, the PTAB issued a decision denying institution of inter partes review.
- The '501 patent was also the subject of another petition for inter partes review filed by Smart Modular Technologies, Inc. on August 23, 2014, in Smart Modular Technologies, Inc. v. Netlist, Inc., Case IPR 2014-01374, previously pending before the PTAB. On April 27, 2016, the PTAB issued a final written decision finding that 13 claims of the '501 patent were not unpatentable.

For at least the above reasons, Applicant believes that all of the references submitted in the IDS forms filed on 2/18/16, 2/17/16, 2/16/16, 12/24/14, 8/10/14, 8/7/14 are relevant to the patentability of the claims in this Application. Applicant would greatly appreciate the Examiner initialing and returning the IDS forms indicating that the information has been considered and made of record herein

The Examiner also states that “some of the non-patent literatures (NPLs) (are) submitted without dates, and/or un-matched dates, un-matched Title, and blurred.” Applicant apologize if this is the case and would appreciate the Examiner pointing out to the undersigned which of the NPLs or listed items are affected so Applicant can make best effort to locate better copies of the affected NPLs or make corrections to the listed items.

Claim Amendment

Claims 14, and 20-25 are cancelled. Claims 1-2, 6, 8-9, 12-13, 16-17, and 19 are amended. New claims 26-35 are presented. No new matter is added. After entry of this amendment, the pending claims are 1-13, 15-19, and 26-35.

Rejection of Claims 1-25 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), Second Paragraph

Claims 1-2, 6, 8-9, 12-13, 16-17, and 19 are amended to address the rejections of claims 1-25 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph.

Specifically:

In claim 1, the terms “operable” and “standard memory protocol” are deleted, and further amendments are made in lines 19-27 to clarify that the self-test data patterns are used to test the memory devices.

In claim 2, the recitation of “each rank having a width of N bits” is deleted.

In claim 6, “JEDEC” is spelled out.

Claim 8 is amended to depend from claim 7 instead of claim 4 to resolve the antecedent basis issue.

In claims 13, the term “operable” is deleted, the recitation of “a respective n-bit section of the at least one N-bit wide rank” is amended to be “a respective n-bit section of the at least one N-bit wide rank of memory devices,” as suggested by the Examiner, the term “memory subsystem” is amended to be “memory module,” a “normal mode” is added as another exemplary mode of the plurality of modes, the term “a test mode” is changed to “a self-test mode,” the typographical error of “,” followed by “:” is corrected, and lines 18-22 are amended to address the issue of clarity raised by the Examiner.

Claim 14 is cancelled.

In claim 16-17, the term “a standard memory protocol” is deleted and “JEDEC” is spelled out, as suggested by the Examiner.

Claim 19 is amended to address the issue of clarity raised by the Examiner.

Claims 20-25 are cancelled.

Therefore, Applicant believes that all issues related to the rejection of claims 1-25 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, have been addressed. The rejection should now be withdrawn.

**Rejection of Claims 1-25 under Pre-AIA 35 U.S.C. 103(a) as Being Unpatentable
over Zuraski in view of Jacob**

Claim 1 as amended recites:

A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus; and

a control module mounted on the circuit board and coupled to the address and control ports of the memory devices; and

wherein the memory module is operable in any of a plurality of modes including a normal mode and a test mode;

wherein, during the normal mode, the control module receives system address and control signals from the system memory controller and provides memory address and control signals to the memory devices according to the system address and control signals, and the data module propagates data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals; and

wherein, during the test mode, the control module generates self-test address and control signals without using test procedure driven by the system memory controller, the control module provides the self-test address and control signals to the address and control ports of the memory devices, and the data module generates self-test data patterns for testing the memory devices, isolates the memory devices from the system memory controller, and transmits the self-test data patterns to the data ports of the memory devices, wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals received from the control module.

Zuraski in view of Jacob does not disclose the memory module in claim 1. First of all, Zuraski does not even disclose a memory module. A memory module is a printed circuit board on which memory integrated circuits (memory devices) are mounted (https://en.wikipedia.org/wiki/Memory_module; also, Specification, paragraph [0024] and FIG. 1). Zuraski, on the other hand, is all about a microprocessor 10, which is not a memory module. A microprocessor is a computer processor which incorporates the functions of a computer's central processing unit (CPU) on a single integrated circuit (IC) (<https://en.wikipedia.org/wiki/Microprocessor>; <http://www.dictionary.com/browse/microprocessor>; <http://www.businessdictionary.com/definition/microprocessor.html>). As such, the microprocessor 10 in Zuraski has **embedded** memory units 16a-16b (Zuraski, FIG. 1, and col. 5, line 50, *emphasis added*). These embedded memory units 16a-b are memory cells embedded in the microprocessor integrated circuit (Zuraski, col. 6, lines 48-52) and do

not include the mechanisms (e.g., the input/output circuits and proper packaging) for mounting on a circuit board (Zuraski, col. 6, lines 48-64). In fact, the reason microprocessors have embedded memory units is to eliminate latency associated with off-chip accesses through an inter-chip bus (see “Embedded memory,” Integrated Circuit Engineering Corporation, Smithsonian, the Chip Collection, <http://smithsonianchips.si.edu/ice/cd/MEMORY97/SEC11.PDF>). Therefore, a person of ordinary skill in the art could not possibly be motivated to adopt the teachings of Jacob by taking the embedded memory units 16 out of Zuraski’s microprocessor 10 and mount them onto a circuit board.

Secondly, Zuraski in view of Jacob does not disclose “a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus.” In rejecting claim 1, the Examiner considers Zuraski’s BIST LU 20a-20b as being analogous to the data module in claim 1 (Office Action, page 10, lines 7-11) and Zuraski’s Test Control Unit 26 as being analogous to the system memory controller in claim 1 (Office Action, page 10, lines 16-19). But, like the embedded memory units 16, the BIST logic units (BIST LU) 20a-20b and the Test Control Unit 26 are all part of an integrated circuit chip and cannot be mounted on a circuit board. Also, there is no system memory bus between Zuraski’s BIST LU 20a-20b and the Test Control Unit 26, as required by claim 1 as amended. As shown in FIG. 16 in Zuraski, the system memory bus 116 is **outside of** the microprocessor 10a and cannot possibly be between the BIST LU 20a-20b and the Test Control Unit 26, which are both part of the microprocessor 10.

Furthermore, Zuraski does not disclosure that “during the normal mode ... the data module propagates data signals between the memory devices and the system memory controller.” In rejection claim 1, the Examiner considers Test Control Unit 26 as the “system memory controller” but ignored the fact that the BIST LU 20a-20b do not propagate data signals between the embedded memory units 16a-16b and the Test Control Unit 26 during normal operation. In fact, the role of the Test Control Unit 26 in Zuraski is to configure boundary scan cells 22 to form a serial “scan chain” surrounding logic 12 during boundary scan testing (Zuraski, col. 7, lines 3-9). “During normal

operation of microprocessor 10, signals upon I/O terminals 24 flow to and from logic 12 unimpeded” (Zuraski, col. 7, lines 1-2). Thus, the Test Control Unit 26 has no defined role during normal operation of the microprocessor 10 and cannot be likened to a system memory controller that communicate data, address and control signals with a memory module.

Moreover, Zuraski in view of Jacob does not disclose “a control module mounted on the circuit board ... wherein during the normal mode, the control module receives system address and control signals from the system memory controller and provides memory address and control signals to the memory devices according to the system address and control signals.” As discussed above, the Master Test Unit 18 in Zuraski, which the Examiner considers to be analogous to the control module in claim 1, is part of an integrated circuit and cannot be mounted on a circuit board. Also, the role of the master test unit 18 is to use programmable test data stored within address/control register 42, configuration register 44, and background register 46 of the special register 15 during the programmable BIST routine (col. 7, lines 64-67). It does not receive any system address and control signals and providing memory address and control signals to any memory devices mounted on a circuit board, as required by claim 1 as amended.

Therefore, claim 1 as amended is patentable over Zuraski in view of Jacob.

Claim 2 as amended, claims 3-5, claim 6 as amended, claim 7, claims 8-9 as amended, claims 10-11, claim 12 as amended, and new claims 26-29 depend from claim 1 as amended and include further limitations in addition to the limitations in claim 1 as amended. Therefore, claim 2 as amended, claims 3-5, claim 6 as amended, claim 7, claims 8-9 as amended, claims 10-11, claim 12 as amended, and new claims 26-29 are patentable for at least the same reasons claim 1 is patentable.

Claim 2 as amended, claims 3-5, claim 6 as amended, claim 7, claims 8-9 as amended, claims 10-11, claim 12 as amended, and new claims 26-29 recite additional limitations that are not disclosed in either Zuraski or Jacob. For example:

- Claim 2 requires that the data module include a plurality of data handlers, each respective data handler being coupled between a

respective n-bit section of the system memory bus and a respective group of one or more memory devices. These features are clearly not disclosed in either Zuraski or Jacob because, as stated above, Zuraski's BIST LU 20 are not coupled to any system memory bus.

- Claim 6 requires that the control module generate the self-test addresses and control signals according to a Joint Electron Devices Engineering Council (JEDEC) standard memory protocol, and the data module transmit the self-test data patterns to the memory devices according to the JEDEC standard memory protocol. These features require the control module and the data module generate memory interface signals with, for example, proper edge relationships based on the JEDEC standard (See, for example, Specification, paragraph [0026, lines 10-15]). These features are clearly not disclosed in either Zuraski or Jacob because Zuraski's embedded memory units 16 are not memory devices with memory interfaces that are compliant with any JEDEC standard.
- Claim 9 requires that the self-test data patterns be transmitted to the memory devices in test data signals that have at least one of a programmable slew rate and a programmable peak value that is programmed to correspond to characteristics of write data signals from the system memory controller. These features require the test data signals generated on the memory module to resemble data signals from the system memory controller, which is coupled to the memory module via a system memory bus. These features are clearly not disclosed in either Zuraski or Jacob because Zuraski's embedded memory units 16 do not communicate data with any system memory controller.
- Claim 12 requires that the plurality of data handlers include physically separate integrated circuit packages, with each respective data handler positioned on the circuit board proximate to the respective group of one or more memory devices. These features are clearly not disclosed

in either Zuraski or Jacob because Zuraski's BIST LU 20 are part of a microprocessor which is itself an integrated circuit and cannot possibly include other physically separate integrated circuit packages.

The above arguments with regard to claim 1 as amended, claim 2 as amended, claims 3-5, claim 6 as amended, claim 7, claims 8-9 as amended, claims 10-11, claim 12 as amended, and new claims 26-29 apply to claim 13 as amended, claim 15, claim 16-17 as amended, claim 18, claim 19 as amended, and new claims 30-35. Therefore, claim 13 as amended, claim 15, claim 16-17 as amended, claim 18, claim 19 as amended, and new claims 30-35 for at least the same reasons.

Claims 14 and 20-25 are cancelled. There rejections thereto are now moot.

Applicant submits that claims 1-13, 15-19, and 26-35 are in condition for allowance and respectfully requests such action. The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any issues.

Respectfully submitted,

Date: July 11, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167

(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C3			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Claims in excess of 20	2202	3	40	120
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				120

Electronic Acknowledgement Receipt

EFS ID:	26318127
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	11-JUL-2016
Filing Date:	29-MAR-2014
Time Stamp:	21:01:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 120
RAM confirmation Number	7380
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2016-07-11_Amendment_NFO A_NETL046C3.pdf	200797	yes	21
			7edc35ae6b325501808a110ed7005aa7f70f0473		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Specification		2	2	
	Claims		3	10	
	Applicant Arguments/Remarks Made in an Amendment		11	21	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30040	no	2
			cff25793a586c0a72d77cfdeeac7a209084eabba2		
Warnings:					
Information:					
Total Files Size (in bytes):			230837		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/229,844	Filing Date 03/29/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	07/11/2016	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 28	Minus	** 25	= 3	x \$40 =	120
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						120	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
DIANA BATES

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
 App. No. : 14/229,844
 Filed : March 29, 2014
 For : A Self-Test Memory Module and
 Method of Operation
 Examiner : CHUNG, PHUNG M
 Art Unit : 2189
 Conf. No. : 1018
 Docket No. : NT006US

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

July 11, 2016

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

**PETITION FOR EXTENSION OF TIME TO FILE RESPONSE TO NON-FINAL
OFFICE ACTION DATED MARCH 9, 2016**

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir or Madam:

Applicant has just submitted an amendment and response to the Non-Final Office Action dated March 9, 2016, in which Applicant neglected to include a petition for a one-month extension of time to file the response or submit the fee for the extension.

Applicant hereby petitions for the one-month extension of time to file the response and submits the fee for the extension.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Respectfully submitted,

Date: July 11, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167

(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C3			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	2251	1	100	100
Miscellaneous:				
Total in USD (\$)				100

Electronic Acknowledgement Receipt

EFS ID:	26318790
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	12-JUL-2016
Filing Date:	29-MAR-2014
Time Stamp:	00:15:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 100
RAM confirmation Number	8026
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-07-11_Extension_NFOA_NETL046C3.pdf	64369	no	1
			746f52caf2f2cd4f5673d310140e98e85ba23cf2		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30288	no	2
			f5af4d0ebe2fe5c3c80b104ce5bf3770b8a8f734		

Warnings:

Information:

Total Files Size (in bytes):			94657
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	1	of	2	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	2009/0103387 A1	04-23-2009	Shau	
	2.	2006/0262586 A1		Solomon, J. et al.	
	3.	2008/0162790 A1			

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	4.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 8,081,536, IPR Case Number IPR2015-01021, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
	5.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 7,881,150, IPR Case Number IPR2015-01020, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
	6.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-01372, Final Written Decision, issued March 9, 2016	
	7.	Inter Partes Review Case No. IPR2014-01372, Petitioner's Request for Rehearing, filed April 6, 2016	
	8.	Inter Partes Review Case No. IPR2014-01372, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
	9.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-01374, Final Written Decision, issued March 9, 2016	
	10.	Inter Partes Review Case No. IPR2014-01374, Petitioner's Request for Rehearing, filed April 6, 2016	
	11.	Inter Partes Review Case No. IPR2014-01374, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
	12.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-00970, Decision Denying Petitioner's Request for Rehearing, issued April 27, 2016	
	13.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January 13, 2016	
	14.	Inter Partes Review Case No. IPR2014-00971, Decision - Petitioner's Request for Rehearing, issued April 27 2016	
	15.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision, reissued April 27, 2016	
	16.	Inter Partes Review Case No. IPR2014-00971, Netlist Inc.'s Notice of Appeal, filed June 29, 2016	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
				Attorney Docket Number	NETL.046C3
Sheet	2	of	2		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	17.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
	18.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
	19.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C3			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	2806	1	90	90
Total in USD (\$)				90

Electronic Acknowledgement Receipt

EFS ID:	26343776
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C3
Receipt Date:	13-JUL-2016
Filing Date:	29-MAR-2014
Time Stamp:	19:19:27
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$90
RAM confirmation Number	16359
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-07-13_IDS2_Cover_NETL046C3.pdf	48435	no	2
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Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2016-07-12_IDS_NETL046C3.pdf	103101	no	2
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This is not an USPTO supplied IDS fillable form					
3	Non Patent Literature	2015-04-07_Petition_IPR2015-01021.pdf	2550553	no	64
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Information:					
4	Non Patent Literature	2015-04-07_Petition_IPR2015-01020.pdf	2621086	no	66
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Warnings:					
Information:					
5	Non Patent Literature	2016-03-09_Final_Written_Decision_372.pdf	380192	no	35
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Information:					
6	Non Patent Literature	2016-04-06_Rehearing_Request_372.pdf	159925	no	12
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Warnings:					

Information:					
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Information:					
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Information:					
9	Non Patent Literature	2016-04-06_Requst_Rehearing_374.pdf	158722	no	12
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Information:					
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Warnings:					
Information:					
11	Non Patent Literature	2016-04-27_Decision_Denying_Rehearing_970.pdf	138255	no	14
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Warnings:					
Information:					
12	Non Patent Literature	2016-01-13_Petitioner_rehearing_request-35.pdf	1188616	no	19
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Information:					
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Information:					

14	Non Patent Literature	2016-04-27_Final_Written_Decision_971.pdf	523712	no	61
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Information:					
15	Non Patent Literature	2016-06-29_Notice_Appeal_971.pdf	1359172	no	125
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Warnings:					
Information:					
16	Non Patent Literature	2015-04-29_Exam_Report_NT001EP.pdf	151211	no	5
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Warnings:					
Information:					
17	Non Patent Literature	11-04-2015_Rsps_ER_NT001EP.pdf	1201884	no	26
			6ce107ca9f466a51d1aef5c2cff9cab7d18f5c7c		
Warnings:					
Information:					
18	Non Patent Literature	2016-05-11_Eng_Trans_NGR_NT001KR.pdf	183171	no	5
			1bd82675735359e2c4e2487f593949917116f0f0		
Warnings:					
Information:					
19	Fee Worksheet (SB06)	fee-info.pdf	30121	no	2
			443a568137e36f1804de450f20519391618b65a0		
Warnings:					
Information:					
Total Files Size (in bytes):			11658005		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT

Applicant	: Netlist, Inc.
App. No.	: 14/229,844
Filed	: March 29, 2014
For	: Memory Subsystem and Method of Operation
Examiner	: Unknown
Art Unit	: 2189
Conf No.	: 1018

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 19 references, including 16 non-patent literature documents, copies of which are submitted herewith.

The Applicant would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

Application No.: 14/229,844
Filing Date: March 29, 2014

Also submitted herewith is the fee set forth in 37 C.F.R. § 1.17(p). The Commissioner is authorized to charge any required fees to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: July 13, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	NETL.046C3	1018
79141	7590	10/20/2016		
Jamie J. Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306			EXAMINER CHUNG, PHUNG M	
			ART UNIT	PAPER NUMBER
			2117	
			NOTIFICATION DATE	DELIVERY MODE
			10/20/2016	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
eofficeaction@appcoll.com
infohub.j@gmail.com

Office Action Summary	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
 - If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
 - Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133).
- Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 7/11/16.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-13,15-19 and 26-35 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1-13,15-19 and 26-35 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 3/29/14 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>See Continuation Sheet</u> . | 4) ☐ Other: _____ |

Continuation of Attachment(s) 2). Information Disclosure Statement(s) (PTO/SB/08), Paper No(s)/Mail Date :7/13/16; 2/18/16, 2/17/16, 2/16/16, 12/24/14, 8/10/14 and 8/7/14 .

DETAILED ACTION

The present application is being examined under the pre-AIA first to invent provisions.

Information Disclosure Statement

An applicant's duty of disclosure of material information is not satisfied by presenting a patent examiner with "a mountain of largely irrelevant data from which he is presumed to have been able, with his expertise and with adequate time, to have found the critical data. It ignores the real world conditions under which examiners work." *Rohm & Haas Co. v. Crystal Chemical Co.*, 722 F.2d 1556, 1573, 220 U.S.P.Q. 289 (Fed. Cir. 1983), *cert. denied*, 469 U.S. 851 (1984). An applicant has a duty to not just disclose pertinent prior art references but to make a disclosure in such way as not to "bury" it within other disclosures of less relevant prior art. *See Golden Valley Microwave Foods Inc. v. Weaver Popcorn Co. Inc.*, 24 U.S.P.Q.2d 1801 (N.D. Ind. 1992); *Molins PLC v. Textron Inc.*, 26 U.S.P.Q.2d 1889, 1899 (D. Del. 1992); *Penn Yan Boats, Inc. v. Sea Lark Boats, Inc. et al.*, 175 U.S.P.Q. 260, 272 (S.D. Fl. 1972). It is unreasonable for Examiner to review all of the cited references thoroughly. By signing the accompanying 1449 forms, Examiner is merely acknowledging the submission of the cited references and indicating that only a cursory review has been made.

Examiner notes that *Therasense, Inc. v. Becton, Dickinson and Co.*, 649 F.3d 1276 (Ct. App. 2011) (en banc) has significantly restricted the infringement defense of inequitable conduct. A defendant must show that the patent in question would not have been issued but for undisclosed information, and that the patentee had the intent to

deceive. Examiner suggests that future Information Disclosure Statements cite only the most relevant/inclusive references or portions thereof.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-13, 15-19 and 26-35 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

As per claim 1, lines 7-8, “a control module mounted on the circuit board and coupled to the address and control ports of the memory devices” the connection and/or relationship between “the control module” and “the system memory controller and/or the system memory bus” are unclear. Appropriate correction and/or clarification are required.

As per claims 9 and 34, “the self-test data patterns be transmitted to the memory devices in test data signals that have at least one of a programmable slew rate and a programmable peak value that is programmed to correspond to characteristics of write data signals from the system memory controller” what is meant by “transmitted...in test

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data signals that have at least one of a programmable slew rate and a programmable peak value”? Appropriate correction and/or clarification are required.

As per claim 13, lines 13-16, “the control module generates first self-test address and control signals and provides the first self-test address and control signals to the memory devices; the control module further generates second self-test address and control signals and provides the second self-test address and control signals to the memory device” the interrelationship and/or interconnection between “the first self-test address and control signals” and “the second self-test address and control signals” are unclear, and during the self-test mode, the connection and/or relationship between “the memory devices” and “the system memory controller and/or system memory bus” are unclear;

lines 17-27, the inconsistent used of “test data patterns” in lines 17-18, “check data patterns” in line 26, and “expected data patterns” in line 27 are confusing. Are they the same or different?

Lines 23-24, “wherein the one or more memory devices output a respective n-bit wide read data signal in response to the second self-test address and control signals” is not clear whether “the one or more memory devices output the respective n-bit wide read data signal in response only to the second self-test address and control signals” or response to both the first and second self-test addresses and control signals”?

Lines 23-27, “wherein the one or more memory devices output a respective n-bit wide read data signal in response to the second self-test address and control signals,

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and wherein the each respective data handler receives the n-bit wide read data signal and check data patterns in the respective n-bit wide read data signal with respect to expected data patterns” is not clear as to what it meant. How the memory devices are tested and/or what to do with the response and/or the results of the first and second self-test are unclear. Appropriate correction and/or clarification are required.

As per claim 15, this claim is an improper dependent claim because claim 14 has been canceled. Appropriate correction is required.

As per claims 2-8, 10-12, 16-19, 26-33 and 35, these claims are also rejected because they dependent upon the rejected base claim.

Response to Amendment

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any

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extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

References Cited by Examiner

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. (See the attached form 892).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday from 8:00am-4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, April Y. Blair can be reached on 571-270-1014. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

Notice of References Cited	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-7,631,153 B2	12-2009	Mahrla; Peter	G06F13/28	365/185.33
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner
G06F3/0683, G11C5/04, G11C29/12, G11C7/1006, G11C7/109, G11C8/12	10/12/16	pc

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
714	718, 724, 733, 734, 736, E11.056, E11.169, 42	2/28/16	pc
365	201	2/28/16	pc
716	136	2/28/16	pc

SEARCH NOTES		
Search Notes	Date	Examiner
EAST-CPC Searched.	10/12/16	pc
Inventor Searched.	10/12/16	pc

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L2	3482	((memory devices) or (memory units) or (memory module)) and ((printed circuit board) or (PCB)) or (circuit board)) and ((data module) or (BIST logic units) or (data handlers) or (data controllers) or (data managements)) and ((control module) or (memory controller) or (master test unit))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:07
L3	198237	(system memory controller) or (system controller) or (memory system controller)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:09
L4	8135	(provid\$3) and ((memory address\$2 and control signal\$1 and data patterns or data) and (memory device\$1 or memory subsystem) and (normal mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:09
L5	19589	("BIST") or ("logic BIST") or (in-circuit test) or ("MBIST") or (built-in self test) or (built-in self-test)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:18
L6	114477	((data module) or (data handler\$1) or (data management\$1) or (data controller) or (data manager\$1) or (controller) or (gate or switch or multiplexer or MUX)) and ((normal mode or normal operation mode or normal mode operation or functional mode) or (self-test mode or test mode or test mode operation))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:19
L8	114	I2 and I3 and I6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:56
L9	11	I8 and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	ADJ	ON	2016/10/12 13:56

			IBM_TDB			
L10	1	I9 and I4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:56
L11	36509	(memory rank\$1) or (rank\$1 or block\$1) same (memory module or memory chip)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:57
L13	9116	(memory address\$2 and control signal\$1) and (normal or functional) adj (mode or operation)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 13:59
L14	15	(self-test address and control signals) and (self-test mode or test mode)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 14:00
L15	3	I12 and (I13 or I14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 14:01
L16	8	I9 and I11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 14:07
L17	3	I16 and(I13 or I14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 14:09
L18	3	I16 and (I13 or I14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 14:09
L19	3	I9 not I16	US-PGPUB; USPAT; USOCR; FPRS; EPO;	ADJ	ON	2016/10/12 15:27

			JPO; DERWENT; IBM_TDB			
L20	2150972	((Lee).inv.) or ((Bhakta).inv.) or ((Choi).inv.) or ((Netlist).asn.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:36
L21	4	I16 and I20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:36
L22	4	I9 and I20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:36
L23	4	I21 and I22	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:36
L24	6	I8 and (I11) and (I13 or I14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:38
L25	114	(I24 or I9 or I16 or I8) and ((circuit board) or (printed circuit board) or (surface) or (substrate))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:40
L26	3	(I24 or I16) and ((circuit board) or (printed circuit board) or (surface) or (substrate)) and (computer system) and (memory module) and (system memory controller) and (system memory bus)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2016/10/12 15:42
L27	1	(13/952599).APP.	US-PGPUB; USOCR	ADJ	ON	2016/10/12 15:45
L28	1	("9128632").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2016/10/12 16:01
L29	8	16 AND ((G06F3/0683 OR G11C5/04 OR G11C29/12 OR G11C7/1006 OR G11C7/109 OR G11C8/12).CPC. OR	US-PGPUB; USPAT; USOCR;	ADJ	ON	2016/10/12 16:08

		(714/718 OR 714/733 OR 714/734 OR 714/42).CCLS. OR (G06F12/00 OR G06F3/00 OR G06F3/06).IPCR.)	FPRS; EPO; JPO; DERWENT; IBM_TDB			
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Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	1	of	5		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	5655113	08-05-1997	Leung	
	2.	5878051	03-02-1999	Sharma	
	3.	5936900	08-10-1999	Hii	
	4.	6321320	11-20-2001	Fleischman	
	5.	6747887	06-08-2004	Halbert et al.	
	6.	6832303	12-14-2004	Tanaka	
	7.	6934205	08-23-2005	Pandey et al.	
	8.	7093066	08-15-2006	Klein	
	9.	7168005	01-23-2007	Adams	
	10.	7203873	04-10-2007	Adams	
	11.	7379316	05-27-2008	Rajan	
	12.	7392442	06-24-2008	Averbuj	
	13.	7464225	12-09-2008	Tsern	
	14.	7865674	01-04-2011	Gower et al.	
	15.	8089795	01-03-2012	Rajan	
	16.	8189328	05-29-2012	Kanapathippillai	
	17.	8233303	07-31-2012	Best	
	18.	20040006729	01-08-2004	Pendurkar	
	19.	20080104352	05-01-2008	Talbot	

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
	20.	Inter Partes Review Case No. IPR2014-00970, Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on June 18, 2014		
	21.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1007 to Petition for Inter Partes Review, “Declaration of Donald Albert, Ph.D,” filed on June 18, 2014		
	22.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2002 to Petition for Inter Partes Review, “Excerpt from Merriam-Webster’s Collegiate Dictionary,” filed on June 18, 2014		
	23.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2003 to Petition for Inter Partes Review, “Excerpt from Webster Handy College Dictionary,” filed on June 18, 2014		
	24.	Inter Partes Review Case No. IPR2014-00970, Patent Owner’s Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on September 24, 2014		

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
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NON PATENT LITERATURE DOCUMENTS			
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	32.	“IEEE Standard Test Access Port and Boundary-Scan Architecture” (“JTAG 2001”, IEEE-SA Standards Board, IEEE Std1149.1-2001, June 14, 2001.	
	33.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2001 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “Webster’s II New College Dictionary,” filed on August 23, 2014	
	34.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2002 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “IEEE Standard Dictionary of Electrical and Electronics Terms,” filed on August 22, 2014.	
	35.	Inter Partes Review Case No. IPR2014-01373, Exhibit 2003 to Petition for Inter Partes Review, U.S. Patent No. 8,001,434, “JEDEC Standard, Fully Buffered DIMM (FBDIMM) DfX Design for Validation and Test,” filed on August 22, 2014.	
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	38.	Inter Partes Review Case No. IPR2014-01374, Petition for Inter Partes Review of Claims 1-35 of U.S. Patent No. 8,001,434, filed on August 23, 2014.	
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	43.	Inter Partes Review Case No. IPR2014-01029, Exhibit 1008 to Petition for Inter Partes Review, "Declaration of Charles J. Neuhauser, Ph.D. under 37 C.F.R. § 1.68," filed on June 24, 2014.	
	44.	Inter Partes Review Case No. IPR2014-01029, Supplemental Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on July 23, 2014.	
	45.	Inter Partes Review Case No. IPR2014-01029, Patent Owner's Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on October 17, 2014.	
	46.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014.	
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	51.	Inter Partes Review Case No. IPR2014-00882, Corrected Petition for Inter Partes Review of U.S. Patent No. 7,881,150, filed on July 8, 2014.	
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	53.	Inter Partes Review Case No. IPR2014-00883, Corrected Petition for Inter Partes Review of U.S. Patent No. 8,081,536, filed on July 8, 2014.	
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				Examiner Name	Unknown /Phung Chung/
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	58.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.'s Invalidity Contentions, dated June 6, 2014		
	59.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits A.1-A.10 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
	60.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits B.1-B.8 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
	61.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
	62.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F.1-F.5 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
	63.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G.1-G.6 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
	64.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated June 6, 2014.		
	65.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Diablo Technologies, Inc.'s Invalidity Contentions, dated June 6, 2014.		
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				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	4	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	73.	6918072	07-12-2005	Micron Technology, Inc.	
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	89.	8250295	8/21/12	Amidi et al.	

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	90.	Bayard "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, vol. 47, Issue 5, pp. 1288-1296.	

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<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE							
Final	Original	02/21/2016	10/17/2016						
	1	v	v						
	2	v	v						
	3	v	v						
	4	v	v						
	5	v	v						
	6	v	v						
	7	v	v						
	8	v	v						
	9	v	v						
	10	v	v						
	11	v	v						
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	15	v	v						
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	35		v						

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	2.	Inter Partes Review Case No. IPR2014-01372, Corrected Petition For Inter Partes Review, filed September 22, 2014	
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	14.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Opposition to Petitioner's Motion to Exclude, filed October 26, 2015	
	15.	Inter Partes Review Case No. IPR2014-01372, Patent Owner's Fourth Amended Exhibit List, filed October 26, 2015	
	16.	Inter Partes Review Case No. IPR2014-01372, IPR2014-01372 SMART Response - PO Motion To Exclude Exhibits, filed October 26, 2015	
	17.	Inter Partes Review Case No. IPR2014-01372, IPR2014-01372 SMART Response - PO Motion To Exclude Reply Portions, filed October 26, 2015	
	18.	Inter Partes Review Case No. IPR2014-01372, Reply In Support of Patent Owner Motion to Exclude, filed November 2, 2015	
	19.	Inter Partes Review Case No. IPR2014-01372, SMART Reply in Support of Motion to Exclude, filed November 2, 2015	

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	20.	Inter Partes Review Case No. IPR2014-01372, Record of Oral Hearing, filed December 18, 2015		
	21.	Inter Partes Review Case No. IPR2014-01372, Ex 1026 - Websters New Collegiate, filed September 10, 2015		
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	23.	Inter Partes Review Case No. IPR2014-01372, Ex 1028 - Portola Packaging Reexam Guidance, filed September 10, 2015		
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	30.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2009, 'Excerpts from Merriam Webster's Collegiate Dictionary, Tenth Edition (2001),' filed December 18, 2014		
	31.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2010, 'Excerpts from the American Webster Hand College Dictionary (1981),' filed December 18, 2014		
	32.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2011, 'Excerpts from Merriam Webster's Collegiate Dictionary, Eleventh Edition (2003),' filed December 18, 2014		
	33.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2012, 'Cross-Examination transcript of Nader Bagherzadeh, Ph.D. , 5/21/2015,' filed June 15, 2015		
	34.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2013, 'Report to the Association for Computing,' filed June 15, 2015		
	35.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2014, 'MorphoSys - Case Study,' filed June 15, 2015		
	36.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2016, 'Declaration of Nadar Baghezadeh, Ph.D. ISO Motion to Stay,' filed June 15, 2015		
	37.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2017, 'Design and Implementation,' filed June 15, 2015		
	38.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2018, 'Deposition of Doanld Alpert, Ph.D., 2/12/15,' filed June 15, 2015		
	39.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2019, 'Declaration of Carl Sechen, Ph.D.,' filed June 15, 2015		

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	41.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2021, 'Excerpts from Digital Design Principles,' filed June 15, 2015	
	42.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2022, 'CMOS LVSI,' filed June 15, 2015	
	43.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2025, 'References reviewed by Examiner,' filed June 15, 2015	
	44.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2026, 'Excerpts from Digital Integrated Circuits,' filed June 15, 2015	
	45.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2027, 'Transcript of the September 30, 2015 Conference Call,' filed October 2, 2015	
	46.	Inter Partes Review Case No. IPR2014-01372, Exhibit 2028, 'Attachment to Oct. 19, 2015 Patent Owner Motion to Exclude,' filed October 19, 2015	
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/PC/	2.	2006/0262586 A1		Solomon, J. et al.	
/PC/	3.	2008/0162790 A1			

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/PC/	5.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 7,881,150, IPR Case Number IPR2015-01020, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
/PC/	6.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-01372, Final Written Decision, issued March 9, 2016	
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/PC/	11.	Inter Partes Review Case No. IPR2014-01374, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
/PC/	12.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-00970, Decision Denying Petitioner's Request for Rehearing, issued April 27, 2016	
/PC/	13.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January 13, 2016	
/PC/	14.	Inter Partes Review Case No. IPR2014-00971, Decision - Petitioner's Request for Rehearing, issued April 27 2016	
/PC/	15.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision, reissued April 27, 2016	
/PC/	16.	Inter Partes Review Case No. IPR2014-00971, Netlist Inc.'s Notice of Appeal, filed June 29, 2016	

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				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189 2117
				Examiner Name	Unknown /Phung Chung/
Sheet	2	of	2	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
/PC/	17.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
/PC/	18.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
/PC/	19.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	

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				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	1	of	2		

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Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6011710	01-04-2000	Wiggers	
	2.	20010008006	07-12-2001	Klein	
	3.	20050257109	11-17-2005	Averbuj	
	4.	20070070669	03-29-2007	Tsern	

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	5.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,516,185, IPR Case Number IPR2014-01029, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	6.	Petition for Inter Partes Review filed on 6/22/2014 for U.S. Patent No. 7,881,150, IPR Case Number IPR2014-00882 and IPR Case Number IPR2014-01011, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	7.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,081,536, IPR Case Number IPR2014-00883, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	8.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,001,434, IPR Case Number IPR2014-00970, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	9.	Petition for Inter Partes Review filed on 6/24/2014 for U.S. Patent No. 8,359,501, IPR Case Number IPR2014-00971, and all associated documents including cited references and expert declarations, available at https://ptabtrials.uspto.gov	
	10.	Altera, ACEX iK, Programmable Logic Device Family, Data Sheet, May 2003, Ver 3.4	
	11.	Horowitz, "The Art of Electronics," Cambridge Univ. Press, 1989, selected pages	
	12.	Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379	
	13.	Jacob, Bruce L.; "Synchronous DRAM Architectures, Organizations, and Alternative Technologies". University of Maryland, Dec. 10, 2002.	
	14.	JEDEC Standard No. 21-C Section 4.5.7, 168 Pin Registered SDRAM DIMM Family, Release 7	
	15.	JEDEC 21-C, Section 4.6.1, 278 Pin Buffered SDRAM DIMM Family	
	16.	JEDEC Standard No. 21-C Section 4.1.2.5, Appendix E, "Specific PD's for Synchronous DRAM (SDRAM)," pp. 1-25,	
	17.	JEDEC Standard, "Fully Buffered DIMM (FBDIMM): DfX Design for Validation and Test," JESD82-28, February 2008	

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	18.	Reese, "Introduction to Logic Synthesis using Verilog HDL," Morgan&Claypool Publishers, 2006, pp. 1-28	

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	1.	PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification JEDEC, Standard No. 21-C, Revision 1.3, Jan. 2002, pages 4.20.4-1	

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	1.	JEDEC Global Standards for the Microelectronics Industry, Standards http://www.jedec.org/standards-documents/about-JEDEC-standards (2014)	
	2.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Response pursuant to 37 C.F.R. § 42.120, filed March 17, 2015	
	3.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Reply to Patent Owner Response, June 8, 2015	
	4.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Request for Oral Argument, filed July 2, 2015	
	5.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Motion for Observation re Cross-Examination of Dr. Donald Alpert, filed July 2, 2015	
	6.	Inter Partes Review Case No. IPR2014-00970, Patent Owner's Second Amended Exhibit List, filed July 2, 2015	
	7.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Request for Oral Argument, filed July 2, 2015	
	8.	Inter Partes Review Case No. IPR2014-00970, Order: Request for Oral Argument, issued July 7, 2015	
	9.	Inter Partes Review Case No. IPR2014-00970, Petitioner Response to Netlist's Motion for Observations on Cross-Examination, filed July 13, 2015	
	10.	Inter Partes Review Case No. IPR2014-00970, Record of Oral Hearing, issued August 24, 2015	
	11.	Inter Partes Review Case No. IPR2014-00970, Final Written Decision - 35 USC 318 and 37 CFR 42.73, issued December 14, 2015	
	12.	Inter Partes Review Case No. IPR2014-00970, Petitioner's Request for Rehearing, filed January 13, 2016	
	13.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1009 to Petitioner's Reply to Patent Owner Response, "Declaration of Donald Alpert PhD," filed June 8, 2015	
	14.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1010 to Petitioner's Reply to Patent Owner Response, "Deposition Transcript of Dr. Carl Sechen," filed June 8, 2015	
	15.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1011 to Petitioner's Reply to Patent Owner Response, "American Heritage Dictionary," filed June 8, 2015	
	16.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1012 to Petitioner's Reply to Patent Owner Response, "Webster's Encyclopedic Unabridged Dictionary," filed June 8, 2015	
	17.	Inter Partes Review Case No. IPR2014-00970, Exhibit 1013 to Petitioner's Reply to Patent Owner Response, "IEEE Standard 1450," filed June 8, 2015	
	18.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2004 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Defendants' Proposed Preliminary Claim Constructions and Extrinsic Evidence, Case No. 13-cv-05889-YGR July 28, 2014," filed March 17, 2015	

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	19.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2005 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Enlarged Figure 3 from Exhibit 1004," filed March 17, 2015	
	20.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2006 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Enlarged Figure 3 from Exhibit 1004 to Petition for Inter Partes Review of U.S. Patent No. 8,001,434, filed on June 18, 2014, 'Huang et al, "An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers," VLSI Design 2011, p. 379," filed March 17, 2015	
	21.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2007 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Color copy of page 35 to Patent Owner's Preliminary Response filed September 24, 2014," filed March 17, 2015	
	22.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2008 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Color copy of page 38 to Patent Owner's Preliminary Response filed September 24, 2014," filed March 17, 2015	
	23.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2009 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from American Heritage Dictionary," filed March 17, 2015	
	24.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2010 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Webster's Third New International Dictionary," filed March 17, 2015	
	25.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2011 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from New Riverside University Dictionary," filed March 17, 2015	
	26.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2012 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Declaration of Carl Sechen, Ph.D.," filed March 17, 2015	
	27.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2013 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Cross-Examination transcript of Donald Alpert, Ph.D., taken February 12, 2015," filed March 17, 2015	
	28.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2014 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Design of High-Performance Microprocessor Circuits," filed March 17, 2015	
	29.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2015 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Logic and Computer Design Fundamentals," filed March 17, 2015	
	30.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2016 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Digital Design, Principles an Practices," filed March 17, 2015	

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	31.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2017 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Excerpts from Digital Logic with Verilog Design," filed March 17, 2015	
	32.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2018 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Decision on Institution of IPR for 8,001,434 (IPR2014-01372)," filed March 17, 2015	
	33.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2019 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Recent Advances in BIST," filed March 17, 2015	
	34.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2020 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Decision on Institution of IPR for 8,359,501 (IPR2014-01375)," filed March 17, 2015	
	35.	Inter Partes Review Case No. IPR2014-00970, Exhibit 2021 to Patent Owner's Response pursuant to 37 C.F.R. § 42.120, "Cross-Examination transcript of Dr. Alpert, June 24, 2015," filed March 17, 2015	
	36.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Response, filed March,17,2015	
	37.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's First Amended Exhibit List, filed March,18,2015	
	38.	Inter Partes Review Case No. IPR2014-00971, Motion to File Corrected Patent Owner Response, filed April,15,2015	
	39.	Inter Partes Review Case No. IPR2014-00971, Corrected Patent Owner Response, filed April,15,2015	
	40.	Inter Partes Review Case No. IPR2014-00971, Petitioner's Reply to Patent Owner Response, filed June,8,2015	
	41.	Inter Partes Review Case No. IPR2014-00971, Patent Owner's Request for Oral Argument, filed July,2,2015	
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	43.	Inter Partes Review Case No. IPR2014-00971, Petitioner Request for Oral Argument, filed July,2,2015	
	44.	Inter Partes Review Case No. IPR2014-00971, Order Request for Oral Argument, filed July,7,2015	
	45.	Inter Partes Review Case No. IPR2014-00971, Petitioner Response to Netlist's Motion for Observations on Cross-Examination, filed July,13,2015	
	46.	Inter Partes Review Case No. IPR2014-00971, Record of Oral Hearing, filed August,24,2015	
	47.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December,14,2015	
	48.	Inter Partes Review Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January,13,2016	

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	49.	Inter Partes Review Case No. IPR2014-00971, Exhibit 1009, 'Declaration of Donald Alpert PhD,' filed June,8,2015	

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Application Serial No: 14/229,844**

Attorney Docket No: NETL.046C3

Customer Number: 79141

Date of Electronic Notification: 10/20/2016

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REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	14/229,844	Filing Date	2014-03-29	Docket Number (if applicable)	N3044.10010US04	Art Unit	2189
First Named Inventor	Hyun Lee			Examiner Name	CHUNG, PHUNG M		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV.

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☒ Amendment/Reply

☐ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 505963

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

Applicant Signature

Doc code: RCEX

Doc description: Request for Continued Examination (RCE)

PTO/SB/30EFS (07-14)

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Signature of Registered U.S. Patent Practitioner			
Signature	Jamie J. Zheng/	Date (YYYY-MM-DD)	2017-03-20
Name	Jamie J. Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

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The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/229,844
Filed : March 29, 2014
For : A Self-Test Memory Module and
Method of Operation
Examiner : CHUNG, PHUNG M
Art Unit : 2189
Conf. No. : 1018
Docket No. : N3044.10010US04

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

March 20, 2017

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND REQUEST FOR CONTINUED EXAMINATION**Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir or Madam:

Applicant submits the following amendments and remarks in response to Final Office Action Dated October 20, 2016 in the above-identified patent application.

Applicant hereby petitions for a 2-month extension of time to respond to the Final Office Action.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Amendments to the Claims: Begin on page 2 of this paper.

Remarks: Begin on page 9 of this paper.

In the Claims:

1. (Currently Amended) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements; and

a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and

wherein the memory module is operable in any of a plurality of modes including a normal mode and a test mode;

wherein, during the normal mode, the control module receives system address and control signals from the system memory controller and provides memory address and control signals to the memory devices according to the system address and control signals, and the data module propagates data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals; and

wherein, during the test mode, the control module generates self-test address and control signals without using any test procedure driven by the system memory controller, the control module provides the self-test address and control signals to the address and control ports of the memory devices, and the data module ~~generates self-test data patterns for testing the memory devices,~~ isolates the memory devices from the system memory controller, and transmits ~~the~~ self-test data patterns to the data ports of the memory devices, wherein the self-test data patterns are generated by the data handler logic elements according to commands from the control module, and wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals received from the control module.

2. (Currently Amended) The memory module of claim 1, wherein the memory devices are arranged in at least one rank, each rank having a set of memory devices

organized in a plurality of groups, each group including one or more memory devices, wherein the system memory bus has a width of N bits and is organized in a plurality of n-bit sections corresponding, respectively, to groups of one or more memory devices in each rank, wherein the data module includes a plurality of data handlers, each respective data handler being coupled between a respective n-bit section of the system memory bus and a respective group of one or more memory devices, the each respective data handler generating respective n-bit wide self-test data patterns and transmitting the respective n-bit wide self-test data patterns to the respective group of one or more memory devices.

3. (Previously Presented) The memory module of claim 2, wherein each memory device is n-bit wide.

4. (Currently Amended) The memory module of claim 2, wherein the control module generates additional self-test address and control signals and provides the additional self-test address and control signals to the address and control ports of the memory devices during the test mode, the additional self-test address and control signals causing the memory devices to output a N-bit wide read data signal, and wherein the each respective data handler receives a respective n-bit section of the N-bit wide read data signal ~~and checks a data pattern in the respective n-bit section of the N-bit wide read data signal with respect to an expected data pattern.~~

5. (Currently Amended) The memory module of claim 4, wherein the each respective data handler is configured to generate expected data pattern ~~is generated in the respective data handler~~ after the control module generates the additional self-test address and control signals and to check the respective n-bit section of the N-bit wide read data signal with respect to the expected self-test data patterns.

6. (Currently Amended) The memory module of claim 1, wherein the ~~the~~ control module generates the self-test addresses and control signals according to a Joint Electron Devices Engineering Council (JEDEC) standard memory protocol, and wherein the data module transmits the self-test data patterns to the memory devices according to the JEDEC standard memory protocol.

7. (Currently Amended) The memory module of claim 5-4, wherein the data module stores failure information when a test failure is indicated by the respective n-bit section of the N-bit wide read data signal data pattern not being in agreement with the expected data pattern.
8. (Previously Presented) The memory module of claim 7, wherein the data module is further configured to report the failure information to the control module.
9. (Cancelled)
10. (Previously Presented) The memory module of claim 1, wherein the data module generates and transmits the self-test data patterns to the memory devices in response to a write command from the control module, and wherein the data module read back test data patterns from the memory devices in response to a read command from the control module.
11. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers operate independently of each other without being in communication with each other, and wherein, during the test mode, the each respective data handler generates n-bit wide self-test data patterns to test a corresponding portion of a memory space of the memory module independently of the other data handlers.
12. (Previously Presented) The memory module of claim 2, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective group of one or more memory devices.
13. (Cancelled)
14. (Cancelled)
15. (Cancelled)
16. (Cancelled)

17. (Cancelled)
18. (Cancelled)
19. (Cancelled)
20. (Cancelled)
21. (Cancelled)
22. (Cancelled)
23. (Cancelled)
24. (Cancelled)
25. (Cancelled)
26. (Previously Presented) The memory module of claim 1, wherein the control module comprises a memory device controller and a test controller, wherein the memory device controller receives the system address and control signals from the system memory controller and the self-test address and control signals from the test controller.
27. (Previously Presented) The memory module of claim 26, wherein the control module further comprises a register coupled to the address and control ports of the memory devices, and wherein the memory device controller sends address and control information from either the system address and control signals or from the self-test address and control signals to the register depending on whether the memory module is in the normal mode or test mode.
28. (Previously Presented) The memory module of claim 2, wherein the each respective data handler comprises a data handler logic element that generates test data patterns and a switch coupled to the data handler logic element and to the respective n-bit section of the system memory bus, and wherein the switch is configured to selectively input to the respective group of one or more memory devices either write data signals

from the system memory controller or test data patterns from the data handler logic element depending on whether the memory module is in the normal mode or test mode.

29. (Currently Amended) The memory module of claim 28, wherein the switch is further configured to receive read data signals from the ~~the~~ respective group of one or more memory devices and sends the read data signals to the system memory controller or to the data handler logic element depending on whether the memory module is in the normal mode or the test mode.

30. (Cancelled)

31. (Cancelled)

32. (Cancelled)

33. (Cancelled)

34. (Cancelled)

35. (Cancelled)

36. (New) The memory module of claim 2, wherein each data handler is coupled to two memory devices in each rank.

37. (New) A memory module accessible in a computer system by a system memory controller via a system memory bus, the memory module comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports, the data ports including a plurality of groups of data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including a plurality of data handlers, wherein each respective data handler is coupled between a respective segment of the system bus and a corresponding group of the data ports, wherein the each respective data handler includes a respective data handler logic element and a switch circuit, wherein the memory module is operable in any of a plurality of modes including a

normal mode and a test mode, wherein the switch circuit is configured to propagate data signals between the respective segment of the system bus and the corresponding group of data ports during the normal mode, and wherein the switch circuit is configured to propagate self-test data patterns between the data handler logic element and the corresponding group of data ports during the test mode; and

a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus, the control module including self-test logic, control logic and a register;

wherein, during the normal mode, the control logic provides system address and control signals from the system memory controller to the register, and the register provides registered system address and control signals to the memory devices; and

wherein, during the test mode, the self-test logic generates self-test address and control signals, the control logic provides the self-test address and control signals to the register, and the register provides registered self-test address and control signals to the address and control ports of the memory devices.

38. (New) The memory module of claim 37, wherein the memory devices are arranged in at least one rank, each rank having a width of N bits and including a set of memory devices organized in a plurality of groups, each group including one or more memory devices, wherein the system memory bus has a width of N bits and is organized in a plurality of n-bit sections corresponding to respective groups of memory devices in each rank, wherein each respective data handler is coupled between a respective n-bit section of the system memory bus and a respective group of one or more memory devices, the each respective data handler generating respective n-bit wide self-test data patterns during the test mode and transmitting the respective n-bit wide self-test data patterns to the respective group of one or more memory devices.

39. (New) The memory module of claim 38, wherein each memory device is n-bit wide and each group of one or more memory devices includes a single memory device.

40. (New) The memory module of claim 38, wherein each group of one or more memory devices includes two memory devices.

41. (New) The memory module of claim 38, wherein the plurality of data handlers operate independently of each other without being in communication with each other, and wherein, during the test mode, the each respective data handler generates n-bit-wide self-test data patterns to test the corresponding group of one or more memory devices independently of the other data handlers.

42. (New) The memory module of claim 38, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective group of one or more memory devices.

43. (New) The memory module of claim 37, wherein the self-test logic generates the self-test addresses and control signals according to a Joint Electron Devices Engineering Council (JEDEC) standard memory protocol.

44. (New) The memory module of claim 37, wherein the each respective data handler is configured to generate and transmit the self-test data patterns to the corresponding group of data ports in response to a write command from the control module.

45. (New) The memory module of claim 37, wherein the each respective data handler is configured to receive the self-test data patterns from the corresponding group of data ports in response to a read command from the control module.

46. (New) The memory module of claim 45, wherein the each respective data handler is configured to check the self-test data patterns with respect to expected data patterns.

47. (New) The memory module of claim 46, wherein the each respective data handler is configured to store failure information when a test failure is indicated by the self-test data pattern not being in agreement with the expected data pattern.

48. (New) The memory module of claim 47, wherein the each respective data handler is configured to report the failure information to the control module.

REMARKS

This amendment is in response to the Final Office Action dated October 20, 2016.
In the Office Action, the Examiner:

- rejected claims 1-13, 15-19 and 26-35 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph.

Request for Continued Examination

Applicant files herewith a request for continued examination

Claim Amendment

Claims 9, 13, 15-19 and 30-35 are cancelled. Claims 1-2, 4-7, 11 and 29 are amended. New claims 36-48 are presented. No new matter is added. After entry of this amendment, the pending claims are 1-8, 10-12, 26-29, and 36-48.

Rejection of Claims 1-13, 15-19 and 26-35 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), Second Paragraph

Claim 9 is cancelled. The amendments to claims 1-2, 4-7 and 11 address the rejections of claims 1-12 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph. Therefore, claim 1 as amended, and its dependent claims, claims 2-12 and 26-29, are now allowable.

Claims 13, 15-19 and 30-35 are cancelled. Their rejections thereto are now moot.

New claims 36-48 do not have the issues raised by the Examiner related to the rejection of claims 1-13, 15-19 and 26-35 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph. Therefore, new claims 36-48 are also allowable.

Applicant submits that claims 1-8, 10-12, 26-29, and 36-48 are in condition for allowance and respectfully requests such action. The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any issues.

Respectfully submitted,

Date: March 20, 2017

/Jamie J. Zheng/

51167

Jamie J. Zheng

(Reg. No.)

Customer No. 79141

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	N3044.10010US04			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 2 months with \$0 paid	1252	1	600	600
Miscellaneous:				
Total in USD (\$)				600

Electronic Acknowledgement Receipt

EFS ID:	28686016
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	N3044.10010US04
Receipt Date:	21-MAR-2017
Filing Date:	29-MAR-2014
Time Stamp:	01:12:30
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 600
RAM confirmation Number	032117INTEFSW01145600
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	2017-03-20_RCE_NETL046C3.pdf	1349914	no	3
			a08c39d4707dc711535bf512918ec47a9b32baf6		
Warnings:					
Information:					
2	Response After Final Action	2017-03-20_Amendment_RCE_NETL046C3.pdf	347759	no	10
			6423dd9276591ff9de0046c0dc69af468eab7229		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30356	no	2
			8fb7989991e0e92ecb64ebaf6daace816f2f804f		
Warnings:					
Information:					
Total Files Size (in bytes):			1728029		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/229,844	Filing Date 03/29/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO**APPLICATION AS FILED – PART I**

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	03/21/2017	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 28	Minus	** 28	= 0	X \$40 =	0
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	X \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'L FEE

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
CAROL BARNES

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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		01 FC : 1801		1,200.00	DA	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	7210059		2007-04-24	Jeddeloh	
	2	6011748		2000-01-04	Lepejian, et al.	
	3	6639820		2003-10-28	Khandekar, et al.	
	4	6317352		2001-11-13	Halbert, et al.	
	5	6108798		2000-08-22	Heidel, et al.	
	6	3588831		1971-06-28	Figueroa	
	7	3618041		1971-11-02	Horikoshi	
	8	6271060		2001-08-07	Zandman, et al.	
If you wish to add additional U.S. Patent citation information please click the Add button.						Add

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

U.S.PATENT APPLICATION PUBLICATIONS							Remove	
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear		
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	1							
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NON-PATENT LITERATURE DOCUMENTS							Remove	
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.						T ⁵
	1	Inter partes review Case No. IPR2017-00561, Petition For Inter Partes Review Of U.S. Patent No. 8,001,434, filed January 5, 2017.						
	2	Inter partes review Case No. IPR2017-00561, Exhibit 1010 to Petition for Inter Partes Review, "EMC and the Printed Circuit Board", filed January 5, 2017.						
	3	Inter partes review Case No. IPR2017-00561, Exhibit 1012 to Petition for Inter Partes Review, "Microsoft Computer Dictionary (5th Ed.)", filed January 5, 2017.						
	4	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "JEDEC PC2-5300 RDIMM PCB Plot", filed January 5, 2017.						

**INFORMATION DISCLOSURE
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Application Number	14229844
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Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

5	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "The American Heritage Dictionary", filed January 5, 2017.
6	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Memory Systems", filed January 5, 2017.
7	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Designing Embedded Hardware", filed January 5, 2017.
8	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Micron DDR3 SDRAM UDIMM Specifications", filed January 5, 2017.
9	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "JEDEC PC2-5300 RDIMM Assembly", filed January 5, 2017.
10	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Micron - Technical Note TN4720. DDR2 (Point-to-Point) Package Sizes and Layout Basics", filed January 5, 2017.
11	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Declaration of Dr. Pinaki Mazumder", filed January 5, 2017.
12	Inter partes review Case No. IPR2017-00562, Petition For Inter Partes Review Of U.S. Patent No. 8,359,501, filed January 3, 2017.
13	Inter partes review Case No. IPR2017-00560, Petition For Inter Partes Review Of U.S. Patent No. 8,689,064, filed January 3, 2017.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jamie J. Zheng/	Date (YYYY-MM-DD)	2017-03-24
Name/Print	Jamie J. Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	28736829
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10010US04
Receipt Date:	24-MAR-2017
Filing Date:	29-MAR-2014
Time Stamp:	19:11:19
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	IPR2017-00561_01-05-2017_IP R-Petition.pdf	12333041	no	80
			401fb18bc4e2e8ccc7e063f9872f7428b34d e123		

Warnings:

Information:					
2	Non Patent Literature	1_1010_IPR2017-00561_EMCan dthePrintedCircuitBoard.pdf	9431297	no	62
			013195a31e288990f600472291ccb2e8cd72e70d		
Warnings:					
Information:					
3	Non Patent Literature	2_1012_IPR2017-00561_Micros oftComputerDictionary5thEd. pdf	757530	no	7
			423ffd1309ce03bc2b9a5aab0b67df230073df94		
Warnings:					
Information:					
4	Non Patent Literature	3_1015_IPR2017-00561-JEDEC- PC2-5300-RDIMM-PCB-Plot.pdf	5031229	no	10
			0955ef3b6f3152eeef818f6d4b8f4d51b41ca05c2		
Warnings:					
Information:					
5	Non Patent Literature	9_1007_IPR2017-00561-TheAmericanHeritageDictionary.pdf	13338494	no	8
			175d2038e40a0cc75a8c36525fc9f299e94cc17a		
Warnings:					
Information:					
6	Non Patent Literature	10_1008_IPR2017-00561_Mem orySystems.pdf	19855239	no	86
			42d3f4f98ea00733cb09708714361fa4e056b318		
Warnings:					
Information:					
7	Non Patent Literature	11_1009_IPR2017-00561_Desig ningEmbeddedHardware.pdf	7784608	no	67
			ee3e3bdf2b1551efb8721e80d67f27a64dccc4ee4		
Warnings:					
Information:					
8	Non Patent Literature	13_1013_IPR2017-00561_Micro n-DDR3SDRAMUDIMMSpecificati ons.pdf	7833891	no	11
			55aa6ce9451439efe7f50e1caa38c6e7804bb8a7c		
Warnings:					
Information:					

9	Non Patent Literature	14_1013_IPR2017-00561_JEDE CPC2-5300RDIMMAsembly. pdf	1207380	no	2
			cbf1620adaaa5be86bc92072c9581f4cfaea 2fc0		
Warnings:					
Information:					
10	Non Patent Literature	24_1013_IPR2017-00561_Decla ration-of-DrMazumder.pdf	18674598	no	127
			ae19e14ef59b14b74894a622e2589b15e0bb 246ed		
Warnings:					
Information:					
11	Information Disclosure Statement (IDS) Form (SB08)	2017-03-24_IDS_N3044-10010 US04.pdf	1036007	no	6
			b51f72ef8b5f59f95f12ae5df846807201d3 180		
Warnings:					
Information:					
12	Non Patent Literature	23_1013_IPR2017-00561_Micro nTechnicalNoteTN4720DDR2P oint-to- PointPackageSizesandLayoutB asics.pdf	7785402	no	12
			4f3f7a2aef07e6490d1dfea6ddfafa66e825 67		
Warnings:					
Information:					
13	Non Patent Literature	IPR2017-00560_01-03-2017_IP R-Petition.pdf	10319523	no	33
			cc3b67e961507c1bde6383a77b76390d0b 31630a		
Warnings:					
Information:					
14	Non Patent Literature	IPR2017-00562_01-03-2017_IP R-Petition.pdf	15650590	no	50
			61dda78dbdf52cf1fb9b111e40bc80ebed0 0b8e7		
Warnings:					
Information:					
Total Files Size (in bytes):			131038829		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	N3044.10010US04	1018

79141 7590 04/04/2017
Jamie J. Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

CHUNG, PHUNG M

ART UNIT	PAPER NUMBER
----------	--------------

2117

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

04/04/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com
infohub.j@gmail.com
d@mabr.com

Office Action Summary	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 3/21/17.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-8, 10-12, 26-29 and 36-48 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☒ Claim(s) 1-8, 10-12, 26-29 and 36 is/are allowed.
- 7) ☒ Claim(s) 37-48 is/are rejected.
- 8) ☐ Claim(s) _____ is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 3/29/14 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 7/13/16 and 3/24/17.
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.
- 4) ☐ Other: _____.

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 3/21/17 has been entered.

Information Disclosure Statement

An applicant's duty of disclosure of material information is not satisfied by presenting a patent examiner with "a mountain of largely irrelevant data from which he is presumed to have been able, with his expertise and with adequate time, to have found the critical data. It ignores the real world conditions under which examiners work." *Rohm & Haas Co. v. Crystal Chemical Co.*, 722 F.2d 1556, 1573, 220 U.S.P.Q. 289 (Fed. Cir. 1983), cert. denied, 469 U.S. 851 (1984). An applicant has a duty to not just disclose pertinent prior art references but to make a disclosure in such way as not to "bury" it within other disclosures of less relevant prior art. See *Golden Valley Microwave Foods Inc. v. Weaver Popcorn Co. Inc.*, 24 U.S.P.Q.2d 1801 (N.D. Ind. 1992); *Molins PLC v. Textron Inc.*, 26 U.S.P.Q.2d 1889, 1899 (D. Del. 1992); *Penn Yan Boats, Inc. v. Sea Lark Boats, Inc.* et al, 175 U.S.P.Q. 260, 272 (S.D. Fl. 1972). It is unreasonable for Examiner to review all of the cited references thoroughly. By signing the accompanying

1449 forms. Examiner is merely acknowledging the submission of the cited references and indicating that only a cursory review has been made.

Examiner notes that *Therasense, Inc. v. Becton, Dickinson and Co.*, 649 F.3d 1276 (Ct. App. 2011) (en banc) has significantly restricted the infringement defense of inequitable conduct. A defendant must show that the patent in question would not have been issued but for undisclosed information, and that the patentee had the intent to deceive. Examiner suggests that future Information Disclosure Statements cite only the most relevant/inclusive references or portions thereof.

Drawings

The drawings are objected to under 37 CFR 1.83(a). The drawings must show every feature of the invention specified in the claims. Therefore, “the control module including self-test logic and control logic” must be shown or the feature(s) canceled from the claim(s). No new matter should be entered.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as “amended.” If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for

consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 37-48 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

As per claim 37, lines 6-16, "a data module..., and wherein the switch is configured to propagate self-test data patterns between the data handler logic element and the corresponding group of data ports during the test mode", and

lines 17-26, "a control module..., wherein, during the test mode, the self-test logic generates self-test address and control signals, the control logic provides the self-test address and control signals to the register, and the register provides registered self-test

Art Unit: 2117

address and control signals to the address and control ports of the memory devices" the relationship and/or connection between the "data module" and the "control module" are unclear and how they test the memory is unclear. How the memory is tested or verified is unclear. Appropriate correction and/or clarification are required.

As per claims 38-48, these claims are also rejected because they dependent upon the rejected base claim.

Claims 1-8, 10-12, 26-29 and 36 are allowable.

References Cited by Examiner

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. (See the attached form 892).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday from 8:00am-4:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an

Art Unit: 2117

interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, April Y. Blair can be reached on 571-270-1014. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

Notice of References Cited	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-6,178,126 B1	01-2001	Kirihata; Toshiaki	G11C29/72	365/200
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		14229844
	Filing Date		2014-03-29
	First Named Inventor	Hyun Lee	
	Art Unit	2117	
	Examiner Name	CHUNG, PHUNG M	
	Attorney Docket Number	N3044.10010US04	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	7210059		2007-04-24	Jeddeloh	
	2	6011748		2000-01-04	Lepejian, et al.	
	3	6639820		2003-10-28	Khandekar, et al.	
	4	6317352		2001-11-13	Halbert, et al.	
	5	6108798		2000-08-22	Heidel, et al.	
	6	3588831		1971-06-28	Figuroa	
	7	3618041		1971-11-02	Horikoshi	
	8	6271060		2001-08-07	Zandman, et al.	
If you wish to add additional U.S. Patent citation information please click the Add button.						Add

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
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U.S.PATENT APPLICATION PUBLICATIONS							Remove	
Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear		
	1							
If you wish to add additional U.S. Published Application citation information please click the Add button.							Add	
FOREIGN PATENT DOCUMENTS							Remove	
Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ² i	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1							
If you wish to add additional Foreign Patent Document citation information please click the Add button							Add	
NON-PATENT LITERATURE DOCUMENTS							Remove	
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.						T ⁵
	1	Inter partes review Case No. IPR2017-00561, Petition For Inter Partes Review Of U.S. Patent No. 8,001,434, filed January 5, 2017.						
	2	Inter partes review Case No. IPR2017-00561, Exhibit 1010 to Petition for Inter Partes Review, "EMC and the Printed Circuit Board", filed January 5, 2017.						
	3	Inter partes review Case No. IPR2017-00561, Exhibit 1012 to Petition for Inter Partes Review, "Microsoft Computer Dictionary (5th Ed.)", filed January 5, 2017.						
	4	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "JEDEC PC2-5300 RDIMM PCB Plot", filed January 5, 2017.						

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

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Attorney Docket Number	N3044.10010US04

5	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "The American Heritage Dictionary", filed January 5, 2017.
6	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Memory Systems", filed January 5, 2017.
7	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Designing Embedded Hardware", filed January 5, 2017.
8	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Micron DDR3 SDRAM UDIMM Specifications", filed January 5, 2017.
9	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "JEDEC PC2-5300 RDIMM Assembly", filed January 5, 2017.
10	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Micron - Technical Note TN4720. DDR2 (Point-to-Point) Package Sizes and Layout Basics", filed January 5, 2017.
11	Inter partes review Case No. IPR2017-00561, Exhibit 1015 to Petition for Inter Partes Review, "Declaration of Dr. Pinaki Mazumder", filed January 5, 2017.
12	Inter partes review Case No. IPR2017-00562, Petition For Inter Partes Review Of U.S. Patent No. 8,359,501, filed January 3, 2017.
13	Inter partes review Case No. IPR2017-00560, Petition For Inter Partes Review Of U.S. Patent No. 8,689,064, filed January 3, 2017.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/PHUNG M CHUNG/	Date Considered	03/29/2017
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
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Attorney Docket Number	N3044.10010US04

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jamie J. Zheng/	Date (YYYY-MM-DD)	2017-03-24
Name/Print	Jamie J. Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
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4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Search Notes 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner
G11C29/12, G11C29/52	3/28/17	PC

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
EAST-CPC Searched.	3/28/17	PC
Inventor Searched.	3/28/17	PC

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant					☐ CPA					☐ T.D.					☐ R.1.47				
CLAIM		DATE																	
Final	Original	02/21/2016	10/17/2016	03/29/2017															
	1	v	v	=															
	2	v	v	=															
	3	v	v	=															
	4	v	v	=															
	5	v	v	=															
	6	v	v	=															
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<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
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☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47									
CLAIM		DATE							
Final	Original	02/21/2016	10/17/2016	03/29/2017					
			37	v					
			38	v					
			39	v					
			40	v					
			41	v					
			42	v					
			43	v					
			44	v					
			45	v					
			46	v					
			47	v					
			48	v					

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1315	(system memory controller) and (computer system) and (memory module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:36
L2	344170	((data module or data handlers or data management) and ((data handler logic element) and (switch or gate or multiplexer or MUX or controller or selector) and (data generation element) and (verification element))) and (((control module or controller module) and ((self-test logic) and (control logic) and (register))))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/03/28 21:36
L3	557	L1 and L2	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:36
L4	521	((memory devices) and (address ports and control ports and data ports)) and L3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/03/28 21:36
L5	373205	(normal mode or operation mode or functional mode) or (test mode)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:36
L6	82	L4 and L5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:36
L7	397738	(select\$3 or enabl\$3 or control\$4 or switch\$3 or gat\$3 or multiplex\$3 or MUX) same (normal mode or normal operation or functional mode) or (test mode or self-test mode or self test mode)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:37

L8	51	L6 and L7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:37
L9	64	((memory module) or (memory subsystem)) and ((system memory controller or memory controller) and (data module) and (memory devices) and (control module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:37
L10	3	L8 and L9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:37
L11	3373	((memory self-test address\$2 and control signal\$1 and test data patterns or test data or test vectors) and (memory device\$1) and (test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:39
L12	7	((control module or controller module) and ((self-test logic) and (control logic) and (register)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:40
L13	9	I8 and (I11 or I12)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:41
L14	3699	((memory devices) or (memory units) or (memory module)) and ((printed circuit board) or (PCB)) or (circuit board) and ((data module) or (BIST logic units) or (data handlers) or (data controllers) or (data managements)) and ((control module) or (memory controller) or (master test unit)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:41
L15	20011	("BIST") or ("logic BIST") or (in-circuit test) or ("MBIST") or (built-in self test) or (built-in self-test)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:41
L16	118153	((data module) or (data handler\$1) or (data management\$1) or (data controller) or (data manager\$1) or (controller) or (gate or switch or	US-PGPUB; USPAT; USOCR; FPRS; EPO;	ADJ	ON	2017/03/28 21:41

		multiplexer or MUX)) and ((normal mode or normal operation mode or normal mode operation or functional mode) or (self-test mode or test mode or test mode operation))	JPO; DERWENT; IBM_TDB			
L17	38094	(memory rank\$1) or (rank\$1 or block\$1) same (memory module or memory chip)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:41
L18	9	I13 and (I14 or I15 or I16 or I17)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:48
L19	8	18 AND ((G11C29/12 OR G11C29/52).CPC. OR (714/718 OR 714/42 OR 714/733 OR 714/799).OCLS. OR (G11C29/12 OR G11C29/10).IPOR.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:51
L20	6175	(compar\$3 or detect\$3 or determin\$3 or identify\$3) and (test failure or fault or fail or error or defect or bad or abnormal or interrupted or test result) and ((data pattern or test data or test pattern or test vector) and (expected or predetermined or known or reference or predefined) adj (data pattern or test data or test vector))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:52
L22	2262298	((Lee).inv.) or ((Bhakta).inv.) or ((Choi).inv.) or ((Netlist).asn.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:55
L23	3	I18 and I22	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:55
L24	1	I20 and I8	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/03/28 21:55

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown/PHUNG M CHUNG/
Sheet	1	of	2	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	2009/0103387 A1	04-23-2009	Shau	
	2.	2006/0262586 A1		Solomon, J. et al.	
	3.	2008/0162790 A1			

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	4.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 8,081,536, IPR Case Number IPR2015-01021, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
	5.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 7,881,150, IPR Case Number IPR2015-01020, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
	6.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-01372, Final Written Decision, issued March 9, 2016	
	7.	Inter Partes Review Case No. IPR2014-01372, Petitioner's Request for Rehearing, filed April 6, 2016	
	8.	Inter Partes Review Case No. IPR2014-01372, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
	9.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-01374, Final Written Decision, issued March 9, 2016	
	10.	Inter Partes Review Case No. IPR2014-01374, Petitioner's Request for Rehearing, filed April 6, 2016	
	11.	Inter Partes Review Case No. IPR2014-01374, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
	12.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-00970, Decision Denying Petitioner's Request for Rehearing, issued April 27, 2016	
	13.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January 13, 2016	
	14.	Inter Partes Review Case No. IPR2014-00971, Decision - Petitioner's Request for Rehearing, issued April 27 2016	
	15.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision, reissued April 27, 2016	
	16.	Inter Partes Review Case No. IPR2014-00971, Netlist Inc.'s Notice of Appeal, filed June 29, 2016	

EXAMINER /PHUNG M CHUNG/

DATE CONSIDERED 03/29/2017

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C./

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown
Sheet	2	of	2	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	17.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
	18.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
	19.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

EXAMINER

/PHUNG M CHUNG/

DATE CONSIDERED

03/29/2017

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re application of	Hyun Lee	Art Unit 2117
Application No.:	14/229,844	
Filed:	March 29, 2014	
Conf. No.:	1018	
For:	Memory Subsystem and Method of Operation	
Examiner:	CHUNG, PHUNG M	
Customer No.:	97149	
Attorney Docket No.:	N3044.10010US04	

REQUEST FOR REFUND UNDER 37 CFR § 1.26

REFUND SECTION

Accounting Division, Office of Finance
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

I. Refund Request

On March 21, 2017, Applicant's counsel authorized a one-time payment of a two month Extension of time (\$600) and a first Request for Continued Examination (\$1,200). Because the Applicant is currently entitled to small entity status, Applicant respectfully requests a refund for the difference of (\$900).

II. Manner of Refund.

Applicant requests that the \$300 refund for two month Extension of time be refunded to the VISA Credit Card used originally to pay the fees and that the \$600 refund for a first Request for Continued Examination be refunded to the deposit account originally used to pay the fees.

Dated this 11th day of April, 2017.

Respectfully submitted,

/Jamie J. Zheng, Reg. no. 51167/

JAMIE J. ZHENG
Registration No. 51,167
Attorney for Applicant
Telephone No. (650) 521-4828

Electronic Acknowledgement Receipt

EFS ID:	28892575
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10010US04
Receipt Date:	11-APR-2017
Filing Date:	29-MAR-2014
Time Stamp:	14:07:07
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Refund Request	2017-04-11_Refund_N3044-10 010US04.pdf	233428	no	2
			8a3dc2168233a784d90b1a32c815559cde0 2a1f6		

Warnings:

Information:	
Total Files Size (in bytes):	233428
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re application of Hyun Lee
Application No.: 14/229,844
Filed: March 29, 2014
Conf. No.: 1018
For: Memory Subsystem and Method of Operation
Examiner: CHUNG, PHUNG M
Customer No.: 97149
Attorney Docket No.: N3044.10010US04

Art Unit
2117

USPTO
RECEIPTS ACCOUNTING
DIVISION
2017 APR 11 AM 8:51

REQUEST FOR REFUND UNDER 37 CFR § 1.26

REFUND SECTION

Accounting Division, Office of Finance
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

I. Refund Request

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Application No. 14/229,844

Attorney Docket Number 14/229,844
Request for Refund Under 37 CFR § 1.26

II. Manner of Refund.

Applicant requests that the \$300 refund for two month Extension of time be refunded to the VISA Credit Card used originally to pay the fees and that the \$600 refund for a first Request for Continued Examination be refunded to the deposit account originally used to pay the fees.

Dated this 11th day of April, 2017.

Respectfully submitted,

/Jamie J. Zheng, Reg. no. 51167/

JAMIE J. ZHENG
Registration No. 51,167
Attorney for Applicant
Telephone No. (650) 521-4828

Electronic Acknowledgement Receipt

EFS ID:	28686016
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	04/17/2017 SSANDARA 00000002 505963 14229844 01 FC:2801 600.00 DA Memory Subsystem and Method of Operation 04/17/2017 SSANDARA 00000001 14229844 01 FC:2252 300.00 DP
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	N3044.10010US04
Receipt Date:	21-MAR-2017
Filing Date:	29-MAR-2014
Time Stamp:	01:12:30
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	VISA
RAM confirmation Number	032117INTEFSW01145600
Deposit Account	
Authorized User	
The Director of the USPTO is hereby authorized to charge indicated fees and credit adjustments as follows: Adjustment date: 04/17/2017 SSANDARA 04/05/2017 CBARNES1 00000002 505963 14229844 01 FC:1801 1200.00 GR Adjustment date: 04/17/2017 SSANDARA 03/21/2017 INTEFSW 00008185 14229844 01 FC:1252 -600.00 DP	

Refund Ref:
04/17/2017

0030194781

Credit Card Refund Total: \$300.00

VISA 600: XXXXXXXXXXXX5877

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/229,844
Filed : March 29, 2014
For : A Self-Test Memory Module and
Method of Operation
Examiner : CHUNG, PHUNG M
Art Unit : 2189
Conf. No. : 1018
Docket No. : N3044.10010US04

CERTIFICATE OF EFS WEB
TRANSMISSION

I hereby certify that this correspondence and
any other attachment noted on the Automated
Acknowledgement Receipt, is being
transmitted from within the Pacific Time zone
to the Commissioner for Patents via the EFS
Web Server on:

September 5, 2017

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT**Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Madam:

Applicant submits the following amendments and remarks in response to the Non-Final Office Action Dated April 4, 2017 in the above-identified patent application.

Applicant hereby petitions for a 2-month extension of time to respond to the Office Action.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5394.

Amendments to the Claims: Begin on page 2 of this paper.

Remarks: Begin on page 15 of this paper.

In the Claims:

1. (Currently Amended) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements; and

a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and

wherein the memory module is operable in any of a plurality of modes including a normal mode and a test mode;

wherein, during the normal mode, the control module ~~receives-is configured to receive~~ system address and control signals from the system memory controller and ~~provides-to output~~ memory address and control signals to the memory devices according to the system address and control signals, and the data module ~~propagates-is configured to propagate~~ data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals; and

wherein, during the test mode, the control module ~~generates-self-test-address-and control signals without using any test procedure driven by the system memory controller; the control module provides-the-is configured to output~~ self-test address and control signals to the address and control ports of the memory devices, and the data module ~~isolates-is configured to isolate~~ the memory devices from ~~being accessed by~~ the system memory controller[,] and ~~to transmit transmits~~ self-test data patterns ~~provided by the data handler logic elements~~ to the data ports of the memory devices; ~~wherein the self-test data patterns are generated by the data handler logic elements according to one or more~~ commands from the control module, and wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals ~~received from the control module~~.

2. (Currently Amended) The memory module of claim 1, wherein the memory devices are arranged in at least one rank and organized in a plurality of memory device groups, each rank of the at least one rank having a set of memory devices across a full bit width of the memory module, organized in a plurality of groups, each respective memory device group of the plurality of memory device groups including respective one or more memory devices in each rank of the at least one rank, wherein each data signal of the data signals propagated by the data module during the normal mode is N-bits wide, system memory bus has a width of N-bits and is organized in a plurality of n-bit sections corresponding, respectively, to groups of one or more memory devices in each rank, wherein the data module includes a plurality of data handlers, each respective data handler of the plurality of data handlers including a respective one of the data handler logic elements, wherein the each respective data handler is configured to propagate a respective n-bit section of the each data signal between the system memory controller and being coupled between a respective n-bit section of the system memory bus and a respective memory device group of the plurality of memory device groups one or more memory devices, n being a fraction of N, and wherein the each respective data handler generating is further configured to output respective n-bit wide self-test data patterns provided by the respective one of the data handler logic elements and transmitting the respective n-bit wide self-test data patterns to the respective memory device group of one or more memory devices.

3. (Currently Amended) The memory module of claim 2, wherein each memory device is n-bit-n-bits wide and wherein the respective memory device group includes a single respective memory device in each rank of the at least one rank.

4. (Currently Amended) The memory module of claim 2, wherein the control module generates is configured to output additional self-test address and control signals and provides the additional self-test address and control signals to the address and control ports of the memory devices during the test mode, the additional self-test address and control signals causing the memory devices in one of the at least one rank to output a N-bit wide read data signal, and wherein the each respective data handler is configured to receive receives a respective n-bit section of the N-bit wide read data signal.

5. (Currently Amended) The memory module of claim 4, wherein the each respective data handler is configured to ~~generate expected data pattern after the control module generates the additional self-test address and control signals and to check the~~ respective n-bit section of the N-bit wide read data signal with respect to the expected self-test data patterns.
6. (Cancelled)
7. (Currently Amended) The memory module of claim 5, wherein the data module ~~stores~~ is configured to store failure information when a test failure is indicated by the respective n-bit section of the N-bit wide read data signal not being in agreement with the expected data pattern.
8. (Previously Presented) The memory module of claim 7, wherein the data module is further configured to report the failure information to the control module.
9. (Cancelled)
10. (Currently Amended) The memory module of claim 1, wherein the data module ~~generates and transmits~~ is configured to output the self-test data patterns to the memory devices in response to a write command from the control module, and wherein the data module is further configured to read back test data patterns from the memory devices in response to a read command from the control module.
11. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers ~~are configured to~~ operate independently of each other without being in communication with each other, and wherein, during the test mode, the each respective data handler ~~is configured to output the respective~~ generates n-bit wide self-test data patterns to test a corresponding portion of a memory space of the memory module independently of the other data handlers.
12. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the

each respective data handler is positioned on the circuit board proximate to the respective ~~memory device group of one or more memory devices.~~

13. (Cancelled)

14. (Cancelled)

15. (Cancelled)

16. (Cancelled)

17. (Cancelled)

18. (Cancelled)

19. (Cancelled)

20. (Cancelled)

21. (Cancelled)

22. (Cancelled)

23. (Cancelled)

24. (Cancelled)

25. (Cancelled)

26. (Currently Amended) The memory module of claim 1, wherein the control module comprises a memory device controller and a test controller, wherein the memory device controller ~~receives-is configured to receive~~ the system address and control signals from the system memory controller and the self-test address and control signals from the test controller.

27. (Currently Amended) The memory module of claim 26, wherein the control module further comprises a register coupled to the address and control ports of the

memory devices, and wherein the memory device controller ~~sends~~ is configured to output address and control information from either the system address and control signals or from the self-test address and control signals to the register depending on whether the memory module is in the normal mode or test mode.

28. (Currently Amended) The memory module of claim 2, wherein the each respective data handler is configured to provide first data paths between the respective memory device group and the system memory controller or second data paths between the respective memory device group and the respective one of the data handler logic elements ~~comprises a data handler logic element that generates test data patterns and a switch coupled to the data handler logic element and to the respective n-bit section of the system memory bus; and wherein the switch is configured to selectively input to the respective group of one or more memory devices either write data signals from the system memory controller or test data patterns from the data handler logic element~~ depending on whether the memory module is in the normal mode or test mode.

29. (Cancelled) The memory module of claim 28, wherein the switch is further configured to receive read data signals from the respective group of one or more memory devices and sends the read data signals to the system memory controller or to the data handler logic element depending on whether the memory module is in the normal mode or the test mode.

30. (Cancelled)

31. (Cancelled)

32. (Cancelled)

33. (Cancelled)

34. (Cancelled)

35. (Cancelled)

36. (Currently Amended) The memory module of claim 2, wherein ~~each memory device is n/2-bit wide and wherein the respective memory device group includes two respective memory devices in each of the at least one rank each data handler is coupled to two memory devices in each rank.~~

37. (Cancelled)

38. (Cancelled)

39. (Cancelled)

40. (Cancelled)

41. (Cancelled)

42. (Cancelled)

43. (Cancelled)

44. (Cancelled)

45. (Cancelled)

46. (Cancelled)

47. (Cancelled)

48. (Cancelled)

49. (New) A memory module operable in a computer system with a system memory controller, the memory module comprising:

a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller;

memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;

a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and

a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks;

wherein the memory module is configurable to perform at least a first operation and a second operation;

wherein, in the first operation, at least one rank of the one or more ranks is configured to be accessed by the system memory controller, and the data handlers are configured to propagate one or more data signals between the at least one rank and the connector;

wherein, in the second operation, the memory devices are not accessed by the system memory controller, and the each respective data handler is configured to provide respective patterns of data signals to the respective group of one or more memory devices based on information received from the control module.

50. (New) The memory module of claim 49, wherein each of the one or more data signals is N-bit wide, and wherein, in the first operation, the each respective data handler is configured to propagate a respective n-bit section of each of the one or more data signals between at least one respective memory device in the at least one rank and the connector, n being a fraction of N; and wherein, in the second operation, the respective patterns of data signals are each n-bits wide.

51. (New) The memory module of claim 50, wherein each memory device is n-bit wide and wherein the respective group of one or more memory devices include a single respective memory device in each of the one or more ranks.

52. (New) The memory module of claim 50, wherein each memory device is $n/2$ -bit wide and wherein the respective group of one or more memory devices include two respective memory devices in each of the one or more ranks.
53. (New) The memory module of claim 50, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective group of one or more memory devices.
54. (New) The memory module of claim 49, wherein the each respective data handler is configured to output the respective patterns of data signals to the respective group of one or more memory devices in response to a write command from the control module.
55. (New) The memory module of claim 54, wherein, in the second operation, the each respective data handler is further configured to receive patterns of data signals from the respective group of one or more memory devices in response to a read command from the control module.
56. (New) The memory module of claim 55, wherein the each respective data handler is further configured to check the patterns of data signals received from the respective group of one or more memory devices with respect to expected data patterns.
57. (New) The memory module of claim 49, wherein the each respective data handler includes a respective data handler logic element configured to output the respective patterns of data signals in the second operation, wherein the each respective data handler is configured to provide first data paths between the respective group of one or more memory devices and the connector in the first operation and to provide second data paths between the respective group of one or more memory devices and the respective data handler logic element in the second operation, the second data paths being different from the first data paths.
58. (New) A memory module operable in a computer system with a system memory controller, the memory module comprising:
a printed circuit board (PCB) having a connector configured to provide electrical

connections for data, address and control signals between the memory module and the system memory controller;

memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;

a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and

a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks;

wherein the memory module is configurable to operate in one of at least a first mode and a second mode;

wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations, and the data handlers are configured to propagate data signals associated with the memory read or write operations between the memory devices and the connector;

wherein, in the second mode, the memory devices are not accessed by the system memory controller, and the each respective data handler is configured to provide respective patterns of data signals to the respective group of one or more memory devices based on information received from the control module.

59. (New) The memory module of claim 58, wherein each of the one or more data signals is N-bit wide, and wherein, when the memory module is operating in the first mode, the each respective data handler is configured to propagate a respective n-bit section of each of the one or more data signals between at least one respective memory device in the at least one rank and the connector, n being a fraction of N; and wherein, when the memory module is operating in the second mode, the respective patterns of data signals are each n-bits wide.

60. (New) The memory module of claim 59, wherein each memory device is n-bit wide and wherein the respective group of one or more memory devices include a single respective memory device in each of the one or more ranks.
61. (New) The memory module of claim 59, wherein each memory device is n/2-bit wide and wherein the respective group of one or more memory devices include two respective memory devices in each of the one or more ranks.
62. (New) The memory module of claim 58, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective group of one or more memory devices.
63. (New) The memory module of claim 58, wherein the each respective data handler is configured to output the respective patterns of data signals to the respective group of one or more memory devices in response to a write command from the control module.
64. (New) The memory module of claim 63, wherein, when the memory module is operating in the second mode, the each respective data handler is further configured to receive patterns of data signals from the respective group of one or more memory devices in response to a read command from the control module.
65. (New) The memory module of claim 64, wherein the each respective data handler is further configured to check the patterns of data signals received from the respective group of one or more memory devices with respect to expected data patterns.
66. (New) The memory module of claim 58, wherein the each respective data handler includes a respective data handler logic element configured to output the respective patterns of data signals, wherein the each respective data handler is configured to provide first data paths between the respective group of one or more memory devices and the connector or second data paths between the respective group of one or more memory devices and the respective data handler logic element depending on whether the memory module is configured to operate in the first mode or the second mode, the second data paths being different from the first data paths.

67. (New) A memory module operable in a computer system with a system memory controller, the memory module comprising:

- a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller;

- memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;

- a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and

- a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks;

- wherein the memory module is configurable to be in one of at least a first configuration and a second configuration;

- wherein, in the first configuration, the memory module is accessible by the system memory controller for memory read and write operations, and the data handlers are configured to propagate data signals associated with the memory read or write operations between the memory devices and the connector;

- wherein, in the second configuration, the memory devices are not accessible by the system memory controller, and the each respective data handler is configured to provide respective patterns of data signals to the respective group of one or more memory devices based on information received from the control module.

68. (New) The memory module of claim 67, wherein each of the one or more data signals is N-bit wide, and wherein, when the memory module is in the first configuration, the each respective data handler is configured to propagate a respective n-bit section of each of the one or more data signals between at least one respective memory device in the

at least one rank and the connector, n being a fraction of N ; and wherein, when the memory module is in the second configuration, the respective patterns of data signals are each n -bits wide.

69. (New) The memory module of claim 68, wherein each memory device is n -bit wide and wherein the respective group of one or more memory devices include a single respective memory device in each of the one or more ranks.

70. (New) The memory module of claim 68, wherein each memory device is $n/2$ -bit wide and wherein the respective group of one or more memory devices include two respective memory devices in each of the one or more ranks.

71. (New) The memory module of claim 67, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is positioned on the circuit board proximate to the respective group of one or more memory devices.

72. (New) The memory module of claim 67, wherein the each respective data handler is configured to output the respective patterns of data signals to the respective group of one or more memory devices in response to a write command from the control module.

73. (New) The memory module of claim 72, wherein, when the memory module is in the second configuration, the each respective data handler is further configured to receive patterns of data signals from the respective group of one or more memory devices in response to a read command from the control module.

74. (New) The memory module of claim 73, wherein the each respective data handler is further configured to check the patterns of data signals received from the respective group of one or more memory devices with respect to expected data patterns.

75. (New) The memory module of claim 67, wherein the each respective data handler includes a respective data handler logic element configured to output the respective patterns of data signals, wherein the each respective data handler is configured to provide first data paths between the respective group of one or more memory devices and the

connector or second data paths between the respective group of one or more memory devices and the respective data handler logic element depending on whether the memory module is in the first configuration or the second configuration, the second data paths being different from the first data paths.

REMARKS

This amendment is in response to the Non-Final Office Action dated April 4, 2017. In the Office Action, the Examiner:

- commented on previously filed information disclosure statements
- objected to drawings;
- allowed claims 1-8, 10-12, 26-29 and 36;
- rejected claims 37-48 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph.

Claim Amendment

Claims 6, 29 and 37-48 are cancelled. Claims 1-5, 7, 10-12, 26-28 and 36 are amended. New claims 49-75 are presented. Support for the amendment and the new claims can be found throughout the Specification and Drawings (e.g., in paragraphs [0025], [0030], [0035], [0040]-[0045]). Therefore, no new matter is added. After entry of this amendment, the pending claims are 1-5, 7-8, 10-12, 26-28, 36, and 49-75.

Allowe Claims

Applicant appreciates the allowance of claims 1-8, 10-12, 26-29 and 36. Claims 1-5, 7, 10-12, 26-28 and 36 are amended for clarity upon further review, and are believed to be allowable as amended.

Objection to Drawings

The drawings are objected to due to claim terms in claim 37. Since claim 37 and its dependent claims are cancelled, the objection is now moot.

Rejection of Claims 37-48 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA),

Second paragraph

Claims 37-48 are cancelled. The rejections thereto are now moot.

New Claims 49-75

New claims 49-75 are believed to be allowable.

Applicant submits that claims 1-5, 7-8, 10-12, 26-28, 36, and 49-75 are in condition for allowance and respectfully requests such action. The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any issues.

Respectfully submitted,

Date: September 5, 2017

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167

(Reg. No.)

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14229844
	Filing Date	2014-03-29
	First Named Inventor	Hyun Lee
	Art Unit	2117
	Examiner Name	CHUNG, PHUNG M
	Attorney Docket Number	N3044.10010US04

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	5995731		1999-11-30	Crouch , et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²ⁱ	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

1	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Patent Owner's Preliminary Response, filed April 10, 2017.
2	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Exhibit 2004 - Excerpts from Microsoft Computer Dictionary Fifth Edition (2002), filed April 10, 2017.
3	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Exhibit 2005 - Qualcomm Announces Sampling of the MSM6250, filed April 10, 2017.
4	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Trial Instituted Document, filed July 7, 2017.
5	Inter Partes Review of U.S. Patent No. 8,359,501, Case No. IPR2017-00562, Patent Owner's Preliminary Response, filed April 10, 2017.
6	Inter Partes Review of U.S. Patent No. 8,359,501, Case No. IPR2017-00562, Trial Instituted Document, filed July 7, 2017.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

***EXAMINER:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

☒ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/JAMIE J. ZHENG/	Date (YYYY-MM-DD)	2017-09-05
Name/Print	JAMIE J. ZHENG	Registration Number	51167

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10010US04			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	2202	12	40	480
INDEPENDENT CLAIMS IN EXCESS OF 3	2201	1	210	210
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 2 months with \$0 paid	2252	1	300	300
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	2806	1	90	90
Total in USD (\$)				1080

Electronic Acknowledgement Receipt

EFS ID:	30276404
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10010US04
Receipt Date:	05-SEP-2017
Filing Date:	29-MAR-2014
Time Stamp:	21:11:07
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1080
RAM confirmation Number	090617INTEFSW21124000
Deposit Account	505394
Authorized User	Brian Gonzalez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2017-09-05_OA-Response_N3044-10010US04.pdf	378965	yes	16
			66fefa4a4b0f61cdb784c98a538f86605076660f		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Claims		2	14	
	Applicant Arguments/Remarks Made in an Amendment		15	16	
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2017-09-05_IDS_N3044-10010US04.pdf	1043028	no	4
			782d533dd2659507e21b5f29bb0ae9882efb5d79		
Warnings:					
Information:					
3	Non Patent Literature	IPR2017-00561_Exhibit2004_2017-04-10.pdf	8459679	no	5
			e6a2f0a6d94465aac5cf4586fe21e294036ae2f60		
Warnings:					
Information:					
4	Non Patent Literature	IPR2017-00561_Exhibit2005_2017-04-10.pdf	102187	no	6
			44677e5a940eff2c58c8dccc5f16804aed3435af4		
Warnings:					
Information:					
5	Non Patent Literature	IPR2017-00561_Patent-Owners-Response_2017-04-10.pdf	2299848	no	82
			bdd6b3da380433196c3fc18ba10e274895504dfa		

Warnings:					
Information:					
6	Non Patent Literature	IPR2017-00561_Trial-Instituted_2017-07-07.pdf	2087593 8d7e6de34703b6fe26a9251885af52a3480cac47	no	34
Warnings:					
Information:					
7	Non Patent Literature	IPR2017-00562_Patent-Owners-Response_2017-04-10.pdf	1742702 0a701efe69a3e4e2cd40805c8bf6bd74a07bdbc	no	63
Warnings:					
Information:					
8	Non Patent Literature	IPR2017-00562_Trial-Instituted_2017-04-10.pdf	2165738 688d6a250192d1b3b60b23697979a4e6e14e4e8e	no	30
Warnings:					
Information:					
9	Fee Worksheet (SB06)	fee-info.pdf	35799 fe32d62cb2d99cdcb51d43b332a2f00ede014f61	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			18315539		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 09/13/2017

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		01 FC : 2202	80.00 DA		

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/229,844	Filing Date 03/29/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO**APPLICATION AS FILED – PART I**

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA		RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A		N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A		N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A		N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*		X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*		X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).				
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))					
* If the difference in column 1 is less than zero, enter "0" in column 2.				TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	09/05/2017	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	* 42	Minus	** 28	= 14		x \$40 =	560
	Independent (37 CFR 1.16(h))	* 4	Minus	***3	= 1		x \$210 =	210
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
	TOTAL ADD'L FEE							770

(Column 1)

(Column 2)

(Column 3)

		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=		X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=		X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
	TOTAL ADD'L FEE							

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
RENEE M. COLLINS

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Alexandria, Virginia 22313-1450
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	N3044.10010US04	1018
79141	7590	12/08/2017	EXAMINER	
Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098			CHUNG, PHUNG M	
			ART UNIT	PAPER NUMBER
			2117	
			NOTIFICATION DATE	DELIVERY MODE
			12/08/2017	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

d@mabr.com
bisraelsen@mabr.com

Office Action Summary	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 9/5/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☒ This action is **FINAL**. 2b) ☐ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1-5, 7, 8, 10-12, 26-29 and 36 is/are pending in the application.
5a) Of the above claim(s) 49-75 is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 1-5, 7-8, 10-12, 26-29 and 36 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 3/29/2014 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☐ Notice of References Cited (PTO-892)
- 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 9/5/2017.
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____.
- 4) ☐ Other: ____.

DETAILED ACTION

The present application is being examined under the pre-AIA first to invent provisions.

Election/Restrictions

Newly submitted claims 49-75 are directed to an invention that is independent or distinct from the invention originally claimed for the following reasons:

Independent claim 1 and its dependent claims thereof, drawn to a memory module..., wherein, during the test mode, the control module is configured to output self-test address and control signals to the address and control ports of the memory devices, and to transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands from the control module, and wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals from the control module, which is different from independent claims 49, 58 and 67 and their dependent claims thereof, drawn to a memory module..., wherein, in the second operation, the each respective data handler is configured to provide respective patterns of data signals to the respective group of one or more memory devices based on information received from the control module.

Since applicant has received an action on the merits for the originally presented invention, this invention has been constructively elected by original presentation for prosecution on the merits. Accordingly, claims 49-75 are withdrawn from consideration

as being directed to a non-elected invention. See 37 CFR 1.142(b) and MPEP § 821.03.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claims 1-5, 7-8, 10-12, 26-29 and 36 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more.

Independent claim 1 recites the limitations of:

“during the normal mode: configured to receive system address and control signal from the system memory controller and to output memory address and control signals to the memory device according to the system address and control signals”, and

“configured to propagate data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals”; and

“during the test mode: configured to output self-test address and control signals to the address and control ports of the memory devices”;

“configured to isolate the memory devices from being accessed by the system memory controller and to transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory device according to one or more commands from the control module, and wherein the self-test data patterns are

received by at least a portion of the memory devices according to the self-test address and control signals from the control module". These limitations are directed to intended use statements, and these limitations are abstract idea without significantly more, similar to Cyberfone, the concept of using categories to organize, store and transmit information; Cap One Financial, the concept of collecting, displaying, and manipulating data; and Content Extraction, the concept of data recognition and storage, were found to be abstract ideas.

Claim 1 does not include additional elements that, when taken alone or in combination, are sufficient to amount to significantly more than the abstract idea because the recited "memory devices", "circuit board", "control ports", "data ports", "system memory bus", "data handler logic elements", "data module", "control module", "system memory controller" are well-known and conventional/generic computer components performing well-known and conventional/generic computer functions.

Dependent claims 2-5, 7-8, 10-12, 26-29, 36 and 36 recite no additional limitation that would amount to significantly more than the abstract idea defined in independent claim 1.

Accordingly, for the reasons provided above, claims 1-5, 7-8, 10-12, 26-29 and 36 are directed to an abstract idea, hence, not patent eligible under 35 USC 101.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-5, 7-8, 10-12, 26-29 and 36 rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

As per claim 1, line 19-29, "wherein, during the test mode, the control module is configured to output self-test address and control signals to the address and control ports of the memory devices, and the data module is configured to isolate the memory devices from being accessed by the system memory controller and transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands from the control module, and wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals from the control module" is not clear whether, during the test mode, the self-test data patterns that are received by at least a portion of the memory devices according to the self-test address and control signals from the control module is actually doing the test for that the at least a portion of the memory devices according to the self-test address and control signals from the control module or just for storage over there? How the at least a portion of the memory device is tested is unclear and what is the result and/or what to with the test result? Appropriate correction and/or clarification are required.

As per claims 2-5, 7-8, 10-12, 26-29 and 36, these dependent claims are rejected with similar problems as discussed above and/or because they depend on the rejected base claim 1.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday from 8:00am-4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, April Y. Blair can be reached on 571-270-1014. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number	14229844
	Filing Date	2014-03-29
	First Named Inventor	Hyun Lee
	Art Unit	2117
	Examiner Name	CHUNG, PHUNG M
	Attorney Docket Number	N3044.10010US04

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	1	5995731		1999-11-30	Crouch , et al.	

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	14229844
Filing Date	2014-03-29
First Named Inventor	Hyun Lee
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	N3044.10010US04

1	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Patent Owner's Preliminary Response, filed April 10, 2017.
2	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Exhibit 2004 - Excerpts from Microsoft Computer Dictionary Fifth Edition (2002), filed April 10, 2017.
3	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Exhibit 2005 - Qualcomm Announces Sampling of the MSM6250, filed April 10, 2017.
4	Inter Partes Review of U.S. Patent No. 8,00,1434, Case No. IPR2017-00561, Trial Instituted Document, filed July 7, 2017.
5	Inter Partes Review of U.S. Patent No. 8,359,501, Case No. IPR2017-00562, Patent Owner's Preliminary Response, filed April 10, 2017.
6	Inter Partes Review of U.S. Patent No. 8,359,501, Case No. IPR2017-00562, Trial Instituted Document, filed July 7, 2017.

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature	/PHUNG M CHUNG/	Date Considered	12/05/2017
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

EAST Search History**EAST Search History (Prior Art)**

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	112002	((data module) or (data handler\$1) or (data management\$1) or (data controller) or (data manager\$1) or (memory controller) or (gate or switch or multiplexer or MUX or selector)) and ((normal mode or normal operation mode or normal mode operation or functional mode or standard read/write mode) or (self-test mode or test mode or test mode operation))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:15
L2	3539	((memory self-test address\$2 and control signal\$1 and test data patterns or test data or test vectors) and (memory device\$1) and (test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:16
L3	3236	I1 and I2	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:22
L4	3189	I3 and ((data handler logic elements) and (self-test data patterns) and (self-test address and control signals))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:22
L5	2022	((memory devices) or (memory units) or (memory module)) and (((printed circuit board) or (PCB)) or (circuit board)) and ((data module) or (BIST logic units) or (data handlers) or (data handler logic elements) or (data controllers) or (data managements)) and ((control module) or (memory controller) or (master test unit))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:25
L6	24	I4 and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:35
L7	532542	((control module or controller module) and ((self-test logic) and (control logic) and (register)))	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2017/12/04 14:37

			EPO; JPO; DERWENT; IBM_TDB			
L8	20	I6 and I7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:37
L9	20	I8 and ((memory devices) and (address ports and control ports and data ports))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:38
L10	3	I9 and ((memory rank\$1) or (rank\$1 or block\$1) same (memory module or memory chip))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:39
L11	72	(memory module) or (memory subsystem)) and ((system memory controller or memory controller) and (data module) and (memory devices) and (control module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:39
L12	3	I9 and I11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:40
L13	2344752	((Lee).inv.) or ((Bhakta).inv.) or ((Choi).inv.) or ((Netlist).asn.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:40
L14	3	I9 and I13	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:41
L16	1386	(system memory controller) and (computer system) and (memory module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2017/12/04 14:43
L19	3	9 AND ((G11C29/10 OR G11C29/12 OR G11C29/52).CPC. OR (714/719 or	US-PGPUB; USPAT;	ADJ	ON	2017/12/04 14:45

		714/718 OR 714/42 OR 714/733 OR 714/734 or 714/30 or 714/799).CCLS. OR (G11C29/12 OR G11C29/10).IPCR.)	USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
L20	3	I16 and I9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:46
L21	3	I10 or I12 or I14 or I20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2017/12/04 14:47

12/ 4/ 2017 2:47:32 PM
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Search Notes 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner
G11C29/10 , G11C29/12, G11C29/52	12/4/2017	PC

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
EAST-CPC Searched.	12/4/2017	PC
Inventor Searched.	12/4/2017	PC

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant						☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE									
Final	Original	02/21/2016	10/17/2016	03/29/2017	12/04/2017						
	1	v	v	=	V						
	2	v	v	=	V						
	3	v	v	=	V						
	4	v	v	=	V						
	5	v	v	=	v						
	6	v	v	=	-						
	7	v	v	=	v						
	8	v	v	=	v						
	9	v	v	-	-						
	10	v	v	=	v						
	11	v	v	=	v						
	12	v	v	=	v						
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	14	v	-	-	-						
	15	v	v	-	-						
	16	v	v	-	-						
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	27		v	=	v						
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	30		v	-	-						
	31		v	-	-						
	32		v	-	-						
	33		v	-	-						
	34		v	-	-						
	35		v	-	-						
			36	=	v						

<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	02/21/2016	10/17/2016	03/29/2017	12/04/2017				
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			38	v	-				
			39	v	-				
			40	v	-				
			41	v	-				
			42	v	-				
			43	v	-				
			44	v	-				
			45	v	-				
			46	v	-				
			47	v	-				
			48	v	-				
			49		N				
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			67		N				
			68		N				
			69		N				
			70		N				
			71		N				
			72		N				

<i>Index of Claims</i> 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected
=	Allowed

-	Cancelled
÷	Restricted

N	Non-Elected
I	Interference

A	Appeal
O	Objected

☐ Claims renumbered in the same order as presented by applicant						☐ CPA		☐ T.D.		☐ R.1.47	
CLAIM		DATE									
Final	Original	02/21/2016	10/17/2016	03/29/2017	12/04/2017						
			73		N						
			74		N						
			75		N						

CERTIFICATION AND REQUEST FOR CONSIDERATION UNDER THE AFTER FINAL CONSIDERATION PILOT PROGRAM 2.0		
Practitioner Docket No.: N3044.10010US04	Application No.: 14/229,844	Filing Date: 03-29-2014
First Named Inventor: Hyun Lee	Title: Memory Subsystem and Method of Operation	
APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS CONSIDERATION UNDER THE AFTER FINAL CONSIDERATION PILOT PROGRAM 2.0 (AFCP 2.0) OF THE ACCOMPANYING RESPONSE UNDER 37 CFR 1.116. 1. The above-identified application is (i) an original utility, plant, or design nonprovisional application filed under 35 U.S.C. 111(a) [a continuing application (<i>e.g.</i>, a continuation or divisional application) is filed under 35 U.S.C. 111(a) and is eligible under (i)], or (ii) an international application that has entered the national stage in compliance with 35 U.S.C. 371(c). 2. The above-identified application contains an outstanding final rejection. 3. Submitted herewith is a response under 37 CFR 1.116 to the outstanding final rejection. The response includes an amendment to at least one independent claim, and the amendment does not broaden the scope of the independent claim in any aspect. 4. This certification and request for consideration under AFCP 2.0 is the only AFCP 2.0 certification and request filed in response to the outstanding final rejection. 5. Applicant is willing and available to participate in any interview requested by the examiner concerning the present response. 6. This certification and request is being filed electronically using the Office's electronic filing system (EFS-Web). 7. Any fees that would be necessary consistent with current practice concerning responses after final rejection under 37 CFR 1.116, <i>e.g.</i>, extension of time fees, are being concurrently filed herewith. [There is no additional fee required to request consideration under AFCP 2.0.] 8. By filing this certification and request, applicant acknowledges the following: • Reissue applications and reexamination proceedings are not eligible to participate in AFCP 2.0. • The examiner will verify that the AFCP 2.0 submission is compliant, <i>i.e.</i>, that the requirements of the program have been met (see items 1 to 7 above). For compliant submissions: ○ The examiner will review the response under 37 CFR 1.116 to determine if additional search and/or consideration (i) is necessitated by the amendment and (ii) could be completed within the time allotted under AFCP 2.0. If additional search and/or consideration is required but cannot be completed within the allotted time, the examiner will process the submission consistent with current practice concerning responses after final rejection under 37 CFR 1.116, <i>e.g.</i>, by mailing an advisory action. ○ If the examiner determines that the amendment does not necessitate additional search and/or consideration, or if the examiner determines that additional search and/or consideration is required and could be completed within the allotted time, then the examiner will consider whether the amendment places the application in condition for allowance (after completing the additional search and/or consideration, if required). If the examiner determines that the amendment does not place the application in condition for allowance, then the examiner will contact the applicant and request an interview. ▪ The interview will be conducted by the examiner, and if the examiner does not have negotiation authority, a primary examiner and/or supervisory patent examiner will also participate. ▪ If the applicant declines the interview, or if the interview cannot be scheduled within ten (10) calendar days from the date that the examiner first contacts the applicant, then the examiner will proceed consistent with current practice concerning responses after final rejection under 37 CFR 1.116.		
Signature /Jamie J. Zheng, Reg. No. 51167/	Date February 8, 2018	
Name (Print/Typed) Jamie J. Zheng	Practitioner Registration No. 51167	
Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.		
☒ * Total of <u>1</u> forms are submitted.		

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/229,844
Filed : March 29, 2014
For : A Self-Test Memory Module and
Method of Operation
Examiner : CHUNG, PHUNG M
Art Unit : 2117
Conf. No. : 1018
Docket No. : N3044.10010US04

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence and any other attachment noted on the Automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web Server on:

February 8, 2018

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT**Mail Stop Amendment**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Madam:

Applicant submits the following amendments and remarks in response to the Final Office Action Dated December 8, 2017 in the above-identified patent application.

Submitted herewith is a Certification and Request for Consideration under the After Final Consideration Pilot Program 2.0.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5394.

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 11 of this paper.

In the Specification:

Please replace the title to the following:

| ~~Memory Subsystem and Method of Operation~~ A SELF-TEST MEMORY MODULE
AND METHOD OF OPERATION

In the Claims:

1. (Currently Amended) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements; and

a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and

wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode ~~normal mode and a test mode~~;

wherein, ~~during the normal mode~~, the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals, and the data module in the first mode is configured to propagate first ~~propagate~~ data signals between the memory devices and the system memory controller, ~~the data signals being transmitted or received by~~ the memory devices receiving or outputting the data signals in response to the first memory address and control signals; and

wherein, ~~during the test mode~~, the control module in the second mode is configured to output ~~self test~~ second memory address and control signals to the address and control ports of the memory devices, and the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit second data signals having ~~self test~~ data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, and wherein ~~the self test data patterns are received by~~ at least a portion of the memory devices are configured to receive the second data signals according to the second memory ~~self test~~ address and control signals from the control module.

2. (Currently Amended) The memory module of claim 1, wherein the memory devices are arranged in at least one rank and organized in a plurality of memory device groups, each rank of the at least one rank having a set of memory devices across a full bit width of the memory module, each respective memory device group of the plurality of memory device groups including respective one or more memory devices in each rank of the at least one rank, wherein each data signal of the first data signals propagated by the data module during the ~~normal~~-first mode is N-bits wide, wherein the data module includes a plurality of data handlers, each respective data handler of the plurality of data handlers including a respective one of the data handler logic elements, wherein the each respective data handler is configured to propagate a respective n-bit section of the each data signal of the first data signals between the system memory controller and a respective memory device group of the plurality of memory device groups, n being a fraction of N, and wherein the each respective data handler is further configured to output a respective n-bit wide section of each of the second data signals having a respective data pattern ~~self test data patterns~~ provided by the respective one of the data handler logic elements to the respective memory device group.

3. (Previously Presented) The memory module of claim 2, wherein each memory device is n-bits wide and wherein the respective memory device group includes a single respective memory device in each rank of the at least one rank.

4. (Currently Amended) The memory module of claim 2, wherein the control module is configured to output ~~additional self test~~ third memory address and control signals to the address and control ports of the memory devices during the ~~test~~-second mode, the ~~additional self test~~ third memory address and control signals causing the memory devices in one of the at least one rank to output a N-bit wide read data signal, and wherein the each respective data handler is configured to receive a respective n-bit section of the N-bit wide read data signal.

5. (Currently Amended) The memory module of claim 4, wherein the each respective data handler is configured to check the respective n-bit section of the N-bit

wide read data signal with respect to ~~expected self-test data patterns~~ a corresponding data pattern stored in the respective data handler logic element.

6. (Cancelled)

7. (Currently Amended) The memory module of claim 5, wherein the data module is configured to store failure information when a test failure is indicated by the respective n-bit section of the N-bit wide read data signal not being in agreement with the ~~expected~~ corresponding data pattern.

8. (Previously Presented) The memory module of claim 7, wherein the data module is further configured to report the failure information to the control module.

9. (Cancelled)

10. (Currently Amended) The memory module of claim 1, wherein the data module is configured to output the second data signals ~~self-test data patterns~~ to the memory devices in response to a write command output from the control module, and wherein the data module is further configured to receive the N-bit wide read data signal ~~read-back test data patterns~~ from the memory devices in response to a read command output from the control module.

11. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers are configured to operate independently of each other without being in communication with each other, ~~and wherein, during the test mode, the each respective data handler is configured to output the respective n-bit wide self-test data patterns to test a corresponding portion of a memory space of the memory module independently of the other data handlers.~~

12. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is ~~positioned~~ disposed on the circuit board at a position corresponding proximate to the respective memory device group.

13. (Cancelled)

14. (Cancelled)

15. (Cancelled)

16. (Cancelled)

17. (Cancelled)

18. (Cancelled)

19. (Cancelled)

20. (Cancelled)

21. (Cancelled)

22. (Cancelled)

23. (Cancelled)

24. (Cancelled)

25. (Cancelled)

26. (Cancelled)

27. (Cancelled)

28. (Currently Amended) The memory module of claim 2, wherein the each respective data handler ~~is configured to provide~~ includes first data paths between the respective memory device group and the system memory controller ~~or and~~ and second data paths between the respective memory device group and the respective one of the data handler logic elements, and wherein the each respective data handler is configured to propagate the respective n-bit section of the each data signal using the first data paths and is configured to transmit the respective n-bit wide section of each of the second data signals using the second data paths ~~depending on whether the memory module is in the normal mode or test mode.~~

- 29. (Cancelled)
- 30. (Cancelled)
- 31. (Cancelled)
- 32. (Cancelled)
- 33. (Cancelled)
- 34. (Cancelled)
- 35. (Cancelled)
- 36. (Previously Presented) The memory module of claim 2, wherein each memory device is $n/2$ -bit wide and wherein the respective memory device group includes two respective memory devices in each of the at least one rank.
- 37. (Cancelled)
- 38. (Cancelled)
- 39. (Cancelled)
- 40. (Cancelled)
- 41. (Cancelled)
- 42. (Cancelled)
- 43. (Cancelled)
- 44. (Cancelled)
- 45. (Cancelled)
- 46. (Cancelled)
- 47. (Cancelled)

48. (Cancelled)
49. (Cancelled)
50. (Cancelled)
51. (Cancelled)
52. (Cancelled)
53. (Cancelled)
54. (Cancelled)
55. (Cancelled)
56. (Cancelled)
57. (Cancelled)
58. (Currently Amended) A memory module operable in a computer system with a system memory controller, the memory module comprising:
- a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller;
 - memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;
 - a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and
 - a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more

memory devices including at least one respective memory device in each of the one or more ranks;

wherein the memory module is configurable to operate in ~~one~~any of at least a first mode and a second mode;

wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations, and the data handlers are configured to propagate one or more data signals associated with the memory read or write operations between the memory devices and the connector;

wherein, in the second mode, the memory devices are not ~~accessed~~accessible by the system memory controller, and the each respective data handler is configured to provide respective data patterns ~~of data signals~~ to the respective group of one or more memory devices based on information ~~received~~output from the control module.

59. (Currently Amended) The memory module of claim 58, wherein each of the one or more data signals is N-bit wide, and wherein, when the memory module is operating in the first mode, the each respective data handler is configured to propagate a respective n-bit section of each of the one or more data signals between at least one respective memory device in the at least one rank and the connector, n being a fraction of N; and wherein, when the memory module is operating in the second mode, the respective data patterns ~~of data signals~~ are each n-bits wide.

60. (Previously Presented) The memory module of claim 59, wherein each memory device is n-bit wide and wherein the respective group of one or more memory devices include a single respective memory device in each of the one or more ranks.

61. (Previously Presented) The memory module of claim 59, wherein each memory device is n/2-bit wide and wherein the respective group of one or more memory devices include two respective memory devices in each of the one or more ranks.

62. (Currently Amended) The memory module of claim 58, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is ~~positioned~~disposed on the circuit board at a position corresponding proximate to the respective group of one or more memory devices.

63. (Currently Amended) The memory module of claim 58, wherein the each respective data handler is configured to output the respective data patterns of data signals to the respective group of one or more memory devices in response to a write command output from the control module.

64. (Currently Amended) The memory module of claim 63, wherein, when the memory module is operating in the second mode, the each respective data handler is further configured to receive ~~patterns of read~~ data signals from the respective group of one or more memory devices in response to a read command output from the control module.

65. (Currently Amended) The memory module of claim 64, wherein the each respective data handler is further configured to check the ~~patterns of read~~ data signals received from the respective group of one or more memory devices with respect to expected-corresponding data patterns stored in the each respective data handler.

66. (Currently Amended) The memory module of claim ~~59-58~~, wherein the each respective data handler includes a respective data handler logic element configured to output the respective patterns of data signals, wherein the each respective data handler is ~~configured to provide~~ includes first data paths between the respective group of one or more memory devices and the connector and ~~or~~ second data paths between the respective group of one or more memory devices and the respective data handler logic element, and wherein the each respective data handler is configured to propagate the respective n-bit section of the each of the one or more data signals using the first data paths and is configured to provide the data patterns using the second data paths depending on whether the memory module is configured to operate in the first mode or the second mode, the second data paths being different from the first data paths.

67. (Cancelled)

68. (Cancelled)

69. (Cancelled)

70. (Cancelled)

71. (Cancelled)

72. (Cancelled)

73. (Cancelled)

74. (Cancelled)

75. (Cancelled)

REMARKS

This amendment is in response to the Final Office Action dated December 8, 2017 (the "Office Action"). In the Office Action, the Examiner:

- withdraw claims 49-75 from consideration as being directed to a non-elected invention;
- rejected claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 101;
- rejected claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph.

Claim Amendment

Claims 26-27, 49-57, and 67-75 are cancelled. Claims 1-2, 4-5, 7, 10-12, 28, 58-59, 62-66 are amended. Support for the amendment can be found throughout the Specification and Drawings, for example, in paragraphs [0008] and [0040]. Therefore, no new matter is added.

Claim 1 is amended to correct a typographical error, to rename some of the claim terms, and to add clarity to some of the claim limitations. Claims 2, 4-5, 7, 10-12, and 28 are amended for consistency with the amendments made in claim 1 and for clarity. Claims 58-66 are also amended for clarity and consistency. Thus, the amendments to the claims do not raise issues that would require further consideration and/or search.

Withdrawal of claims 49-75 from consideration as being directed to a non-elected invention

The Office Action states that claims 49-75 are directed to an invention that is independent or distinct from the invention originally claimed and withdrew these claims from consideration. Applicant respectfully disagree. As laid out in the following table, claim 58 is clearly directed to the same subject matter, a memory module, and the limitations in claim 58 can be mapped to corresponding limitations in claims 1 and 2. Therefore, the withdrawal of at least claims 58-66 from consideration was improperly made and reconsideration of these claims are respectfully requested. Claims 49-57 and 67-75 are cancelled without prejudice to expedite prosecution.

Claim 58	Claims 1 and 2
A memory module operable in a computer system with a system memory controller, the memory module comprising:	(Claim 1) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:
a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller memory devices mounted on the PCB	(Claim 1) memory devices mounted on a circuit board, the memory devices having address and control ports and data ports
and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module	(Claim 2) wherein the memory devices are arranged in at least one rank ... each rank of the at least one rank having a set of memory devices across a full bit width of the memory module
a control module mounted on the PCB and electrically coupled to the connector and to the memory devices,	(Claim 1) a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus
wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and	(Claim 1) wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output memory address and control signals to the memory devices according to the system address and control signals, wherein the control module in the second mode is configured to output second memory address and control signals to the address and control ports of the memory devices
a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector,	(Claim 1) a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements (Claim 2) wherein the data module includes a plurality of data handlers,... wherein the each respective data handler is configured to propagate a respective n-bit section of the each data signal between the system memory controller and a respective memory device group of the plurality of memory device groups, n being a fraction of N, and wherein the each respective data handler is further configured to output respective n-bit wide self-test data patterns provided by the respective one of the data handler logic elements to the respective memory device group

Claim 58	Claims 1 and 2
the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks	(Claim 2) wherein the memory devices are ... organized in a plurality of memory device groups, ... each respective memory device group of the plurality of memory device groups including respective one or more memory devices in each rank of the at least one rank
wherein the memory module is configurable to operate in any of at least a first mode and a second mode	(Claim 1) wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode
wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations,	(Claim 1) wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output memory address and control signals to the memory devices according to the system address and control signals,
and the data handlers are configured to propagate one or more data signals associated with the memory read or write operations between the memory devices and the connector,	(Claim 1) and the data module in the first mode is configured to propagate data signals between the memory devices and the system memory controller,
wherein, in the second mode, the memory devices are not accessible by the system memory controller,	(Claim 1) wherein the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller
and the each respective data handler is configured to provide respective data patterns to the respective group of one or more memory devices based on information output from the control module.	(Claim 1) the data module in the second mode is configured to ... transmit second data signals having data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, (Claim 2) wherein the each respective data handler is further configured to output a respective n-bit wide section of each of the second data signals having a respective data pattern provided by the respective one of the data handler logic elements to the respective memory device group.

Rejection of claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 101

The Office Action rejected claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 101 as being directed to a judicial exception (i.e., law of nature, a natural phenomenon, or

abstract idea) without significantly more. In rejection claim 1 under 35 U.S.C. 101, the Office Action separates the functional limitations of:

"during the normal mode: configured to receive system address and control signal from the system memory controller and to output memory address and control signals to the memory device according to the system address and control signals", and

"configured to propagate data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals"; and

"during the test mode: configured to output self-test address and control signals to the address and control ports of the memory devices";

"configured to isolate the memory devices from being accessed by the system memory controller and to transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory device according to one or more commands from the control module, and wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals from the control module"

from their corresponding structures, and states that:

"These limitations are directed to intended use statements, and these limitations are abstract idea without significantly more, similar to Cyberfone, the concept of using categories to organize, store and transmit information; Cap One Financial, the concept of collecting, displaying, and manipulating data; and Content Extraction, the concept of data recognition and storage, were found to be abstract ideas."

Applicant respectfully disagrees.

First of all, claim 1 claims a memory module comprising memory devices, a data module, a control module, and other structural and functional features. It has long been established that "[f]eatures of an apparatus may be recited either structurally or functionally. In re Schreiber, 128 F.3d 1473, 1478, 44 USPQ2d 1429, 1432 (Fed. Cir. 1997)." See MPEP 2114. "A claim term is functional when it recites a feature 'by what it does rather than by what it is' (e.g., as evidenced by its specific structure or specific ingredients). In re Swinehart, 439 F.2d 210, 212, 169 USPQ 226, 229 (CCPA 1971). There is nothing inherently wrong with defining some part of an invention in functional terms. Functional language does not, in and of itself, render a claim improper. Id." See MPEP § 2173.05(g).

The above-cited limitations of claim 1 use the term “configured to” to recite some of the functional features in the claimed memory module. It has been established that “configured to” in claims requires a structure that is more than just capable of performing a function, but is actually made so that it can be used to perform the function without further modification. *Aspex Eyewear, Inc. v. Marchon Eyewear, Inc.* (Fed. Cir. 2012) Therefore, claim 1 as amended recites a memory module operable in any of a plurality of modes including a first mode and a second mode, the memory module comprising, among other things, a data module and a control module configured to perform the respective functions recited in the claims depending on which mode the memory module is in. The recitation of these functions are not intended use statements, as alleged in the Office Action. Instead, these functions are what the data module and the control module are capable of doing because of how they are designed and made, and are therefore meaningful limitations thereof.

Applicant is baffled by the statement in the Office Action that claim 1 is “similar to Cyberfone, the concept of using categories to organize, store and transmit information; Cap One Financial, the concept of collecting, displaying, and manipulating data; and Content Extraction, the concept of data recognition and storage.” It has been warned by the courts that the claims of a patent should not be described at a “high level of abstraction” and untethered from the language of the claims, all but ensuring that the exceptions to Section 101 swallow the rule. *Enfish v. Microsoft*, 822 F.3d 1327 (Fed. Cir. 2016) The Federal Circuit ruling in *Enfish* is a reminder that patent claims must be read as a whole when analyzing eligibility under Section 101, and care must be taken not to simply water-down the claims to a high-level, abstract concept.

The Office Action, in rejecting claim 1 under 35 U.S.C. 101, not only oversimplifies claim 1 but also severely mischaracterizes it. Claim 1 does not recite anything close to “using categories to organize, store and transmit information,” or “collecting, displaying, and manipulating data,” or “data recognition and storage” None of these concepts is even remotely similar to what is claimed in claim 1. Claim 1 is NOT about data processing. It claims an apparatus with specifically structured hardware components that are designed and made to be configurable to perform specific hardware

functions, such as, for example, “to propagate data signals between the memory devices and the system memory controller,” “to isolate the memory devices from being accessed by the system memory controller,” and “to transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands from the control module,” etc.

The Office Action further errs by stating that “the recited ‘memory devices’, ‘circuit board’, ‘control ports’, ‘data ports’, ‘system memory bus’, ‘data handler logic elements’, ‘data module’, ‘control module’, ‘system memory controller’ are well-known and conventional/generic computer components performing well-known and conventional/generic computer functions.”

As provided in MPEP 2114, “[i]f an examiner concludes that a functional limitation is an inherent characteristic of the prior art, then to establish a prima case of anticipation or obviousness, the examiner should explain that the prior art structure inherently possesses the functionally defined limitations of the claimed apparatus. In re Schreiber, 128 F.3d at 1478, 44 USPQ2d at 1432. See also Bettcher Industries, Inc. v. Bunzl USA, Inc., 661 F.3d 629, 639-40, 100 USPQ2d 1433, 1440 (Fed. Cir. 2011).”

Here, the Office Action presents no prior art memory module with the structural and functional features recited in claim 1. The allegation that the memory module of claim 1 and its various components are well-known and conventional/generic computer components performing well-known and conventional/generic computer functions are therefore baseless.

At the time the present application was filed, there were no well-known and conventional/generic memory modules operable in any of a plurality of modes including a first mode and a second mode and including a data module coupled between the data ports of the memory devices and the system memory bus, wherein the data module in the first mode is configured to propagate data signals between the memory devices and the system memory controller, and wherein the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit data patterns provided by the data handler logic elements to the

data ports of the memory devices according to one or more commands output from the control module. Therefore, the data module cannot be well-known and conventional/generic. The same can be said about other components of the memory module of claim 1.

Therefore, the rejection of claim 1 and its dependent claims, claims 2-5, 7-8, 10-12, 28 and 36 under 35 U.S.C. 101 should be withdrawn.

Rejection of Claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), Second paragraph

The Office Action rejected claim 1 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, citing line 19-29 of claim 1 and stating:

“[It] is not clear whether, during the test mode, the self-test data patterns that are received by at least a portion of the memory devices according to the self-test address and control signals from the control module is actually doing the test for that the at least a portion of the memory devices according to the self-test address and control signals from the control module or just for storage over there? How the at least a portion of the memory device is tested is unclear and what is the result and/or what to with the test result?”

It is within the knowledge of those skilled in the art that memory modules are commonly tested by transmitting test data to their memory devices, which can be configured to receive and store the test data according to address and control signals. The test data are then read back from the memory devices for comparison. See, for example, US6829728B2 and US6928593B1, which are cited in the Information Disclosure Statements filed on February 15, 2016 in the present application.

Claim 1 recites a memory module that improves upon the prior art by being operable in any of a plurality of modes including a first mode and a second mode, and by including, among other things, a control module and a data module configured to perform certain functions depending on whether the memory module is in the first mode or second mode. For example, the data module in the first mode “is configured to propagate first data signals between the memory devices and the system memory controller”, and in the second mode is configured “to isolate the memory devices from being accessed by the

system memory controller and to transmit second data signals having data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module.” These limitations are consistent with the teachings of the Specification. (See, for example, paragraphs [0025]-[0026] and [0041]-[0042]).

Thus, claim 1 is an apparatus claim with structural and functional limitations that set forth the subject matter which the inventor or a joint inventor regards as the invention and that particularly point out and distinctly define the metes and bounds of the subject matter to be protected by the patent grant. Infringement can be proven by showing that a memory module include the structural and functional limitations clearly recited in the claim. Therefore, the requirement of 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, is met.

Furthermore, 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, does not require that claim 1 include details on “what is the result and/or what to with the test result,” as inquired in the Office Action. Answers to these inquiries can be found in the Specification. (See, for example, paragraphs [0042]-0045])

The above arguments apply to claims 2-5, 7-8, 10-12, 28 and 36.

Therefore, the rejection of claims 1-5, 7-8, 10-12, 28 and 36 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, should be withdrawn.

Applicant submits that claims 1-5, 7-8, 10-12, 28, 36, and 49-75 are in condition for allowance and respectfully requests such action. Applicant respectfully requests for a telephone interview with Examiner under the After Final Consideration Pilot Program 2.0 to resolve any remaining issues.

Respectfully submitted,

Date: February 8, 2018

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Electronic Acknowledgement Receipt

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Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	After Final Consideration Program Request	2018-02-08_AFCP_N3044-10010US04.pdf	226777	no	2
			99941d1b691499ec82aab51c1cf82275cc3daaba		

Warnings:

Information:					
2		2018-02-08_OA-Response_N3044-10010US04.pdf	489711	yes	19
			38c405c55b2803ec3711f7b1c9bf84674b026863		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Response After Final Action		1	1	
	Specification		2	2	
	Claims		3	11	
	Applicant Arguments/Remarks Made in an Amendment		12	19	
Warnings:					
Information:					
Total Files Size (in bytes):			716488		
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 14/229,844		Filing Date 03/29/2014		☐ To be Mailed			
ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$40 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$210 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	02/08/2018	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 21	Minus	** 28	= 0			x \$50 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0			x \$230 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
AMENDMENT		(Column 1)		(Column 2)		(Column 3)					
		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$ 0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$ 0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))											
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								myrtle B leigh			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	N3044.10010US04	1018
79141	7590	02/14/2018	EXAMINER	
Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098			CHUNG, PHUNG M	
			ART UNIT	PAPER NUMBER
			2117	
			NOTIFICATION DATE	DELIVERY MODE
			02/14/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

d@mabr.com
bisraelsen@mabr.com

Advisory Action Before the Filing of an Appeal Brief	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

--The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

THE REPLY FILED 08 February 2018 FAILS TO PLACE THIS APPLICATION IN CONDITION FOR ALLOWANCE.

NO NOTICE OF APPEAL FILED

1. ☐ The reply was filed after a final rejection. No Notice of Appeal has been filed. To avoid abandonment of this application, applicant must timely file one of the following replies: (1) an amendment, affidavit, or other evidence, which places the application in condition for allowance;
 (2) a Notice of Appeal (with appeal fee) in compliance with 37 CFR 41.31; or (3) a Request for Continued Examination (RCE) in compliance with 37 CFR 1.114 if this is a utility or plant application. Note that RCEs are not permitted in design applications. The reply must be filed within one of the following time periods:

a) ☐ The period for reply expires _____ months from the mailing date of the final rejection.

b) ☒ The period for reply expires on: (1) the mailing date of this Advisory Action; or (2) the date set forth in the final rejection, whichever is later. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of the final rejection.

c) ☐ A prior Advisory Action was mailed more than 3 months after the mailing date of the final rejection in response to a first after-final reply filed within 2 months of the mailing date of the final rejection. The current period for reply expires _____ months from the mailing date of the prior Advisory Action or SIX MONTHS from the mailing date of the final rejection, whichever is earlier.

Examiner Note: If box 1 is checked, check either box (a), (b) or (c). ONLY CHECK BOX (b) WHEN THIS ADVISORY ACTION IS THE FIRST RESPONSE TO APPLICANT'S FIRST AFTER-FINAL REPLY WHICH WAS FILED WITHIN TWO MONTHS OF THE FINAL REJECTION. ONLY CHECK BOX (c) IN THE LIMITED SITUATION SET FORTH UNDER BOX (c). See MPEP 706.07(f).

Extensions of time may be obtained under 37 CFR 1.136(a). The date on which the petition under 37 CFR 1.136(a) and the appropriate extension fee have been filed is the date for purposes of determining the period of extension and the corresponding amount of the fee. The appropriate extension fee under 37 CFR 1.17(a) is calculated from: (1) the expiration date of the shortened statutory period for reply originally set in the final Office action; or (2) as set forth in (b) or (c) above, if checked. Any reply received by the Office later than three months after the mailing date of the final rejection, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

NOTICE OF APPEAL

2. ☐ The Notice of Appeal was filed on _____. A brief in compliance with 37 CFR 41.37 must be filed within two months of the date of filing the Notice of Appeal (37 CFR 41.37(a)), or any extension thereof (37 CFR 41.37(e)), to avoid dismissal of the appeal. Since a Notice of Appeal has been filed, any reply must be filed within the time period set forth in 37 CFR 41.37(a).

AMENDMENTS

3. ☒ The proposed amendments filed after a final rejection, but prior to the date of filing a brief, will not be entered because

a) ☒ They raise new issues that would require further consideration and/or search (see NOTE below);

b) ☐ They raise the issue of new matter (see NOTE below);

c) ☐ They are not deemed to place the application in better form for appeal by materially reducing or simplifying the issues for appeal; and/or

d) ☐ They present additional claims without canceling a corresponding number of finally rejected claims.

NOTE: _____. (See 37 CFR 1.116 and 41.33(a)).

4. ☐ The amendments are not in compliance with 37 CFR 1.121. See attached Notice of Non-Compliant Amendment (PTOL-324).

5. ☐ Applicant's reply has overcome the following rejection(s): _____.

6. ☐ Newly proposed or amended claim(s) _____ would be allowable if submitted in a separate, timely filed amendment canceling the non-allowable claim(s).

7. ☒ For purposes of appeal, the proposed amendment(s): (a) ☒ will not be entered, or (b) ☐ will be entered, and an explanation of how the new or amended claims would be rejected is provided below or appended.

AFFIDAVIT OR OTHER EVIDENCE

8. ☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

9. ☐ The affidavit or other evidence filed after final action, but before or on the date of filing a Notice of Appeal will not be entered because applicant failed to provide a showing of good and sufficient reasons why the affidavit or other evidence is necessary and was not earlier presented. See 37 CFR 1.116(e).

10. ☐ The affidavit or other evidence filed after the date of filing the Notice of Appeal, but prior to the date of filing a brief, will not be entered because the affidavit or other evidence failed to overcome all rejections under appeal and/or appellant fails to provide a showing of good and sufficient reasons why it is necessary and was not earlier presented. See 37 CFR 41.33(d)(1).

11. ☐ The affidavit or other evidence is entered. An explanation of the status of the claims after entry is below or attached.

REQUEST FOR RECONSIDERATION/OTHER

12. ☒ The request for reconsideration has been considered but does NOT place the application in condition for allowance because:
The underline portions of claims 1, 2, 4, 5, 7, 10, 11, 12, 28, 58, 59, 62, 63, 64, 65 and 66 would require further consideration and search.

13. ☐ Note the attached Information *Disclosure Statement*(s). (PTO/SB/08) Paper No(s). _____

14. ☐ Other: _____.

STATUS OF CLAIMS

15. The status of the claim(s) is (or will be) as follows:

Claim(s) allowed: _____.

Claim(s) objected to: _____.

Claim(s) rejected: 1-5, 7-8, 10-12, 26-29 and 36 (See the Final Office Action).

Claim(s) withdrawn from consideration: _____.

/Phung M. Chung/
Primary Examiner

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL (Submitted Only via EFS-Web)

Application Number	14229844	Filing Date	2014-03-29	Docket Number (if applicable)	N3044.10010US04	Art Unit	2117
First Named Inventor	Hyun Lee			Examiner Name	CHUNG, PHUNG M		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV.

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☒ Amendment/Reply

☐ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☐ Other _____

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to Deposit Account No 505394

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

Applicant Signature

Doc code: RCEX

Doc description: Request for Continued Examination (RCE)

PTO/SB/30EFS (07-14)

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Signature of Registered U.S. Patent Practitioner			
Signature	JAMIE J. ZHENG/	Date (YYYY-MM-DD)	2018-02-15
Name	Jamie Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

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Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/229,844
Filed : March 29, 2014
For : A Multi-Mode Memory Module
with Data Handlers
Examiner : CHUNG, PHUNG M
Art Unit : 2117
Conf. No. : 1018
Docket No. : N3044.10010US04

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence and
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Web Server on:

February 15, 2018

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT AND REQUEST FOR CONTINUED EXAMINATION**Mail Stop Amendment**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Madam:

The following amendments and remarks are in response to the Final Office Action
Dated December 8, 2017 in the above-identified patent application, and replace the
amendment filed on February 8, 2018, which was not entered.

Submitted herewith is a Request for Continued Examination.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit
Account No. 50-5394.

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 13 of this paper.

In the Specification:

Please replace the title to the following:

| ~~Memory Subsystem and Method of Operation~~ A Multi-Mode Memory Module with Data
Handlers

In the Claims:

1. (Currently Amended) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements; and

a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and

wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode ~~normal mode and a test mode~~;

wherein, ~~during the normal mode~~, the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals, and the data module in the first mode is configured to propagate one or more first ~~propagate~~ data signals between the memory devices and the system memory controller, the one or more first data signals being transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals; and

wherein, ~~during the test mode~~, the control module in the second mode is configured to output ~~self test~~ second memory address and control signals to the address and control ports of the memory devices, and the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit one or more second data signals including ~~self test~~ data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, and wherein ~~the self test data patterns are received by~~ at least a portion of the memory devices are configured to receive the one or more second data signals according to the second memory ~~self test~~ address and control signals from the control module.

2. (Currently Amended) The memory module of claim 1, wherein the memory devices are arranged in at least one rank and organized in a plurality of memory device groups, each rank of the at least one rank having a set of memory devices across a full bit width of the memory module, each respective memory device group of the plurality of memory device groups including respective one or more memory devices in each rank of the at least one rank, wherein each data signal of the one or more first data signals propagated by the data module during the ~~normal~~-first mode is N-bits wide, wherein the data module includes a plurality of data handlers, each respective data handler of the plurality of data handlers including a respective one of the data handler logic elements, wherein the each respective data handler is configured to propagate a respective n-bit section of the each data signal of the one or more first data signals between the system memory controller and a respective memory device group of the plurality of memory device groups, n being a fraction of N, and wherein the each respective data handler is further configured to output a respective n-bit wide section of each of the one or more second data signals including one or more respective n-bit-wide data patterns ~~self test data patterns~~ provided by the respective one of the data handler logic elements to the respective memory device group.

3. (Previously Presented) The memory module of claim 2, wherein each memory device is n-bits wide and wherein the respective memory device group includes a single respective memory device in each rank of the at least one rank.

4. (Currently Amended) The memory module of claim 2, wherein the control module is configured to output ~~additional self test~~-third memory address and control signals to the address and control ports of the memory devices during the ~~test~~-second mode, the ~~additional self test~~-third memory address and control signals causing the memory devices in one of the at least one rank to output at least one N-bit wide read data signal, and wherein the each respective data handler is configured to receive a respective n-bit section of each of the at least one N-bit wide read data signal.

5. (Currently Amended) The memory module of claim 4, wherein the each respective data handler is configured to check the respective n-bit section of each of the

at least one N-bit wide read data signal with respect to ~~expected self test data patterns~~ a corresponding data pattern stored in the respective data handler logic element.

6. (Cancelled)

7. (Currently Amended) The memory module of claim 5, wherein the data module is configured to store failure information when a test failure is indicated by the respective n-bit section of each of the at least one N-bit wide read data signal not being in agreement with the ~~expected-corresponding~~ data pattern.

8. (Previously Presented) The memory module of claim 7, wherein the data module is further configured to report the failure information to the control module.

9. (Cancelled)

10. (Currently Amended) The memory module of claim 1, wherein the data module is configured to output the one or more second data signals ~~self test data patterns~~ to the memory devices in response to a write command output from the control module, and wherein the data module is further configured to receive the at least one N-bit wide read data signal ~~read back test data patterns~~ from the memory devices in response to a read command output from the control module.

11. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers are configured to operate independently of each other without being in communication with each other, ~~and wherein, during the test mode, the each respective data handler is configured to output the respective n-bit wide self test data patterns to test a corresponding portion of a memory space of the memory module independently of the other data handlers.~~

12. (Currently Amended) The memory module of claim 2, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is ~~positioned-disposed~~ on the circuit board at a position corresponding proximate to the respective memory device group.

13. (Cancelled)

14. (Cancelled)

15. (Cancelled)

16. (Cancelled)

17. (Cancelled)

18. (Cancelled)

19. (Cancelled)

20. (Cancelled)

21. (Cancelled)

22. (Cancelled)

23. (Cancelled)

24. (Cancelled)

25. (Cancelled)

26. (Cancelled)

27. (Cancelled)

28. (Currently Amended) The memory module of claim 2, wherein the each respective data handler ~~is configured to provide~~ includes first data paths between the respective memory device group and the system memory controller ~~or and~~ and second data paths between the respective memory device group and the respective one of the data handler logic elements, and wherein the each respective data handler is configured to propagate the respective n-bit section of the each data signal of the one or more first data signals using the first data paths, and is further configured to transmit the respective n-bit wide section of each of the one or more second data signals using the second data paths depending on whether the memory module is in the normal mode or test mode.

29. (Cancelled)
30. (Cancelled)
31. (Cancelled)
32. (Cancelled)
33. (Cancelled)
34. (Cancelled)
35. (Cancelled)
36. (Previously Presented) The memory module of claim 2, wherein each memory device is $n/2$ -bit wide and wherein the respective memory device group includes two respective memory devices in each of the at least one rank.
37. (Cancelled)
38. (Cancelled)
39. (Cancelled)
40. (Cancelled)
41. (Cancelled)
42. (Cancelled)
43. (Cancelled)
44. (Cancelled)
45. (Cancelled)
46. (Cancelled)
47. (Cancelled)

48. (Cancelled)
49. (Cancelled)
50. (Cancelled)
51. (Cancelled)
52. (Cancelled)
53. (Cancelled)
54. (Cancelled)
55. (Cancelled)
56. (Cancelled)
57. (Cancelled)
58. (Currently Amended) A memory module operable in a computer system with a system memory controller, the memory module comprising:
- a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller;
 - memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;
 - a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and
 - a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more

memory devices including at least one respective memory device in each of the one or more ranks;

wherein the memory module is configurable to operate in ~~one~~any of at least a first mode and a second mode;

wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations, and the data handlers are configured to propagate one or more data signals associated with the memory read or write operations between the memory devices and the connector;

wherein, in the second mode, the memory devices are not accessed by the system memory controller, and the each respective data handler is configured to provide respective data patterns ~~of data signals~~ to the respective group of one or more memory devices based on information ~~received~~output from the control module.

59. (Currently Amended) The memory module of claim 58, wherein each of the one or more data signals is N-bit wide, and wherein, when the memory module is operating in the first mode, the each respective data handler is configured to propagate a respective n-bit section of each of the one or more data signals between at least one respective memory device in the at least one rank and the connector, n being a fraction of N; and wherein, when the memory module is operating in the second mode, the respective data patterns ~~of data signals~~ are each n-bits wide.

60. (Previously Presented) The memory module of claim 59, wherein each memory device is n-bit wide and wherein the respective group of one or more memory devices include a single respective memory device in each of the one or more ranks.

61. (Previously Presented) The memory module of claim 59, wherein each memory device is n/2-bit wide and wherein the respective group of one or more memory devices include two respective memory devices in each of the one or more ranks.

62. (Currently Amended) The memory module of claim 58, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is ~~positioned~~disposed on the circuit board at a position corresponding proximate to the respective group of one or more memory devices.

63. (Currently Amended) The memory module of claim 58, wherein the each respective data handler is configured to output the respective data patterns of data signals to the respective group of one or more memory devices in response to a write command output from the control module.

64. (Currently Amended) The memory module of claim 63, wherein, when the memory module is operating in the second mode, the each respective data handler is further configured to receive read data patterns of data signals from the respective group of one or more memory devices in response to a read command output from the control module.

65. (Currently Amended) The memory module of claim 64, wherein the each respective data handler is further configured to check the read data patterns of data signals received from the respective group of one or more memory devices with respect to expected-corresponding data patterns stored in the each respective data handler.

66. (Currently Amended) The memory module of claim 59-58, wherein the each respective data handler includes a respective data handler logic element configured to output the respective patterns of data signals, wherein the each respective data handler is configured to provide includes first data paths between the respective group of one or more memory devices and the connector and ~~or~~ second data paths between the respective group of one or more memory devices and the respective data handler logic element, and wherein the each respective data handler is configured to propagate the respective n-bit section of the each of the one or more data signals using the first data paths, and is further configured to provide the respective data patterns using the second data paths depending on whether the memory module is configured to operate in the first mode or the second mode, the second data paths being different from the first data paths.

67. (Cancelled)

68. (Cancelled)

69. (Cancelled)

- 70. (Cancelled)
- 71. (Cancelled)
- 72. (Cancelled)
- 73. (Cancelled)
- 74. (Cancelled)
- 75. (Cancelled)
- 76. (New) The memory module of claim 1, wherein the memory devices are arranged one or more ranks, each rank of the one or more ranks having a set of memory devices across a full bit width of the memory module, and wherein the second data signals are received by a selected rank of the one or more ranks according to the second memory address and control signals from the control module.
- 77. (New) The memory module of claim 2, wherein the second data signals are received by a selected rank of the one or more ranks according to the second memory address and control signals from the control module.
- 78. (New) The memory module of claim 1, wherein the data patterns provided by the data handler logic elements are based on information that the data handler logic elements receive from the control module.
- 79. (New) The memory module of claim 2, wherein the respective n-bit-wide data pattern is based on information the respective one of the data handler logic elements receive from the control module.
- 80. (New) The memory module of claim 1, wherein the control module is further configured to provide the data patterns to the data module.
- 81. (New) The memory module of claim 28, wherein the each respective data handler is further configured to isolate the first data paths in the second mode.

82. (New) The memory module of claim 58, wherein, in the second mode, the each respective data handler is further configured to isolate the respective group of one or more memory devices from the connector.

83. (New) The memory module of claim 59, wherein, in the second mode, the each respective data handler is further configured to isolate the respective group of one or more memory devices from the connector.

84. (New) The memory module of claim 58, wherein the memory devices are configured to receive or output the one or more data signals in response to corresponding address and control signals output from the control module.

85. (New) The memory module of claim 59, wherein the respective group of one or more memory devices are configured to receive or output the respective n-bit section of each of the one or more data signals in response to corresponding address and control signals output from the control module.

86. (New) The memory module of claim 59, wherein the each respective data handler is configured to output the respective data patterns to the respective group of one or more memory devices in response to a write command output from the control module.

87. (New) The memory module of claim 58, wherein the memory devices are configured to receive the data patterns provided by the data handlers in response to corresponding address and control signals output from the control module.

88. (New) The memory module of claim 59, wherein the respective group of one or more memory devices are configured to receive or output the respective data patterns in response to corresponding address and control signals output from the control module.

REMARKS

This amendment is in response to the Final Office Action dated December 8, 2017 (the "Office Action"). In the Office Action, the Examiner:

- withdraw claims 49-75 from consideration as being directed to a non-elected invention;
- rejected claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 101;
- rejected claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph.

Claim Amendment

Claims 26-27, 49-57, and 67-75 are cancelled. Claims 1-2, 4-5, 7, 10-12, 28, 58-59, 62-66 are amended. New claims 76-88 are added. Support for the amendment can be found throughout the Specification and Drawings, for example, in paragraphs [0008] and [0040]. Support for the new claims can be found throughout the Specification and Drawings, for example, in paragraphs [0025], [0040]-[0041], and [0050]. Therefore, no new matter is added.

Withdrawal of claims 49-75 from consideration as being directed to a non-elected invention

The Office Action states that claims 49-75 are directed to an invention that is independent or distinct from the invention originally claimed and withdrew these claims from consideration. Applicant respectfully disagree. As laid out in the following table, claim 58 is clearly directed to the same subject matter, a memory module, and the limitations in claim 58 can be mapped to corresponding limitations in claims 1 and 2. Therefore, the withdrawal of at least claims 58-66 from consideration was improperly made and reconsideration of these claims are respectfully requested. Claims 49-57 and 67-75 are cancelled without prejudice to expedite prosecution.

Claim 58	Claims 1 and 2
A memory module operable in a computer system with a system memory controller, the memory module comprising:	(Claim 1) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

Claim 58	Claims 1 and 2
a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller memory devices mounted on the PCB	(Claim 1) memory devices mounted on a circuit board, the memory devices having address and control ports and data ports
and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module	(Claim 2) wherein the memory devices are arranged in at least one rank ... each rank of the at least one rank having a set of memory devices across a full bit width of the memory module
a control module mounted on the PCB and electrically coupled to the connector and to the memory devices,	(Claim 1) a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus
wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and	(Claim 1) wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output memory address and control signals to the memory devices according to the system address and control signals, wherein the control module in the second mode is configured to output second memory address and control signals to the address and control ports of the memory devices
a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector,	(Claim 1) a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements (Claim 2) wherein the data module includes a plurality of data handlers,... wherein the each respective data handler is configured to propagate a respective n-bit section of the each data signal between the system memory controller and a respective memory device group of the plurality of memory device groups, n being a fraction of N, and wherein the each respective data handler is further configured to output respective n-bit wide self-test data patterns provided by the respective one of the data handler logic elements to the respective memory device group
the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks	(Claim 2) wherein the memory devices are ... organized in a plurality of memory device groups, ... each respective memory device group of the plurality of memory device groups

Claim 58	Claims 1 and 2
	including respective one or more memory devices in each rank of the at least one rank
wherein the memory module is configurable to operate in any of at least a first mode and a second mode	(Claim 1) wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode
wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations,	(Claim 1) wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output memory address and control signals to the memory devices according to the system address and control signals,
and the data handlers are configured to propagate one or more data signals associated with the memory read or write operations between the memory devices and the connector,	(Claim 1) and the data module in the first mode is configured to propagate one or more first data signals between the memory devices and the system memory controller,
wherein, in the second mode, the memory devices are not accessible by the system memory controller,	(Claim 1) wherein the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller
and the each respective data handler is configured to provide respective data patterns to the respective group of one or more memory devices based on information output from the control module.	(Claim 1) the data module in the second mode is configured to ... transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, (Claim 2) wherein the each respective data handler is further configured to output a respective n-bit wide section section of each of the one or more second data signals including one or more respective n-bit-wide data patterns provided by the respective one of the data handler logic elements to the respective memory device group.

Rejection of claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 101

The Office Action rejected claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 101 as being directed to a judicial exception (i.e., law of nature, a natural phenomenon, or abstract idea) without significantly more. In rejection claim 1 under 35 U.S.C. 101, the Office Action separates the functional limitations of:

"during the normal mode: configured to receive system address and control signal from the system memory controller and to output memory address and control signals to the memory device according to the system address and control signals", and

"configured to propagate data signals between the memory devices and the system memory controller, the data signals being transmitted or received by the memory devices in response to the memory address and control signals"; and

"during the test mode: configured to output self-test address and control signals to the address and control ports of the memory devices";

"configured to isolate the memory devices from being accessed by the system memory controller and to transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory device according to one or more commands from the control module, and wherein the self-test data patterns are received by at least a portion of the memory devices according to the self-test address and control signals from the control module"

from their corresponding structures, and states that:

“These limitations are directed to intended use statements, and these limitations are abstract idea without significantly more, similar to Cyberfone, the concept of using categories to organize, store and transmit information; Cap One Financial, the concept of collecting, displaying, and manipulating data; and Content Extraction, the concept of data recognition and storage, were found to be abstract ideas.”

Applicant respectfully disagrees.

First of all, claim 1 claims a memory module comprising memory devices, a data module, a control module, and other structural and functional features. It has long been established that “[f]eatures of an apparatus may be recited either structurally or functionally. In re Schreiber, 128 F.3d 1473, 1478, 44 USPQ2d 1429, 1432 (Fed. Cir. 1997).” See MPEP 2114. “A claim term is functional when it recites a feature ‘by what it does rather than by what it is’ (e.g., as evidenced by its specific structure or specific ingredients). In re Swinehart, 439 F.2d 210, 212, 169 USPQ 226, 229 (CCPA 1971). There is nothing inherently wrong with defining some part of an invention in functional terms. Functional language does not, in and of itself, render a claim improper. Id.” See MPEP § 2173.05(g).

The above-cited limitations of claim 1 use the term “configured to” to recite some of the functional features in the claimed memory module. It has been established that “configured to” in claims requires a structure that is more than just capable of performing a function, but is actually made so that it can be used to perform the function without further modification. *Aspex Eyewear, Inc. v. Marchon Eyewear, Inc.* (Fed. Cir. 2012) Therefore, claim 1 as amended recites a memory module operable in any of a plurality of modes including a first mode and a second mode, the memory module comprising, among other things, a data module and a control module configured to perform the respective functions recited in the claims depending on which mode the memory module is in. The recitation of these functions are not intended use statements, as alleged in the Office Action. Instead, these functions are what the data module and the control module are capable of doing because of how they are designed and made, and are therefore meaningful limitations thereof.

Applicant is baffled by the statement in the Office Action that claim 1 is “similar to Cyberfone, the concept of using categories to organize, store and transmit information; Cap One Financial, the concept of collecting, displaying, and manipulating data; and Content Extraction, the concept of data recognition and storage.” It has been warned by the courts that the claims of a patent should not be described at a “high level of abstraction” and untethered from the language of the claims, all but ensuring that the exceptions to Section 101 swallow the rule. *Enfish v. Microsoft*, 822 F.3d 1327 (Fed. Cir. 2016) The Federal Circuit ruling in *Enfish* is a reminder that patent claims must be read as a whole when analyzing eligibility under Section 101, and care must be taken not to simply water-down the claims to a high-level, abstract concept.

The Office Action, in rejecting claim 1 under 35 U.S.C. 101, not only oversimplifies claim 1 but also severely mischaracterizes it. Claim 1 does not recite anything close to “using categories to organize, store and transmit information,” or “collecting, displaying, and manipulating data,” or “data recognition and storage” None of these concepts is even remotely similar to what is claimed in claim 1. Claim 1 is NOT about data processing. It claims an apparatus with specifically structured hardware components that are designed and made to be configurable to perform specific hardware

functions, such as, for example, “to propagate data signals between the memory devices and the system memory controller,” “to isolate the memory devices from being accessed by the system memory controller,” and “to transmit self-test data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands from the control module,” etc.

The Office Action further errs by stating that “the recited ‘memory devices’, ‘circuit board’, ‘control ports’, ‘data ports’, ‘system memory bus’, ‘data handler logic elements’, ‘data module’, ‘control module’, ‘system memory controller’ are well-known and conventional/generic computer components performing well-known and conventional/generic computer functions.”

As provided in MPEP 2114, “[i]f an examiner concludes that a functional limitation is an inherent characteristic of the prior art, then to establish a prima case of anticipation or obviousness, the examiner should explain that the prior art structure inherently possesses the functionally defined limitations of the claimed apparatus. In re Schreiber, 128 F.3d at 1478, 44 USPQ2d at 1432. See also Bettcher Industries, Inc. v. Bunzl USA, Inc., 661 F.3d 629, 639-40, 100 USPQ2d 1433, 1440 (Fed. Cir. 2011).”

Here, the Office Action presents no prior art memory module with the structural and functional features recited in claim 1. The allegation that the memory module of claim 1 and its various components are well-known and conventional/generic computer components performing well-known and conventional/generic computer functions are therefore baseless.

At the time the present application was filed, there were no well-known and conventional/generic memory modules operable in any of a plurality of modes including a first mode and a second mode and including a data module coupled between the data ports of the memory devices and the system memory bus, wherein the data module in the first mode is configured to propagate data signals between the memory devices and the system memory controller, and wherein the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit data patterns provided by the data handler logic elements to the

data ports of the memory devices according to one or more commands output from the control module. Therefore, the data module cannot be well-known and conventional/generic. The same can be said about other components of the memory module of claim 1.

Therefore, the rejection of claim 1 and its dependent claims, claims 2-5, 7-8, 10-12, 28 and 36 under 35 U.S.C. 101 should be withdrawn.

Rejection of Claims 1-5, 7-8, 10-12, 26-29 and 36 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), Second paragraph

The Office Action rejected claim 1 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, citing line 19-29 of claim 1 and stating:

“[It] is not clear whether, during the test mode, the self-test data patterns that are received by at least a portion of the memory devices according to the self-test address and control signals from the control module is actually doing the test for that the at least a portion of the memory devices according to the self-test address and control signals from the control module or just for storage over there? How the at least a portion of the memory device is tested is unclear and what is the result and/or what to with the test result?”

It is within the knowledge of those skilled in the art that memory modules are commonly tested by transmitting test data to their memory devices, which can be configured to receive and store the test data according to address and control signals. The test data are then read back from the memory devices for comparison. See, for example, US6829728B2 and US6928593B1, which are cited in the Information Disclosure Statements filed on February 15, 2016 in the present application.

Claim 1 recites a memory module that improves upon the prior art by being operable in any of a plurality of modes including a first mode and a second mode, and by including, among other things, a control module and a data module configured to perform certain functions depending on whether the memory module is in the first mode or second mode. For example, the data module in the first mode “is configured to propagate first data signals between the memory devices and the system memory controller”, and in the second mode is configured “to isolate the memory devices from being accessed by the

system memory controller and to transmit second data signals having data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module.” These limitations are consistent with the teachings of the Specification. (See, for example, paragraphs [0025]-[0026] and [0041]-[0042]).

Thus, claim 1 is an apparatus claim with structural and functional limitations that set forth the subject matter which the inventor or a joint inventor regards as the invention and that particularly point out and distinctly define the metes and bounds of the subject matter to be protected by the patent grant. Infringement can be proven by showing that a memory module include the structural and functional limitations clearly recited in the claim. Therefore, the requirement of 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, is met.

Furthermore, 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, does not require that claim 1 include details on “what is the result and/or what to with the test result,” as inquired in the Office Action. Answers to these inquiries can be found in the Specification. (See, for example, paragraphs [0042]-0045])

The above arguments apply to claims 2-5, 7-8, 10-12, 28 and 36.

Therefore, the rejection of claims 1-5, 7-8, 10-12, 28 and 36 under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, should be withdrawn.

New claims 76-81 depend from claim 1 as amended and include further limitations in addition to the limitations in claim 1 as amended. Therefore, new claims 76-81 are patentable for at least the same reasons claim 1 as amended is patentable.

New claims 82-88 depend from claim 58 as amended and include further limitations in addition to the limitations in claim 58 as amended. Therefore, new claims 82-88 are patentable for at least the same reasons claim 58 as amended is patentable.

In light of the amendments and remarks, Applicant submits that claims 1-5, 7-8, 10-12, 28, 36, and 49-88 are in condition for allowance and respectfully requests such

action. The Examiner is invited to call the undersigned to resolve any remaining issues.

Respectfully submitted,

Date: February 15, 2018

/Jamie J. Zheng/

51167

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(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	Memory Subsystem and Method of Operation			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10010US04			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
RCE- 2ND AND SUBSEQUENT REQUEST	2820	1	950	950
Total in USD (\$)				950

Electronic Acknowledgement Receipt

EFS ID:	31809122
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	Memory Subsystem and Method of Operation
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10010US04
Receipt Date:	15-FEB-2018
Filing Date:	29-MAR-2014
Time Stamp:	20:19:52
Application Type:	Utility under 35 USC 111(a)

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Deposit Account	505394
Authorized User	Brian Gonzalez

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37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	2018-02-15_RCE_N3044-10010US04.pdf	1349893	no	3
			d0580a7ca75a094748399a338343157a94237eb4		
Warnings:					
Information:					
2		2018-02-15_OA-Response_N3044-10010US04.pdf	499059	yes	21
			71095e26f32504fa5c478a038684dda29822ee32		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment Submitted/Entered with Filing of CPA/RCE		1	1	
	Specification		2	2	
	Claims		3	12	
	Applicant Arguments/Remarks Made in an Amendment		13	21	
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30642	no	2
			a9eaff4dce12c352a8c035c1a4f7e61d9ff51ab9		
Warnings:					
Information:					
Total Files Size (in bytes):			1879594		

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New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/229,844	Filing Date 03/29/2014	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO**APPLICATION AS FILED – PART I**

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

	02/15/2018	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	* 34	Minus	** 28	= 6	X \$50 =	300
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$230 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	300

(Column 1)

(Column 2)

(Column 3)

		CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

SLIE
VIKKI GRAY

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	N3044.10010US04	1018
79141	7590	03/01/2018	EXAMINER	
Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098			CHUNG, PHUNG M	
			ART UNIT	PAPER NUMBER
			2117	
			NOTIFICATION DATE	DELIVERY MODE
			03/01/2018	ELECTRONIC

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The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

d@mabr.com
bisraelsen@mabr.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	

All participants (applicant, applicant's representative, PTO personnel):

(1) Phung M. Chung. (3)_____.

(2) Jamie Zheng. (4)_____.

Date of Interview: 23 February 2018.

Type: ☒ Telephonic ☐ Video Conference
☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☐ No.
If Yes, brief description: _____.

Issues Discussed ☐101 ☒112 ☐102 ☐103 ☒Others
(For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1-5, 7-8, 10-12, 26-29 and 36, and 58-59,76-88.

Identification of prior art discussed: _____.

Substance of Interview

(For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

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☐ Attachment

/Phung M. Chung/
Primary Examiner, Art Unit 2117

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

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Paragraph (b)

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The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
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Examiner to Check for Accuracy

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**UNITED STATES DEPARTMENT OF COMMERCE
U.S. Patent and Trademark Office**

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Alexandria, Virginia 22313-1450

APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR / PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
14/229,844	29 March, 2014	LEE ET AL.	N3044.10010US04

Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098		EXAMINER	
		Phung M. Chung	
		ART UNIT	PAPER
		2117	20180223

DATE MAILED:

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

Please find the attached Applicant Initiated Interview Summary.

/Phung M. Chung/
Primary Examiner
Art Unit: 2117

<i>Applicant-Initiated Interview Summary</i>	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	

All participants (applicant, applicant's representative, PTO personnel):

(1) Phung M. Chung. (3)_____.

(2) Jamie Zheng. (4)_____.

Date of Interview: 23 February 2018.

Type: ☒ Telephonic ☐ Video Conference
☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☐ No.
If Yes, brief description: _____.

Issues Discussed ☐101 ☒112 ☐102 ☐103 ☒Others
(For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1-5, 7-8, 10-12, 26-29 and 36, and 58-59,76-88.

Identification of prior art discussed: _____.

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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	N3044.10010US04	1018
79141	7590	04/10/2018	EXAMINER	
Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098			CHUNG, PHUNG M	
			ART UNIT	PAPER NUMBER
			2117	
			NOTIFICATION DATE	DELIVERY MODE
			04/10/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

d@mabr.com
bisraelsen@mabr.com

<i>Examiner-Initiated Interview Summary</i>	Application No. 14/229,844	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	

All participants (applicant, applicant's representative, PTO personnel):

(1) Phung M. Chung. (3) ____.

(2) Jamie Zheng. (4) ____.

Date of Interview: 30 March 2018.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☐ No.
 If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☐102 ☐103 ☒Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 58.

Identification of prior art discussed: _____.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Discussed during the interview that the data module of claim 58 should be amended to include data handler logic elements. (See the attached Examiner's amendment).

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☐ Attachment

/Phung M. Chung/ Primary Examiner, Art Unit 2117	
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The present application is being examined under the pre-AIA first to invent provisions.

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. Applicant's submission filed on 2/15/18 has been entered.

EXAMINER'S AMENDMENT

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed in response to this Quayle Action to ensure consideration of such an amendment.

Authorization for this examiner's amendment was given in an interview with attorney Jamie Zheng on 3/30/2018.

The application has been amended as follows:

Claim 58. (Currently Amended) A memory module operable in a computer system with a system memory controller, the memory module comprising:

a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller;

memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;

a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and

a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers includes a data handler logic element, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks;

wherein the memory module is configurable to operate in any of at least a first mode and a second mode;

wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations, and the data handlers are configured to propagate one or more data signals associated with the memory read or write operations between the memory devices and the connector;

wherein, in the second mode, the memory devices are not accessed by the system memory controller, and the data handler logic element in the each respective

data handler is configured to provide respective data patterns to the respective group of one or more memory devices based on information output from the control module.

Claims 1-5, 7-8, 10-12, 28, 36, 58-3, 66, 82-83, and 85-88 are allowable.

Ex parte Quayle Action

This application is in condition for allowance except for the following formal matters:

Claim Objections

Claims 64-65 and 84 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

As per claim 64, line 2, "includes a respective data handler logic element", for a suggestion, it should be changed to - - includes the respective data handler logic element- -. Appropriate correction and/or clarification are required.

As per claim 65, this claim is also objected to because it depends on the objected claim 64.

As per claim 84, line 3, "address and control signals output form the control module" for a suggestion, it should be changed to - - address and control signals output [form] from the control module. Appropriate correction and/or clarification are required.

Prosecution on the merits is closed in accordance with the practice under *Ex parte Quayle*, 25 USPQ 74, 453 O.G. 213, (Comm'r Pat. 1935).

A shortened statutory period for reply to this action is set to expire **TWO MONTHS** from the mailing date of this letter.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday from 8:00am-4:30pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, April Y. Blair can be reached on 571-270-1014. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

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☐ Attachment

/Phung M. Chung/ Primary Examiner, Art Unit 2117	
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	547199	((control module or controller module) and ((self-test logic) and (control logic) and (register)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 08:49
L2	18698	I1 and (((data module) or (data hander\$1) or (data management\$1) or (data controller) or (data manager\$1) or (memory controller) or (gate or switch or multiplexer or MUX or selector)) and ((normal mode or normal operation mode or normal mode operation or functional mode or standard read/write mode or first mode) and (selt-test mode or test mode or test mode operation or second mode)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 08:58
L3	6	I2 and ((memory devices) and (address port and control ports and data port))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 09:01
L4	3	I2 and ((data handler logic elements) and (self-test data patterns or data patterns) and ((self-test address or address) and (control signals)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 09:03
L5	6	I3 not I4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 09:09
L6	6	(system memory controller) and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 09:10
L7	6	I6 and ((memory self-test address\$2 and control signal\$1 and test data patterns or test data or test vectors or data patterns or test patterns) and	US-PGPUB; USPAT; USOCR; FPRS;	OR	ON	2018/03/31 09:42

		(memory device\$1) and (test mode or second mode))	EPO; JPO; DERWENT; IBM_TDB			
L8	1	I7 and ((memory rank\$1) or (rank\$1 or block\$1) same (memory module or memory chip))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 09:43
L10	2442486	((Lee).inv.) or ((Bhakta).inv.) or ((Choi).inv.) or ((Netlist).asn.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 10:38
L14	6	I7 and ((issolat\$3 or disconnect\$3 or disabl\$3 or swith\$3 or control\$4 or gat\$3 or deselect\$3) and (memory devices near2 accessed near2 system memory controller) and (second mode or test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 10:39
L17	0	I10 and ((issolat\$3 or disconnect\$3 or disabl\$3 or swith\$3 or control\$4 or gat\$3 or deselect\$3) and (memory devices near2 accessed near2 system memory controller) and (second mode or test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 10:40
L18	187112	I10 and ((issolat\$3 or disconnect\$3 or disabl\$3 or swith\$3 or control\$4 or gat\$3 or deselect\$3) and (memory devices near2 accessed near2 system memory controller) and (second mode or test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 10:41
L22	1208	I18 and I2	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 10:42
L24	1102	I22 and ((memory self-test address\$2 and control signal\$1 and test data patterns or test data or test vectors or data patterns or test patterns) and (memory device\$1) and (test mode or second mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/03/31 10:43
L25	3	I24 and ((data handler logic elements) and (self-test data patterns or data patterns) and ((self-test address or address) and (control signals)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/03/31 10:43
L26	3	25 AND ((G11C29/10 OR G11C29/12).CPC. OR (714/719 OR	US-PGPUB; USPAT;	ADJ	ON	2018/03/31 10:51

	714/718 OR 714/733 OR 714/734 OR 714/736 OR 714/819 OR 365/201).OCLS. OR (G11C29/00).IPCR.)	USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
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3/ 31/ 2018 10:52:19 AM
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Search Notes 	Application/Control No. 14229844	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner
G11C29/10, G11C29/12	3/31/18	PC

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

SEARCH NOTES		
Search Notes	Date	Examiner
EAST-CPC Searched.	3/31/18	PC
Inventor Searched.	3/31/18	PC

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
G11C	29/10, 29/12	3/31/18	PC

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PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/229,844
Filed : March 29, 2014
For : A Multi-Mode Memory Module
With Data Handlers
Examiner : CHUNG, PHUNG M
Art Unit : 2117
Conf. No. : 1018
Docket No. : 129980-5053-US

CERTIFICATE OF EFS WEB
TRANSMISSION

I hereby certify that this correspondence and
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Web Server on:

May 2, 2018

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT

Mail Stop Amendment

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Madam:

The following amendments and remarks are in response to the Ex Parte Quayle
Action Dated April 10, 2018 in the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit
Account No. 50-0310 (order No. 129980-5053-US).

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 12 of this paper.

IN THE SPECIFICATION:

Please replace the title to the following:

~~A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS~~ MULTI-MODE
MEMORY MODULE WITH DATA HANDLERS

In the Claims:

1. (Previously Presented) A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:

memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;

a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements; and

a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and

wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode;

wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals, and the data module in the first mode is configured to propagate one or more first data signals between the memory devices and the system memory controller, the one or more first data signals being transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals; and

wherein the control module in the second mode is configured to output second memory address and control signals to the address and control ports of the memory devices, and the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, and wherein at least a portion of the memory devices are configured to receive the one or more second data signals according to the second memory address and control signals from the control module.

2. (Previously Presented) The memory module of claim 1, wherein the memory devices are arranged in at least one rank and organized in a plurality of memory device groups, each rank of the at least one rank having a set of memory devices across a full bit width of the memory module, each respective memory device group of the plurality of memory device groups including respective one or more memory devices in each rank of the at least one rank, wherein each data signal of the one or more first data signals propagated by the data module during the first mode is N-bits wide, wherein the data module includes a plurality of data handlers, each respective data handler of the plurality of data handlers including a respective one of the data handler logic elements, wherein the each respective data handler is configured to propagate a respective n-bit section of the each data signal of the one or more first data signals between the system memory controller and a respective memory device group of the plurality of memory device groups, n being a fraction of N, and wherein the each respective data handler is further configured to output a respective n-bit wide section of each of the one or more second data signals including one or more respective n-bit-wide data patterns provided by the respective one of the data handler logic elements to the respective memory device group.
3. (Previously Presented) The memory module of claim 2, wherein each memory device is n-bits wide and wherein the respective memory device group includes a single respective memory device in each rank of the at least one rank.
4. (Previously Presented) The memory module of claim 2, wherein the control module is configured to output third memory address and control signals to the address and control ports of the memory devices during the second mode, the third memory address and control signals causing the memory devices in one of the at least one rank to output at least one N-bit wide read data signal, and wherein the each respective data handler is configured to receive a respective n-bit section of each of the at least one N-bit wide read data signal.
5. (Previously Presented) The memory module of claim 4, wherein the each respective data handler is configured to check the respective n-bit section of each of the

at least one N-bit wide read data signal with respect to a corresponding data pattern stored in the respective data handler logic element.

6. (Cancelled)

7. (Previously Presented) The memory module of claim 5, wherein the data module is configured to store failure information when a test failure is indicated by the respective n-bit section of each of the at least one N-bit wide read data signal not being in agreement with the corresponding data pattern.

8. (Previously Presented) The memory module of claim 7, wherein the data module is further configured to report the failure information to the control module.

9. (Cancelled)

10. (Previously Presented) The memory module of claim 1, wherein the data module is configured to output the one or more second data signals to the memory devices in response to a write command output from the control module, and wherein the data module is further configured to receive the at least one N-bit wide read data signal from the memory devices in response to a read command output from the control module.

11. (Previously Presented) The memory module of claim 2, wherein the plurality of data handlers are configured to operate independently of each other without being in communication with each other.

12. (Previously Presented) The memory module of claim 2, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is disposed on the circuit board at a position corresponding to the respective memory device group.

13. (Cancelled)

14. (Cancelled)

15. (Cancelled)

16. (Cancelled)
17. (Cancelled)
18. (Cancelled)
19. (Cancelled)
20. (Cancelled)
21. (Cancelled)
22. (Cancelled)
23. (Cancelled)
24. (Cancelled)
25. (Cancelled)
26. (Cancelled)
27. (Cancelled)
28. (Previously Presented) The memory module of claim 2, wherein the each respective data handler includes first data paths between the respective memory device group and the system memory controller and second data paths between the respective memory device group and the respective one of the data handler logic elements, and wherein the each respective data handler is configured to propagate the respective n-bit section of the each data signal of the one or more first data signals using the first data paths, and is further configured to transmit the respective n-bit wide section of each of the one or more second data signals using the second data paths.
29. (Cancelled)
30. (Cancelled)

- 31. (Cancelled)
- 32. (Cancelled)
- 33. (Cancelled)
- 34. (Cancelled)
- 35. (Cancelled)
- 36. (Previously Presented) The memory module of claim 2, wherein each memory device is $n/2$ -bit wide and wherein the respective memory device group includes two respective memory devices in each of the at least one rank.
- 37. (Cancelled)
- 38. (Cancelled)
- 39. (Cancelled)
- 40. (Cancelled)
- 41. (Cancelled)
- 42. (Cancelled)
- 43. (Cancelled)
- 44. (Cancelled)
- 45. (Cancelled)
- 46. (Cancelled)
- 47. (Cancelled)
- 48. (Cancelled)
- 49. (Cancelled)

50. (Cancelled)

51. (Cancelled)

52. (Cancelled)

53. (Cancelled)

54. (Cancelled)

55. (Cancelled)

56. (Cancelled)

57. (Cancelled)

58. (Previously Presented) A memory module operable in a computer system with a system memory controller, the memory module comprising:

a printed circuit board (PCB) having a connector configured to provide electrical connections for data, address and control signals between the memory module and the system memory controller;

memory devices mounted on the PCB and arranged in one or more ranks, each respective rank of the one or more ranks having a respective set of memory devices across a full bit-width of the memory module;

a control module mounted on the PCB and electrically coupled to the connector and to the memory devices, wherein the control module is configured to output address and control signals to the memory devices based on information received via the connector; and

a plurality of data handlers mounted on the PCB, wherein each respective data handler of the plurality of data handlers includes a data handler logic element, wherein each respective data handler of the plurality of data handlers is electrically coupled between a respective group of one or more memory devices and the connector, the respective group of one or more memory devices including at least one respective memory device in each of the one or more ranks;

wherein the memory module is configurable to operate in any of at least a first mode

and a second mode;

wherein, in the first mode, the memory module is configured to be accessed by the system memory controller for memory read and write operations, and the data handlers are configured to propagate one or more data signals associated with the memory read or write operations between the memory devices and the connector;

wherein, in the second mode, the memory devices are not accessed by the system memory controller, and the data handler logic element in the each respective data handler is configured to provide respective data patterns to the respective group of one or more memory devices based on information output from the control module.

59. (Previously Presented) The memory module of claim 58, wherein each of the one or more data signals is N-bit wide, and wherein, when the memory module is operating in the first mode, the each respective data handler is configured to propagate a respective n-bit section of each of the one or more data signals between at least one respective memory device in the at least one rank and the connector, n being a fraction of N; and wherein, when the memory module is operating in the second mode, the respective data patterns are each n-bits wide.

60. (Previously Presented) The memory module of claim 59, wherein each memory device is n-bit wide and wherein the respective group of one or more memory devices include a single respective memory device in each of the one or more ranks.

61. (Previously Presented) The memory module of claim 59, wherein each memory device is $n/2$ -bit wide and wherein the respective group of one or more memory devices include two respective memory devices in each of the one or more ranks.

62. (Previously Presented) The memory module of claim 58, wherein the plurality of data handlers include physically separate integrated circuit packages, and wherein the each respective data handler is disposed on the circuit board at a position corresponding to the respective group of one or more memory devices.

63. (Currently Amended) The memory module of claim 58, wherein the each respective data handler is configured to output the respective data patterns to the

respective group of one or more memory devices in response to one or more write commands~~a write command~~ output from the control module.

64. (Currently Amended) The memory module of claim 63, wherein, when the memory module is operating in the second mode, the each respective data handler is further configured to receive read data patterns from the respective group of one or more memory devices in response to one or more read commands~~a read command~~ output from the control module.

65. (Previously Presented) The memory module of claim 64, wherein the each respective data handler is further configured to check the read data patterns received from the respective group of one or more memory devices with respect to corresponding data patterns stored in the each respective data handler.

66. (Currently Amended) The memory module of claim 59, ~~wherein the each respective data handler includes a respective data handler logic element configured to output the respective patterns of data signals,~~ wherein the each respective data handler includes first data paths between the respective group of one or more memory devices and the connector, and further includes second data paths between the respective group of one or more memory devices and the respective data handler logic element, and wherein the each respective data handler is configured to propagate the respective n-bit section of the each of the one or more data signals using the first data paths, and is further configured to provide the respective data patterns using the second data paths.

67. (Cancelled)

68. (Cancelled)

69. (Cancelled)

70. (Cancelled)

71. (Cancelled)

72. (Cancelled)

73. (Cancelled)
74. (Cancelled)
75. (Cancelled)
76. (Currently Amended) The memory module of claim 1, wherein the memory devices are arranged in one or more ranks, each rank of the one or more ranks having a set of memory devices across a full bit width of the memory module, and wherein the second data signals are received by a selected rank of the one or more ranks according to the second memory address and control signals from the control module.
77. (Previously Presented) The memory module of claim 2, wherein the second data signals are received by a selected rank of the one or more ranks according to the second memory address and control signals from the control module.
78. (Previously Presented) The memory module of claim 1, wherein the data patterns provided by the data handler logic elements are based on information that the data handler logic elements receive from the control module.
79. (Currently Amended) The memory module of claim 2, wherein the respective n-bit-wide data pattern is based on information which the respective one of the data handler logic elements ~~receive~~ receives from the control module.
80. (Previously Presented) The memory module of claim 1, wherein the control module is further configured to provide the data patterns to the data module.
81. (Previously Presented) The memory module of claim 28, wherein the each respective data handler is further configured to isolate the first data paths in the second mode.
82. (Previously Presented) The memory module of claim 58, wherein, in the second mode, the each respective data handler is further configured to isolate the respective group of one or more memory devices from the connector.

83. (Previously Presented) The memory module of claim 59, wherein, in the second mode, the each respective data handler is further configured to isolate the respective group of one or more memory devices from the connector.

84. (Currently Amended) The memory module of claim 58, wherein the memory devices are configured to receive or output the one or more data signals in response to corresponding address and control signals output ~~form~~ from the control module.

85. (Previously Presented) The memory module of claim 59, wherein the respective group of one or more memory devices are configured to receive or output the respective n-bit section of each of the one or more data signals in response to corresponding address and control signals output from the control module.

86. (Previously Presented) The memory module of claim 59, wherein the each respective data handler is configured to output the respective data patterns to the respective group of one or more memory devices in response to a write command output from the control module.

87. (Previously Presented) The memory module of claim 58, wherein the memory devices are configured to receive the data patterns provided by the data handlers in response to corresponding address and control signals output form the control module.

88. (Currently Amended) The memory module of claim 59, wherein the respective group of one or more memory devices are configured to receive ~~or output~~ the respective data patterns in response to corresponding address and control signals output from the control module.

REMARKS

This amendment is in response to the the Ex Parte Quayle Action Dated April 10, 2018 (the “Office Action”). In the Office Action, the Examiner:

- allowed claims 1-12, 28, 36, 58-63, 66, 76-83 and 85-88;
- objected to claims 64, 65 and 84.

Applicant would like to thank the Examiner for the telephone interview on March 30, 2018, and appreciates the allowance of claims 1-12, 28, 36, 58-63, 66, 76-83 and 85-88. In this amendment, claims 63-64, 66, 76, 79, 84 and 88 are amended. The amendments in these claims are to correct minor issues and to address the objections in the Office Action. Therefore, no new matter is added.

A replacement title is submitted to correct a minor issue.

In light of the amendments and remarks, Applicant submits that the present application is in condition for allowance and respectfully requests such action. The Examiner is invited to call the undersigned to resolve any remaining issues.

Respectfully submitted,

Date: May 2, 2018

/Jamie J. Zheng/

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51167
(Reg. No.)

DB2/ 33228822.1

Electronic Acknowledgement Receipt

EFS ID:	32517796
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5053US
Receipt Date:	02-MAY-2018
Filing Date:	29-MAR-2014
Time Stamp:	20:19:30
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980_5053US_AMEND.pdf	136719	yes	13
			1626be9124e8f06cbf44068f0e0f873e6bc7a930		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Response after Ex Parte Quayle Action	1	1
	Specification	2	2
	Claims	3	12
	Applicant Arguments/Remarks Made in an Amendment	13	13

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				Electronically filed May 17, 2018	
				Application Number	14/229,844
				Filing Date	March 29, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2117
Examiner Name	Chung, Phung M				
Sheet	1	of	3	Attorney Docket Number	129980-5053-US

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
		6,130,852	10-10-2000	Ohtani et al.	
		6,333,873 B1	12-25-2001	Kumanoya et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
		Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)				

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners' Updated Exhibit List, filed February 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Updated Exhibit List, filed February 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Hearing Transcript, filed March 22, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Termination Decision Document, filed May 3, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1034, 'Petitioners Demonstratives,' filed February 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2016, 'Patent Owner's Demonstratives,' filed February 9, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners Reply, filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Opposition to Petitioners' Motion to Exclude, filed March 14, 2018

Examiner's Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	14/229,844
				Filing Date	March 29, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2117
Examiner Name	Chung, Phung M				
Sheet	2	of	3	Attorney Docket Number	129980-5053-US

		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Opposition to Patent Owner's Motion to Exclude, filed March 14, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Brief in Response to Show Cause Order, filed March 20, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Submission Regarding Order to Show Cause, filed March 20, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, PETITIONERS' REPLY TO PATENT OWNER'S OPPOSITION TO MOTION TO EXCLUDE, filed March 21, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Reply in Support of Its Motion to Exclude, filed March 21, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Motion to Strike, filed March 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Opposition to Patent Owner's Motion to Strike, filed April 4, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Order To Show Cause Conduct of the Proceedings, filed March 13, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Request for Oral Argument 37 C.F.R. sec 42.70, filed March 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Hearing Transcript, filed April 18, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Conduct of the Proceeding, filed May 2, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1034, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 70 (Fed. Cir. Dec. 21, 2017),' filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1036, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 43 (Fed. Cir. Apr. 6, 2017),' filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1028, 'SanDisk Corp. v. Netlist Inc., IPR2014-00970, Final Written Dec.,' filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1035, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 31 (Fed. Cir. Jan. 6, 2017),' filed February 7, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners Reply, filed February 7, 2018

Examiner's Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	14/229,844
				Filing Date	March 29, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2117
				Examiner Name	Chung, Phung M
Sheet	3	of	3	Attorney Docket Number	129980-5053-US

		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Request for Oral Argument, filed February 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Order To Show Cause Conduct of the Proceedings, filed March 13, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Opposition to Petitioners' Motion to Exclude, filed March 14, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Opposition to Patent Owner's Motion to Exclude, filed March 14, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Brief in Response to Show Cause Order, filed March 20, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Submission Regarding Order to Show Cause, filed March 20, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Reply to Patent Owner's Opposition to Motion to Exclude, filed March 21, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Reply in Support of Its Motion to Exclude, filed March 21, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Motion to Strike, filed March 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Opposition to Patent Owner's Motion to Strike, filed April 4, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Hearing Transcript, filed April 18, 2018

Examiner's Signature		Date Considered	
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Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Gary Scott Williams/Edith Fuentes			
Attorney Docket Number:	129980-5053US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	2806	1	120	120
Total in USD (\$)				120

Electronic Acknowledgement Receipt

EFS ID:	32657888
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Gary Scott Williams/Edith Fuentes
Filer Authorized By:	Gary Scott Williams
Attorney Docket Number:	129980-5053US
Receipt Date:	17-MAY-2018
Filing Date:	29-MAR-2014
Time Stamp:	21:09:46
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 120
RAM confirmation Number	051818INTEFSW00005863500310
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	PTAB-IPR2017-00560-27-Petitioners-Updated-Exhibit-List.pdf	607700	no	5
			afb2acccee1fd8c1e5bdd650f2cc46f316e6f61a		
Warnings:					
Information:					
2	Non Patent Literature	PTAB-IPR2017-00560-28-Patent-Owners-Updated-Exhibit-List.pdf	548883	no	4
			f3ae8404ee3a2cff1125923aa714c2ae2c7e4cca		
Warnings:					
Information:					
3	Non Patent Literature	PTAB-IPR2017-00560-29-Hearing-Transcript_Part1.pdf	19464674	no	50
			e7f9a406383d68e57b573c6928d79152532f9c83		
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4	Non Patent Literature	PTAB-IPR2017-00560-29-Hearing-Transcript_Part2.pdf	20099510	no	50
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Warnings:					
Information:					
5	Non Patent Literature	PTAB-IPR2017-00560-29-Hearing-Transcript_Part3.pdf	5924619	no	16
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Warnings:					
Information:					
6	Non Patent Literature	PTAB-IPR2017-00560-30-Termination-Decision-Documnet.pdf	6281844	no	18
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Warnings:					
Information:					

7	Non Patent Literature	PTAB-IPR2017-00560-Exhibit-1034-Petitioners-Demonstratives.pdf	20468190	no	51
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8	Non Patent Literature	PTAB-IPR2017-00560-Exhibit-2016-Patent-Owners-Demonstratives.pdf	21386231	no	50
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9	Non Patent Literature	2018-02-07-IPR2017-00561-18-PetitionersReply.pdf	8690696	no	33
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10	Non Patent Literature	2018-02-28-IPR2017-00561-21-Petitioners-Motion-toExclude.pdf	1950633	no	9
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11	Non Patent Literature	2018-02-28-IPR2017-00561-23-Patent-Owner-Motion-to-Exclude.pdf	520930	no	4
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Information:					
12	Non Patent Literature	2018-03-14-IPR2017-00561-25-Patent-Owner-Opposition-to-Petitioners-Motion-to-Exclude.pdf	3893471	no	13
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13	Non Patent Literature	2018-03-04-IPR2017-00561-26-Petitioners-Opposition-to-Patent-Owner-Motion-to-Exclude.pdf	1274779	no	6
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14	Non Patent Literature	2018-03-20-IPR2017-00561-27-Petitioners-Brief-in-Response-to-Show-Cause-Order.pdf	1951343	no	8
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15	Non Patent Literature	2018-03-20-IPR2017-00561-28-Patent-Owner-Submission-Re-Order-to-Show-Cause.pdf	1968564	no	8
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16	Non Patent Literature	2018-03-21-IPR2017-00561-29-PETITIONERS-REPLY-PATENT-OWNERS-OPPOSITION-MOTION-TO-EXCLUDE.pdf	2170814	no	9
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Information:					
17	Non Patent Literature	2018-03-21-IPR2017-00561-30-Patent-Owner-Reply-ISO-Its-Motion-to-Exclude.pdf	721047	no	4
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Information:					
18	Non Patent Literature	2018-03-28-IPR2017-00561-32-Patent-Owners-Motion-to-Strike.pdf	1373588	no	8
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Information:					
19	Non Patent Literature	2018-04-04-IPR2017-00561-33-Petitioners-Opposition-to-Patent-Owner-Motion-to-Strike.pdf	1943597	no	8
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20	Non Patent Literature	2018-03-13-IPR2017-00561-24-Order-To-Show-Cause-Conduct-of-the-Proceedings.pdf	1149754	no	5
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Information:					

21	Non Patent Literature	2018-03-28-IPR2017-00561-31-Request-for-Oral-Argument-37-CFR-sec-42-70.pdf	1653702	no	6
			463f471184c88848cc7138c9bef36ccc6b8256e4		
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22	Non Patent Literature	2018-04-18-IPR2017-00561-34-Hearing-Transcript.pdf	25770340	no	68
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23	Non Patent Literature	2018-05-02-IPR2017-00561-35-Conduct-of-the-Proceeding.pdf	703821	no	3
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24	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1034-Netlist-SanDisk-No-2016-02274.pdf	135860	no	1
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25	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1036-Netlist-SanDisk-2016-02274_Part1.pdf	14118094	no	40
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26	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1036-Netlist-SanDisk-2016-02274_Part2.pdf	15632677	no	40
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27	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1036-Netlist-SanDisk-2016-02274_Part3.pdf	416410	no	2
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28	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1028-SanDisk-v-Netlist-IPR2014-00970-Final-Written-Dec.pdf	25761248	no	63
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29	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1035-Netlist-SanDisk-2016-02274_Part1.pdf	16079058	no	40
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30	Non Patent Literature	PTAB-IPR2017-00561-Exhibit-1035-Netlist-SanDisk-2016-02274_Part2.pdf	17203971	no	37
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31	Non Patent Literature	2018-02-07-IPR2017-00562-18-Petitioners-Reply.pdf	5945183	no	25
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32	Non Patent Literature	201802-28-IPR2017-00562-21-Petitioners-Motion-to-Exclude.pdf	762426	no	4
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33	Non Patent Literature	2018-02-28-IPR2017-00562-22-Petitioners-Request-Oral-Argument.pdf	579041	no	4
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34	Non Patent Literature	2018-02-28-IPR2017-00562-23-Patent-Owner-Motion-to-Exclude.pdf	486496	no	3
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35	Non Patent Literature	2018-03-13-IPR2017-00562-24-Order-To-Show-Cause-Conduct-of-Proceedings.pdf	1152960	no	5
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36	Non Patent Literature	2018-03-14-IPR2017-00562-25-Patent-Owners-Opposition-to-Petitioners-Motion-to-Exclude.pdf	2271027	no	8
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37	Non Patent Literature	2018-03-04-IPR2017-00562-26-Petitioners-Opposition-Patent-Owner-Motion-to-Exclude.pdf	1095891	no	5
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38	Non Patent Literature	2018-03-20-IPR2017-00562-27-Petitioners-Brief-Response-to-Show-Cause-Order.pdf	1967751	no	8
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39	Non Patent Literature	2018-03-20-IPR2017-00562-28-Patent-Owners-Submission-Re-Order-to-Show-Cause.pdf	1968826	no	8
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41	Non Patent Literature	2018-03-21-IPR2017-00562-30-Patent-Owners-Reply-in-Support-of-Its-Motion-to-Exclude.pdf	718160	no	4
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42	Non Patent Literature	2018-03-28-IPR2017-00562-32-Patent-Owners-Motion-to-Strike.pdf	1954524	no	8
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43	Non Patent Literature	2018-04-04-IPR2017-00562-33-Petitioners-Opposition-to-Patent-Owners-Motion-to-Strike.pdf	1917019	no	8
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44	Non Patent Literature	2018-04-06-IPR2017-00562-34-Hearing-Transcript.pdf	25770329	no	68
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45		129980-5053US_IDS.pdf	476338	yes	7
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	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	4	
	Information Disclosure Statement (IDS) Form (SB08)		5	7	
Warnings:					
Information:					
46	Fee Worksheet (SB06)	fee-info.pdf	30292	no	2
			96779cf9e44e1853fe2903ab2c7675d7af911190		
Warnings:					
Information:					
Total Files Size (in bytes):			288169628		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Lee et al.	Confirmation No.:	1018
Serial No.:	14/229,844	Art Unit:	2117
Filed:	March 29, 2014	Examiner:	Chung, Phung M.
For:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS	Attorney Docket No.:	129980-5053-US

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:
 - 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
 - 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☐ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
 - 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
 - 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.

3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

- 4a. ☒ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4c. ☒ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5053-US).

- 4d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
5. ☒ Certification Statement (applicable if Item 3a or Item 4a is checked)
- (Check either Item 5a, 5b, 5c or 5d)*
- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☒ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☐ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).
- (Check appropriate Items 7a, 7b and/or 7c)*
- 7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.
- 7b. ☐ Copies of publications listed on Form PTO-1449 from prior application Serial No. , filed on , of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7c. ☐ Copies of the publications listed on the attached Form PTO-1449 that were not previously cited in prior application Serial No. , filed on , are provided herewith.

8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on .
9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:
- (Check Item 9a, 9b, or 9c)
- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5053-US).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: May 17, 2018



Jamie J. Zheng
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000

51,167
(Reg. No.)

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
				Examiner Name	<i>Chung, Phung M</i>
Sheet	1	of	4	Attorney Docket Number	<i>129980-5053-US</i>

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code ²	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		6,069,407	05-30-2000	Hamzehdoost	
		6,396,136 B2	05-28-2002	Kalidas et al.	
		6,541,872 B1	04-01-2003	Schrock et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

1.97(i)

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Preliminary Response, filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Trial Instituted Document, filed May, 15, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Response, filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners' Reply, filed December, 21, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners Motion to Exclude, filed January, 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, filed January, 23, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Opposition to Petitioners' Motion to Exclude, filed January, 23, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners' Response to Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, filed January, 30, 2018

Examiner's Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
Examiner Name	<i>Chung, Phung M</i>				
Sheet	<i>2</i>	of	<i>4</i>	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners Reply to Patent Owner's Opposition to Motion to Exclude, filed January, 30, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1003, 'Declaration of Dr. Mazumder,' filed January, 3, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2001, 'US Provisional Patent Application Serial No. 60/456,451 (Averbuj),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2002, 'Prosecution file history of Application 10/630,480 (Averbuj),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2004, 'Excerpts from Microsoft Computer Dictionary Fifth Edition (2002),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2005, 'Qualcomm Announces Sampling of the MSM6250,' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2007, 'Deposition transcript of Dr. Mazumder,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2008, 'Search on Amazon website for memory module,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2006, 'Excerpts from Testing and Testable Design of High-Density,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2014, 'JEDEC Standard, JESD205 (2007),' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2010, 'Declaration of R. Jacob Baker, Ph.D., P.E.,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2011, 'Kingston press release from Oct. 11, 2005,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2012, 'IPR2014-00882, Paper 33, FWD,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2013, 'Excerpts from Merriam-Webster's Dictionary (2006),' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2015, 'Excerpts from Memory Systems by Jacob (2007),' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1026, 'Deposition Transcript of R. Jacob Baker,' filed December, 21, 2017

Examiner's Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
Examiner Name	<i>Chung, Phung M</i>				
Sheet	3	of	4	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1029, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 69-2 (Fed. Cir. Nov. 14, 2017),' filed December, 21, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1030, 'JEDEC Standard Document JESD84-A41: Embedded Multimedia Card (eMMC) Product Standard, Standard Capacity (June 2007),' filed December, 21, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Preliminary Response, filed April, 10, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Response, filed November, 21, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Trial Instituted Document, filed July, 7, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1015, 'JEDEC PC2-5300 RDIMM PCB Plot,' filed January, 3, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1003, 'Declaration of Dr. Mazumder.,' filed January, 3, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2001, 'U.S. Provisional Patent Application Serial No. 60/456,451 (Averbuj),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2002, 'Prosecution file history of Application Serial No. 10/630,480 (Averbuj),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2010, 'Declaration of R. Jacob Baker, Ph.D., P.E.,' filed November, 21, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2012, 'Understanding DRAM, IBM application,' filed November, 21, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2013, '2016-2666, U.S. CAFC, IPR2014-01374 FWD affirmed,' filed November, 21, 2017
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2014, '2016-2274, U.S. CAFC, IPR2014-00971 FWD affirmed,' filed November, 21, 2017
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Preliminary Response, filed April, 10, 2017
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Trial Instituted Document, filed July, 7, 2017
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Response, filed November, 21, 2017
Examiner's Signature		Date Considered

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
Examiner Name	<i>Chung, Phung M</i>				
Sheet	4	of	4	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, EXHIBITS 1003, 'Declaration of Dr. Mazumder.,' filed January, 3, 2017
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, EXHIBITS 2010, 'Declaration of R. Jacob Baker, Ph.D., P.E.,' filed November, 21, 2017

Examiner's Signature		Date Considered	
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Electronic Acknowledgement Receipt

EFS ID:	32658274
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Gary Scott Williams/Edith Fuentes
Filer Authorized By:	Gary Scott Williams
Attorney Docket Number:	129980-5053US
Receipt Date:	17-MAY-2018
Filing Date:	29-MAR-2014
Time Stamp:	22:47:27
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	PTAB-IPR2017-00560-6-Patent-Owners-Preliminary-Response.pdf	13539390	no	47
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9	Non Patent Literature	PTAB-IPR2017-00560-Exhibit-1003-Declaration-of-Dr-Mazumder.pdf	15340881	no	56
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21	Non Patent Literature	PTAB-IPR20170056025PetitionersResponsePatentOwnersListingNewArguments-EvidencePetitionersReply.pdf	874136	no	4
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22	Non Patent Literature	PTAB-IPR2017-00560-2011-Kingston-press-release-from-10-11-2005.pdf	1346939	no	3
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23	Non Patent Literature	PTAB-IPR2017-00560-Exhibit-2012-IPR2014-00882-Paper-33-FWD.pdf	19261610	no	51
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	Transmittal Letter	1	4
	Information Disclosure Statement (IDS) Form (SB08)	5	8

Warnings:

Information:

Total Files Size (in bytes):	524198209
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Lee et al.	Confirmation No.:	1018
Serial No.:	14/229,844	Art Unit:	2117
Filed:	March 29, 2014	Examiner:	Chung, Phung M.
For:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS	Attorney Docket No.:	129980-5053-US

**INFORMATION DISCLOSURE STATEMENT
SUBMITTED UNDER 37 C.F.R. § 1.97(i)**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information that the applicant considers cumulative with respect to references already of record in the subject U.S. patent application or otherwise not material to the examination of the application. It is requested that the references cited herein be made of record in the application in accordance with rule 37 C.F.R. §1.97(i).

1. Enclosures accompanying this Information Disclosure Statement are:
 - 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
 - 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☐ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
 - 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
 - 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.

3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b or 3c)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check either Item 4a and 4b or 4c)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

5. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(i) as it does not comply with § 1.97(a)-(d), Applicants request that this Information Disclosure Statement be placed in the file by the office.

6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.

7. ☐ This application is a continuing application under 37 C.F.R. §1.53(b) or (d).

(Check appropriate Items 7a, 7b and/or 7c)

7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.

7b. ☐ Copies of publications listed on Form PTO-1449 from prior application Serial No. _____, filed on _____, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).

7c. ☐ Copies of the publications listed on Form PTO-1449 were not previously cited in prior application Serial No. _____, filed on _____, and are provided herewith.

8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)

8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on _____. A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on _____.

9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:

(Check Item 9a, 9b, or 9c)

9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.

9b. ☐ set forth in the application.

9c. ☐ enclosed as an attachment hereto.

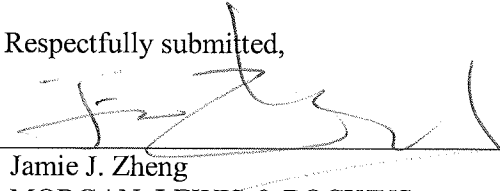
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5053-US).

11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a

search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: May 17, 2018


Jamie J. Zheng

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

(650) 843-4000

51,167

(Reg. No.)

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed August 28, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
				Examiner Name	<i>Chung, Phung M</i>
Sheet	1	of	1	Attorney Docket Number	<i>129980-5053-US</i>

PATENT DOCUMENTS						
Examiner Initials	Cite No.	Document Number Number - Kind Code ²	Publication Date MM-DD- YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	
FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
NON PATENT LITERATURE DOCUMENTS						
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published				
		Inter Partes Review of US Patent No. 8,001,434 B1, Case No. IPR2017-00561, Final Written Decision, filed July 5, 2018				
		Inter Partes Review of US Patent No. 8,359,501 B1, Case No. IPR2017-00562, Final Written Decision, filed July 5, 2018				

Examiner's Signature		Date Considered	
-------------------------	--	--------------------	--

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Gary Scott Williams/Edith Fuentes			
Attorney Docket Number:	129980-5053US			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	2806	1	120	120
Total in USD (\$)				120

Electronic Acknowledgement Receipt

EFS ID:	33567201
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Gary Scott Williams/Edith Fuentes
Filer Authorized By:	Gary Scott Williams
Attorney Docket Number:	129980-5053US
Receipt Date:	28-AUG-2018
Filing Date:	29-MAR-2014
Time Stamp:	20:38:28
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 120
RAM confirmation Number	082918INTEFSW00006073500310
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980-5053US_IDS.pdf	263304	yes	5
			a1156e5c2d7a0d1f69286cb6366a2c1fc41c4134		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	4	
	Information Disclosure Statement (IDS) Form (SB08)		5	5	
Warnings:					
Information:					
2	Non Patent Literature	IPR2017-00561_Final-Written-Decision_2018-07-05.pdf	21722067	no	63
			a22ffec4ed9d17fb89c1ee2d441f5f2a19bb0ea6		
Warnings:					
Information:					
3	Non Patent Literature	IPR2017-00562_Final-Written-Decision_2018-07-05.pdf	17753724	no	52
			1849199b37baac4cb38b0cfe030c7a9d8be1e4e7		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	30289	no	2
			673fa6f45a0041baf3cab7a0950d0fbeb664e592		
Warnings:					
Information:					
Total Files Size (in bytes):			39769384		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Lee et al.	Confirmation No.:	1018
Serial No.:	14/229,844	Art Unit:	2117
Filed:	March 29, 2014	Examiner:	Chung, Phung M.
For:	A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS	Attorney Docket No.:	129980-5053-US

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:
 - 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
 - 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☐ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
 - 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
 - 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.

3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

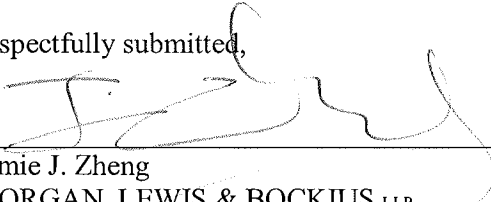
- 4a. ☒ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4c. ☒ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5053-US).

- 4d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
5. ☒ Certification Statement (applicable if Item 3a or Item 4a is checked)
- (Check either Item 5a, 5b, 5c or 5d)*
- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☒ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☐ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☐ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).
- (Check appropriate Items 7a, 7b and/or 7c)*
- 7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.
- 7b. ☐ Copies of publications listed on Form PTO-1449 from prior application Serial No. , filed on , of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7c. ☐ Copies of the publications listed on the attached Form PTO-1449 that were not previously cited in prior application Serial No. , filed on , are provided herewith.

8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on .
9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:
- (Check Item 9a, 9b, or 9c)
- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5053-US).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: August 28, 2018



Jamie J. Zheng
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000

51,167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 10/16/2018
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER

CHUNG, PHUNG M

ART UNIT

PAPER NUMBER

2111

DATE MAILED: 10/16/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	129980-5053US	1018

TITLE OF INVENTION: A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$500	\$0.00	\$0.00	\$500	01/16/2019

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 10/16/2018
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	129980-5053US	1018

TITLE OF INVENTION: A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$500	\$0.00	\$0.00	\$500	01/16/2019

EXAMINER	ART UNIT	CLASS-SUBCLASS
CHUNG, PHUNG M	2111	714-719000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	129980-5053US	1018
79141	7590	10/16/2018	EXAMINER	
Morgan Lewis Bockius LLP (PA)-J. Zheng			CHUNG, PHUNG M	
1400 Page Mill Road			ART UNIT	
Palo Alto, CA 94304			PAPER NUMBER	
			2111	
DATE MAILED: 10/16/2018				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/229,844	Applicant(s) Lee et al.	
	Examiner Phung M Chung	Art Unit 2111	AIA Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment dated on 5/2/18.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-5, 7-8, 10-12, 28, 36, 58-66 and 76-88, which are now as 1-34. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file areply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>5/17/18, 5/17/18 and 8/28/18.</u> 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☒ Examiner's Amendment/Comment 6. ☐ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/PHUNG M CHUNG/ Primary Patent Examiner, Art Unit 2111	10/4/18
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Notice of Pre-AIA or AIA Status

The present application is being examined under the pre-AIA first to invent provisions.

Information Disclosure Statement (IDS)

Due to the large number of Non Patent Literatures cited in the IDS dated on 5/17/18, 5/17/18 and 8/28/18. Examiner is unable to review all of the cited references thoroughly. Examiner is merely acknowledging the submission of the cited references and indicating that only a cursory review has been made.

/PHUNG M CHUNG/
Primary Patent Examiner, Art Unit 2111

<i>Search Notes</i> 	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner Phung M Chung	Art Unit 2111

CPC - Searched*		
Symbol	Date	Examiner
G11C29/10, G11C29/12	10/04/2018	PC

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

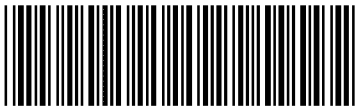
US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
EAST-CPC Searched.	10/04/2018	PC
Inventors Searched.	10/04/2018	PC
NPL Searched.	10/04/2018	PC

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G11C	29/10, 29/12	10/04/2018	PC

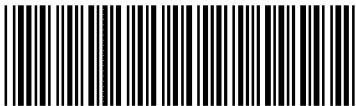
/PHUNG M CHUNG/ Primary Patent Examiner, Art Unit 2111	10/4/18
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Issue Classification 	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner Phung M Chung	Art Unit 2111

CPC			
Symbol			Type
G11C	/	29	/ 10 F 2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/	/			

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 34	
/PHUNG M CHUNG/ Primary Examiner, Art Unit 2111 (Primary Examiner) _____ (Date) 04 October 2018		O.G. Print Claim(s) 1	O.G. Print Figure 1

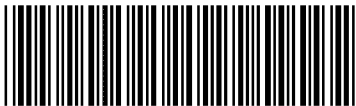
<i>Issue Classification</i> 	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner Phung M Chung	Art Unit 2111

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G11C	/	29	/ 00
NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	34	
/PHUNG M CHUNG/ Primary Examiner, Art Unit 2111	04 October 2018	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 14/229,844	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner Phung M Chung	Art Unit 2111

☐ Claims renumbered in the same order as presented by applicant
 ☒ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
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3	3	-	14	-	25	12	36	-	47	19	58	-	69	18	80
4	4	-	15	-	26	-	37	-	48	20	59	-	70	11	81
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-	6	-	17	10	28	-	39	-	50	22	61	-	72	24	83
6	7	-	18	-	29	-	40	-	51	28	62	-	73	33	84
7	8	-	19	-	30	-	41	-	52	29	63	-	74	25	85
-	9	-	20	-	31	-	42	-	53	30	64	-	75	26	86
15	10	-	21	-	32	-	43	-	54	31	65	16	76	34	87
8	11	-	22	-	33	-	44	-	55	23	66	13	77	27	88

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 34	
/PHUNG M CHUNG/ Primary Examiner, Art Unit 2111 (Primary Examiner) _____ (Date) 04 October 2018		O.G. Print Claim(s) 1	O.G. Print Figure 1

EAST Search History

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L28	0	(memory module AND computer system AND system memory controller AND system memory bus AND memory devices AND circuit board AND data module AND control module AND data handler logic elements AND address AND control ports AND data ports AND first operation mode AND second operation mode AND first data signals AND first memory address AND control signals AND second data singals AND second memory address AND control signals).clm.	US-PGPUB; USPAT	ADJ	ON	2018/10/04 17:10

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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	14/229,844
				Filing Date	March 29, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2117
				Examiner Name	Chung, Phung M
Sheet	1	of	4	Attorney Docket Number	129980-5053-US

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code ²	Publication Date MM-DD- YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		6,069,407	05-30-2000	Hamzehdoost	
		6,396,136 B2	05-28-2002	Kalidas et al.	
		6,541,872 B1	04-01-2003	Schrock et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

1.97(i)

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Preliminary Response, filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Trial Instituted Document, filed May, 15, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Response, filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners' Reply, filed December, 21, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners Motion to Exclude, filed January, 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, filed January, 23, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Opposition to Petitioners' Motion to Exclude, filed January, 23, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners' Response to Patent Owner's Listing of New Arguments and Evidence in Petitioners' Reply, filed January, 30, 2018

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
Examiner Name	<i>Chung, Phung M</i>				
Sheet	2	of	4	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners Reply to Patent Owner's Opposition to Motion to Exclude, filed January, 30, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1003, 'Declaration of Dr. Mazumder,' filed January, 3, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2001, 'US Provisional Patent Application Serial No. 60/456,451 (Averbuj),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2002, 'Prosecution file history of Application 10/630,480 (Averbuj),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2004, 'Excerpts from Microsoft Computer Dictionary Fifth Edition (2002),' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2005, 'Qualcomm Announces Sampling of the MSM6250,' filed April, 10, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2007, 'Deposition transcript of Dr. Mazumder,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2008, 'Search on Amazon website for memory module,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2006, 'Excerpts from Testing and Testable Design of High-Density,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2014, 'JEDEC Standard, JESD205 (2007),' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2010, 'Declaration of R. Jacob Baker, Ph.D., P.E.,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2011, 'Kingston press release from Oct. 11, 2005,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2012, 'IPR2014-00882, Paper 33, FWD,' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2013, 'Excerpts from Merriam-Webster's Dictionary (2006),' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2015, 'Excerpts from Memory Systems by Jacob (2007),' filed October, 31, 2017
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1026, 'Deposition Transcript of R. Jacob Baker,' filed December, 21, 2017

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
Examiner Name	<i>Chung, Phung M</i>				
Sheet	3	of	4	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1029, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 69-2 (Fed. Cir. Nov. 14, 2017),' filed December, 21, 2017	
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1030, 'JEDEC Standard Document JESD84-A41: Embedded Multimedia Card (eMMC) Product Standard, Standard Capacity (June 2007),' filed December, 21, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Preliminary Response, filed April, 10, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Response, filed November, 21, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Trial Instituted Document, filed July, 7, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1015, 'JEDEC PC2-5300 RDIMM PCB Plot,' filed January, 3, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1003, 'Declaration of Dr. Mazumder.,' filed January, 3, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2001, 'U.S. Provisional Patent Application Serial No. 60/456,451 (Averbuj),' filed April, 10, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2002, 'Prosecution file history of Application Serial No. 10/630,480 (Averbuj),' filed April, 10, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2010, 'Declaration of R. Jacob Baker, Ph.D., P.E.,' filed November, 21, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2012, 'Understanding DRAM, IBM application,' filed November, 21, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2013, '2016-2666, U.S. CAFC, IPR2014-01374 FWD affirmed,' filed November, 21, 2017	
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 2014, '2016-2274, U.S. CAFC, IPR2014-00971 FWD affirmed,' filed November, 21, 2017	
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Preliminary Response, filed April, 10, 2017	
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Trial Instituted Document, filed July, 7, 2017	
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Response, filed November, 21, 2017	
Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
Examiner Name	<i>Chung, Phung M</i>				
Sheet	4	of	4	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, EXHIBITS 1003, 'Declaration of Dr. Mazumder.,' filed January, 3, 2017
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, EXHIBITS 2010, 'Declaration of R. Jacob Baker, Ph.D., P.E.,' filed November, 21, 2017

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed August 28, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
				Examiner Name	<i>Chung, Phung M</i>
Sheet	1	of	1	Attorney Docket Number	<i>129980-5053-US</i>

PATENT DOCUMENTS						
Examiner Initials	Cite No.	Document Number Number - Kind Code ²	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	
FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
NON PATENT LITERATURE DOCUMENTS						
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published				
		Inter Partes Review of US Patent No. 8,001,434 B1, Case No. IPR2017-00561, Final Written Decision, filed July 5, 2018				
		Inter Partes Review of US Patent No. 8,359,501 B1, Case No. IPR2017-00562, Final Written Decision, filed July 5, 2018				

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				Electronically filed May 17, 2018	
				Application Number	14/229,844
				Filing Date	March 29, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2117
Examiner Name	Chung, Phung M				
Sheet	1	of	3	Attorney Docket Number	129980-5053-US

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code ²	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		6,130,852	10-10-2000	Ohtani et al.	
		6,333,873 B1	12-25-2001	Kumanoya et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ - Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Petitioners' Updated Exhibit List, filed February 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Patent Owner's Updated Exhibit List, filed February 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Hearing Transcript, filed March 22, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, Termination Decision Document, filed May 3, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 1034, 'Petitioners Demonstratives,' filed February 9, 2018
		Inter Partes Review of US Patent No.8689064, Case No. IPR2017-00560, EXHIBITS 2016, 'Patent Owner's Demonstratives,' filed February 9, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners Reply, filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Opposition to Petitioners' Motion to Exclude, filed March 14, 2018

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				Electronically filed May 17, 2018	
				Application Number	14/229,844
				Filing Date	March 29, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2117
Examiner Name	Chung, Phung M				
Sheet	2	of	3	Attorney Docket Number	129980-5053-US

		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Opposition to Patent Owner's Motion to Exclude, filed March 14, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Brief in Response to Show Cause Order, filed March 20, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Submission Regarding Order to Show Cause, filed March 20, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, PETITIONERS' REPLY TO PATENT OWNER'S OPPOSITION TO MOTION TO EXCLUDE, filed March 21, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Reply in Support of Its Motion to Exclude, filed March 21, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Patent Owner's Motion to Strike, filed March 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Petitioners' Opposition to Patent Owner's Motion to Strike, filed April 4, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Order To Show Cause Conduct of the Proceedings, filed March 13, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Request for Oral Argument 37 C.F.R. sec 42.70, filed March 28, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Hearing Transcript, filed April 18, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, Conduct of the Proceeding, filed May 2, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1034, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 70 (Fed. Cir. Dec. 21, 2017),' filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1036, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 43 (Fed. Cir. Apr. 6, 2017),' filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1028, 'SanDisk Corp. v. Netlist Inc., IPR2014-00970, Final Written Dec.,' filed February 7, 2018
		Inter Partes Review of US Patent No.8001434, Case No. IPR2017-00561, EXHIBITS 1035, 'Netlist, Inc. v. SanDisk LLC, No. 2016-02274, ECF No. 31 (Fed. Cir. Jan. 6, 2017),' filed February 7, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners Reply, filed February 7, 2018

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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DB2/ 33277349.1

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /P.M.C/

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed May 17, 2018</i>	
				Application Number	<i>14/229,844</i>
				Filing Date	<i>March 29, 2014</i>
				First Named Inventor	<i>Hyun Lee</i>
				Art Unit	<i>2117</i>
				Examiner Name	<i>Chung, Phung M</i>
Sheet	3	of	3	Attorney Docket Number	<i>129980-5053-US</i>

		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Request for Oral Argument, filed February 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Motion to Exclude, filed February 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Order To Show Cause Conduct of the Proceedings, filed March 13, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Opposition to Petitioners' Motion to Exclude, filed March 14, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Opposition to Patent Owner's Motion to Exclude, filed March 14, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Brief in Response to Show Cause Order, filed March 20, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Submission Regarding Order to Show Cause, filed March 20, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Reply to Patent Owner's Opposition to Motion to Exclude, filed March 21, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Reply in Support of Its Motion to Exclude, filed March 21, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Patent Owner's Motion to Strike, filed March 28, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Petitioners' Opposition to Patent Owner's Motion to Strike, filed April 4, 2018
		Inter Partes Review of US Patent No.8359501, Case No. IPR2017-00562, Hearing Transcript, filed April 18, 2018

Examiner's Signature	/PHUNG M CHUNG/	Date Considered	10/04/2018
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1355011	(memory devices near3 circuit board) and (address near1 control near2 data ports)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:30
L2	127088	((data module) with (data handler logic elements)) near3 (circuit board) near4 ((data ports near2 memory devices) and (system memory bus))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:32
L3	26378	I1 and I2	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:36
L4	806	(control module) near3 (circuit board) near4 (data module) near2 (address and control ports) near2 (memory devices) near3 (system memory bus)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:38
L5	221	I3 and I4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:39
L6	3309487	(control module near2 first mode) near3 (receive system address and control signals) near2 (system memory controller)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:42
L7	118280	I6 and ((control module near2 first mode) and (output\$4 adj first memory address and control signals) near2 (memory devices) near3 (system address and control signals))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 10:44
L8	350838	(data module near2 first mode) near3	US-PGPUB;	OR	ON	2018/10/04

		(propagate near3 first data signals) near2 (memory devices and system memory controller)	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			10:58
L9	3069	I8 and ((data module near2 first mode) and (first data signals) near1 (transmitted or received) near4 (portion near2 memory devices) near4 (first memory address and control signals))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:03
L10	111	I5 and I7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:08
L11	10	I9 and I10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:08
L12	275591	(control module near2 second mode) near3 (output second memroy address and control singals) near2 (address and control ports near2 memory device)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:09
L13	804779	(data module near2 second mode) near3 (isolate near1 memory devices near2 accessed near2 system memory controller) and (transmit nar3 second data signals near1 data patterns near3 data handler logic elements near2 data ports neae2 memory devices) near5 (commands output near2 control module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:12
L14	682878	I13 and ((second mode) and ((portion near2 memory devices near3 receive near4 second data signals) near3 (second memory address and control signal near2 control module)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:15
L15	2	I11 and I12	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 11:16
L16	42122	I12 and I14	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	2018/10/04 11:19

			JPO; DERWENT; IBM_TDB			
L17	38	I10 and I16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 12:07
L18	2	I11 and I16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 12:07
L19	2	I17 and I9	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 12:09
L20	33	I17 and I8	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 12:09
L21	2538623	((Lee).inv.) or ((Bhakta).inv.) or ((Choi).inv.) or ((Netlist).asn.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/10/04 12:10
L22	2	I20 and I21	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 12:10
L25	33	I20 and ((issolat\$3 or disconnect\$3 or disabl\$3 or switch\$3 or control\$4 or gat\$3 or deselect\$3 or demultiplex\$3 or (deMUX\$3)) and (memory devices near2 accessed near2 system memory controller) and (second mode or test mode))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2018/10/04 12:13
L27	4	I25 AND ((G11C29/10 OR G11C29/12).CPC. OR (714/719 OR 714/718 OR 714/733 OR 714/734 OR 714/736 OR 714/819 OR 365/201).CCLS. OR (G11C29/00).IPCR.)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2018/10/04 16:20

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/229,844	03/29/2014	2111	2190	129980-5053US	20	3

CONFIRMATION NO. 1018
CORRECTED FILING RECEIPT

79141
Morgan Lewis Bockius LLP (PA)-J. Zheng
1400 Page Mill Road
Palo Alto, CA 94304



Date Mailed: 11/02/2018

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/745,790 01/19/2013 PAT 8689064
which is a CON of 13/183,253 07/14/2011 PAT 8359501
which is a CON of 12/422,925 04/13/2009 PAT 8001434
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: No

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 04/21/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/229,844**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

MULTI-MODE MEMORY MODULE WITH DATA HANDLERS

Preliminary Class

714

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

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this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	129980-5053US	1018
79141	7590	11/08/2018		
Morgan Lewis Bockius LLP (PA)-J. Zheng			EXAMINER	
1400 Page Mill Road			CHUNG, PHUNG M	
Palo Alto, CA 94304				
			ART UNIT	PAPER NUMBER
			2111	
			NOTIFICATION DATE	DELIVERY MODE
			11/08/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

padocketingdepartment@morganlewis.com
vskliba@morganlewis.com

**UNITED STATES DEPARTMENT OF COMMERCE****U.S. Patent and Trademark Office**

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P.O. Box 1450

Alexandria, Virginia 22313-1450

APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR/ PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
14/229,844	03/29/2014	Lee et al.	129980-5053US

Morgan Lewis Bockius LLP (PA)-J. Zheng 1400 Page Mill Road Palo Alto, CA 94304		EXAMINER	
		Phung M Chung	
		ART UNIT	PAPER
		2111	20181104

DATE MAILED: _____

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

Please find the attached form 1449 dated 10/20/2016.

/PHUNG M CHUNG/
Primary Patent Examiner, Art Unit 2111

11/4/2018

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2125 2117
				Examiner Name	Unknown /Phung Chung/
				Attorney Docket Number	NETL.046C3
Sheet	1	of	2		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
/PC/	1.	2009/0103387 A1	04-23-2009	Shau	
/PC/	2.	2006/0262586 A1	11/23/2006	Solomon, J. et al.	
/PC/	3.	2008/0162790 A1	07/03/2006	IM	

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
/PC/	4.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 8,081,536, IPR Case Number IPR2015-01021, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
/PC/	5.	Petition for Inter Partes Review filed on 4/7/2015 for U.S. Patent No. 7,881,150, IPR Case Number IPR2015-01020, and all associated documents including cited references and expert declarations, available at https://ptab.uspto.gov/?utm_campaign=subscriptioncenter&utm_content=&utm_medium=email&utm_name=&utm_source=govdelivery&utm_term=#/login	
/PC/	6.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-01372, Final Written Decision, issued March 9, 2016	
/PC/	7.	Inter Partes Review Case No. IPR2014-01372, Petitioner's Request for Rehearing, filed April 6, 2016	
/PC/	8.	Inter Partes Review Case No. IPR2014-01372, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
/PC/	9.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-01374, Final Written Decision, issued March 9, 2016	
/PC/	10.	Inter Partes Review Case No. IPR2014-01374, Petitioner's Request for Rehearing, filed April 6, 2016	
/PC/	11.	Inter Partes Review Case No. IPR2014-01374, Decision Denying Petitioner's Request for Rehearing, issued July 11, 2016	
/PC/	12.	Inter Partes Review of U.S. Patent No. 8,001,434, IPR Case No. IPR2014-00970, Decision Denying Petitioner's Request for Rehearing, issued April 27, 2016	
/PC/	13.	Inter Partes Review of U.S. Patent No. 8,359,501, IPR Case No. IPR2014-00971, Petitioner's Request for Rehearing, filed January 13, 2016	
/PC/	14.	Inter Partes Review Case No. IPR2014-00971, Decision - Petitioner's Request for Rehearing, issued April 27 2016	
/PC/	15.	Inter Partes Review Case No. IPR2014-00971, Final Written Decision, reissued April 27, 2016	
/PC/	16.	Inter Partes Review Case No. IPR2014-00971, Netlist Inc.'s Notice of Appeal, filed June 29, 2016	

EXAMINER	/Phung Chung/	DATE CONSIDERED	10/11/2016
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* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/229,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189 2117
				Examiner Name	Unknown /Phung Chung/
Sheet	2	of	2	Attorney Docket Number	NETL.046C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
/PC/	17.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
/PC/	18.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
/PC/	19.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	

EXAMINER /Phung Chung/

DATE CONSIDERED

10/11/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	129980-5053US	1018
79141	7590	11/20/2018		
Morgan Lewis Bockius LLP (PA)-J. Zheng			EXAMINER	
1400 Page Mill Road			CHUNG, PHUNG M	
Palo Alto, CA 94304				
			ART UNIT	PAPER NUMBER
			2111	
			NOTIFICATION DATE	DELIVERY MODE
			11/20/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

padocketingdepartment@morganlewis.com
vskliba@morganlewis.com

**UNITED STATES DEPARTMENT OF COMMERCE****U.S. Patent and Trademark Office**

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P.O. Box 1450

Alexandria, Virginia 22313-1450

APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR/ PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
14/229,844	03/29/2014	Lee et al.	129980-5053US

Morgan Lewis Bockius LLP (PA)-J. Zheng 1400 Page Mill Road Palo Alto, CA 94304		EXAMINER	
		Phung M Chung	
		ART UNIT	PAPER
		2111	20181113

DATE MAILED: _____

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

Please find the attached Applicant Initiated Interview Summary.

/PHUNG M CHUNG/
Primary Patent Examiner, Art Unit 2111

11/13/18

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiners responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicants correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,-
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicants record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiners version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, Interview Record OK on the paper recording the substance of the interview along with the date and the examiners initials.

Electronically filed December 20, 2018

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	1018
Serial No.:	14/229,844	Art Unit:	2111
Filed:	March 29, 2014	Examiner:	CHUNG, PHUNG M
For:	<i>MULTI-MODE MEMORY MODULE WITH DATA HANDLERS</i>	Attorney Docket No.:	129980-5053-US

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

SUBSTANCE OF INTERVIEW

The Applicant thanks Examiner Phung M Chung for the opportunity to discuss this application in a telephonic interview with Applicant's counsel, Jamie J. Zheng (51,167) on November 13, 2018. In the interview, the Applicant and the Examiner discussed the status of the application and IDS filed on August 28, 2018.

Respectfully submitted,

Date: December 20, 2018

/Jamie J. Zheng/

51,167

Jamie J. Zheng

(Reg. No.)

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

(650) 843-4000

Electronic Acknowledgement Receipt

EFS ID:	34664475
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	MULTI-MODE MEMORY MODULE WITH DATA HANDLERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5053US
Receipt Date:	20-DEC-2018
Filing Date:	29-MAR-2014
Time Stamp:	21:41:13
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant summary of interview with examiner	129980_5053US_INTERVIEW.pdf	104987	no	1
			468c3d87d7ea2792792d045e18d1903df15ad772		

Warnings:

Information:	
Total Files Size (in bytes):	104987
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	

PART B - FEE(S) TRANSMITTAL

Electronically filed January 15, 2019

Complete and send this form with applicable fee(s), to: Mail

Mail Stop ISSUE FEE
 Commissioner of Patents
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
 (571) 273-2885

or Fax

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advanced orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS

MORGAN, LEWIS & BOCKIUS LLP
 1400 Page Mill Road
 Palo Alto, CA 94304

Note: A certificate of mailing below can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing.

Certificate of Mailing

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)_____
(Signature)_____
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTY'S DOCKET NO.	CONFIRMATION NO.
14/229,844	03/29/2014	Hyun Lee	129980-5053-US	1018

TITLE OF INVENTION A MULTI-MODE MEMORY MODULE WITH DATA HANDLERS

APPLN. TYPE	ENTITY STATUS	ISSUE FEE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	1000	0	1000	01/16/2019

EXAMINER	ART UNIT	CLASS-SUB CLASS
CHUNG, PHUNG M	2111	714-719000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).
☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list
 (1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
 (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed

1. **Morgan, Lewis & Bockius LLP**
 2. _____
 3. _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE:
 Netlist, Inc.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)
 Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

☒ Issue Fee
☐ Advanced Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above.)

☐ A check in the amount of the fee(s) enclosed
☐ Payment by credit card. Form PTO-2038 is attached.
☒ The Director is hereby authorized to charge the required fee(s), or credit any overpayment, to Deposit Account Number 50-0310 (order no. 129980-5053-US).

5. Change in Small Entity Status (from status indicated above)

☐ a. Applicant certifying micro entity status. See 37 CFR 1.29.

NOTE: Absent a valid certification of Micro Entity (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

☐ Applicant asserting small entity status. See 37 CFR 1.27.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

☐ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be sign in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Jamie J. Zheng/

Date: January 15, 2019

Printed Name: Jamie J. Zheng

Registration No. 51,167

Electronic Patent Application Fee Transmittal

Application Number:	14229844			
Filing Date:	29-Mar-2014			
Title of Invention:	MULTI-MODE MEMORY MODULE WITH DATA HANDLERS			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/S. Olivier			
Attorney Docket Number:	129980-5053US			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1000	1000

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	34863766
Application Number:	14229844
International Application Number:	
Confirmation Number:	1018
Title of Invention:	MULTI-MODE MEMORY MODULE WITH DATA HANDLERS
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5053US
Receipt Date:	15-JAN-2019
Filing Date:	29-MAR-2014
Time Stamp:	19:03:00
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	011619INTEFSW00006315500310
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	129980_5253US_Issue_Fee.pdf	112531	no	1
			8fc9f9f7a1ccc3b000babe3d26af5f1189924c16		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30066	no	2
			3477a71f1a03a19943b0a3a19fe2351771563f23		

Warnings:

Information:

Total Files Size (in bytes):			142597
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	3	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS

Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
Change(s) applied to document, /D.W./ 11/1/2018	53.	5394037	02-28-1995	Lattice Semiconductor Corporation	Josephson et al.
	54.	5430335	07-04-1995	Oki Electric Industry Co., Ltd.	Tanoi
	55.	5525917	06-11-1996	Altera Corporation	Wong et al.
	56.	5732241	03-24-1998	Mos Electronics, Corp.	Chan
	57.	5841296	11-24-1998	Xilinx, Inc.	Churcher et al.
	58.	5914543	06-22-1999	Dallas Semiconductor Corporation	Scherpenberg et al.
	59.	6000048	12-07-1999	Cirrus Logic, Inc.	Krishna et al.
	60.	6044481	03-28-2000	Artisan Components, Inc.	Kornachuk et al.
	61.	6070217	05-30-2000	International Business Machines Corporation	Connolly et al.
	62.	6169696	01-02-2001	Micron Technology, Inc.	Bissey
	63.	6194959	02-27-2001	Kabushiki Kaisha Toshiba	Kamoshida et al.
	64.	6216240	04-10-2001	Samsung Electronics Co., Ltd.	Won et al.
	65.	6467056	10-15-2002	Hitachi, Ltd.	Satou et al.
	66.	6560740	05-06-2003	Advanced Micro Devices, Inc.	Zuraski Jr. et al.
	67.	6614702	09-02-2003	Samsung Electronics Co., Ltd.	Lee
	68.	6681358	01-20-2004	Lsi Logic Corporation	Karimi et al.
	69.	6721150	04-13-2004	Advanced Micro Devices, Inc.	Guerrero Jr. et al.
	70.	6812869	11-02-2004	Lattice Semiconductor Corporation	Rahman et al.
	71.	6829728	12-07-2004	Wu-Tung Cheng	
	72.	6910162	06-21-2005	Kingston Technology Corp.	Co et al.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

EXAMINER

/Phung Chung/

DATE CONSIDERED

10/17/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	2	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS

Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
Change(s) applied to document, /D.W./ 11/1/2018	30.	3660675	05-02-1972	Honeywell Inc.	Andrews Jr.
	31.	3757235	09-04-1973	Shell Oil Co.	McCormick et al.
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	43.	4987321	01-22-1991	Eastman Kodak Company	Toohy
	44.	5033048	07-16-1991	Digital Equipment Corporation	Pierce et al.
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	47.	5173906	12-22-1992	Dreibelbis Jeffrey H	
	48.	5222066	06-22-1993	Motorola, Inc.	Gula et al.
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	50.	5304856	04-19-1994	At&T Bell Laboratories	Rainal
	51.	5337254	08-09-1994	Hewlett-Packard Company	Knee et al.
	52.	5359235	10-25-1994	Digital Equipment Corporation	Coyle et al.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

EXAMINER

/Phung Chung/

DATE CONSIDERED

10/17/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	Unknown /Phung Chung/
Sheet	4	of	5	Attorney Docket Number	NETL.046C3

US PATENT DOCUMENTS

Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
Change(s) applied to document, /D.W./ 11/1/2018	73.	6918072	07-12-2005	Micron Technology, Inc.	Cowles et al.
	74.	6928024	08-09-2005	Infineon Technologies Ag	Pfeiffer et al.
	75.	6928593	08-09-2005	Intel Corporation	Halbert et al.
	76.	6930509	08-16-2005	Stmicroelectronics Limited	Banik
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	79.	7053470	05-30-2006	Azul Systems, Inc.	Sellers et al.
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	86.	7774667	08-10-2010	Renesas Technology Corp.	Saito et al.
	87.	7957218	06-07-2011	Freescale Semiconductor, Inc.	Welker
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	89.	8250295	8/21/12	Amidi et al.	

NON PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	90.	Bayard "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, vol. 47, Issue 5, pp. 1288-1296.	

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

EXAMINER

/Phung Chung/

DATE CONSIDERED

10/17/2016

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/299,844
				Filing Date	03/29/2014
				First Named Inventor	Hyun Lee
				Art Unit	2189
Examiner Name	Unknown /Phung Chung/				
Attorney Docket Number	NETL.046C3				
Sheet	1	of	5		

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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	3.	20020131535	09-19-2002	Huber Brian W.	
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	5.	20030098742	05-29-2003	Nakagawa Osamu Samuel	
	6.	20030126346	07-03-2003	Kuo Sung H.	
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	17.	20060144015	07-06-2006	Cash John W Iii	
	18.	20060147217	07-06-2006	Hahin Jayne C	
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	20.	20060242458	10-26-2006	Daniel Feldman	
	21.	20060262586	11-23-2006	Solomon Jeffrey C	
	22.	20060271748	11-30-2006	Intel Corporation	Jain et al.
	23.	20070030814	02-08-2007	Samsung Electronics Co., Ltd.	Shin et al.
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	26.	20070109707	05-17-2007	International Rectifier Corporation	Honda
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	29.	20070223296	09-27-2007	Christopher Miller	

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

EXAMINER /Phung Chung/

DATE CONSIDERED 10/17/2016

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UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/229,844	02/26/2019	10217523	129980-5053US	1018

79141 7590 02/06/2019
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 179 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Netlist, Inc., Irvine, CA;
Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

**The United States Patent and Trademark Office
PATENT TRIAL AND APPEAL BOARD**



A petition has been filed in Patent Number 10,217,523 on August 21, 2020.

The Case Number is IPR2020-01421.

To view the documents filed in this petition, go to <https://ptab.uspto.gov/#/login>

Click on Search PTAB and enter the Patent Number or the Trial or Case Number and select the Search button.

Questions regarding this notice should be directed to the Patent Trial and Appeal Board at 571-272-7822.

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court WESTERN DISTRICT OF TEXAS, WACO DIVISION on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO. 6:20-cv-00525	DATE FILED 6/15/2020	U.S. DISTRICT COURT WESTERN DISTRICT OF TEXAS, WACO DIVISION
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX, INC. and SK HYNIX AMERICA INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 10,217,523	2/26/2019	NETLIST, INC.
2		
3		
4		
5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
 Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC. and SK HYNIX AMERICA INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2020-01042 (Patent 10,474,595 B2)
IPR2020-01044 (Patent 9,858,218 B1)
IPR2020-01421 (Patent 10,217,523 B1)¹

Before BRYAN F. MOORE, JON M. JURGOVAN, SHEILA F. McSHANE
and KARA L. SZPONDOWSKI, *Administrative Patent Judges*.²

PER CURIAM.

TERMINATION
Due to Settlement After Institution of Trial
35 U.S.C. § 317, 37 C.F.R. § 42.74

¹ We exercise our discretion to issue one order for all of the above captioned proceedings. The proceedings have not been consolidated, and the parties are not authorized to use this caption format.

² This is not an expanded panel of the Board. It is a listing of all Judges on the panels of the above-listed proceedings.

IPR2020-01042 (Patent 10,474,595 B2)
IPR2020-01044 (Patent 9,858,218 B1)
IPR2020-01421 (Patent 10,217,523 B1)

I. INTRODUCTION

With the Board's authorization, Petitioner SK hynix Inc. and SK hynix America Inc. ("Petitioner") and Netlist, Inc. ("Patent Owner") (collectively "the Parties") filed in each of the above proceedings a Joint Motion to Withdraw Petition for *Inter Partes* Review in each of the above-identified proceedings ("Joint Motion"). Paper 22.³ In support of the Joint Motion, the Parties filed in each proceeding a corresponding Confidential Agreement, Ex. 2035 ("Settlement Agreement"),⁴ as well as a corresponding Joint Request to Keep Agreement Confidential and Separate pursuant to 37 C.F.R. § 42.74(c), Paper 23 ("Joint Request").⁵

II. DISCUSSION

In each Joint Motion, the Parties represent that Petitioner and Patent Owner have resolved their disputes and request that "the Board terminate this proceeding in its entirety." Joint Motion 1–2. The Parties represent that the Agreement is "a true copy of any agreement [and] understanding . . . made in connection with, or in contemplation of, the withdrawal of the Petition for the proceeding under 37 CFR 42.74(b)." *Id.* at 1. The Parties

³ For purposes of expediency, we cite to Papers filed in IPR2020-01044, unless otherwise noted. The Parties filed a similar Motion in IPR2020-01042 (Paper 23) and IPR2020-01421 (Paper 14). Although the Parties were authorized to file a Joint Motion to Terminate the Proceedings under 37 C.F.R. § 42.72, the Parties stylize their Motion as a Motion to Withdraw Petition for *Inter Partes* Review. We treat the Motion as a Joint Motion to Terminate the proceedings under 35 U.S.C. § 317(a) and 37 C.F.R. § 42.72.

⁴ For purposes of expediency, we cite to Exhibits filed in IPR2020-01044. The Settlement Agreement was filed in IPR2020-01042 as Exhibit 2035 and in IPR2020-01421 as Exhibit 2022.

⁵ The Parties filed a similar Request in IPR2020-01042 (Paper 24) and IPR2020-01421 (Paper 15).

IPR2020-01042 (Patent 10,474,595 B2)
IPR2020-01044 (Patent 9,858,218 B1)
IPR2020-01421 (Patent 10,217,523 B1)

further represent that they have “filed a joint motion to dismiss with prejudice” the concurrent district court litigation regarding the patents at issue, and “do not anticipate further litigation between them concerning the . . . patent[s].” *Id.* Accordingly, the Parties jointly request termination of these proceedings. *Id.* at 2.

The Board generally expects that a case “will terminate after the filing of a settlement agreement, unless the Board has already decided the merits.” Consolidated Trial Practice guide, 86 (Nov. 2019) (Consolidate TPG”); *see* 35 U.S.C. § 317(a); 37 C.F.R. § 42.72. Here, although the Board has instituted *inter partes* review in each of the above-identified proceedings, the Board has not decided the merits of these proceedings. Under these circumstances, therefore, we determine that it is appropriate to terminate these proceedings as to the Parties, without rendering a final written decision.

In each Joint Request, the Parties request that the Settlement Agreement be treated as business confidential information and be kept separate from the files of the patent involved in these *inter partes* proceedings. Joint Request 1. After reviewing the Settlement Agreement, we find that the Settlement Agreement contains confidential business information regarding the terms of settlement. We determine that good cause exists to treat the Settlement Agreement as business confidential information pursuant to 37 C.F.R. § 42.74(c).

This Order does not constitute a final written decision pursuant to 35 U.S.C. § 318(a).

IPR2020-01042 (Patent 10,474,595 B2)
IPR2020-01044 (Patent 9,858,218 B1)
IPR2020-01421 (Patent 10,217,523 B1)

III. ORDER

In consideration of the foregoing, it is hereby

ORDERED that the Joint Motion to Terminate each proceeding is *granted*, and the above-identified proceedings are *terminated*; and

FURTHER ORDERED that the Joint Request to Keep Agreement Confidential and Separate in each proceeding is *granted*, and the Settlement Agreement shall be kept separate from the file of Patents; 10,474,595 B2, 9,858,218 B1, and 10,217,523 B1 and made available only to Federal Government agencies on written request, or to any person on a showing of good cause, pursuant to 37 C.F.R. § 42.74(c).

IPR2020-01042 (Patent 10,474,595 B2)
IPR2020-01044 (Patent 9,858,218 B1)
IPR2020-01421 (Patent 10,217,523 B1)

FOR PETITIONER:

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FOR PATENT OWNER:

William Meunier
Serge Subach
Matthew S. Galica
Andrew H. DeVoogd
MINTZ, LEVIN, COHN, FERRIS,
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msgalica@mintz.com
ahdevoogd@mintz.com

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court _____ for the District of Delaware _____ on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO.	DATE FILED 10/15/2021	U.S. DISTRICT COURT for the District of Delaware
PLAINTIFF SAMSUNG ELECTRONICS CO., LTD. and SAMSUNG SEMICONDUCTOR, INC.		DEFENDANT NETLIST, INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 10,217,523	2/26/2019	Netlist, Inc. (upon information and belief)
2 10,474,595	11/12/2019	Netlist, Inc. (upon information and belief)
3 9,858,218	1/2/2018	Netlist, Inc. (upon information and belief)
4 7,619,912	11/17/2009	Netlist, Inc. (upon information and belief)
5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
 Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00063
Patent 10,217,523 B1

Before BRYAN F. MOORE, JON M. JURGOVAN, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION
Granting Institution of *Inter Partes* Review
35 U.S.C. § 314(a)

I. INTRODUCTION

A. Background

Samsung Electronics Co., Ltd. (“Petitioner”)¹ filed a Petition requesting *inter partes* review of claims 1–34 (“the challenged claims”) of U.S. Patent No. 10,217,523 B1 (Ex. 1001, “the ’523 patent”) pursuant to 35 U.S.C. §§ 311–319, along with the supporting Declaration of Dr. Vivek Subramanian. Paper 1 (“Pet.”); Ex. 1003. Netlist, Inc. (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 7 (“Prelim. Resp.”). With authorization (Paper 8), Petitioner filed a Reply to Patent Owner’s Preliminary Response (Paper 11 (“Pet. Reply”)), with Patent Owner filing a Sur-Reply (Paper 12 (“PO Sur-Reply”)).

We have authority under 35 U.S.C. § 314(a), which provides that an *inter partes* review may not be instituted “unless . . . the information presented in the petition . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.”

For the reasons that follow, we determine that Petitioner has demonstrated that there is a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. For the reasons set forth below, and pursuant to 35 U.S.C. § 314, we institute an *inter partes* review of claims 1–34 of the ’523 patent.

¹ Petitioner identifies itself and Samsung Semiconductor, Inc. as another real party-in-interest. Pet. 1.

B. Related Proceedings

At the time of the Petition filing, Petitioner indicated *Samsung Electronics Co., Ltd. et al. v Netlist, Inc.*, No. 1:21-cv-01453 (D. Del. filed Oct. 15, 2021) (“Related Case”) involves the ’523 patent. Pet. 1. Petitioner also identified several other related matters. Pet. 1–3. At the time of the filing of Mandatory Notices, Patent Owner also identified several related matters. Paper 5, 3–5 (Notices).

The Board previously instituted trial in IPR2020-01421 on the same claims of the ’523 Patent and on the same grounds presented here, but with a different petitioners, SK hynix Inc. and SK hynix America Inc. *See* Ex. 1043. That IPR was terminated after institution because a settlement was reached. Ex. 1044.

C. The ’523 Patent

The ’523 patent is titled “Multi-Mode Memory Module with Data Handlers” and issued on February 26, 2019, from an application filed on March 29, 2014. Ex. 1001, codes (22), (45), (54).

The ’523 patent is directed to a self-testing memory module for testing a plurality of memory devices mounted thereon. Ex. 1001, 5:4–27. Figure 2, reproduced below, is a block diagram illustrating component blocks of memory module 12 and memory controller 14. Ex. 1001, 5:28–34, 8:41–62, 9:22–42, Fig. 2.

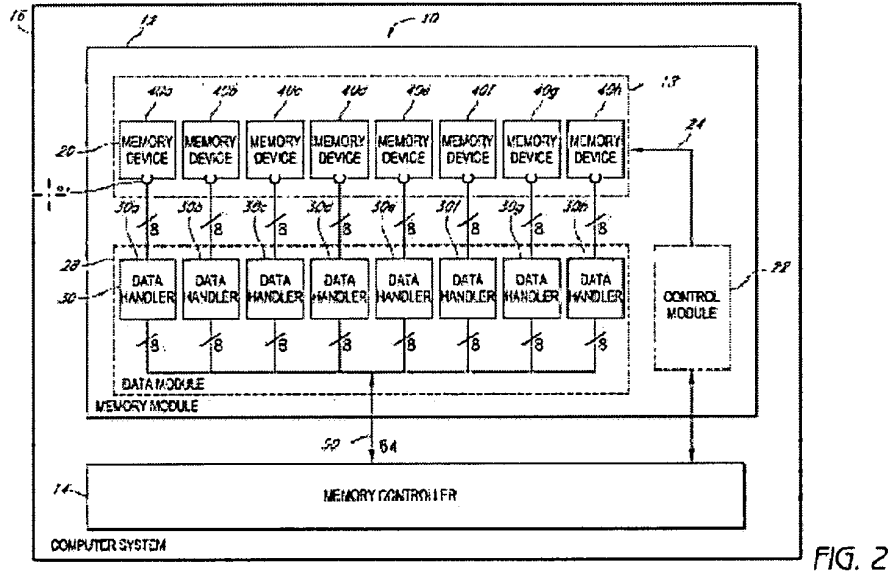


FIG. 2

Figure 2, above, shows memory module 12 includes control module 22 that generates address and control signals for testing memory devices 20 and data module 28 that includes a plurality of distributed data handlers 30 that are each located in proximity to corresponding memory device 20 and act as a buffer between memory device 20 and system memory controller 14. Data handlers 30 of data module 28 generate test patterns to write to memory devices 20 and compare test patterns read from memory devices 20 to the written patterns to identify faults. *Id.*

Figure 3, reproduced below, provides additional detail regarding the control module, data handlers and their components and interconnections. Ex. 1001, Fig. 3.

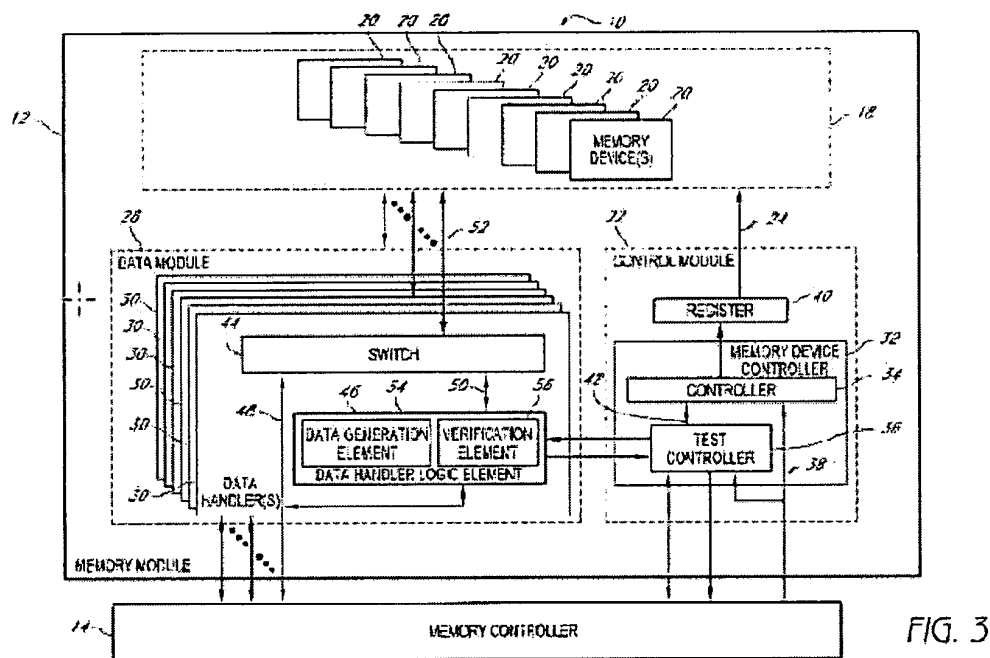


FIG. 3

As shown in Figure 3, above, shows additional control module 22, data handlers 30, and memory devices 20 and certain components thereof.

Challenged claims 1 and 19 of the '523 patent are independent.

Claim 1, an illustrative independent claim of the '523 patent, is reproduced below, with sub-paragraphing added to the limitations for reference purposes.

1. [a] A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:
 - [b] memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;
 - [c] a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements;
 - [d] a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and

[e] wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode;

[f] [i] wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals, and [ii] the data module in the first mode is configured to propagate one or more first data signals between the memory devices and the system memory controller, [iii] the one or more first data signals being transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals; and

[g] [i] wherein the control module in the second mode is configured to output second memory address and control signals to the address and control ports of the memory devices, and [ii] the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, and [iii] wherein at least a portion of the memory devices are configured to receive the one or more second data signals according to the second memory address and control signals from the control module.

Ex. 1001, 16:55–17:30.

D. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of claims 1–34 of the '523 patent on the following grounds:

Claims Challenged	35 U.S.C. §	Reference(s)
1–34	103(a)	Ellsberry, ² Jeddeloh ³

² US Pub. No. 2006/0277355 A1, published December 7, 2006. Ex. 1005.

Claims Challenged	35 U.S.C. §	Reference(s)
1–34	103(a)	Ellsberry, Jeddeloh, Averbuj ⁴
14, 17–34	103(a)	Ellsberry, Jeddeloh, Lee ⁵
14, 17–34	103(a)	Ellsberry, Jeddeloh, Averbuj, Lee

Pet. 5.

II. DISCRETIONARY DENIAL UNDER § 314(a)

A. Overview

Patent Owner requests that we exercise our discretion under 35 U.S.C. § 314(a) to deny the Petition under *Apple Inc. v. Fintiv, Inc.*, IPR2020-00019, Paper 11 (PTAB Mar. 20, 2020) (precedential) (“*Fintiv*”). Prelim. Resp. 43–45.

In assessing whether to exercise such discretion, the Board weighs the following factors:

1. whether the court granted a stay or evidence exists that one may be granted if a proceeding is instituted;
2. proximity of the court’s trial date to the Board’s projected statutory deadline for a final written decision;
3. investment in the parallel proceeding by the court and the parties;
4. overlap between issues raised in the petition and in the parallel proceeding;
5. whether the petitioner and the defendant in the parallel proceeding are the same party; and
6. other circumstances that impact the Board’s exercise of discretion, including the merits.

³ US 7,310,752 B2, issued December 18, 2007, claiming priority to Application No. 10/660,844, filed on Sept 12, 2003. Ex. 1006.

⁴ US Pub. No. 2005/0257109 A1, published November 17, 2005. Ex. 1007.

⁵ US Pub. No. 2006/0095817 A1, published May 4, 2006. Ex. 1008.

Fintiv at 6. Recognizing that “there is some overlap among these factors” and that “[s]ome facts may be relevant to more than one factor,” the Board “takes a holistic view of whether efficiency and integrity of the system are best served by denying or instituting review.” *Id.*

As identified above, the Related Case, which is a declaratory judgment action involving the ’523 patent, is pending in the District of Delaware. *See* Prelim. Resp. 44.

We address each *Fintiv* factor below.

B. Factor 1 – Stay of Related Litigation Proceeding

Patent Owner asserts a stay is unlikely given that an IPR against another patent asserted in the Related Case was filed within a week of the Preliminary Response. *Id.* We decline to attempt to predict how the District Court would proceed should a stay be requested by either of the parties. Accordingly, we find that this factor neither favors nor weighs against discretionary denial of *inter partes* review.

C. Factor 2 — Proximity of Court’s Trial Date

Patent Owner admits that “[t]he district of Delaware has not yet set a trial date.” *Id.* Accordingly, this factor weighs against exercising our discretion to deny *inter partes* review.

D. Factor 3 — Investment in the Parallel Proceeding

Patent Owner admits “[t]he parties have invested in pleading and motion practice in the parallel proceeding, although claim construction and merits determinations have not been reached.” *Id.* at 44–45. Accordingly, in view of the significant work that remains to be completed in the Related Case, this factor weighs against exercising our discretion to deny *inter partes* review.

E. Factor 4 — Overlap With Issues Raised in Parallel Proceeding

Patent Owner admits that “Petitioner does not challenge the validity of the ’523 Patent in the parallel proceedings.” *Id.* at 45. Nevertheless, “Petitioner has agreed to a *Sotera*-type stipulation to avoid any overlap with the Delaware action” Pet. Reply 2. Consequently, we find that this factor weighs against exercising our discretion to deny *inter partes* review.

F. Factor 5 — Commonality of Parties in Parallel Proceedings

Patent Owner asserts that the Petitioner is a party to the Related Case. Prelim. Resp. 45. In *Huawei Tech. Co. v. WSOU Inv., LLC*, IPR2021-00225, the Board found “this factor favors denial if trial precedes the Board’s Final Written Decision and favors institution if the opposite is true.” Paper 11 at 14 (PTAB June 14, 2021) (internal quotation marks omitted). Given that no trial date has been set in the Related Case, we find that this factor weighs against exercising our discretion to deny *inter partes* review.

G. Factor 6 — Other Circumstances

Patent Owner argues that we should consider that the “Petitioner here has not met its burden of showing that the asserted prior art and/or arguments were not already considered by the Office in the ’523 Patent’s prosecution, much less demonstrated a material error by the Office.” *Id.* Those issues are more relevant to an evaluation of discretionary denial under 325(d). Thus, this factor is neutral.

H. Conclusion

The *Fintiv* factors weigh strongly against exercising our discretion to deny *inter partes* review. Thus, based on our holistic assessment of the *Fintiv* factors, we decline to exercise our discretion under 35 U.S.C. § 314(a) to deny *inter partes* review.

III. DISCRETIONARY DENIAL UNDER § 325(d)

Patent Owner contends that we should deny institution under 35 U.S.C. § 325(d) because the Petition reuses the same art and arguments that were previously presented to the Office. Prelim. Resp. 41. Patent Owner asserts “Petitioner has the burden to establish why the art and proposed grounds present something *new* and *nonduplicative* of what was before the Office during prosecution, yet failed to do so.” *Id.* at 42. Specifically, Patent Owner argues “Petitioner has not shown that Ellsberry, Jeddelloh, or the arguments based thereon, are not substantially the same as the art and arguments the Office considered during prosecution, and Averbuj was even expressly considered and made of record during prosecution.” *Id.* at 42–43. Petitioner argues that we should not exercise discretion under § 325(d) to deny institution because Petitioner does not have any specified burden under § 325(d) and “during prosecution the Examiner specifically complained about the ‘unreasonable’ number of prior-art references that Patent Owner was dumping into the record without any explanation of their relevance.” Pet. Reply. 1–2 (citing Ex. 1002, 226).

Applicable Framework

Pursuant to 35 U.S.C. § 325(d), the Director may elect not to institute a proceeding if the challenge to the patent is based on matters previously presented to the Office. Section 325(d) states, in relevant part, “[in] determining whether to institute or order a proceeding under this chapter, chapter 30, or chapter 31, the Director may take into account whether, and reject the petition or request because, the same or substantially the same prior art or arguments previously were presented to the Office.” The Board

uses the following two-part framework in determining whether to exercise discretion under § 325(d) to deny institution:

- (1) whether the same or substantially the same art previously was presented to the Office or whether the same or substantially the same arguments previously were presented to the Office;
- and (2) if either condition of the first part of the framework is satisfied, whether the petitioner has demonstrated that the Office erred in a manner material to the patentability of challenged claims.

Advanced Bionics, LLC v. MED-EL Elektromedizinische Geräte GmbH, IPR2019-01469, Paper 6 at 8 (PTAB Feb. 3, 2020) (precedential).

In applying the two-part framework, we consider the following non-exclusive factors:

- (a) the similarities and material differences between the asserted art and the prior art involved during examination;
- (b) the cumulative nature of the asserted art and the prior art evaluated during examination;
- (c) the extent to which the asserted art was evaluated during examination, including whether the prior art was the basis for rejection;
- (d) the extent of the overlap between the arguments made during examination and the manner in which Petitioner relies on the prior art or Patent Owner distinguishes the prior art;
- (e) whether Petitioner has pointed out sufficiently how the Examiner erred in its evaluation of the asserted prior art; and
- (f) the extent to which additional evidence and facts presented in the Petition warrant reconsideration of the prior art or arguments.

Becton, Dickinson & Co. v. B. Braun Melsungen AG, IPR2017-01586, Paper 8 at 17–18 (PTAB Dec. 15, 2017) (precedential as to § III.C.5, first paragraph). Factors (a), (b), and (d) relate to whether the art or arguments presented in the Petition are the same or substantially the same as those

previously presented to the Office. *Advanced Bionics*, Paper 6 at 10. Factors (c), (e), and (f) relate to whether the petitioner has demonstrated a material error by the Office. *Id.* If a condition of the first part of the framework is satisfied and the petitioner fails to make a showing of material error, the Director generally will exercise discretion not to institute *inter partes* review. *Advanced Bionics*, Paper 6 at 9.

As to the burden, Patent Owner argues “Petitioner *always* ‘has the burden from the onset to show with particularity why the patent it challenges is unpatentable’ ***and why the Office should use its discretion to institute.*** *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363, 1367 (Fed. Cir. 2016); 35 U.S.C. § 312(a)(3) [(“*Harmonic*”)]; *Dynamic Drinkware, LLC v. Nat’l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (no burden on patent owner) [(“*Drinkware*”).” Nevertheless, neither *Drinkware* nor *Harmonic* discusses or professes to address whether the burden on Petitioner with regard to its patent challenge applies to the Board’s ability to deny institution for discretionary reasons. We find that the second part of the *Advanced Bionics* test requires that Petitioner point out sufficiently how the Examiner erred;⁶ however, the first part of the *Advanced Bionics* test does not assign a specific burden or responsibility on Petitioner to initially disprove that the same or substantially the same art at issue was previously presented to the Office. Thus, we disagree with Patent Owner that “Petitioner has the burden to establish why the art and proposed grounds present something *new* and *nonduplicative* of what was before the Office during prosecution.” *See* Prelim. Resp. 42.

⁶ We do not need to decide here if this is correctly characterized as a “burden.”

Relevant Procedural History

The Averbuj reference asserted in the IPR was cited in an IDS during prosecution of the '523 patent. Ex. 1002, 61, 255. We do not find, and Patent Owner does not cite to, any substantive discussion of Averbuj in the prosecution history.

Part One of Advanced Bionics Framework

In the first part of the *Advanced Bionics* framework, we consider whether the same or substantially the same art or arguments previously were presented to the Office. *Advanced Bionics*, Paper 6 at 8. *Becton, Dickinson* factors (a), (b), and (d) inform the analysis. *Id.* at 9–10.

Patent Owner contends that Averbuj was expressly considered during prosecution of the '523 patent. Prelim. Resp. 42. As noted above, Averbuj was submitted in an IDS. Ex. 1002, 61, 255. During prosecution, the Examiner stated “[i]t is unreasonable for Examiner to review all of the cited references thoroughly” and that “only a cursory review [of the cited references] has been made.” Ex. 1002, 226.

Given the facts above, because Averbuj was not presented in combination with the other references asserted here and is only used in two of the four grounds, the combination of references asserted in the Petition do not qualify as art that was “previously presented” to the Office. Thus, we find that the combination of references asserted in the Petition, were not previously presented to the Office.

Additionally, Patent Owner does not assert or explain sufficiently that Ellsberry, Jeddeloh, and/or Lee are cumulative of any reference cited during prosecution of the '523 patent. Prelim. Resp. 41–43. On this record, we are

not persuaded that Ellsberry, Jeddeloh, and Lee are cumulative of any reference cited during prosecution of the '523 patent.

Accordingly, the first condition of the first part of the *Advanced Bionics* framework is not satisfied, and we need not consider whether Petitioner has demonstrated that the Office erred in a manner material to the patentability of challenged claims.

Conclusion

After evaluating the *Becton, Dickinson* factors within the *Advanced Bionics* framework, and based on the particular circumstances here, we conclude that the combination of references asserted in the Petition, were not previously presented to the Office. Thus, the same or substantially the same prior art as in the Petition was not presented previously to the Office. Accordingly, we determine that discretionary denial of the Petition under 35 U.S.C. § 325(d) is not appropriate.

IV. ANALYSIS

A. Level of Ordinary Skill in the Art

Petitioner asserts that a person of ordinary skill in the art would have a “Bachelor’s degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.” Pet. 6.

Patent Owner does not dispute Petitioner’s characterization or offer its own. Accordingly, for purposes of this decision, we adopt Petitioner’s characterization of the level of ordinary skill, which we determine is consistent with the '523 patent written description and the asserted prior art.

B. Claim Construction

For cases like this one, where the petition for *inter partes* review was filed after November 13, 2018, the Board interprets claim terms in accordance with the standard used in federal district court in a civil action involving the validity or infringement of a patent. *See* 37 C.F.R. § 42.100(b) (2019). Under the principles set forth by our reviewing court, the “words of a claim ‘are generally given their ordinary and customary meaning,’” as would be understood by a person of ordinary skill in the art in question at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312 (Fed. Cir. 2005) (en banc) (quoting *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996)). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1015, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17).

Petitioner offers constructions of several terms which are not disputed by Patent Owner. Pet. 13–15. Nevertheless, we determine that it is not necessary to provide an express interpretation of any of the terms of the claims at this juncture. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017); *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999) (“[O]nly those terms need be construed that are in controversy, and only to the extent necessary to resolve the controversy.”).

C. Legal Background

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness.⁷ *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

*D. Alleged Obviousness of Claims 1–34
Over Ellsberry and Jeddeloh*

Petitioner contends that claims 1–34 are rendered obvious by the combination of Ellsberry and Jeddeloh. Pet. 34–96. To support its contentions, Petitioner provides explanations as to how Ellsberry and Jeddeloh teach each claim limitation. *Id.* Petitioner also relies upon the Subramanian Declaration (Ex. 1003) to support its positions. Patent Owner presents no issues or arguments specific to reading the teachings of references on the limitations, rather Patent Owner concentrates its disputes on the motivation to combine the prior art and alleges hindsight in the combination. *See generally* Prelim. Resp. Patent Owner also relies upon the Brogioli Declaration (Ex. 2001) to support its positions.

⁷ Patent Owner presents no objective indicia in the Preliminary Response. *See generally* Prelim. Resp.

We begin our discussion with a brief summary of Ellsberry and Jeddeloh and then address the evidence and arguments presented.

1. Ellsberry (Ex. 1005)

Ellsberry relates to “a device, system, and method for expanding the memory capacity of a memory module.” Ex. 1005, code (57). “A control unit and memory bank switch are mounted on a memory module to selectively control write and/or read operations to/from memory devices communicatively coupled to the memory bank switch.” *Id.* “By selectively routing data to and from the memory devices, a plurality of memory devices may appear as a single memory device to the operating system.” *Id.*

Figure 2 of Ellsberry, shown below, “illustrates a block diagram of a capacity-expanding memory system 200 according to one embodiment.” Ex. 1005 ¶ 28. System 200 has DIMM interface 202 that couples to a memory socket and communication bus over which data, memory addresses, commands, and control information are transmitted. *Id.* The capacity-expanding feature of the invention is accomplished by a combination of control unit 204 and one or more memory bank switches 206, 208. *Id.*

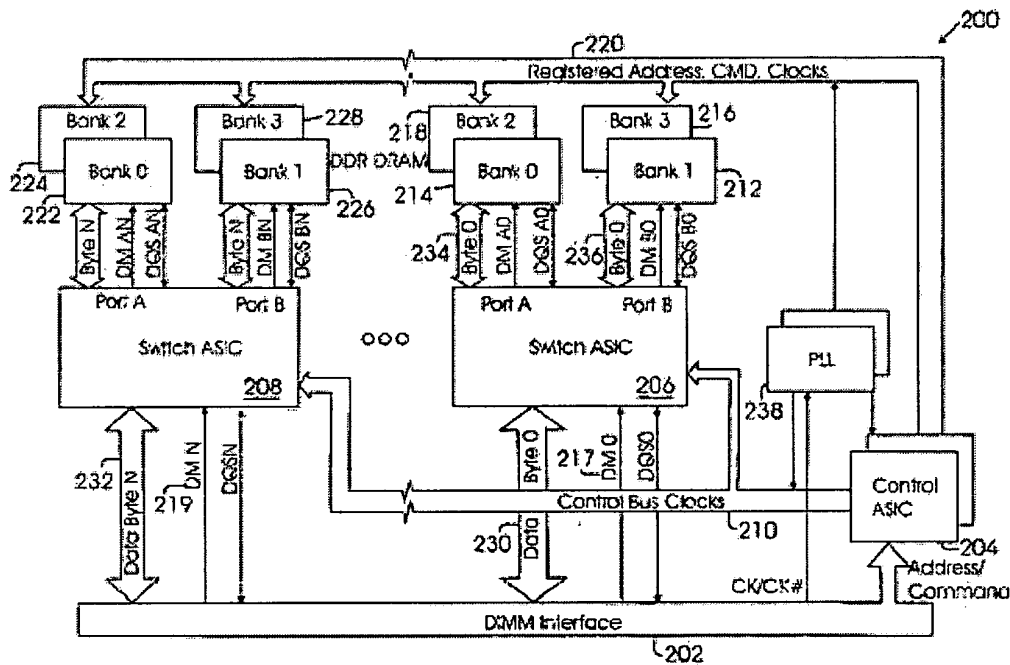


Fig. 2

Figure 2, above, illustrates a block diagram of a capacity-expanding memory device.

Figure 2 of Ellsberry illustrates system 200 with control ASIC 204 that receives addresses and commands from DIMM interface 202 and generates corresponding control signals on bus 210 and addresses on bus 220 to selectively connect memory banks 212–228 to DIMM interface 202 via switch ASICs 206, 208. Ex. 1005 ¶¶ 28–29.

Specifically, control unit 204 receives memory addresses and commands from DIMM interface 202 and controls switches 206 and 208 via control bus 210 to indicate how data from DIMM interface 202 should be received from and/or stored in memory banks 212–228. Ex. 1005 ¶¶ 29–30, Fig. 2. Control unit 204 also generates address and command information on address bus 220 to access memory banks 212–228. *Id.* Switches 206 and

208 receive data from, or provide data to, DIMM interface 202 via data buses 230 and 232. *Id.*

2. Jeddeloh (Ex. 1006)

Jeddeloh relates to “[a] memory module [that] includes several memory devices coupled to a memory hub [that] includes several link interfaces coupled to . . . a self-test module.” Ex. 1006, code (57).

Jeddeloh’s Figure 3, reproduced below, depicts each of the following self-test elements as a “functional block,” teaching that each block is “conventional, and can be implemented using well-known techniques and circuitry” (*id.* at 9:46–51):

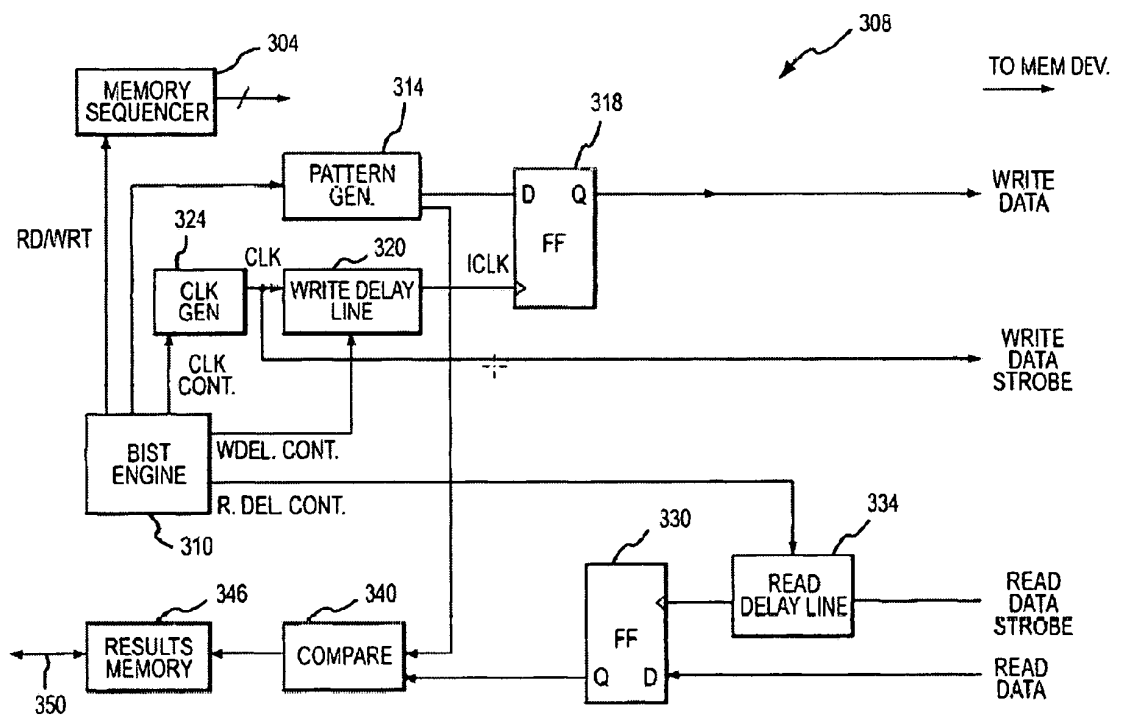


FIG.3

Figure 3, above, illustrates a block diagram of a self-test device.

Figure 3 above, shows BIST engine 310 and memory sequencer 304 that generate and distribute address and command/control signals during testing, and pattern generator 314, compare circuit 340, and results memory 346 that generate, distribute, and compare test patterns, and store fault information. *Id.* at 9:57–67, 10:30–51. Jeddeloh also discloses use of a SMBus to communicate with off-module components, for instance, a test apparatus can use it to “set memory testing parameters and receive test results.” *Id.* at 8:26–53, 10:47–51.

2. *Analysis*

a. Motivation to Combine

Petitioner contends it would have been obvious to modify the teachings of Ellsberry with Jeddeloh’s teachings to include self-test functionality. Pet. 25–34. As background to the discussion motivation to combine, below we discuss Petitioner’s proposed combination of Ellsberry and Jeddeloh.

Ellsberry discloses a memory module that separates circuitry into handling address and control signals and handling data signals as represented in Petitioner’s annotated Figure 2 below. Pet. 27 (citing Ex. 1005, Fig. 2).

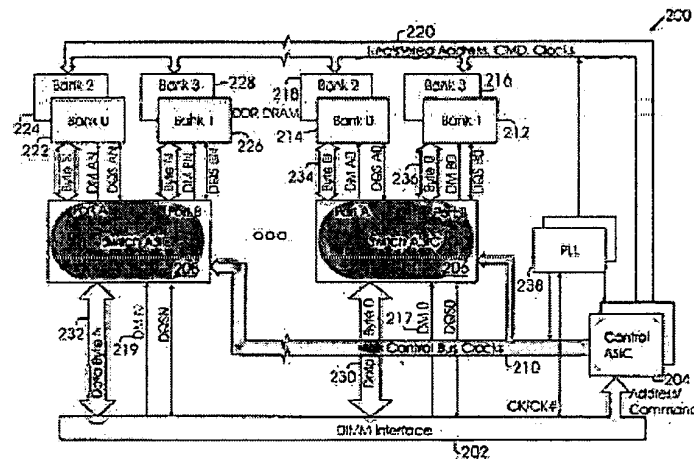


Fig. 2

Petitioner's annotated Figure 2 of Ellsberry above shows address/control signals are handled by Control ASIC 204 (grey) and data signals are handled by a plurality of distributed Switch ASICs 206 thru 208 (purple). *Id.* at 26–27. Jeddelloh discloses a “functional block diagram” of self-test circuitry in Figure 3 (annotated). *Id.* at 29 (citing Ex. 1006, Fig. 3).

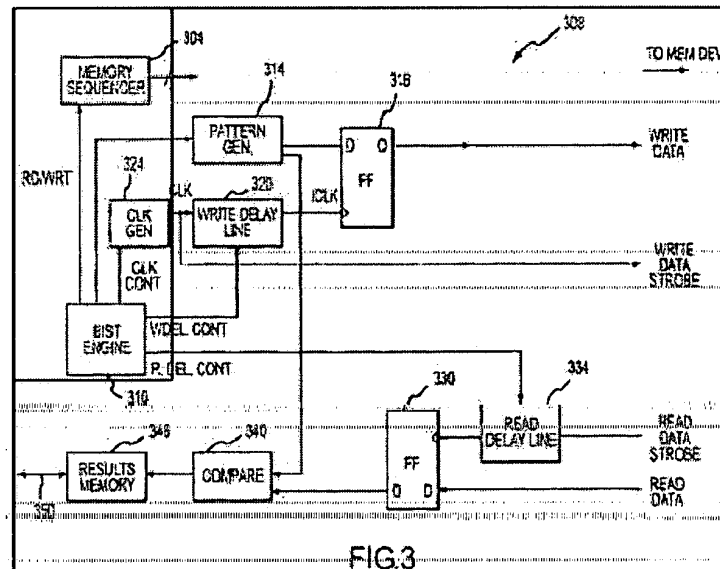


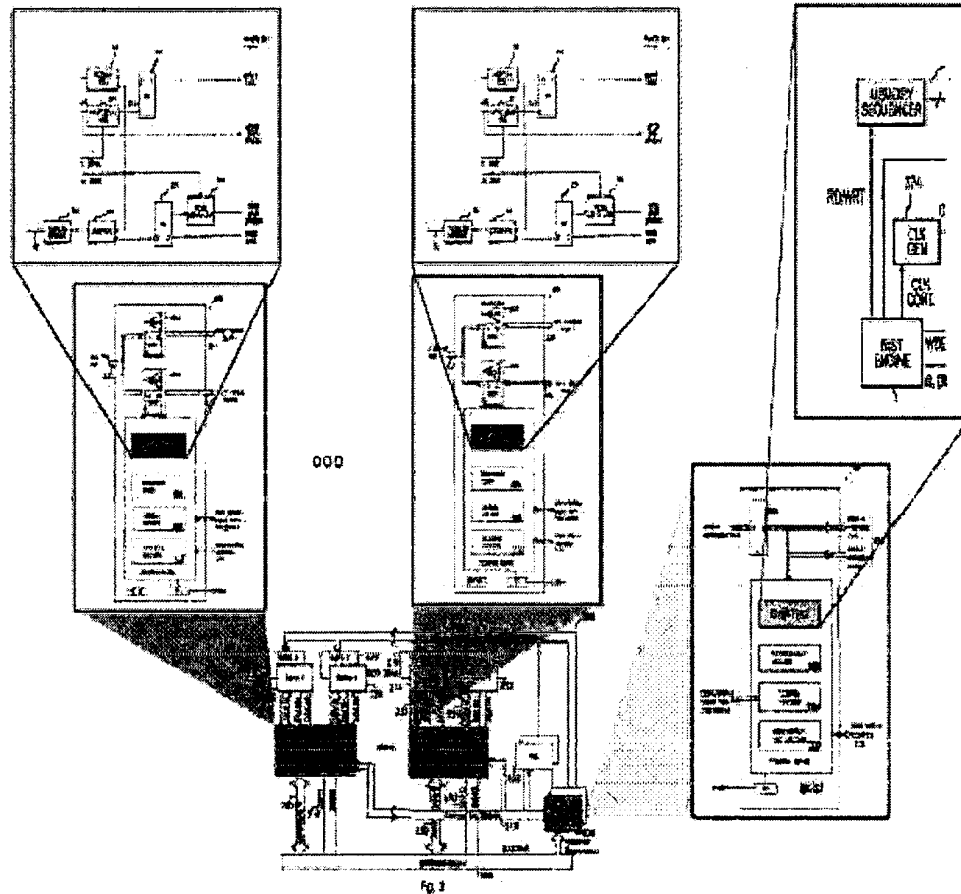
FIG 3

Petitioner's annotated Figure 3 above shows a BIST engine 310, clock generator 324, and memory sequencer 304 (in grey) that generate and distribute address and command/control signals during testing, and pattern

generator 314, compare circuit 340, and results memory 346 (in purple) that generate, distribute, and compare test patterns, and store fault information.

Id.

In Petitioner's combination, the self-test functionality of Jeddeloh would be divided between circuitry for handling address/control signals and circuitry for handling data signals. Self-test address/control signal circuitry would be implemented in Ellsberry's Control ASIC to send commands to self-test data signal circuitry implemented in Ellsberry's distributed Switch ASICs and address and control information to the memory devices in order to carry out the testing functionality described in Jeddeloh. *Id.* at 29–30. This combination is represented by Petitioner's annotated combination of Figure 2 of Ellsberry and Figure 3 of Jeddeloh, reproduced below. *Id.* at 30 (citing Ex. 1006, 9:46–51, Fig. 3).



As represented in Petitioner's annotated combination of Figure 2 of Ellsberry and Figure 3 of Jeddeloh, above, to carry out self-test functionality, the self-test circuitry in the modified Control ASIC would send address/control signals to the memory devices (*e.g.*, over the bus of Ellsberry Figure 2) and command signals to the Switch ASICs (*e.g.*, over bus 210 of Ellsberry Figure 2), as Petitioner contends is taught by Ellsberry and Jeddeloh. *Id.* at 30–31 (citing Ex. 1005 ¶ 56, Fig. 13; Ex. 1006, 10:30–51; Ex. 1003 ¶ 143). Additionally, to enter test mode and carry out the self-test functionality, the memory module would first receive mode commands (*e.g.*, over a maintenance bus), as Petitioner contends is taught by Jeddeloh. *Id.* (citing Ex. 1006, 8:26–53, 10:3–6).

For purposes of this Decision, and based on Petitioner's citations to specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Ellsberry with Jeddeloh's teachings to include self-test functionality. *Id.* at 31–37. For example, Petitioner asserts:

Ellsberry and Jeddeloh[] are analogous art to the 523 Patent because each is in the same field of endeavor – memory module design and architecture, EX1001, 1:32-33; EX1005, Abstract; EX1006, 3:61-4:19, and each is reasonably pertinent to the 523 Patent's addressed problem – improving memory modules by architecture design and/or testing, EX1001, 1:32-33; EX1005, Abstract; EX1006, 3:61-4:19. EX1003, ¶145.

Also, a Skilled Artisan would have considered the Combined System to be an arrangement of old elements each performing its known functions (Ellsberry's module acting as computer storage and Jeddeloh[]'s self-test functionality performing self-test on that module) and yielding what one would expect from the arrangement (a memory module with self-test). Ellsberry and Jeddeloh[] each employ known signaling and testing techniques found in many prior art systems. EX1006, 9:46-51. Combining them would therefore have been well within a Skilled Artisan's abilities, would not have resulted in any unpredictable results and could have been readily accomplished without undue experimentation and with a reasonable expectation of success. EX1003, ¶¶146-147.

Id. at 31–32.

Additionally, as to why a testing mode would be added to Ellsberry, Petitioner contends there are several specific reasons (“Reasons for Testing”). *Id.* at 32–34; Ex. 1003 ¶¶ 149–53. For example, Petitioner first contends, quoting a non-asserted reference, that “[a]s chips become faster in frequency, there is a greater need for hardware to include self-test logic.” *Id.*

at 32 (citing Ex. 1018, 1:14–18). Thus, Petitioner contends that Ellsberry discloses that its modules are JEDEC compliant and JEDEC standards at the time required self-testing. *Id.* (citing Ex. 1005 ¶ 50, claims 10, 19; Ex. 1016, 1–2, 25–28, 55–66; Ex. 1017, 2–11). Second, Petitioner contends that because Ellsberry’s Control ASIC configures the memory devices on the module “rather than [being] directly programmed by the host system,” a person of ordinary skill would have been motivated to have a built in testing capability in the Control ASIC itself. *Id.* at 33 (citing Ex. 1005 ¶ 44; Ex. 1003 ¶ 150). Third, Petitioner contends that Jeddeloh emphasizes the need for testing circuitry in memory modules because of the increasing significance of signal timing. *Id.* (citing Ex. 1006, 2:41–51, 3:47–52). Fourth, Petitioner contends that because Ellsberry describes that its Control ASIC configures the memory devices, one of ordinary skill would be motivated to add testing and, further, because Ellsberry has a “Mode pin,” to operate in other modes, a skilled artisan would have been motivated to provide functionality to switch between additional modes. *Id.* at 33–34 (citing Ex. 1005 ¶ 33; Ex. 1003 ¶ 152). Fifth, Petitioner also contends “a Skilled Artisan would have been motivated to add testing in a mode separate from normal operating mode because the purpose of testing was to confirm the module would operate appropriately in normal mode.” *Id.* at 34 (citing Ex. 1006, 2:43–50, 3:47–53; Ex. 1016, 1–2, 25–28 55–66; Ex. 1017, 2–11. Ex. 1003 ¶ 153). Sixth, Petitioner would add testing mode because “normal operation where the host sends read/write commands to the Control ASIC would interfere with such testing.” *Id.*

Finally, as to the motivation to distribute Jeddeloh's components into different parts of Ellsberry, Petitioner contends there are several reasons, including:

- First, such distribution is consistent with Ellsberry's architecture. (Pet. 35);
- Second, such a distribution of self-test functionality is consistent with how each of Ellsberry's Switch ASICs already included logic elements and connections for implementing data transactions in 8-bit portions (Pet. 35–36);
- Third, a Skilled Artisan would have been motivated to make such a design because other references teach Skilled Artisans to place such self-test functionality in distributed data buffers similar to Ellsberry's Switch ASICs (Pet. 36 (citing Ex. 1019 ¶¶ 77, 97, Figs. 5, 18; Ex. 1020, 15:57–65, 18:63–19:23, 35:10–37:1, Figs. 5, 18, 36; Ex. 1007 ¶¶ 49–50, 53, 55–57, Figs. 1, 6–7; Ex. 1025, 30));
- Fourth, an additional reference, Lepejian, teaches Skilled Artisans that, in order to reduce busing area, pattern generators should be distributed to each of the Switch ASICs. (Pet. 36 (citing Ex. 1021, 3:45–53));
- Fifth, “a Skilled Artisan would have been motivated to place self-test pattern generator functionality, such as Jeddeloh[]'s, in Ellsberry's Switch ASICs in order to place it close to its associated memory devices, thereby simplifying the wiring plan on the circuit board” (Pet. 36–37); and
- Sixth, quoting another reference, Zimmerman, it was known that “functionality shown embodied in a single integrated circuit or functional block may be implemented using multiple cooperating circuits or blocks, or vice versa.” (Pet. 37 (citing Ex. 1026, 7:43–46, Ex. 1027, 7:7–14)).

Pet. 34–37; Ex. 1003 ¶¶ 154–65.

In response, Patent Owner argues that Ellsberry fails to disclose the claimed “second mode” [or testing mode], as recited in claims 1 and 19, because it would not have been obvious to one of ordinary skill in the art to modify Ellsberry to provide for the “second mode.” Prelim Resp. 10.

Specifically, Patent Owner argues

First, the Petition fails to show that a POSITA would have been motivated in the first place to combine Ellsberry, which is directed to memory emulation, with Jeddeloh, which is directed to a central hub architecture with separate self-test module. EX2001, ¶¶39-44. Second, a POSITA would not disassemble parts of Jeddeloh’s self-test module and distribute it across the Ellsberry’s memory module architecture because doing so directly conflicts with, and is taught away by, the teachings of both Ellsberry and Jeddeloh. EX2001, ¶¶45-58. Finally, the Grounds fail because the Combined System is not functional as proposed, and modifying the proposed combination so that would have functioned in the manner required by the claims would have been beyond the technical ability of a POSITA. EX2001, ¶¶ 59-69.

Id.; *see also id.* at 11–40. As explained below, in general, these arguments do not sufficiently undermine Petitioner’s contentions but, in some instances, we determine that these issues are better resolved during trial.

More specifically, Patent Owner argues “Petitioner proposes that the centralized self-test module of Jeddeloh be disassembled and spread across different circuits within Ellsberry. However, the architectures of Ellsberry and Jeddeloh are antithetical to one another, and a POSITA would not have found it obvious to combine them or have known how to accomplish such a feat.” Prelim. Resp. 25.

Patent Owner asserts that “the references [cited by Petitioner’s declarant] show what a POSITA could have done, but do not show or teach

that said POSITA would have actually been motivated to redundantly distribute parts of Jeddeloh's architecture across Ellsberry's switch ASIC." Prelim. Resp. 30.

Patent Owner argues that "in order to create a *functional* switch ASIC [] Petitioner further alleges (without pointing to any particular reference) that a POSITA would have implemented a MUX/DEMUX circuit in order to manage the data convergence resulting from Ellsberry's originally-disclosed data paths and Petitioner's newly-concocted data path." *Id.* at 21. Patent Owner further argues "the architectures of Ellsberry and Jeddeloh are antithetical to one another" because for example, "Jeddeloh's self-test module is placed *behind* (relative to memory devices) its memory interface components," as opposed to in "new data paths between the memory controller and the memory devices in Ellsberry" as in Petitioner's combination. *Id.* at 25, 29. Specifically, Patent Owner explains "Jeddeloh's self-test module needs Jeddeloh's memory interface components to ensure that the nature of the signals sent and received by memory controller 280 correspond to the nature of the signals that memory devices are capable of sending and receiving." *Id.* at 29. Patent Owner also argues "Petitioner wholly ignores the fact that Jeddeloh's self-test module *requires* its memory interfaces to interact with memory devices, and provides no explaining how a POSITA would include said memory interfaces into Ellsberry. Ultimately, Patent Owner asserts "this combination alone would not have resulted in an operable system, and a POSITA would not have had the technical skill to make it operable without undue experimentation." *Id.* at 24.

Patent Owner also relies on U.S. Patent No. 8,154,901 (the '901 patent) for the proposition that adding a multiplexer would "present

transmission line discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with electrical lengths longer than $\frac{1}{4}$ of the wavelength of the operational frequency or the signal transition rate)” and “signal reflections may occur, changing wave characteristics and degrading system performance.” *Id.* at 33–34 (citing Ex 2002, 1:40–56). Thus, Patent Owner’s argument focuses on motivation to combine, teaching away, and the ability of one of ordinary skill to make the combination.

Petitioner contends “in the Combined System data travels over two different data paths in the two different modes. Because each path merges into the same port on the Switch ASIC, only one path can be used at a time, using for instance a MUX/DEMUX as was well-known, otherwise there could be data collisions.” Pet. 59 (citing Ex. 1003 ¶ 242).

Petitioner cites to its declarant who testifies extensively on how the combined system could be implemented and relies on several non-asserted references as support for that testimony. Ex. 1003 ¶¶ 258–59 (citing Exs. 1006, 1007, 1026, 1035, 1030 (pin citations omitted)).

Although we find that Petitioner has provided sufficient evidence in support of the proposed combination at this juncture, the parties’ evidence, particularly the declarants’ testimony, presents a classic battle of the experts, which we determine would be more fully addressed during a trial. Patent Owner and Petitioner will have the opportunity to cross-examine Dr. Subramanian and Dr. Brogioli, respectively, during the course of the trial.

As such, we invite the parties to further address the combination of references during the trial. For example, the parties should address the issue of whether the new data paths created with Jeddeloh’s self-test module

placed behind (relative to memory devices) its memory interface components would have been within the ability of one of ordinary skill to achieve. Although this issue was raised in IPR2020-01421, the instant Petitioner here, Samsung Electronics, has not had the opportunity to respond to Patent Owner's new declaration, which was not part of the record of IPR2020-01421. *See* Exs. 1042, 1043. Given Patent Owner's suggestion that such a combination may be possible (i.e. what could have been done) (Prelim. Resp. 30), even if beyond the skill of an ordinary artisan, we decline to deny institution based on the alleged problems with achieving the combination at this stage.⁸

Patent Owner also argues "Ellsberry and Jeddeloh are directed to different problems in the art." Prelim. Resp. 11. Specifically, Patent Owner argues "Ellsberry is directed to a distributed memory architecture and is exclusively focused on expanding a memory module's capacity by using multiple ranks of memory devices and activating and deactivating the memory devices in real time" and "Jeddeloh is directed to a memory module with a central hub architecture with a separate self-test module that is not directly connected with the memory devices." *Id.* at 11–12. Patent Owner also asserts "[t]here is no other problem that Ellsberry seeks to resolve,

⁸ Patent Owner asserts that, unlike IPR2020-01421, Patent Owner "provides new evidence to rebut those arguments, supported by expert declaration testimony." Prelim. Resp. 20 n.4. Although Patent Owner presents some new arguments (especially regarding production testing), we did not determine that the new arguments undermined Petitioner's contentions sufficiently. Additionally, to the extent that Patent Owner's declarant repeated arguments presented in the prior preliminary response in IPR2020-01421, further explanation of those arguments rather than just repetition of the arguments would have been helpful.

indeed, Ellsberry makes no mention of any need or desire to implement any form of testing on the memory module for any purpose.” *Id.* at 12 (citing Ex. 1005; Ex. 2001, ¶¶ 40, 30). Finally, Patent Owner asserts, “[i]n sharp contrast, the ’523 Patent and the challenged claims are directed to system integration and initialization—i.e., to testing during standard operation of the device—and include a ‘first mode’ or normal mode in which the module operates as normal within a computer system, such as a personal computer system,” which according to Patent Owner “is a completely different type of ‘testing’ than disclosed in Jeddelloh.” *Id.* at 13.

As an initial matter, the distinction between testing in Jeddelloh and the ’523 patent may be relevant to whether Jeddelloh is analogous art, but Patent Owner does not present an analogous art argument. *See id.* at 13–14.

Additionally, as to whether Ellsberry and Jeddelloh are designed to serve completely different purposes from one another, it is not correct, as Patent Owner suggests, “that a person of ordinary skill attempting to solve a problem will be led only to those elements of prior art designed to solve the same problem.” *KSR*, 550 U.S. at 1742. To the contrary, the Court in *KSR* explained, “familiar items may have obvious uses beyond their primary purposes, and in many cases a person of ordinary skill will be able to fit the teachings of multiple patents together like pieces of a puzzle. . . . A person of ordinary skill is also a person of ordinary creativity, not an automaton.” *Id.* An ordinarily skilled artisan may be motivated to pursue the desirable properties taught by one prior art reference, even if that means foregoing the benefits taught by another prior art reference. *See In re Urbanski*, 809 F.3d 1237, 1244 (Fed. Cir. 2016); *see also Medichem, S.A. v. Rolabo, S.L.*, 437 F.3d 1157, 1165 (Fed. Cir. 2006) (“[A] given course of

action often has simultaneous advantages and disadvantages, and this does not necessarily obviate motivation to combine.”). Thus, this argument does not sufficiently undermine Petitioner’s contentions.

Patent Owner also argues Petitioner’s specific motivations to add self-test to Ellsberry, i.e. Reason for Testing, are insufficient. Patent Owner argues, as to Petitioner’s first argument, Petitioner’s reliance on the statement that “[a]s chips become faster in frequency, there is a greater need for hardware to include self-test logic” (Pet. 32) fails on the basis that “just because a device has a fast clock speed does not mean a BIST must be included *on the module itself*” and “just because Bravo may have found it advantageous to add self-test logic on a relatively simple system would not have meant a POSITA would have believed it to be advantageous to add the same logic on a complex, distributed system like Ellsberry.” Prelim. Resp. 15–16. However, the statement of motivation relied on by Petitioner applies more widely than Patent Owner’s limited view. Thus, this argument does not sufficiently undermine Petitioner’s contentions.

Patent Owner also argues, as to Petitioner’s third argument, that Petitioner’s reliance on the fact that the purpose of testing is to confirm the module would operate appropriately in normal mode (Pet. 31) fails because “Petitioner’s argument ignores the fact that Jeddeloh’s disclosure is in the context of *production* testing, which is different from the system integration operational testing disclosed in the claims of the ’523 Patent.” *Id.* at 12, 13, 17. We note that Patent Owner’s declarant states that “[t]he claims of the ’523 Patent . . . are directed to . . . testing, during operation once the module has been integrated into an end-computer system.” Ex. 2001 ¶ 26. However, Petitioner’s declarant does not provide further support or

explanation for this statement. Additionally, Patent Owner's citations to Ex. 1001, 7:36–39, 13:59–62 and claim 1 do not sufficiently support the contention that the '523 Patent is limited to production testing. Additionally, Petitioner has not had the opportunity to respond to this distinction. Although this issue may be shown to have some merit, at this stage, we decline to discount Petitioner's reliance on the statements in Jeddeloh to provide motivation to combine based on this argument.

Patent Owner also asserts “the proposed combination put forward by Petitioner actually has the potential to *increase* signal timing issues, vitiating Petitioner's proposed motivation to combine.” Prelim. Resp. 17. We further determine that even if timing issues are increased, a person of ordinary skill can weigh the benefits of testing against the timing issues created. *Winner Int'l Royalty Corp. v. Wang*, 202 F.3d 1340, 1349, n.8 (Fed. Cir. 2000) (competing benefits do not weigh against obviousness, rather, the benefits, both lost and gained, in combining the teachings of the prior art should be weighed against one another). Therefore, this argument does not sufficiently undermine Petitioner's contentions.

“There is no requirement that the prior art contain an express suggestion to combine known elements to achieve the claimed invention. Rather, the suggestion to combine may come from the prior art, as filtered through the knowledge of one skilled in the art.” *DyStar Textilfarben GmbH & Co. Deutschland KG v. C.H. Patrick Co.*, 464 F.3d 1356, 1361 (Fed. Cir. 2006) (citation omitted). The need for self-test in faster chips and the need for self-test to confirm the operation of normal mode in chips is something one of ordinary skill can take into account, even if Jeddeloh and Ellsberry are not specifically included in the support upon which Petitioner relies.

We generally agree, however, with Patent Owner’s arguments that: 1) as to Petitioner’s first argument, Ellsberry’s mention of JEDEC compliance is not specific to self-test; 2) as to Petitioner’s second and fourth argument, that Ellsberry’s silence as to self-test does not suggest adding self-test; 3) and also as to Petitioner’s fifth and sixth argument, that the existence of a “Mode” does not provide motivation to add any and all modes. Prelim. Resp. 14, 17–18.

Patent Owner further argues that each of Petitioner’s motivations fails because Petitioner is relying on hindsight. *Id.* at 20. Because we are persuaded, at this stage of the proceeding, by some of Petitioner’s proposed bases of motivations, we do not agree that Petitioner is relying on hindsight.

Patent Owner also argues that there are some instances of teaching away in the record. Prelim. Resp. 21–25, 33–40. A teaching away requires a reference to actually criticize, discredit, or otherwise discourage the claimed solution. *See In re Fulton*, 391 F.3d 1195, 1201 (Fed. Cir. 2004) (“The prior art's mere disclosure of more than one alternative does not constitute a teaching away from any of these alternatives because such disclosure does not criticize, discredit, or otherwise discourage the solution claimed.”).

More specifically, Patent Owner argues Petitioner’s “insertion of the MUX/DEMUX into the data paths in Ellsberry’s memory bank switches” is “exactly what Ellsberry is designed to avoid.” Prelim. Resp. 33. Patent Owner relies on the ’901 patent, which is incorporated by reference in the ’523 patent, which states

the multiplexer and/or demultiplexer 12 can present transmission line discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with electrical

lengths longer than $\frac{1}{4}$ of the wavelength of the operational frequency or the signal transition rate). In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

Id. at 33–34 (quoting Ex. 2002, 1:44–56). Patent Owner further asserts this is “why Jeddeloh placed its self-test module behind its memory interfaces 270a-d. It is also exactly why Ellsberry chose to avoid using FET switches (e.g., MUX/DEMUX) to choose between two different banks of its memory devices.” *Id.* at 34. We agree the ’901 patent language may appear to disparage the use of multiplexers in certain situations; however, the linkage between the ’901 patent statement and the configurations in Ellsberry and Jeddeloh is not sufficiently explained at this stage of the proceeding. We invite additional briefing to address this issue during trial. Additionally, the ’901 patent statement uses permissive language such as “may cause” which suggest that, although such a configuration may have been disfavored, it may have been a viable option. Merely teaching an alternative or equivalent method does not teach away from the use of a claimed method. *See In re Mouttet*, 686 F.3d 1322, 1334 (Fed. Cir. 2012) (citations omitted); *In re Dunn*, 349 F.2d 433, 438 (CCPA 1965).

Patent Owner additionally asserts that “Ellsberry went to great lengths to design two approaches of activating/deactivating the memory banks ‘depending on two different memory addressing schemes (e.g., column or row addressing).’” Prelim. Resp. at 34. Specifically, Patent Owner argues “[t]his was done to avoid the need to control the data paths using any

hardwired switching mechanisms.” *Id.* (citing Ex. 1005 ¶ 33; Ex. 2001 ¶ 48). In fact, in the background of the invention, as quoted by Patent Owner, Ellsberry warns when

[a] high-speed switching system, field effect transistor (FET) switches, is employed to select a data line associated with each level of a stacked memory module to reduce the loading effect on the data lines in memory access. The problem with this scheme is that while FET switches have a fast propagation delay, their switch time is too slow and imprecise to reliably comply with industry standards, such as the Joint Engineering Device Engineering Council (JEDEC) standards, used in many memory applications.

Ex. 1005 ¶ 9.

Nevertheless, Petitioner asserts “[a] Skilled Artisan would have understood that the bidirectional drivers 402 and 404 perform a multiplexing/demultiplexing function depending on the direction of driving. EX1003, ¶¶ 110-111. It is this use of bidirectional drivers which solve the problem of the prior-art FET switches described in the background of Ellsberry. *Id.*” Pet. 18 n.2. In other words, Petitioner suggests “multiplexing/demultiplexing function[s]” are in fact contemplated by Ellsberry. Patent Owner suggests that any multiplexer would be a type of FET Switch. Prelim. Resp. 34 (citing Ex. 2001 ¶¶ 47–48). Patent Owner also suggests JEDEC standards compliance is a reason that would prevent a person of ordinary skill from pursuing this combination. *Id.* For similar reasons as above, Patent Owner also suggests such a combination would be

beyond the ability of one of ordinary skill to have a reasonable expectation of success in achieving.⁹ *Id.* at 36–40.

As to this complex technology, and at this juncture, we considered and credited Dr. Subramanian’s testimony regarding the feasibility of combination as well as Dr. Brogioli’s testimony regarding the unfeasibility of combination at this juncture. Ex. 1003 ¶ 242; *see also Wyers*, 616 F.3d at 1240 n.5 (Fed. Cir. 2010) (“expert testimony may be critical, for example, to establish ... the existence (or lack thereof) of a motivation to combine references”). Although Petitioner has provided sufficient evidence in support of its assertions, we cannot fully resolve this conflict between the parties’ characterization of multiplexing functions at this stage and on this record. Again, we invite additional briefing on any linkage between the problem asserted in the background of the invention and the two alleged solutions.

⁹ Patent Owner uses the term undue experimentation. Prelim. Resp. 36. Undue experimentation is part of the enablement inquiry. *See Impax Labs., Inc. v. Aventis Pharms. Inc.*, 545 F.3d 1312, 1314 (Fed. Cir. 2008) (“[A] prior art reference must enable one of ordinary skill in the art to make the invention without undue experimentation.”). A reference, however, qualifies as prior art in determining obviousness, independent of enablement. *In re Antor Media Corp.*, 689 F.3d 1282, 1292 (Fed. Cir. 2012); *see also In re Morsa*, 713 F.3d 104, 111–12 (Fed. Cir. 2013) (remanding for proper enablement analysis in anticipation rejection, but affirming obviousness rejection based on the same prior art). Thus, we do not engage in enablement analyses in our obviousness determination. Instead, we interpret Patent Owner’s undue-experimentation argument as an assertion of no reasonable expectation of success.

Thus, for the reasons above, for purposes of this decision, and based on Petitioner's citations to specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Ellsberry with Jeddeloh's teachings to include self-test functionality. Pet. 31–37.

a. Independent claim 1

i. Limitation [a] (Preamble)

Petitioner contends that Ellsberry teaches a memory module accessible in a computer system by a system memory controller via a system memory bus, as recited in the preamble of claim 1. Pet. 38. Petitioner relies upon Ellsberry's disclosure of "memory module 106," "processing unit 102," that is "coupled to a memory module 106 to retain or store information," and that Ellsberry's memory module is accessible by the memory controller "*via a system memory bus*" as reflected by path 110 (Figure 1), corresponding DIMM interface with data buses 230–232 (Figure 2), and corresponding edge interface (Figures 5 and 6). *Id.* (citing Ex. 1005 ¶¶ 11, 17, 23, 26–28, 47, 51, Figs. 1, 2, 5, 6; Ex. 1015, 6; Ex. 1003 ¶¶ 171–175).

We have reviewed the record, and find that Petitioner provides sufficient evidence that the prior art teaches the preamble and limitation [a] of claim 1.¹⁰ Patent Owner makes no arguments contesting the prior art teachings of these claim limitations. *See generally* Prelim. Resp.

¹⁰ We need not determine whether the preamble of claim 1 is limiting because Petitioner has sufficiently shown that the combination of Ellsberry and Jeddeloh teaches the preamble at this juncture. *See Nidec*, 868 F.3d at 1017.

ii. Limitation [b]

Petitioner contends that limitation [b] of claim 1 is taught by Ellsberry's disclosure DRAMs which are mounted on a circuit board. Pet. 38–39 (citing Ex. 1005 ¶¶ 3, 30, 32, 47, 49, 51, 54, 56, Figs 5, 6, 11, 13; Ex. 1003 ¶ 180). Petitioner contends that Ellsberry discloses “*address and control ports*” and “*data ports*” coupled to the address and control lines from the Control ASICs (ACA/ACB/ACAA/ACBB, CS0A-CS1BB) and data lines from the Switch ASIC (DQA/DQB), respectively. *Id.* at 39–40 (citing Ex. 1005 ¶ 30; Ex. 1003 ¶¶ 181–82).

We have reviewed the record, and find that Petitioner provides sufficient evidence that Ellsberry teaches limitation 1[b]. Patent Owner makes no other arguments contesting the prior art teachings of this claim limitation. *See generally* Prelim. Resp.

iii. Limitation [c]

Petitioner asserts that Ellsberry's nine Switch ASICs (“D”) in Figures 5 and 6 collectively comprise “*a data module*,” correspond to Switch ASICs 206-208 (Figure 2) (mounted on a circuit board), and pass data between the memory devices and the memory bus/memory controller. Pet. 40–42 (citing Ex. 1005 ¶¶ 3, 28–30, 47, 49, 51, Fig. 2; Ex. 1003 ¶¶ 186–87). Petitioner further contends that Ellsberry's Switch ASICs are “*coupled between the data ports of the memory devices and the system memory bus*” because Figure 2 shows them coupled between DIMM interface 202 and the “*data ports of the memory devices*” (212–228). *Id.* at 42 (citing Ex. 1005 ¶¶ 11, 28–30, 47, Figs. 5, 6, 11, 13; Ex. 1015, 6; Ex. 1003 ¶ 188). Petitioner contends Ellsberry's “*data module*” (Switch ASICs) in all embodiments “*includ[es] data handler logic elements*” located in the “control block” of

Figure 4 and, in the combined system, the “*data handler logic elements*” would also include logic elements implementing the functionality of Jeddeloh’s pattern generator and compare circuitry. *Id.* at 42–43 (citing Ex. 1005 ¶ 45; Ex. 1006, 9:44–10:67; Ex. 1003 ¶¶ 136–165, 190).

We have reviewed the record, and find that Petitioner provides sufficient evidence that Ellsberry teaches limitation 1[c]. Patent Owner makes no arguments contesting the prior art teachings of this claim limitation. *See generally* Prelim. Resp.

iv. Limitation [d]

Petitioner contends that Ellsberry teaches limitation [d] by its disclosure of a Control ASIC (“*control module*”), that is “*coupled to the data module*” (Switch ASICs) via bus 210, “*coupled to... the address and control ports of the memory devices*” via bus 220, as depicted in greater detail in, and “*coupled to... the system memory bus*” via DIMM interface 202 (Figure 2), corresponding to the edge interface of Figure 5 and 6. Pet. 44 (citing Ex. 1005 ¶¶ 28–32, 47–52, 54, 56, Fig. 11, 13; Ex. 1015, 6; Ex. 1003 ¶¶ 194–99). Petitioner further contends that in the First Embodiment,¹¹ the “*control module*” is depicted in Figure 5 as “C” and in Figure 13 as Control Unit ASIC 1302 and in the Second Embodiment, the “*control module*” comprises two “control units 604” (individually or collectively) with one depicted as “C” in Figure 6 and shown in Figure 11 as Control Unit ASICs 1102/1104 and as depicted in Figures 5 and 6, the Control ASIC(s) is/are

¹¹ Petitioner asserts “FIGS. 2, 5, and 13 of Ellsberry correspond to a ‘First Embodiment’ utilizing four ranks, each composed of nine 8-bit memory devices for a total bit width of 72. FIGS. 2, 6, and 11 correspond to a ‘Second Embodiment’ utilizing pairs of 4-bit memory devices to implement the same bit width. EX1003, ¶108.” Pet. 22.

“mounted on the circuit board.” *Id.* at 45–46 (citing Ex. 1005 ¶¶ 3, 47, 49, 51, Figs. 5, 6; 1003 ¶¶ 194–95).

We have reviewed the record, and find that Petitioner provides sufficient evidence that Ellsberry teaches limitation 1[d]. Patent Owner makes no arguments contesting the prior art teachings of this claim element. *See generally* Prelim. Resp.

v. Limitation [e]

Petitioner asserts that the combination of Ellsberry and Jeddeloh teaches limitation 1[e] by Ellsberry’s disclosure of a normal mode used to read/write DRAMs in response to controller commands and, in the proposed combination, by Ellsberry operating in a test mode as taught by Jeddeloh. Pet. 46–48 (citing Ex. 1005 ¶¶ 41, 42, Fig. 8A; Ex. 1003 ¶¶ 203–10).

We have reviewed the record and find that Petitioner provides sufficient evidence that the combination of Ellsberry and Jeddeloh teaches limitation 1[e]. Patent Owner makes no arguments contesting the prior art teachings of this claim limitation. *See generally* Prelim. Resp.

vi. Limitation [f.i]

Petitioner contends that Ellsberry teaches limitation [f.i] by its disclosure Control ASIC (“*control module*”), in the first (normal) mode, “*receive[s] system address and control signals from the system memory controller*” over the DIMM interface (Figure 2), which corresponds to the edge interface of Figures 5 and 6. Pet. 48–49 (citing Ex. 1005 ¶¶ 29, 35, 36, 45, 47, 51, Figs. 2, 5, 6, 8A, 11, 13, code (57); Ex. 1003 ¶¶ 168–77, 193–201, 218). Petitioner also contends the Control ASIC in the first mode is also configured to output address and command signals (“*first memory address and control signals*”) to the memory devices on bus 220 according

to the received system signals, for example, in Figures 3, 11, and 13. *Id.* at 45. Additionally, Petitioner contends Ellsberry also lists “*first memory address and control signals*” in Figure 8A, such as READ/WRITE/NOP, and describes operations relating thereto. *Id.* at 50–51 (Ex. 1005 ¶¶ 10, 11, 33, 41, 42, Figs. 8A, 11, 13; Ex. 1003 ¶ 219).

We have reviewed the record, and find that Petitioner provides sufficient evidence that Ellsberry teaches limitation 1[f.i]. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

vii. Limitation [f.ii]

Petitioner contends that Ellsberry teaches limitation [f.ii] by its disclosure of, in Figure 2, Switch ASICs (“*data module*”) that “*propagate*” 8-bit sections of “*first data signals*” between “*the memory devices*” (DRAM) and DIMM interface (which is connected to “*the system memory controller*”) via 8-bit buses (green), and, in Figure 4, details of the Switch ASIC bus connections. Pet. 52 (citing Ex. 1005 ¶¶ 29–30, 45; Ex. 1003 ¶¶ 168–177, 185–192). Petitioner further contends that Figures 5 and 13, as well as 6 and 11, contain corresponding depictions of Switch ASICs that “*propagate*” respective 8-bit sections of 72-bit wide “*data signals between the memory devices and the system memory controller,*” with Figures 5 and 6 each showing nine switch ASICs coupled between an edge interface (corresponding to DIMM interface) and four sets of nine 8-bit DRAMs (504 & 512) or pairs of 4-bit DRAMs, and Figure 13 showing each Switch ASIC in Figure 5 receives 8-bit signals from the system memory controller (DQ) and passes it to one of four DRAMs via 8-bit signal lines (DQA/DQB), and with Figure 11 functioning like Figure 13, but utilizing pairs of 4-bit

DRAMs instead of single 8-bit DRAMs, and two Control ASICs instead of one. *Id.* at 52–55 (Ex. 1005 ¶¶ 2, 47–51; Ex. 1003 ¶¶ 99–104, 226–232).

We have reviewed the record, and find that Petitioner provides sufficient evidence that Ellsberry teaches limitation 1[f.ii]. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

viii. Limitation [f.iii]

Petitioner contends that Ellsberry teaches limitation [f.iii] by its disclosure of “*first data signals*” are “*transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals*” such as a write or read command received over bus 220 in Figure 2. Pet. 55–56. In Ellsberry’s First and Second Embodiments, for example, the Control ASIC “maps a received logical address to a physical address corresponding to the particular memory bank configuration employed. It also directs commands to the memory banks to indicate which memory bank should be operational and which one should be passive (do nothing).” *Id.* (citing Ex. 1005 ¶¶ 11, 31, Fig. 2; Ex. 1003 ¶¶ 211–222, 236–237).

Specifically, Petitioner contends that “when row/bank mode addressing is used, read and write commands are sent only to the targeted memory device [and it performs the read/write] ... [whereas a] NOP (no-operation) command is sent to the non-targeted memory device,” allowing the Control ASIC to “control data to and from” the memory devices “without the delays caused by otherwise disabling” the non-targeted devices and Ellsberry’s memory devices receiving a NOP command do not

transmit/receive data because NOP commands are not registered. *Id.* at 56 (Ex. 1005 ¶¶ 33, 42; Ex. 1028, 14; Ex. 1003 ¶¶ 238, 239).

We have reviewed the record, and find that Petitioner provides sufficient evidence that Ellsberry teaches limitation 1[f.iii]. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

ix. Limitation [g.i]

Petitioner contends that the proposed combination teaches limitation [g.i] in that during test mode in the Combined System, the modified Control ASIC (“*control module*”) would “*output second memory address and control signals to the address and control ports of the memory devices,*” similar to how Ellsberry’s Control ASIC outputs address/control signals during normal operation. Pet. 56 (citing Ex. 1006, 9:51–10:2, 10:24–29, Fig. 3; Ex. 1003 ¶¶ 243–50).

We have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry and Jeddeloh teaches limitation 1[g.i]. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

x. Limitation [g.ii]

Petitioner contends that the combined system teaches limitation [g.ii] in that there are two sources of data for the memory devices: (i) the system memory controller (during normal mode) and (ii) the pattern generator/compare circuit functionality in each Switch ASIC (during test mode), thus, each source communicates data along a different path in the Switch ASICs, so that “*the system memory controller*” is “*isolate[d]*” from the “*memory devices*” during test mode (“*second mode*”) according to

commands from the Control ASIC (“*control module*”). Pet. 57–60 (citing, *inter alia*, Ex. 1003 ¶ 255–59). Additionally, according to Petitioner, during test mode, the pattern generator functionality of the modified Switch ASICs “*transmit[s] one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices,*” similar to how Ellsberry's Switch ASICs transmit data during normal operation. *Id.* at 60–61 (citing, *inter alia*, Ex. 1003 ¶¶ 273–75).

We have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry and Jeddeloh teaches limitation 1 [g.ii]. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

xi. Limitation [g.iii]

Petitioner contends that the combined system teaches limitation [g.iii] in that the modified Control ASIC outputs address/control signals (“*second memory address and control signals*”) to the ports of the memory devices 240. Pet. 61 (citing Ex. 1006, 9:51–56, 9:56–60, 9:66–10:2, cl. 7, Fig. 3; Ex. 1003, 279–80).

For example, Petitioner contends Jeddeloh’s memory sequencer 304 “generates properly timed signals for controlling the operation of the memory devices 240” as shown in Figure 2 (Ex. 1006, 9:57–60), “the self-test module 300 can also vary the timing and rate at which control and address signals are applied to the memory devices 240 along with the CLK signal or other timing signal” (*id.*, 10:24–28), and Figure 3 further shows a “RD/WRT” signal from the BIST engine 310 to the memory sequencer 304, confirming that signals output from memory sequencer 304 during testing include signals for the memory devices, sent via Ellsberry’s bus 220 in the

combined system, to configure them to receive test data (“*second data signals*”) (*id.*, Fig. 3). Pet. 61 (citing Ex. 1003 ¶¶ 279–80).

We have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry and Jeddeloh teaches limitation 1[g.iii]. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

xii. Conclusion

On this record, we are persuaded that Petitioner has demonstrated a reasonable likelihood of prevailing on its assertion that claim 1 would have been obvious over Ellsberry.

c. Independent Claim 19

Independent claim 19 is an apparatus claim with limitations that parallel most of those in claim 1. *Compare* Ex. 1001, 16:54–17:30 *with id.* at 19:4–43. Petitioner additionally contends that the “*PCB*” of claim 19 is equivalent to Ellsberry’s “substrate 502,” and the “*connector*” is equivalent to Ellsberry’s DIMM interface and edge interface. Pet. 92–97 (citing Ex. 1005 ¶¶ 2, 11, 27, 28, 47, 51, Figs. 1, 2, 5, 6, 11, 13; Ex. 1015, 6, 29, 66; Ex. 1003 ¶¶ 168–241, 254, 282–333, 348–53, 354–62, 365–71, 428–31, 467–85, 540–45, 548–53, 555–57, 560, 562–66, 568–71). Petitioner asserts that, as applied in element [1.f.ii], Ellsberry’s system memory controller connects to the memory module via Ellsberry’s DIMM interface (Figure 2) (“*connector*”)/edge interface (Figures 5 and 6) (“*connector*”) for “*read or write operations*” over the bidirectional data signal lines to each Switch ASIC (“*data handlers*”) shown in Figures 2 and 4. *Id.* at 95–96 (Ex. 1003 ¶¶ 564–566).

We have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry and Jeddeloh teaches the limitations of claim 19. Patent Owner makes no arguments contesting the prior art teachings of the limitations of this claim. *See generally* Prelim. Resp.

On this record, we are persuaded that Petitioner has demonstrated a reasonable likelihood of prevailing on its assertion that claim 19 would have been obvious over the combination of Ellsberry and Jeddeloh.

e. Dependent Claims 2–18, 20–34

Petitioner asserts that the combination of Ellsberry and Jeddeloh teaches the limitations of dependent claims 2–18 and 20–34. Pet. 62–93, 97–104. We have reviewed the record and find that Petitioner provides sufficient evidence that the combination of Ellsberry and Jeddeloh teaches the limitations of these dependent claims. Patent Owner makes no arguments contesting the prior art teachings of the limitations of these claims. *See generally* Prelim. Resp.

On this record, we are persuaded that Petitioner has demonstrated a reasonable likelihood of prevailing on its assertion that claims 2–18 and 20–34 would have been obvious over the combination of Ellsberry and Jeddeloh.

*E. Alleged Obviousness of Claims 1–34
Over Ellsberry, Jeddeloh, and Averbuj*

Petitioner contends that claims 1–34 would have been obvious over the combination of Ellsberry, Jeddeloh, and Averbuj. Pet. 104–10. To support its contentions, Petitioner provides explanations as to how Ellsberry, Jeddeloh, and Averbuj teach each claim limitation. *Id.*

Averbuj (Ex. 1007) “is directed to a distributed, hierarchical built-in self-test (BIST) architecture for” memory modules. Ex. 1007 ¶ 7. Averbuj discloses a memory interface used for self-test functionality. *Id.*, Fig. 6. Multiplexers 45 and 46 in the memory interface 41 are used to merge and select between multiple signal paths. *Id.* ¶ 48. During BIST mode, the memory interface 41 receives test data signals from sequencer 8, and either provides the same data to its memory device as is, or generates its own data by modifying the signals from the sequencer. *Id.* ¶¶ 11, 48–50, 55–56, Figs. 6–7.

Petitioner states that Averbuj is asserted “[t]o the extent [Patent Owner] argues the modified Switch ASICs in test mode do not satisfy that the ‘*configured to isolate*’ limitations of claims 1, 11, 32.” Pet. 104. For example, Petitioner contends “Averbuj discloses a modular BIST architecture for memory modules, including a memory interface with two data paths, one from the system memory controller (‘programmable processor’) for normal operations (green, DATA) and one from a pattern generator during test operations (blue, BIST_DATA_T), to be merged and selected between for provision to the DRAM.” Pet. 105. Additionally, Petitioner contends Averbuj discloses using MUX 45 for that purpose, that “isolates” the DRAM “from normal functionality . . . in BIST mode” in response to the BIST_EN command from a sequencer. *Id.*

We determined that the combination of Ellsberry and Jeddelloh were sufficient to show a reasonable likelihood of success as to claim 1–34. We have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry, Jeddelloh, and Averbuj teach the limitations of claims 1–34, and sufficient rationale to combine the prior art

has been provided. Patent Owner presents no issues or arguments specific to the asserted obviousness contentions beyond motivation to combine Ellsberry and Jeddeloh discussed above. *See generally* Prelim. Resp.

On this record, we are persuaded that Petitioner has demonstrated a reasonable likelihood of prevailing on its assertion that claims 1–34 would have been obvious over the combination of Ellsberry, Jeddeloh, and Averbuj.

*F. Alleged Obviousness of Claims 14 and 17–34
Over Ellsberry, Jeddeloh, and Lee*

Petitioner contends that claims 14 and 17–34 would have been obvious over the combination of Ellsberry, Jeddeloh, and Lee. Pet. 110–12. To support its contentions, Petitioner provides explanations as to how Ellsberry, Jeddeloh, and Lee teach each claim limitation. *Id.*

Lee (Ex. 1008) discloses “a method of testing a memory module, including receiving a test pattern and an associated input mode.” Ex. 1008 ¶ 15, code (57). The test pattern and input mode “may [be] received through a system management bus (SMBUS).” Ex. 1008 ¶¶ 31, 62–63. After receiving the test pattern, a pattern generator in the memory module may either generate a test pattern based on it and the input mode, or output that received test pattern without modification. *Id.* ¶¶ 36–38, 64.

Petitioner asserts Lee “[t]o the extent one argues the Combined System does not satisfy the requirements of claims 14 and 17–18, or element [19.h]’s similar ‘*based on information output from the control module*’ limitation, it also would have been obvious to implement the Combined System in such a manner in further view of Lee.” Pet. 110. For example, Petitioner contends Lee discloses techniques of generating test pattern data

based on control module information received off-module over a maintenance bus, such as an SMBUS. Pet. 111 (citing Ex. 1008 ¶¶ 15, 16, 36–38, 41–48, 64, Fig. 3; Ex. 1003 ¶¶ 490, 534, 575;). Patent Owner presents no issues or arguments specific to the asserted obviousness contentions beyond the motivation to combine Ellsberry and Jeddeloh discussed above. *See generally* Prelim. Resp.

We determined that the combination of Ellsberry and Jeddeloh were sufficient to show a reasonable likelihood of success as to claims 14 and 17–34. Further, we have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry, Jeddeloh, and Lee teaches the limitations of claims 14 and 17–34, and sufficient rationale to combine the prior art has been provided. Patent Owner makes no additional arguments contesting the prior art teachings of the limitations of these claims or the rationale to combine as to Ellsberry, Jeddeloh, and Lee. *See generally*, Prelim. Resp.

On this record, we are persuaded that Petitioner has demonstrated a reasonable likelihood of prevailing on its assertion that claims 1–34 would have been obvious over the combination of Ellsberry, Jeddeloh, and Lee.

*G. Alleged Obviousness of Claims 14 and 17–34
Over Ellsberry, Jeddeloh, Averbuj, and Lee*

Petitioner contends that claims 14 and 17–34 would have been obvious over the combination of Ellsberry, Jeddeloh, Averbuj, and Lee. Pet. 113. To support its contentions, Petitioner provides explanations as to how Ellsberry, Jeddeloh, Averbuj, and Lee teach each claim limitation. *Id.*

Petitioner contends “[t]he ‘*configured to isolate*’ limitation of claim 1, as incorporated into claims 14 and 17–18, would have been obvious in view

of Averbuj” and “Element [19.h]’s ‘*not accessed*’ limitation would have been obvious in view of Averbuj, . . . and the ‘*based on information*’ limitation would have been obvious in view of Lee,” based on the contentions in the other grounds of the Petition. Pet. 113. Patent Owner presents no issues or arguments specific to the asserted obviousness contentions beyond motivation to combine Ellsberry and Jeddeloh discussed above. *See generally* Prelim. Resp.

We determined that the combination of Ellsberry and Jeddeloh were sufficient to show a reasonable likelihood of success as to claim 1–34. Furthermore, we have reviewed the record, and find that Petitioner provides sufficient evidence that the combination of Ellsberry, Jeddeloh, Averbuj, and Lee teach the limitations of claims 14 and 17–34, and sufficient rationale to combine the prior art has been provided. Patent Owner makes no additional arguments contesting the prior art teachings of the limitations of these claims or the rationale to combine as to Ellsberry, Jeddeloh, Averbuj, and Lee. *See generally* Prelim. Resp.

On this record, we are persuaded that Petitioner has demonstrated a reasonable likelihood of prevailing on its assertion that claims 14 and 17–34 would have been obvious over the combination of Ellsberry, Jeddeloh, Averbuj, and Lee.

IV. CONCLUSION

Based on the arguments and evidence presented in the Petition, the Preliminary Response, Reply, and Sur-Reply and accompanying exhibits, we have determined there is a reasonable likelihood Petitioner would prevail with respect to at least one claim challenged in the Petition. We conclude that the threshold has been met for instituting *inter partes* review, and we

institute on all challenged claims and all grounds. *See SAS Inst., Inc. v. Iancu*, 138 S. Ct. 1348, 1354 (2018); *PGS Geophysical AS v. Iancu*, 891 F.3d 1354, 1360 (Fed. Cir. 2018). We have not made a final determination on claim construction or as to the patentability of any of the challenged claims. Our final determination will be based on the record as fully developed during trial.

V. ORDER

Accordingly, it is:

ORDERED that pursuant to 35 U.S.C. § 314(a), an *inter partes* review is instituted as to challenged claims 1–34 of the '523 patent for all grounds raised in the Petition; and

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial, which commences on the entry date of this Order.

IPR2022-00063
Patent 10,217,523 B1

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SAMSUNG ELECTRONICS CO., LTD.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

IPR2022-00063
Patent 10,217,523 B1

Before JON M. JURGOVAN, SHEILA F. McSHANE, and
KARA L. SZPONDOWSKI, *Administrative Patent Judges*.

SZPONDOWSKI, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

We instituted an *inter partes* review of claims 1–34 of U.S. Patent No. 10,217,523 B1 (Ex. 1001, “the ’523 patent”), in response to a Petition (Paper 1, “Pet.”) filed by Samsung Electronics Co., Ltd. (“Petitioner”). Paper 13 (“Dec.”). During the trial, Netlist, Inc. (“Patent Owner”) filed a Response to the Petition (Paper 28, “PO Resp.”), Petitioner filed a Reply (Paper 31, “Reply”), and Patent Owner filed a Sur-reply (Paper 34, “Sur-reply”). In addition, both parties filed Motions to Exclude Evidence (Papers 43 (“PO Mot. Excl.”), 44 (“Pet. Mot. Excl.”)), Oppositions to the motions (Papers 45 (“PO Opp. Excl.”), 46 (“Pet. Opp. Excl.”)), and Replies to the Oppositions (Papers 48 (“Pet. Reply Excl.”), 50 (“PO Opp. Excl.”)).

An oral hearing was held on February 1, 2023, and a copy of the transcript was entered into the record. Paper 52 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6. This Decision is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the claims on which we instituted trial. Based on the complete record, Petitioner has not shown, by a preponderance of the evidence, that claims 1–34 of the ’523 patent are unpatentable.

II. BACKGROUND

A. *Real Parties in Interest*

Petitioner identifies Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. as real parties in interest. Pet. 1. Patent Owner identifies itself as the sole real party in interest. Paper 3, 3.

B. *Related Matters*

Petitioner indicates *Samsung Electronics Co., Ltd. et al. v. Netlist, Inc.*, No. 1:21-cv-01453 (D. Del. filed Oct. 15, 2021) involves the ’523

patent. Pet. 1. Petitioner and Patent Owner also identify many other related matters. Pet. 1–3; Paper 3, 3–5; Paper 5.

The Board previously instituted trial in IPR2020-01421 on the same claims of the '523 patent and on the same grounds presented here, but with different petitioners. *See* Ex. 1043. That IPR was terminated after institution because settlement was reached. Ex. 1044.

C. The '523 Patent (Ex. 1001)

The '523 patent is titled “Multi-Mode Memory Module with Data Handlers” and issued on February 26, 2019, from an application filed on March 29, 2014. Ex. 1001, codes (22), (45), (54).

The '523 patent is directed to a self-testing memory module for testing a plurality of memory devices mounted thereon. Ex. 1001, 5:4–27. Figure 2, reproduced below, is a block diagram illustrating component blocks of memory module 10 and memory controller 14. *Id.* at 5:28–34, 8:41–62, 9:22–42, Fig. 2.

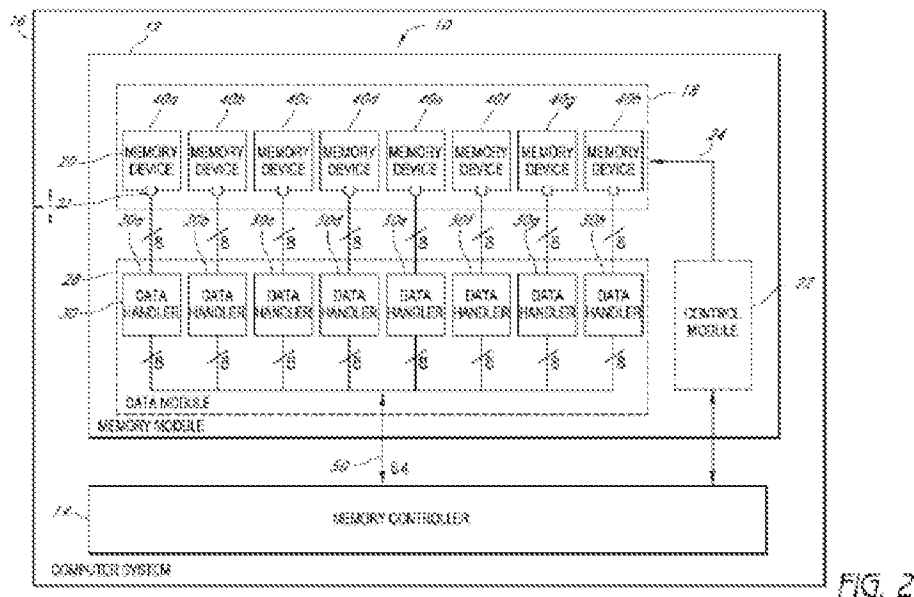


Figure 2, above, shows memory module 10 includes control module 22 that generates address and control signals for testing memory devices 20 and data module 28 that includes a plurality of distributed data handlers 30. Ex. 1001, 5:12–16. Data handlers 30 of data module 28 generate test patterns to write to memory devices 20 and compare test patterns read from memory devices 20 to the written patterns to identify faults. *Id.* at 5:28–34.

Figure 3, reproduced below, provides additional detail regarding the control module, data handlers and their components and interconnections. Ex. 1001, Fig. 3.

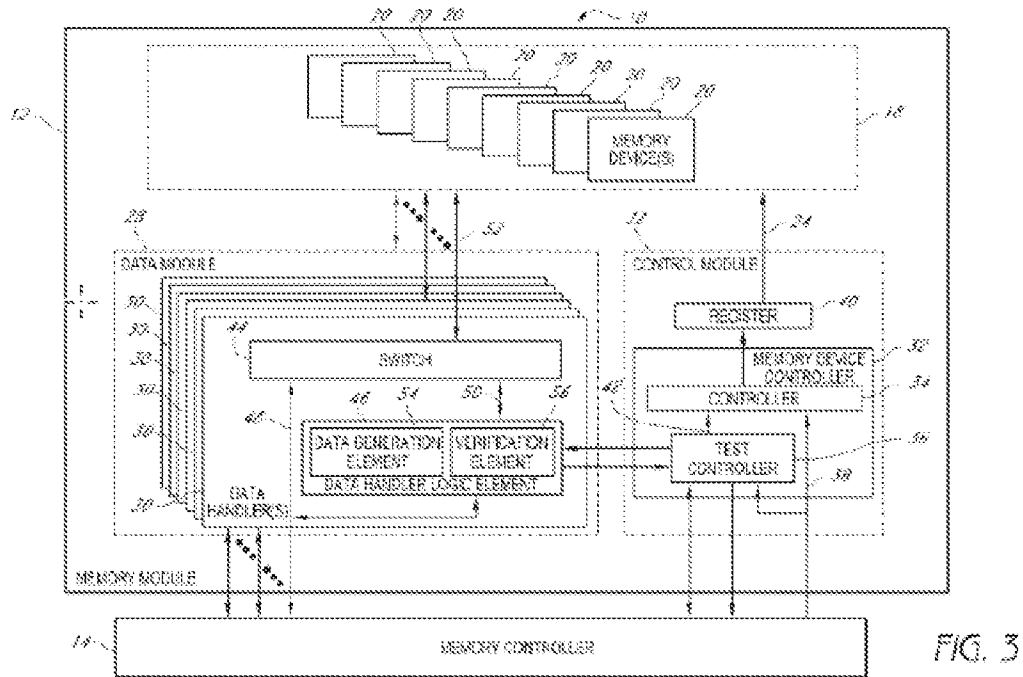


Figure 3, above, shows additional control module 22, data module 28, plurality of memory devices 18, and certain components thereof. Ex. 1001, 9: 32–36, Fig. 3. Control module 22 includes memory device controller 32 with memory controller 34 and test controller 36. *Id.* at 9:38–40, Fig. 3. Data module 28 includes data handlers 30, switch 44, and data handler logic

element 46, which includes data generation element 54 and verification element 56. *Id.* at 10:7–21, 10:38–40.

“In one embodiment, the memory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36.” *Id.* at 9:45–49. “The control module 22 of certain embodiments is configured to selectively input to the address and control ports of the plurality memory devices 18 either the address control signals 38 from the system memory controller 14 or the address and control signals 42 from the control module 22 (e.g., from the test controller 36).” *Id.* at 9:49–55. “For example, the memory device controller 34 may send either the signals 38 from the system memory controller 14 or, alternatively, the signals 42 from the test controller 36, to the register 40 depending on whether the memory module 10 is in normal (non-test) mode or in a test mode, respectively.” *Id.* at 9:55–60. “The test controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28.” *Id.* at 9:63–66.

“The data module 28 . . . may be in communication with one or more of the memory devices 20, the control module 22, and the memory controller.” *Id.* at 10:7–10. “Each of the data handlers 30 of certain embodiments comprises a switch 44,” which “may include a data multiplexer/demultiplexer.” *Id.* at 10:13–15. “In certain embodiments, the switch 44 is configured to selectively input to the corresponding plurality of data ports either data signals 48 from the system memory controller 14 or data signals 50 from the data handler logic elements 46.” *Id.* at 10:17–21.

“The switch 44 of certain embodiments may further be configured to receive data signals 52 (e.g., during a read operation) from the plurality of memory devices 18 and to propagate the data signals 52 to the data handler logic element 46 and/or the memory controller 14.” *Id.* at 10:21–25. “In some embodiments, for example, the switch 44 selectively inputs the data signals 48 to be written to the plurality of memory devices 18 from the system memory controller 14 when the memory module 10 is a normal (non-test mode) mode and, alternatively, inputs the data signals 50 from the data handler logic element 46 during a test mode.” *Id.* at 10:26–31.

D. Illustrative Claims

Petitioner challenges claims 1–34 of the ’523 patent. Pet. 5–6. Claims 1 and 19 are independent. Ex. 1001, 16:55–17:30, 19:4–45. Claim 1, reproduced below with brackets noting Petitioner’s identifiers, is illustrative of the subject matter of the challenged claims:

1. [a] A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:
 - [b] memory devices mounted on a circuit board, the memory devices having address and control ports and data ports;
 - [c] a data module mounted on the circuit board and coupled between the data ports of the memory devices and the system memory bus, the data module including data handler logic elements;
 - [d] a control module mounted on the circuit board and coupled to the data module, the address and control ports of the memory devices, and the system memory bus; and
 - [e] wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode;
 - [f] [i] wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory

address and control signals to the memory devices according to the system address and control signals, and [ii] the data module in the first mode is configured to propagate one or more first data signals between the memory devices and the system memory controller, [iii] the one or more first data signals being transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals; and

[g] [i] wherein the control module in the second mode is configured to output second memory address and control signals to the address and control ports of the memory devices, and [ii] the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller and to transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to one or more commands output from the control module, and [iii] wherein at least a portion of the memory devices are configured to receive the one or more second data signals according to the second memory address and control signals from the control module.

Ex. 1001, 16:55–17:30.

E. Prior Art and Asserted Grounds

Petitioner asserts that claims 1–34 of the '523 patent are unpatentable based on the following grounds:

Claims Challenged	35 U.S.C. § ¹	Reference(s)
1–34	103(a)	Ellsberry, ² Jeddeloh ³
1–34	103(a)	Ellsberry, Jeddeloh, Averbuj ⁴
14, 17–34	103(a)	Ellsberry, Jeddeloh, Lee ⁵
14, 17–34	103(a)	Ellsberry, Jeddeloh, Averbuj, Lee

Pet. 5–6.

In addition, Petitioner relies on the Declarations of Dr. Vivek Subramanian (Exs. 1003, 1061). Patent Owner relies on the Declarations of Dr. Michael Brogioli (Exs. 2001, 2004). Deposition transcripts have been entered into the record for Dr. Subramanian (Exs. 2005, 2016) and Dr. Brogioli (Ex. 1064).

III. ANALYSIS

A. Legal Standards

A claim is unpatentable under 35 U.S.C. § 103(a) if “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the

¹ The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), amended 35 U.S.C. § 103, and was effective on March 16, 2013. Because the ’523 patent has a filing date before the effective date of the applicable AIA amendments, we refer to the pre-AIA versions of 35 U.S.C. § 103.

² US Pub. No. 2006/0277355 A1, published December 7, 2006. Ex. 1005.

³ US 7,310,752 B2, issued December 18, 2007, claiming priority to Application No. 10/660,844, filed on Sept 12, 2003. Ex. 1006.

⁴ US Pub. No. 2005/0257109 A1, published November 17, 2005. Ex. 1007.

⁵ US Pub. No. 2006/0095817 A1, published May 4, 2006. Ex. 1008.

invention was made to a person having ordinary skill in the art to which said subject matter pertains.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations.⁶ *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

A patent claim “is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. An obviousness determination requires finding “both ‘that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.’” *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–68 (Fed. Cir. 2016); *see KSR*, 550 U.S. at 418. Further, an assertion of obviousness “cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *KSR*, 550 U.S. at 418; *In re NuVasive, Inc.*, 842 F.3d 1376, 1383 (Fed. Cir. 2016) (a finding of a motivation to combine “must be supported by a ‘reasoned explanation’”).

B. Level of Ordinary Skill in the Art

Petitioner contends a person of ordinary skill in the art at the time of the invention would have a “Bachelor’s degree in electrical engineering,

⁶ The parties did not present evidence relating to objective indicia of nonobviousness.

computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.” Pet. 6. Petitioner contends that this level of ordinary skill in the art was adopted by the Board in IPR2014-00970, which challenged U.S. Patent No. 8,001,434 (“’434 patent”). *Id.* The ’523 patent is a continuation of the ’434 patent. Ex. 1001, code (63).

In our Institution Decision, we preliminarily adopted Petitioner’s unopposed⁷ proposed level of ordinary skill in the art. Paper 13, 14. In the Response, Patent Owner does not challenge Petitioner’s characterization of the level of ordinary skill in the art, and Dr. Brogioli testifies that “I accept this proposed level of ordinary skill in the art.” PO Resp. 6; *see* Ex. 2004 ¶ 22. Nothing in the full record persuades us that our preliminary finding as to the level of ordinary skill in the art was incorrect. Accordingly, we maintain our adoption of Petitioner’s proposed level of ordinary skill in the art, as consistent with the level of ordinary skill in the art reflected by the prior art of record. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

C. Claim Construction

We construe each claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent,” the same standard used to construe the claim in a civil action. 37 C.F.R. § 42.100(b) (2020). Under this standard, the words of a claim generally are given their

⁷ Patent Owner did not challenge the proposed level of ordinary skill in the art in the Preliminary Response. Paper 7, 5.

“ordinary and customary meaning,” which is the meaning the term would have to a person of ordinary skill at the time of the invention, in the context of the entire patent including the specification. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

For the terms “configured to” and “operate independently,” Petitioner offers the Board’s previous construction in IPR2014-00970, and further states that the Board declined to construe these terms in IPR2020-01421. Pet. 13. For the term “rank,” Petitioner offers a construction, and also contends that the Board declined to construe this term in IPR2020-01421. *Id.* However, for each of the foregoing three terms, Petitioner also contends that “[e]ither way, the prior art invalidates the 523 Patent.” *Id.* Petitioner contends that the term “mode” should be given its plain and ordinary meaning, and also offers the construction adopted by the district court in the Western District of Texas, and further contends that the Board declined to construe this term in IPR2020-01421. *Id.* at 14. For the terms “data handler,” “data handler logic element[s],” and “data module,” Petitioner provides the constructions adopted by the district court in the Western District of Texas. *Id.* at 14–15.

Although in Response, Patent Owner states “[t]o the extent Petitioner or its expert have proffered or applied constructions inconsistent with the plain and ordinary meaning of these terms, Patent Owner explicitly disagrees with those constructions,” in the Sur-reply, Patent Owner states that “the Board needn’t construe any disputed terms because . . . they do not bear on the issues in dispute.” PO Resp. 4; Sur-reply 1.

We agree with Patent Owner that it is not necessary to provide an express interpretation of any terms for purposes of this Decision. *See*

Realtime Data, LLC v. Iancu, 912 F.3d 1368, 1375 (Fed. Cir. 2019) (“The Board is required to construe ‘only those terms . . . that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

D. Ground 1: Obviousness Based on Ellsberry and Jeddeloh

Petitioner contends claims 1–34 would have been obvious over the combination of Ellsberry and Jeddeloh. Pet. 26–104. After reviewing the entire record developed at trial, we determine that Petitioner has not shown, by a preponderance of the evidence, that claims 1–34 are unpatentable over Ellsberry and Jeddeloh.

1. Ellsberry (Ex. 1005)

Ellsberry relates to “a device, system, and method for expanding the memory capacity of a memory module.” Ex. 1005, code (57). “A control unit and memory bank switch are mounted on a memory module to selectively control write and/or read operations to/from memory devices communicatively coupled to the memory bank switch.” *Id.* “By selectively routing data to and from the memory devices, a plurality of memory devices may appear as a single memory device to the operating system.” *Id.*

Figure 2 of Ellsberry, reproduced below, “illustrates a block diagram of a capacity-expanding memory system 200 according to one embodiment.” *Id.* ¶ 28. System 200 has DIMM (dual inline memory module) interface 202 that couples to a memory socket and communication bus over which data, memory addresses, commands, and control information are transmitted. *Id.* The capacity-expanding feature of the invention is accomplished by a combination of control unit 204 and one or more memory bank switches 206, 208. *Id.*

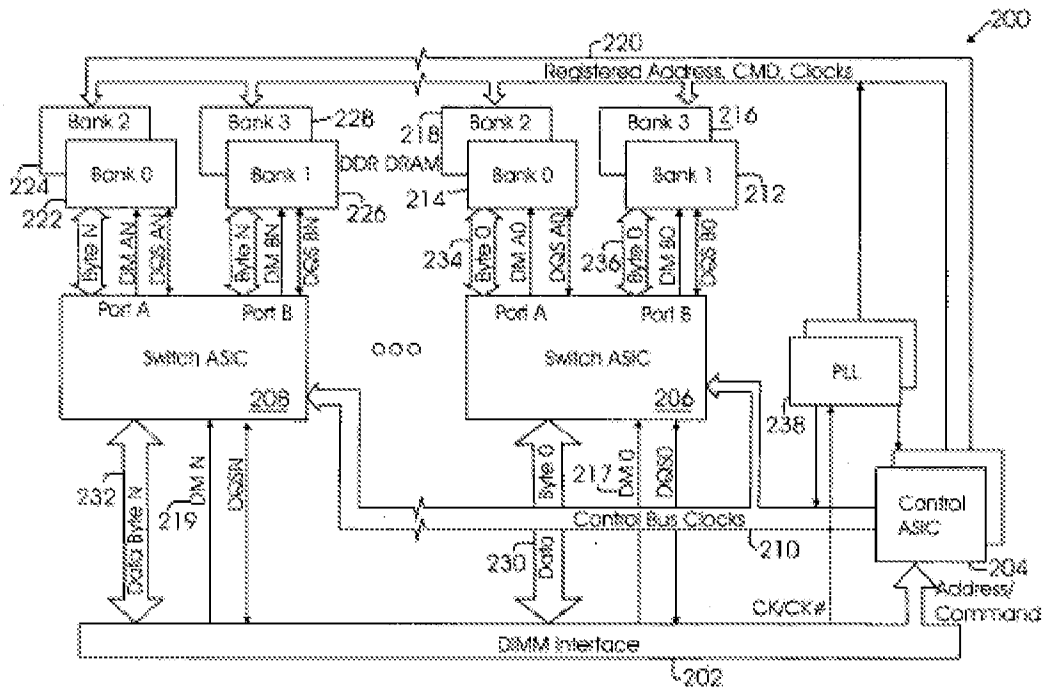


Fig. 2

Figure 2 of Ellsberry, above, illustrates system 200 with control ASIC 204 that receives addresses and commands from DIMM interface 202 and generates corresponding control signals on bus 210 and addresses and command information on bus 220 to selectively connect memory banks 212–228 to DIMM interface 202 via switch ASICs 206, 208. *Id.* ¶¶ 28–29.

Specifically, control unit 204 receives memory addresses and commands from DIMM interface 202 and controls switches 206 and 208 via control bus 210 to indicate how data from DIMM interface 202 should be received from and/or stored in memory banks 212–228. *Id.* ¶¶ 29–30, Fig. 2. Control unit 204 also generates address and command information on address bus 220 to access memory banks 212–228. *Id.* Switches 206 and 208 receive data from, or provide data to, DIMM interface 202 via data buses 230 and 232. *Id.*

Control unit 404 and switches 206 and 208 are further illustrated in Figures 3 and 4. Figure 3, reproduced below, “illustrates a block diagram of an address and command processing system 300” which “may be implemented as part of the control unit 204.” *Id.* ¶ 39.

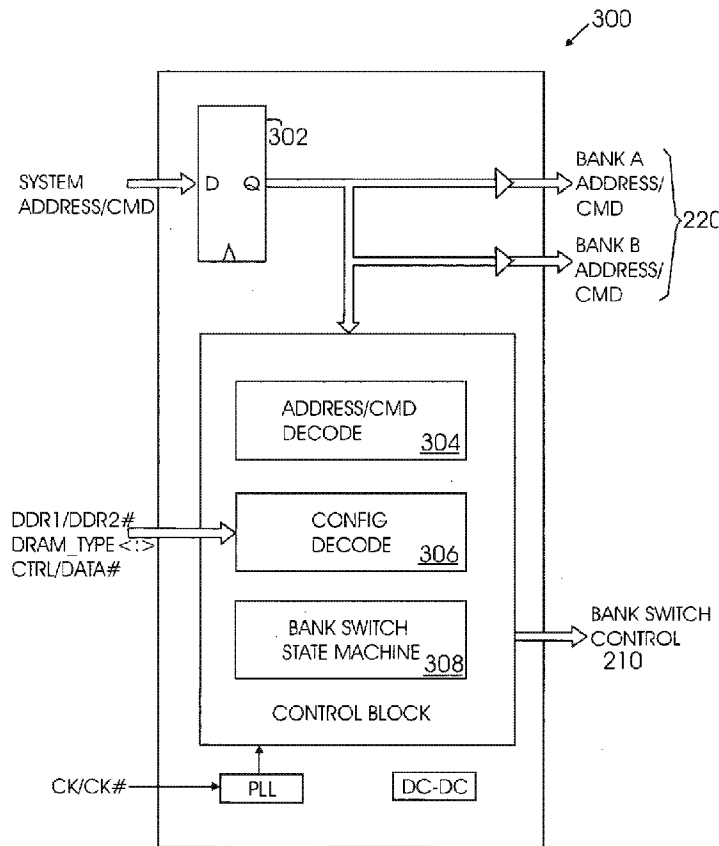


Fig. 3

Figure 3, above, depicts that “[m]emory addresses and command information are received from the DIMM interface 202, buffered in a register 302 and sent to all memory banks . . . over address bus 220.” *Id.* In addition, “[t]he memory address and command information is also decoded 304 and memory configuration information 306 is determined.” *Id.* “A bank switch state machine 308 then determines which memory bank should be activated or accessed.” *Id.* “The state machine 308 sends control

information to the memory bank switches 206 & 208 via the control bus 210 to indicate which memory banks should be activated/deactivated or accessed.” *Id.*

Figure 4, reproduced below, “illustrates a block diagram of a data processing system 400,” which “may be implemented as part of the memory bank switch 206.” *Id.* ¶ 45.

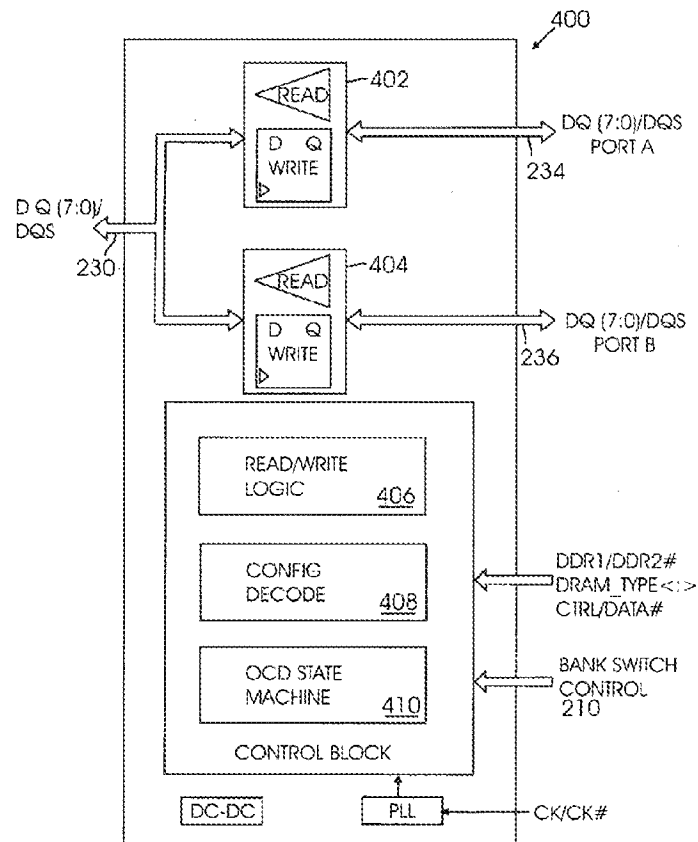


Fig. 4

Figure 4, above, depicts that “[d]ata is transmitted from the DIMM interface 202 via the data bus 230 to bidirectional signal drivers 402 & 404 that transmit and receive data over separate data busses 234 and 236 to the different sets of memory banks.” *Id.* “A read/write logic unit 406 determines whether data is being read from or written to the memory devices

(e.g., 212).” *Id.* “Memory configuration information 408 is obtained from the control unit.” *Id.*

2. Jeddelloh (Ex. 1006)

Jeddelloh relates to “[a] memory module [that] includes several memory devices coupled to a memory hub [that] includes several link interfaces coupled to . . . a self-test module.” Ex. 1006, code (57).

Jeddelloh’s Figure 2, reproduced below, shows an embodiment of a memory hub 200.

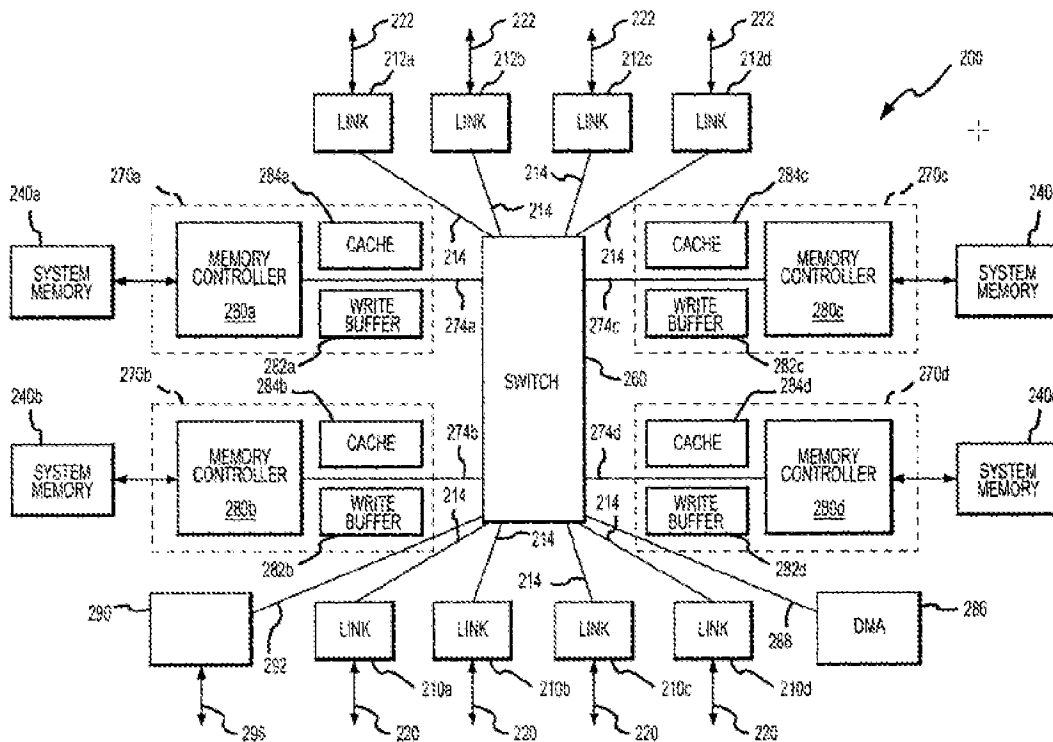


FIG.2

Figure 2, above, depicts memory hub 200 coupled to memory devices 240a-d. Link interfaces 210a-d and 212a-d are coupled to switch 260, and switch 260 is further coupled to memory interfaces 270a-d, which are, in turn, coupled to memory devices 240a-d. Each of the memory interfaces

270a-d includes a memory controller 280a-d, a write buffer 282a-d, and a cache memory unit 284a-d. Self-test module 290 is coupled to the switch 260 through a test bus 292.

Jeddeloh's Figure 3, reproduced below, illustrates a self-test module 308:

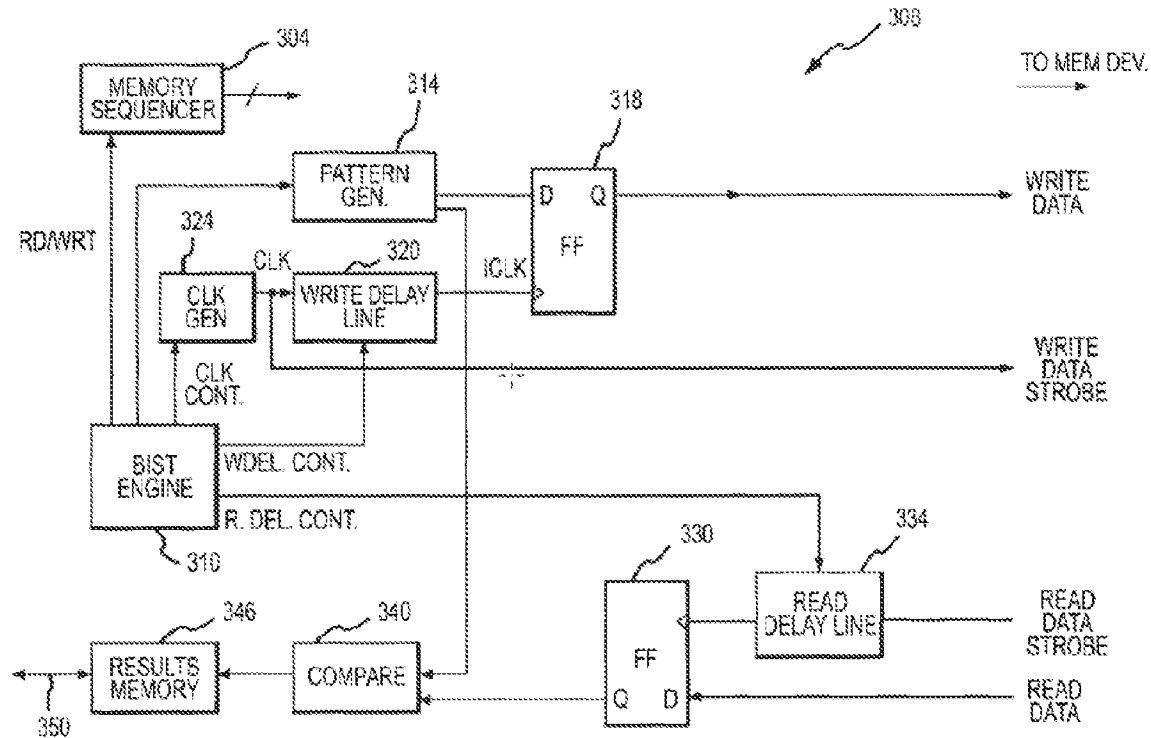


FIG.3

Figure 3 above, shows self-test module 308, including BIST (built-in self test) engine 310 and memory sequencer 304 that generate and distribute address and command/control signals during testing, and pattern generator 314, compare circuit 340, and results memory 346 that generate, distribute, and compare test patterns, and store fault information. *Id.* at 9:57–67, 10:30–51. Jeddeloh states that Figure 3 “is a functional block diagram representative of a suitable self-test module,” and “[t]he functional blocks shown . . . are conventional, and can be implemented using well known

techniques and circuitry.” *Id.* at 9:45–50. Jeddeloh also discloses use of a SMBus to communicate with off-module components, for instance, a test apparatus can use it to “set memory testing parameters and receive test results.” *Id.* at 8:26–53, 10:47–51.

3. Analysis of Independent Claim 1

Petitioner asserts that Ellsberry alone or Ellsberry combined with Jeddeloh teaches all the limitations in claim 1, and that one of ordinary skill in the art would have been motivated to combine Ellsberry and Jeddeloh. Pet. 26–61. Petitioner relies on Ellsberry to teach limitations 1[a], 1[b], 1[d], 1[f][i], 1[f][ii], 1[f][iii], and on the combination of Ellsberry and Jeddeloh to teach limitations 1[c], 1[e], and 1[g][i], 1[g][ii], and 1[g][iii]. *Id.* at 37–61. Petitioner contends it would have been obvious to modify Ellsberry with Jeddeloh’s teachings to include self-test functionality, and to distribute Jeddeloh’s test circuit functionality among Ellsberry’s Control ASIC and Switch ASICs. *Id.* at 25–37.

a) Petitioner’s Proposed Combination

Petitioner contends that Ellsberry discloses a memory module (limitation 1[a]) that separates circuitry into handling address and control signals and handling data signals, as shown in Petitioner’s annotated Figure 2 below. Pet. 26–27, 38 (citing Ex. 1005, Fig. 2).

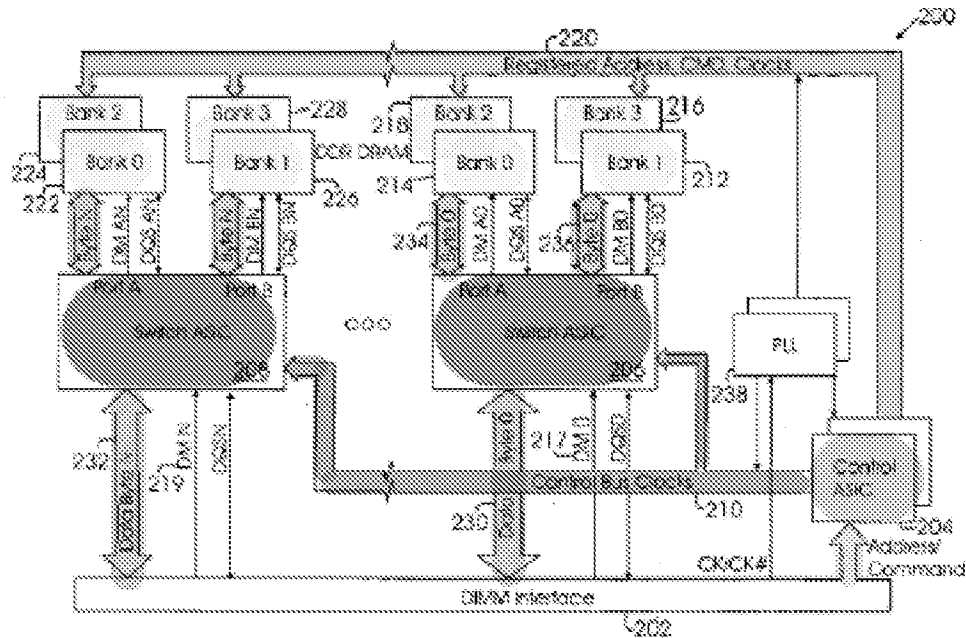


Fig. 2

Petitioner’s annotated Figure 2 of Ellsberry above shows address/control signals are handled by Control ASIC 204 (grey) and data signals are handled by a plurality of distributed Switch ASICs 206 thru 208 (purple). *Id.* at 26–27. According to Petitioner, Control ASIC 204 is coupled to Switch ASICs (purple) via bus 210 (pink), and ASIC 204 is also coupled to the address and control ports of the memory devices (DRAMs) (yellow) via bus 220 (blue). *Id.* at 44. Petitioner asserts that Control ASIC 204 receives system address and control signals over DIMM interface 202 (orange), and outputs address and command signals to the memory devices (yellow) on bus 220 (blue). *Id.* at 48–49. Petitioner further contends that the Switch ASICs (purple) pass data between the memory devices and the memory bus/memory controller (green). *Id.* at 40.

Petitioner contends that Ellsberry’s memory devices (DRAMs) teach the claimed “memory devices” (limitation 1[b]), Ellsberry’s Switch ASICs

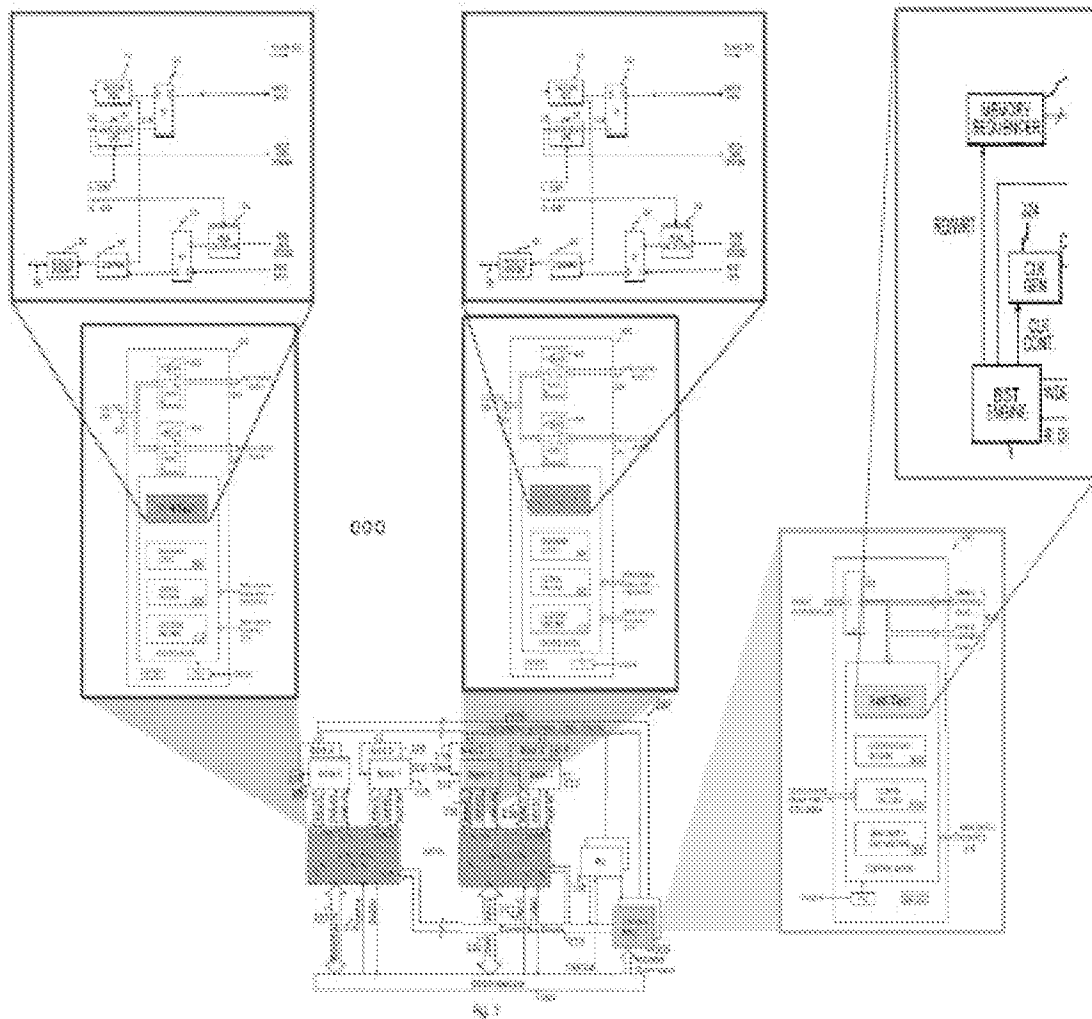
DIMM interface 202 and outputs address and command signals (“*first memory address and control signals*”) to the memory devices on bus 220 (limitations 1[f][i], 1[f][iii]). *Id.* at 48–51, 55. Petitioner contends that Ellsberry’s Switch ASICs “*propagate*” 8-bit sections of “*first data signals*” between “*the memory devices*” (DRAM) and DIMM interface 202 (connected to the “*system memory controller*”) via buses (limitations 1[f][ii], 1[f][iii]). *Id.* at 52–56.

Petitioner contends that, as combined with Jeddeloh, Ellsberry operates in a test (second) mode (limitations 1[e], 1[g][i], 1[g][ii], 1[g][iii]). *Id.* at 48, 56–62. According to Petitioner, the proposed combination is “straightforward”: Ellsberry’s “memory modules would be modified to include a test mode implementing self-test functionality, such as disclosed in” Jeddeloh. *Id.* at 26. In the proposed combination, Jeddeloh’s self-test module is divided between circuitry for handling address/control signals and circuitry for handling data signals. *Id.* at 28. Specifically, “[s]elf-test address/control signal circuitry would be implemented in Ellsberry’s Control ASIC to send commands to self-test data signal circuitry implemented in Ellsberry’s distributed Switch ASICs and address and control information to the memory devices in order to carry out the testing functionality described in” Jeddeloh. *Id.* (emphasis omitted).

Petitioner contends that the functionality of Jeddeloh’s BIST engine 310, clock generator 324, and memory sequencer 304 would be incorporated into Ellsberry’s Control ASIC 204, and the functionality of Jeddeloh’s pattern generator 314, compare circuit 340, and results memory 346 would be incorporated into Ellsberry’s distributed Switch ASICs. *Id.* at 29–30. So, for example, in the proposed combination, the recited “data handler logic

elements” in limitation 1[c] would also include “logic elements implementing the functionality of [Jeddeloh’s] pattern generator and compare circuitry.” *Id.* at 43.

The proposed combination is shown in Petitioner’s annotated compilation of Figures 2, 3, and 4 of Ellsberry and Figure 3 of Jeddeloh, reproduced below. *Id.* at 30 (citing Ex. 1006, 9:46–51, Fig. 3).



Petitioner’s annotated compilation of Figures 2, 3, and 4 of Ellsberry and Figure 3 of Jeddeloh, above, shows the following: (1) added “Self-Test” functionality to the control blocks in Ellsberry’s Figure 3 (Control ASIC)

and Figure 4 (Switch ASIC); (2) within the new “Self-Test” functionality of the Switch ASIC, Petitioner places the purple annotated sections of Jeddeloh’s Figure 3 (i.e., pattern generator 314, compare circuit 340, and results memory 346); and (3) within the new “Self-Test” functionality of the Control ASIC, Petitioner places the grey annotated sections of Jeddeloh’s Figure 3 (i.e., BIST engine 310, clock generator 324, and memory sequencer 304). *See* Pet. 29–30.

Petitioner asserts that to enter test mode and carry out the self-test functionality in the proposed combination, the memory module would receive mode commands (e.g., over a maintenance bus), as taught by Jeddeloh. *Id.* at 30 (citing Ex. 1006, 8:26–53, 10:3–6). Petitioner contends that the modified Control ASIC would “*output second memory address and control signals to the address and control ports of the memory devices*” (limitation 1[g][i]), “similar to how [Ellsberry’s] Control ASIC outputs address/control signals during normal operation.” *Id.* at 56 (citing Ex. 1003 ¶¶ 243–250; Ex. 1006, Fig. 3, 9:51–10:2, 10:24–29). That is, “[t]o carry out that self-test functionality, the self-test circuitry in the modified Control ASIC would send address/control signals to the memory devices (e.g., over Bus 220 of Ellsberry Figure 2).” *Id.* at 30–31. Petitioner contends that in test mode, the address/control signals are generated from Jeddeloh’s memory sequencer 304 in the modified Control ASIC and output to the ports of the memory devices 204 (limitations 1[g][i], 1[g][iii]). *Id.* at 61.

Petitioner further contends that “the self-test circuitry in the modified Control ASIC would send . . . command signals to the Switch ASICs (e.g., over Bus 210 of Ellsberry Figure 2)” (limitation 1[g][ii]). *Id.* at 30–31 (citing Ex. 1005 ¶ 56, Fig. 13; Ex. 1006, 10:30–51; Ex. 1003 ¶ 143)

(emphasis omitted). Petitioner contends that the command signals are generated from Jeddeloh's BIST engine in the modified Control ASIC, and therefore teach "*one or more commands output from the control module.*" (limitation 1[g][ii]). *Id.* at 60.

Petitioner contends that in the proposed combination, "there are two sources of data for the memory devices: (i) the system memory controller (during normal mode) and (ii) the pattern generator/compare circuit functionality in each Switch ASIC (during test mode)." *Id.* at 57. For limitation [g][ii], Petitioner asserts that "[d]uring test mode, the pattern generator functionality of the modified Switch ASICs '*transmit[s] one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices,*' similar to how [Ellsberry's] Switch ASICs transmit data during normal operation." *Id.* at 60 (citing Ex. 1003 ¶¶ 273–274; Ex. 1006, Fig. 3, code (54), 10:3–23). According to Petitioner, "[e]ach source communicates data along a different path in the Switch ASICs, so that '*the system memory controller*' is '*isolate[d]*' from the '*memory devices*' during test mode ('*second mode*') according to commands from the Control ASIC ('*control module*')." *Id.* at 57 (citing Ex. 1003 ¶ 255).

Petitioner asserts that "in normal mode [e.g., limitations 1[f][ii], 1[f][iii]], data from the system memory controller will travel over DIMM interface 202 and data buses 230-232, through the Switch ASICs, and over a port bus (e.g., 234 or 236)." *Id.* at 57. According to Petitioner, "within each Switch ASIC, the data travels from the data bus to one of two output ports, such as Port A, according to commands from the Control ASIC." *Id.* Petitioner's annotated Fig. 2, reproduced below, depicts such a path:

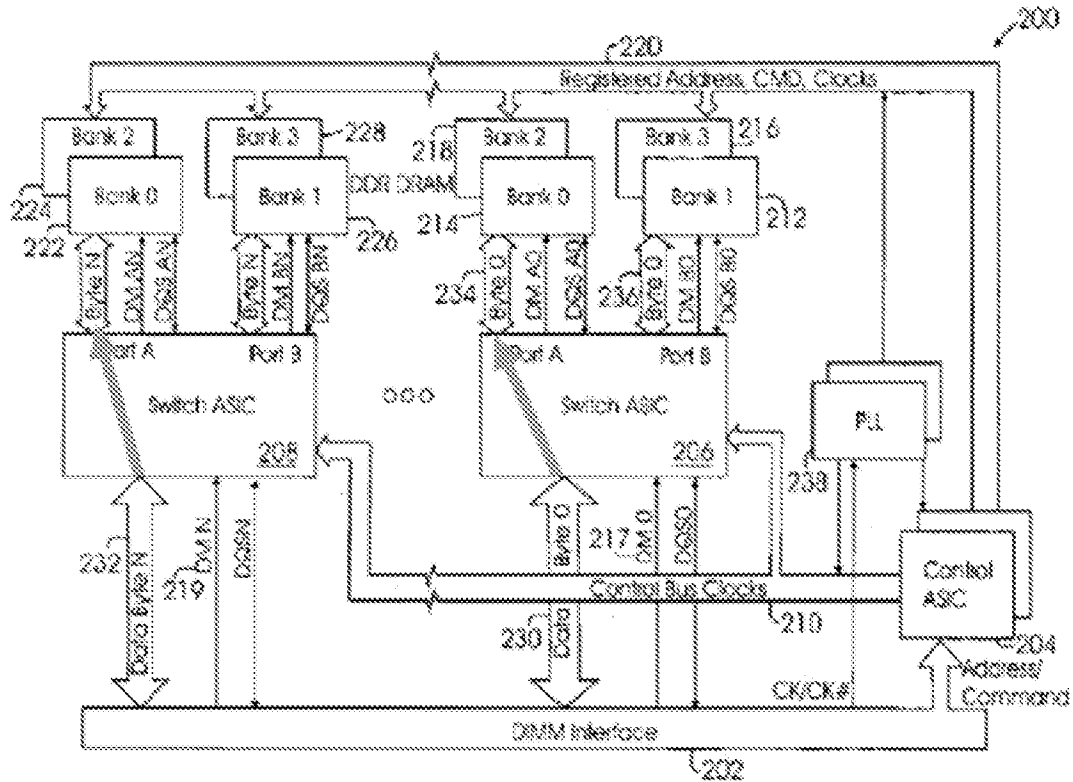
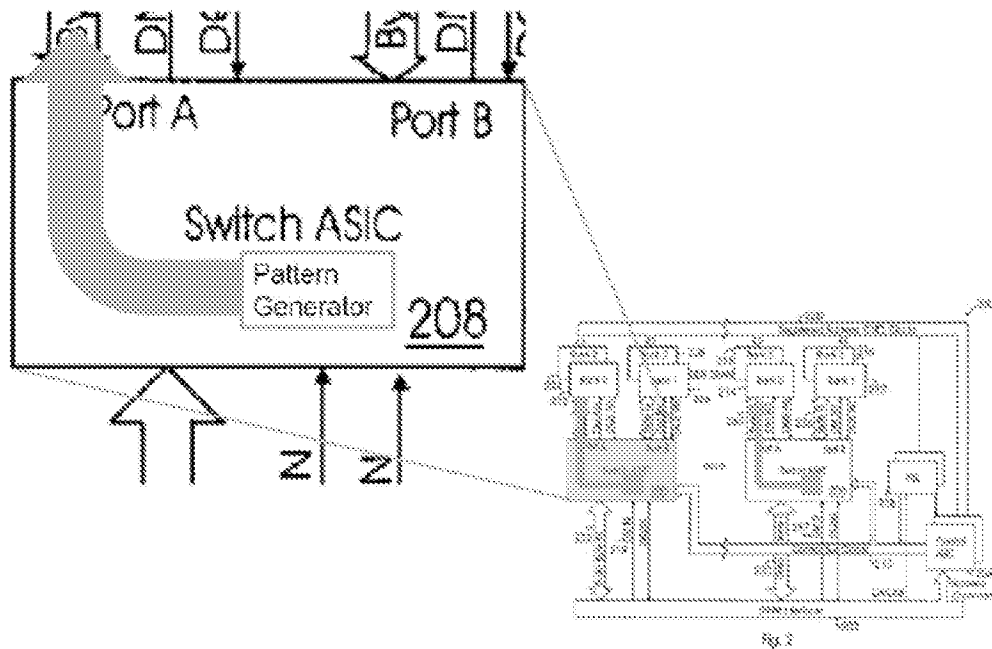


Fig. 2

Id. at 58 (citing Ex. 1003 ¶ 256). Petitioner’s annotated Figure 2, above, shows a diagonal green arrow from data bus 232, through Switch ASIC 208, and up to Port A of Bank 0. The same annotation is shown for data bus 230, through Switch ASIC 206, and up to Port A of Bank 0.

Petitioner contends that in the proposed combination, “[i]n test mode, [e.g., limitations 1[g]ii], 1[g][iii]], each Switch ASIC will output data from its pattern generator to the applicable output port, with the Control ASIC signaling the Switch ASIC (*‘according to one or more commands output from the control module’*) whether to output test data or normal data just as the Control ASIC sends command signals during normal operations.” *Id.* at 58 (citing Ex. 1003 ¶ 257; Ex. 1005, Figs. 2–3, ¶¶ 30–31). Petitioner provides annotated Figure 2, reproduced below, depicting that path.



Id. at 59 (citing Ex. 1003 ¶ 257). Petitioner’s annotated Figure 2, above, shows Figure 2 on the right, and enlarged Switch ASIC 208 on the left, with “Pattern Generator” added inside Switch ASIC 208, and an orange arrow from “Pattern Generator” to Port A of Bank 0.

Petitioner asserts that “in the Combined System data travels over two different data paths in the two different modes. Because each path merges into the same port on the Switch ASIC, only one path can be used at a time, using for instance a MUX/DEMUX as was well-known, otherwise there could be data collisions.” *Id.* (citing Ex. 1003 ¶ 258). In the Reply brief, Petitioner further asserts that “the proposed combination would use a MUX to isolate the *address/control* signals . . . just like the data signals.” Reply 10 (citing Ex. 1061 ¶¶ 701, 741, 754).

Petitioner asserts for limitation 1[g][iii] that the modified Control ASIC generates address/control signals with memory sequencer 304 that are output to the ports of the memory devices 240, and command signals with

BIST engine 310, to Switch ASICs, which generate “*second data signals*” via pattern generator 314. *Id.* at 61 (citing Ex. 1006, Fig. 3, 9:51–60, 9:66–10:2, 10:24–28, cl. 7; Ex. 1003 ¶¶ 279–280. Therefore, according to Petitioner, “the Control ASIC would gain the functionality of generating its own address/command signals when in a self-test mode, and the Switch ASICs would gain the functionality of generating their own data signals when in self-test mode.” Reply 24; *see also* Pet. 58–61.

(1) Petitioner’s Rationale for the Combination

Petitioner asserts that:

[A] Skilled Artisan would have considered the Combined System to be an arrangement of old elements each performing its known functions (Ellsberry’s module acting as computer memory and Jeddeloh[]’s self-test functionality performing self-test on that module) and yielding what one would expect from the arrangement (a memory module with self-test). Ellsberry and Jeddeloh[] each employ known signaling and testing techniques found in many prior art systems. EX1006, 9:46-51. Combining them would therefore have been well within a Skilled Artisan’s abilities, would not have resulted in any unpredictable results and could have been readily accomplished without undue experimentation and with a reasonable expectation of success. EX1003, ¶¶146-147.

Pet. 32 (emphasis omitted). Petitioner additionally provides purported rationale why a person of ordinary skill in the art would be motivated to (1) add the testing mode to Ellsberry and (2) divide the self-test functionality of Jeddeloh’s Figure 3 between Ellsberry’s Control ASIC and distributed Switch ASICs, set forth below. *Id.* at 32–37.

(a) Motivation to Add Testing to Ellsberry

Petitioner contends there are several reasons why a person of ordinary skill in the art would be motivated to add a testing mode to Ellsberry. *Id.* at 32–34; Ex. 1003 ¶¶ 149–53.

First, Petitioner contends, quoting U.S. Patent No. 7,181,659, a non-asserted reference, that “[a]s chips become faster in frequency, there is a greater need for hardware to include self-test logic.” *Id.* at 32 (citing Ex. 1018, 1:14–18). Moreover, Petitioner contends that Ellsberry discloses that its modules are JEDEC compliant and JEDEC standards at the time required self-testing. *Id.* (citing Ex. 1005 ¶ 50, claims 10, 19; Ex. 1016, 1–2, 25–28, 55–66; Ex. 1017, 2–11).

Second, Petitioner contends that because Ellsberry’s Control ASIC configures the memory devices on the module “rather than [being] directly programmed by the host system,” a person of ordinary skill would have been motivated to have a built-in testing capability in the Control ASIC itself. *Id.* at 33 (citing Ex. 1005 ¶ 44; Ex. 1003 ¶ 150).

Third, Petitioner contends that Jeddeloh emphasizes the need for testing circuitry in memory modules because of the increasing significance of signal timing. *Id.* (citing Ex. 1006, 2:41–51, 3:47–52).

Fourth, Petitioner contends that because Ellsberry describes that its Control ASIC configures the memory devices, one of ordinary skill would be motivated to add testing and, further, because Ellsberry has a “Mode” pin, to operate in other modes, a skilled artisan would have been motivated to

provide functionality to switch between additional modes. *Id.* at 33–34 (citing Ex. 1005 ¶ 33; Ex. 1003 ¶ 152).

Fifth, Petitioner also contends a person of ordinary skill in the art “would have been motivated to add testing in a mode separate from normal operating mode because the purpose of testing was to confirm the module would operate appropriately in normal mode.” *Id.* at 34 (citing Ex. 1006, 2:43–50, 3:47–53; Ex. 1016, 1–2, 25–28 55–66; Ex. 1017, 2–11. Ex. 1003 ¶ 153).

Sixth, Petitioner would add testing mode because “normal operation where the host sends read/write commands to the Control ASIC would interfere with such testing.” *Id.*

(b) Motivation to Distribute Jeddeloh’s Components into Ellsberry

Petitioner contends there are several reasons why a person of ordinary skill in the art would be motivated to divide the self-test functionality of Jeddeloh’s Figure 3 between Ellsberry’s Control ASIC and Switch ASICs. Pet. 34–37.

First, Petitioner contends that such distribution is consistent with Ellsberry’s architecture, and how Ellsberry handles address/control signal mapping in the Control ASICs and data signal routing in the Switch ASICs. Pet. 35. According to Petitioner, a person of ordinary skill in the art would have been motivated to divide the added functionality from Jeddeloh in a similar manner in order to simplify the design and minimize changes to Ellsberry. *Id.* (citing Ex. 1003 ¶¶ 155–156).

Second, Petitioner contends such a distribution of self-test functionality is consistent with how each of Ellsberry’s Switch ASICs

already included logic elements and connections for implementing data transactions in 8-bit portions. *Id.* at 35–36. According to Petitioner, a person of ordinary skill in the art “would have understood that additional logic functionality related to communicating data to and from data devices, such as that of [Jeddeloh], could be readily and efficiently included in the Switch ASICs, and configured to receive command signals from the added [Jeddeloh] testing-related functionality in the modified Control ASIC and report results thereto, without undue experimentation.” *Id.* at 36 (citing Ex. 1003 ¶ 158).

Third, Petitioner contends that a person of ordinary skill in the art would have been motivated to make such a design because other references teach placement of such self-test functionality in distributed data buffers similar to Ellsberry’s Switch ASICs. *Id.* (citing Ex. 1019⁸ ¶¶ 77, 97, Figs. 5, 18; Ex. 1020,⁹ 15:57–65, 18:63–19:23, 35:10–37:1, Figs. 5, 18, 36; Ex. 1007 ¶¶ 49–50, 53, 55–57, Figs. 1, 6–7; Ex. 1025, 30).

Fourth, Petitioner contends that an additional reference, Lepejian,¹⁰ teaches that, in order to reduce busing area, pattern generators should be distributed to each of the Switch ASICs. *Id.* (citing Ex. 1021, 3:45–53; Ex. 1003 ¶ 162).

Fifth, Petitioner contends that a person of ordinary skill in the art “would have been motivated to place self-test pattern generator functionality, such as [Jeddeloh’s], in [Ellsberry’s] Switch ASICs in order to place it close to its associated memory devices, thereby simplifying the

⁸ Tsern, US Patent Pub. 2007/0070669 A1 (published Mar. 29, 2007).

⁹ Shaeffer et al., US 7,562,271 B2 (issued July 14, 2009).

¹⁰ Lepejian et al., US 6,011,748 (issued Jan. 4, 2000).

wiring plan on the circuit board, reducing capacitance/propagation delay of the data lines, and reducing the inductance/crosstalk of the data lines,” *Id.* at 36–37 (citing Ex. 1002, 6–7; Ex. 1023, 2:27–3:6, Ex. 1024, 23–25, 207; Ex. 1009, 52, 54; Ex. 1003 ¶ 163) (internal citations omitted).

Finally, quoting another reference, Zimmerman,¹¹ Petitioner argues it was known that “functionality shown embodied in a single integrated circuit or functional block may be implemented using multiple cooperating circuits or blocks, or vice versa.” *Id.* at 37 (citing Ex. 1026, 7:43–46; Ex. 1027, 7:7–14; Ex. 1003 ¶ 165).

b) Patent Owner’s Arguments

Patent Owner contends that Petitioner has failed to show that a person of ordinary skill in the art would be motivated to combine Ellsberry and Jeddeloh in the manner required to achieve the claims of the ’523 patent. PO Resp. 35–70. Specifically, Patent Owner argues that Petitioner “provides no motivation to combine elements of Ellsberry and its *distributed* architecture with and Jeddeloh and its *centralized* architecture in the way the ’523 [p]atent claims require,” and thus, “Petitioner’s arguments are based only on hindsight.” *Id.* at 37 (citing Ex. 2004 ¶ 122); *see also id.* at 39–43. Patent Owner contends that Petitioner’s expert, Dr. Subramanian, admitted that he was “unaware of any prior memory module with a similar architecture to Ellsberry *ever* having incorporated self-test functionality as the ’523 [p]atent does, despite its general architecture being known in the art *for years.*” *Id.* at 36 (citing Ex. 2005, 47:20–48:8, 48:14–17, 52:1–7, 148:14–19; Ex. 2004 ¶¶ 118, 120–121); *see also id.* at 54–55.

¹¹ Zimmerman, US 7,177,211 B2 (issued Feb. 13, 2007).

Patent Owner also argues that Petitioner’s combination “goes directly against the explicit teachings of Jeddeloh” because it distributes “pieces of Jeddeloh’s centralized self-test logic differently . . . across disparate circuits and placing them in direct communication with the memory devices” and without Jeddeloh’s memory interface components. *Id.* at 46. Patent Owner further argues that Petitioner’s combination requires a MUX/DEMUX circuit in each of Ellsberry’s switch ASICs, which is not consistent with the teachings in Ellsberry or Jeddeloh, and further would introduce memory timing issues that Jeddeloh tries to avoid. *Id.* at 47–48.

In addition, Patent Owner argues that Petitioner’s arguments regarding motivation to add testing to Ellsberry are deficient for various reasons. *Id.* at 49–53. Patent Owner also argues that Petitioner’s contentions regarding motivation to distribute pieces of Jeddeloh’s self-test module to separate parts of Ellsberry are likewise deficient. *Id.* at 54–59.

Patent Owner further argues that Petitioner has not established that a person of ordinary skill in the art would have had a reasonable expectation of success in achieving the claimed invention from the combination because, according to Patent Owner, the combined system would not result in a functional device and would have required multiple further modifications and engineering. *Id.* at 59–60. Patent Owner also argues that “Petitioner has identified a relatively low level of experience as being ordinary in the field,” and the challenges encountered by the combination “would have been beyond such a [person of ordinary skill in the art’s] ability to reasonably resolve them.” *Id.* at 60 (citing Ex. 2004 ¶¶ 83–84); *see also id.* at 61–70.

c) Analysis

For the reasons set forth below, we agree with Patent Owner that Petitioner has not explained persuasively why one of ordinary skill in the art, absent hindsight and use of the '523 patent as a roadmap, would have been motivated to combine Ellsberry and Jeddeloh in the manner proposed. *See In re Fritch*, 972 F.2d 1260, 1266 (Fed. Cir. 1992) (citing *In re Gorman*, 933 F.2d 982, 987 (Fed. Cir. 1991)) (“It is impermissible to use the claimed invention as an instruction manual or ‘template’ to piece together the teachings of the prior art so that the claimed invention is rendered obvious.”). Specifically, we determine that Petitioner has not provided sufficient rationale establishing why a person of ordinary skill in the art would have been motivated to divide the self-test functionality of Jeddeloh’s Figure 3 between the Control ASIC and Switch ASICs of Ellsberry and with a reasonable expectation of success in doing so.¹² *See KSR*, 550 U.S. at 418 (establishing that an invention would have been obvious requires “a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements *in the way* the claimed new invention does”) (emphasis added).

Petitioner’s proposed combination requires (1) adding a test mode with self-test functionality to Ellsberry; (2) modifying Jeddeloh’s self-test

¹² Because we determine that Petitioner has not provided sufficient reasoning why a person of ordinary skill in the art would have been motivated to divide the self-test functionality of Jeddeloh’s Figure 3 between the Control ASIC and Switch ASICs of Ellsberry with a reasonable expectation of success, we need not reach whether Petitioner has sufficiently established motivation to add testing to Ellsberry with a reasonable expectation of success.

module, i.e., dividing the self-test functionality and related circuitry shown in Jeddeloh's Figure 3; (3) incorporating the divided self-test functionality and related circuitry in Ellsberry's Control ASIC and Switch ASICs; and (4) adding MUX/DEMUX to each of Ellsberry's Switch ASICs, and further using a MUX to isolate the address/control signals. Pet. 25–37; Reply 10. As a result of incorporating the divided self-test functionality and related circuitry in Ellsberry's Control ASIC and Switch ASICs, Petitioner further asserts that the Control ASIC would generate its own address/control/command signals when in a self-test mode, and the Switch ASICs would generate their own data signals when in self-test mode. Reply 24; Pet. 56–61. That is, according to Petitioner, while in self-test mode, the added memory sequencer functionality in the Control ASIC would generate address and control signals, the added BIST functionality in the Control ASIC would generate command signals, and the added pattern generator/compare circuit functionality in the Switch ASIC would generate data signals. Pet. 56–61. Petitioner's changes therefore require multiple levels of modifications in order to purportedly teach the claimed second mode (limitation 1[e]), and the functionality of the claimed second mode (limitations 1[c], 1[g][1], 1[g][ii], and 1[g][iii]).

A number of facts are undisputed. There is no dispute that Ellsberry's distributed architecture was well known. PO Resp. 36; Ex. 2004 ¶¶ 56–57, 127; Reply 18. There is no dispute that Ellsberry does not teach testing. PO Resp. 36; Ex. 2004 ¶ 63; Pet. 32 (describing asserted motivations to add a testing mode to Ellsberry); Ex. 2005, 64:23–25. There is no dispute that Jeddeloh employs a centralized architecture that is different from Ellsberry's distributed architecture. PO Resp. 39; Reply 14. There is no dispute that

neither reference teaches distributing self-test functionality as in the claimed invention. PO Resp. 36, 38; Pet. 34 (describing dividing the self-test functionality of Jeddeloh); Reply 36 (“the Petition is not asserting anticipation, so it is irrelevant whether any single reference discloses the combination of elements at issue”). Moreover, both Dr. Subramanian and Dr. Brogioli testified that they were not aware of any memory module with a similar architecture to Ellsberry ever having incorporated self-test functionality as configured in the claimed invention, despite Ellsberry’s distributed architecture being well known in the art. Ex. 2004 ¶¶ 18, 126–127, 143; Ex. 2005, 47:20–48:8, 48:14–17, 52:1–7, 148:14–19.

Although Petitioner contends that the combination of Ellsberry and Jeddeloh is merely an arrangement of old elements, each performing the same function (Pet. 31–32), Petitioner has not identified persuasive evidence that supports these contentions. Petitioner identifies Ellsberry’s memory module and Jeddeloh’s self-test functionality, but Petitioner has not identified any “old element” where self-test functionality is divided in the manner in which Petitioner proposes to divide Jeddeloh’s functionality. In other words, this is not a situation where each element is purportedly taught, in some manner, by one of the references, and the elements are merely rearranged and combined. *KSR*, 550 U.S. at 418 (in an obviousness analysis, one must “determine whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue.”); *Sundance, Inc. v. DeMonte Fabricating Ltd.*, 550 F.3d 1356, 1366–1367 (Fed. Cir. 2008) (a combination of prior art is “more likely to be obvious where it ‘simply arranges old elements with each performing the same function it had been known to perform’ and yields no more than one would expect from

such an arrangement.”). Here, the proposed combination involves modifying *both* references in order to reach the claimed invention, and in particular, to purportedly teach the specific requirements of limitations 1[c], 1[e], 1[g][i], 1[g][ii], and 1[g][iii].

Petitioner’s first, second, fifth, and sixth reasons to distribute Jeddeloh’s functionality into Ellsberry all rely on Ellsberry’s distributed architecture as a basis for providing motivation to divide the added functionality and circuitry from Jeddeloh, i.e., in manner consistent with Ellsberry’s architecture. We first address these rationales. Both parties identify the different architectures of Ellsberry and Jeddeloh as a basis for their arguments, but Petitioner contends that these differences provide motivation to combine the references, whereas Patent Owner contends the opposite – that these differences do not support a motivation to combine.¹³

Petitioner relies on Ellsberry’s distributed architecture as the rationale to divide Jeddeloh’s self-test module circuitry, but Ellsberry itself does not teach a testing mode. *See* Reply 36 (Ellsberry’s “distributed architecture *itself* provides a motivation for a [person of ordinary skill in the art] to distribute the BIST functionality from” Jeddeloh). Although the motivation

¹³ In the Reply, Petitioner refers to Ellsberry as an LRDIMM architecture and Jeddeloh as an FBDIMM architecture. *See generally* Reply. Patent Owner contends this is a new argument, and that neither Ellsberry nor Jeddeloh mention LRDIMM or FBDIMM, and that the LRDIMM and FBDIMM module types postdate both references. Sur-reply 10. We agree that neither Ellsberry nor Jeddeloh mention LRDIMM or FBDIMM. *See id.* Regardless of how the architectures of Ellsberry and Jeddeloh are labelled, we agree with Patent Owner that the question is whether Ellsberry would be combined with Jeddeloh, not whether LRDIMM would be combined with FBDIMM. *Id.* at 11. We, therefore, focus our analysis on the disclosures in Ellsberry and Jeddeloh.

to combine may come from the references themselves, *WMS Gaming, Inc. v. International Game Tech.*, 184 F.3d 1339, 1355 (Fed. Cir. 1999), there still must be evidence that “the skilled artisan, confronted with the same problems as the inventor and with no knowledge of the claimed invention, would select the elements from the cited prior art references for combination in the manner claimed.” *In re Rouffet*, 149 F.3d 1350, 1357 (Fed. Cir. 1998); *see also In re Werner Kotzab*, 217 F.3d 1365, 1371, (Fed. Cir. 2000) (“[A] rejection cannot be predicated on the mere identification ... of individual components of claimed limitations. Rather, particular findings must be made as to the reason the skilled artisan, with no knowledge of the claimed invention, would have selected these components for combination in the manner claimed.”). To that end, we agree with Patent Owner that Petitioner’s proposed rationale for the combination appears circular, and based on hindsight. *See* Sur-reply 17; *Monarch Knitting Mach. Corp. v. Sulzer Morat GmbH*, 139 F.3d 877, 880 (Fed. Cir. 1998) (“[d]efining the problem in terms of its solution reveals improper hindsight in the selection of the prior art relevant to obviousness”).

That is, we are not persuaded that simply because Ellsberry has a distributed architecture, that, with no knowledge of the claimed invention, one of ordinary skill in the art would select Jeddelloh’s self-test module, divide that self-test module functionality, implement it into Ellsberry’s Control ASIC and Switch ASICs, and then make the extensive further modifications required to Ellsberry that are necessary to achieve the claimed invention. As the various modifications required to teach the claims (particularly, limitations 1[c], 1[e], 1[g][i], 1[g][ii], and 1[g][iii]), become more complex and untethered to the design of Ellsberry and Jeddelloh, the

influence of impermissible hindsight bias emerges, as further discussed below.

Petitioner’s contentions that the proposed combination is “straightforward because it would not require any new data or control paths and thus would not create any new timing issues” because it would take advantage of the existing data and control paths of Ellsberry’s module, are not persuasive because they oversimplify the changes required, as discussed further below. *See* Reply 14–15, 19–20 (“implementing self-test functionality in either [a centralized or distributed] architecture would have been straightforward, because a [person of ordinary skill in the art] could simply use the **existing** buffers for data and address/command on the memory module to add self-test functionality without any **new** data or control paths”); Ex. 1061 ¶¶ 664–666. For similar reasons, we do not find persuasive Petitioner’s arguments that “the proposed combination is much more straightforward than alternative implementations proposed by the [’523 patent] (but not explained in any detail) that would require **new** structures and/or signal paths.” Reply 15 (citing Ex. 1061 ¶ 688). Nor are we persuaded that the necessary modifications to both references would “simplify the design and minimize the changes to Ellsberry, and thereby likely avoid design problems and produce a more reliable system,” as Dr. Subramanian testifies. *See* Ex. 1003 ¶¶ 155, 158 (emphasis omitted).

Modifying Jeddelloh’s self-test module to divide the functionality and circuitry creates the need to continue modifying the various components of Ellsberry in order to achieve the claimed invention. For example, the modification requires that Ellsberry’s Control ASIC must generate address/control/command signals in a new self-test mode (limitations 1[e],

1[g][i], 1[g][ii], 1[g][iii]), and similarly, Ellsberry's Switch ASICs must generate data signals in a new self-test mode (limitations 1[c], 1[e], 1[g][ii], 1[g][iii]. Reply 24; Pet. 56–61. Furthermore, Ellsberry's Switch ASICs require a MUX/DEMUX in order to handle the different data paths in normal and new self-test mode (limitation 1[e], 1[g][ii]). *See* Pet. 59. Thus, in order to meet each of these claim limitations that require a new testing mode to be added to Ellsberry, Petitioner simply continues to modify the references in a way that conveniently results in the same configuration as the claimed invention. That is, the claims require that the testing data and address/control logic is separated between the data module and the control module. Although the wiring and signaling rerouting itself may not be complex, the combination of functionality and circuitry that requires modification to achieve the correct timing and synchronization results from the knowledge of the particular claim limitations, not any motivation to combine asserted by Petitioner. *See* Ex. 2004 ¶¶ 82–83.

Petitioner cites to supporting testimony from Dr. Subramanian on the combination, but Dr. Subramanian's testimony is unavailing because it essentially repeats what is stated in the Petition, and states that Ellsberry and Jeddeloh “employ known signaling and testing techniques found in many prior art systems,” and combining Ellsberry and Jeddeloh “would therefore have been well within a Skilled Artisans's abilities, would not have resulted in any unpredictable results and could have been readily accomplished without undue experimentation and with a reasonable expectation of success.” Pet. 32 (citing Ex. 1003 ¶¶ 146–147). We find this testimony to be conclusory, at a minimum, particularly in light of the relatively low level

of skill in the art advanced by Petitioner. Further, as discussed below, Dr. Subramanian's testimony is misguided.

We find more persuasive Patent Owner's contentions that "[p]ortions of Jeddeloh's self-test module cannot be disassembled and distributed across Ellsberry's memory module as proposed by Petitioner without creating new pathways or changing how existing components work, which creates timing issues and issues synchronizing the various signals sent to and received from the memory devices." PO Resp. 62–63 (citing Ex. 2004 ¶ 83). For example, Dr. Brogioli explains that "two new circuits have been created using components disclosed in Jeddeloh and deployed to a distributed, and markedly different architecture than Jeddeloh. Not only are new pathways required within Ellsberry's architecture that are not contemplated by Jeddeloh, but these pathways in some instances require being bidirectional." Ex. 2004 ¶ 83. We find this testimony credible, given the nature and extent of the modifications proposed by Petitioner.

This is exemplified in Petitioner's modifications to Ellsberry's Control ASIC in order to teach limitations 1[e], 1[g][i], 1[gii], and 1[g][iii]. Dr. Subramanian testifies that Jeddeloh's "self-test functionality would be implemented in Ellsberry such that the address and command signals are generated in Ellsberry's control unit instead of receiving those address and command signals from the system bus." Ex. 1061 ¶ 665; *see also id.* ¶ 678 (modification involves "adding an address/command generator to Ellsberry's Control ASIC").

Patent Owner argues that the Control ASIC in Petitioner's combined system "would be unable to properly access the memory devices in a 'second' mode," i.e., a testing mode. PO Resp. 66 (citing Ex. 2004 ¶ 97).

Patent Owner argues that any address and control signals originating from the modified Control ASIC's Control Block¹⁴ during test mode will bypass register 302, a necessary component of the Control ASIC that helps control the timing of the signals received from the system. PO Resp. 66; Ex. 2004 ¶¶ 96–97. Dr. Brogioli testifies that “register 302 buffers address and command information received from the DIMM interface,” so if testing mode is used, “signals would not pass through register 302 and hence the timing would change, and thus the functioning.” Ex. 2004 ¶ 96 (citing Ex. 1005 ¶ 39). According to Dr. Brogioli, “without register 302, the carefully synchronized data and address/control signals would no longer be synchronized.” *Id.*

For convenience, Petitioner's modified Fig. 3 of Ellsberry is reproduced below:

¹⁴ Patent Owner contends that without Jeddeloh's memory controller, which is not part of the proposed combination, the proposed combination is unable to generate address signals during test mode in accordance with limitation [1.g.i]. PO Resp. 28–30. Petitioner contends that Jeddeloh's memory sequencer 304 generates address signals for testing because it “receives read and write commands” and then “generates properly timed signals for *controlling* the operation of the memory device.” Reply 5 (citing Ex. 1006, 9:58–59); *see also id.* at 3–6. Petitioner contends that Jeddeloh omitted address and control signal lines from Figure 3 “to avoid unnecessarily obscuring the present invention.” *Id.* at 6 (citing Ex. 1006, 9:51–54, 9:61–64; Ex. 1061 ¶ 736). We find that the cited disclosure from Jeddeloh better supports Petitioner's position.

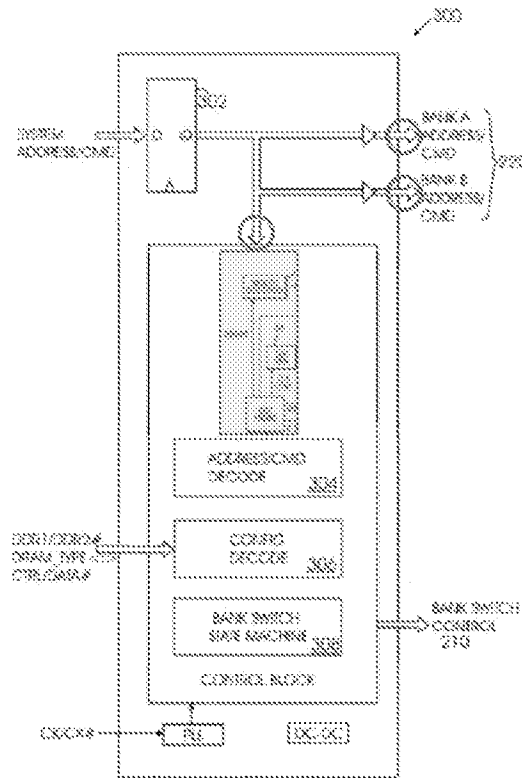


Fig. 3

Reply 40. Petitioner’s modified Figure 3, depicted above, shows Ellsberry’s Figure 3 with the proposed added functionality from Jeddelloh (BIST engine 310, clock generator 324, and memory sequencer 304) in gray in the Control Block. A directional right arrow is depicted on the top left, labelled “system address/cmd” going into register 302, with two single directional right arrows coming out of register 302 and to the right into bank A and bank B address cmd 220 and a single directional down arrow into the Control Block that are labelled with red circles. In the Control Block are four blocks: the added functionality from Jeddelloh in gray, address/cmd decode 304, config decode 306, and bank switch state machine 308. There is a directional right arrow coming out of the Control Block to bank switch control 210.

Patent Owner contends “the single-directional arrow from register 302

to the control block and to bus 220 faces the wrong direction” and “the link would have to be bidirectional before the [combined system] would be able to generate any useful test signals.” PO Resp. 67 (citing Ex. 2004 ¶¶ 97–98). Dr. Brogioli testifies that a person of ordinary skill in the art “would need to account for the change in signal timing resulting from the test address/control signals skipped register 302 in order to synchronize the data and address/control signals.” Ex. 2004 ¶ 98. According to Dr. Brogioli, a person of ordinary skill in the art “would then be met with the problem that if you make the unidirectional arrows shown above bidirectional, that would introduce issues over who is controlling the line (the system or control block), and how.” *Id.* Dr. Brogioli further testifies that a person of ordinary skill in the art would “not be familiar with how to manage a bidirectional interconnect in this context, and how to effectively control the end points that are intended to transmit data over the bidirectional interconnect, in a shared manner, at the speeds that Ellsberry contemplates.” *Id.*

Petitioner responds that Patent Owner’s arguments “misinterpret the normal operation” of Ellsberry. Reply 39 (citing Ex. 1061 ¶¶ 724–731). According to Petitioner, “the connection to the ‘Control Block’ . . . is ***already*** bidirectional, such that signals output on bus 220 come from the ‘Control Block’ in the Control ASIC, not directly from register 302, which is how the Control ASIC can output different signals than it receives.” *Id.* at 39–40 (citing Ex. 1061 ¶¶ 726, 729). Petitioner contends that “in order for the Control Block to output signals on bus 220 in normal mode, that connection must already be bidirectional,” and “[t]he same is true in the proposed combination,” as shown in Petitioner’s annotated Figure 3, above.

Id. at 40 (citing Ex. 1061 ¶ 729).

Dr. Subramanian testifies that “[i]n normal mode, the role of register 302 is to provide address and command signals from the system bus that are *input* to the Control Block, which decodes that information and, in response, outputs the signals to control the memory bank switches (using 210) and memory devices (using 220).” Ex. 1061 ¶ 726 (citing Ex. 1005, Figs. 3, 8B, 9, ¶¶ 12, 32, 44). Dr. Subramanian, therefore, contends that “[t]he signals from the Control ASIC come from the ‘Control Block,’ not directly from register 302, and thus register 302 does not affect the timing of the *output* from the ‘Control Block,’ since the timing is set by the Control Block, not by the register 302.” *Id.* Dr. Subramanian testifies that a person of ordinary skill in the art would have understood “that the arrows in Figure 3 are not indicating universal unidirectional communication from the register 302 into the Control Block and to bus 220 as Dr. Brogioli argues, but also include signals coming out from the Control Block going to bus 220, meaning the connection to the Control Block is already bidirectional.” *Id.* at ¶ 729. Dr. Subramanian further testifies that a person of ordinary skill in the art would have understood “how to route signals to implement self-test functionality inside an ASIC or similar integrated circuit, as evidenced by the prior art.” Ex. 1061 ¶ 730 (citing Ex. 1006, Figs. 1–3; Ex. 1016 at 3, 25–28; Ex. 1058 at 3, 26–29).

We find that the disclosure in Ellsberry better supports Patent Owner’s arguments. *See* Sur-reply 22–26. Central to the dispute is whether Ellsberry’s Figure 3 discloses bidirectional communication, i.e., whether there are already signals coming *from* the Control Block *to* bus 220. Patent Owner and Dr. Brogioli rely on paragraph 39 of Ellsberry, which describes

Figure 3. Sur-reply 22–26; Ex. 2004 ¶ 96. Paragraph 39 of Ellsberry states:

The command processing system 300 controls physical bank selection and bank switching direction. *Memory addresses and command information are received from the DIMM interface 202, buffered in a register 302 and sent to all memory banks (e.g., Bank 0, Bank 1, Bank 2 and Bank 3) over address bus 220. The memory address and command information is also decoded 304 and memory configuration information 306 (e.g., DRAM type, etc.) is determined. The memory configuration information may be determined from preset information. A bank switch state machine 308 then determines which memory bank should be activated or accessed. In one embodiment of the invention, this state machine 308 is a logical translation table that maps a primary space address to a secondary space address based on the memory configuration present. For example, the state machine 308 may be the address mapping table illustrated in FIGS. 7A-F, which was previously described. The state machine 308 sends control information to the memory bank switches 206 & 208 via the control bus 210 to indicate which memory banks should be activated/deactivated or accessed.*

Ex. 1005 ¶ 39 (emphasis added). In other words, Ellsberry teaches that memory address and command information is received from DIMM interface 202, buffered in register 302, and sent to memory banks over address bus 220. It further teaches that the memory address and command information is also decoded (i.e., address/cmd decode block 304 in the control block) to determine memory configuration information (i.e., config decode block 306), and that state machine 308 determines which memory bank switch should be activated or accessed and sends control information to the memory bank switches 206 and 208 over control bus 210. This disclosure in Ellsberry does not disclose that any information is sent *from* the Control Block and *to* bus 220; rather, it only indicates that information is

provided *to* the Control Block and then *to* bus 210. The citations from Ellsberry that Dr. Subramanian relies upon do not state otherwise. For example, paragraph 32 of Ellsberry states that “[a]lthough address information may be sent to all memory banks over address bus 220, the memory bank switch 206 and/or control unit 204 determine which memory devices or banks are accessed.” Petitioner does not persuasively rebut Patent Owner’s arguments or Dr. Brogioli’s testimony as to the disclosure in paragraph 39. We, accordingly, afford more weight to Dr. Brogioli’s testimony on this point, as well as his testimony describing the extent of the modifications required of the proposed combination. *See, e.g.,* Ex. 2004 ¶¶ 83, 98.

As discussed above, Petitioner contends that the proposed combination involves using the same data control mechanisms and pathways (e.g., Reply 24–27, 37), and Dr. Subramanian testifies that adding the self-test functionality to Ellsberry “does not require any new timing or routing at the module level.” Ex. 1061 ¶¶ 678, 682; *see* Reply 25. However, Dr. Subramanian admits that the proposed combination requires that Ellsberry’s Switch ASICs and Control ASIC must be redesigned to add the relevant functionality from Jeddeloh. *See* Ex. 2016, 153:18–154:17, 158:13–23, 160:17–162:7. This is consistent with Dr. Brogioli’s testimony that “two new circuits have been created using components disclosed in Jeddeloh and deployed to a distributed, markedly different architecture than Jeddeloh.” Ex. 2004 ¶ 83.

In addition to the modifications to the Control ASIC, Dr. Subramanian testifies that modifications are required to the Switch ASIC, “such that the write data signals are generated in Ellsberry’s memory bank switches instead

of receiving write data from the system bus, and the memory bank switches then handle the data transfers the same way, and with the same timing, as before.” Ex. 1061 ¶ 665 (emphasis omitted); *see id.* ¶ 678 (“a data generator as taught by Jeddeloh is added to each memory bank switch in Ellsberry to provide data signals in a self-test mode instead of providing the data signals from the system memory bus when in a normal mode”). Dr. Subramanian further testifies that Ellsberry had “already solved the timing and synchronization issues between its control unit and the distributed memory bank switches by following the timing requirements of the relevant JEDEC standards.” Ex. 1061 ¶ 684 (citing Ex. 1005 ¶¶ 46, 50). As discussed, the modification to Ellsberry also requires adding a MUX to isolate the data signals in the Switch ASIC and the address/control signals in the Control ASIC. Pet. 59, Reply 10. According to Dr. Subramanian, “the prior art confirms that a [person of ordinary skill in the art] would have known how to use a multiplexer or demultiplexer . . . without any timing and signal issues.” Ex. 1061 ¶ 701; *see id.* ¶¶ 741, 754, Ex. 1003 ¶ 258.

In our view, Petitioner minimizes the extent of the redesign required to make the proposed combination, and due to that, we find Dr. Brogioli’s testimony to be more credible. *See, e.g.*, Ex. 2004 ¶¶ 81–86. For example, we find persuasive Dr. Brogioli’s testimony that “[w]hen attempting to adapt circuitry and functions meant for one type of architecture for use in a different architecture, a [person of ordinary skill in the art] would have understood that technical problems would arise, such as issues coordinating signals between functional units that were intended to remain united, and issues managing shared buses or data lines, for example.” Ex. 2004 ¶ 82. We also find credible Dr. Brogioli’s testimony that a person of ordinary skill

in the art “would have been met with substantial challenges coordinating data and address signaling circuitry of Jeddeloh once you split up and distributed it into Ellsberry’s specific architecture.” *Id.* at ¶ 83. Dr. Brogioli testifies “[n]ot only are new pathways required within Ellsberry’s architecture that are not contemplated by Jeddeloh, but these pathways in some instances require being bidirectional,” as was shown above in connection with the modifications to the Control ASIC. *Id.* We afford weight to Dr. Brogioli’s testimony that “Jeddeloh’s self-test module cannot be disassembled and distributed across Ellsberry’s memory module as proposed by Petitioner without creating new pathways or changing how existing components work” in view of the extent of the modifications required. *Id.*

Dr. Subramanian’s testimony that Ellsberry “already solved the timing and synchronization issues between its control unit and the distributed memory bank switches by following the timing requirements of the relevant JEDEC standards,” merely supports that Ellsberry, standing alone and without the proposed modifications, is JEDEC compatible, not that Ellsberry “solved the timing and synchronization issues between its control unit and the distributed memory bank switches” in the proposed combination with the proposed modifications to Ellsberry, including the addition of self-test functionality from Jeddeloh. *See* Ex. 1061 ¶ 684.

The record supports that the proposed modifications were informed by the requirements of the claims. The fact that a person of ordinary skill in the art “could” make these changes does not inform us that a person of ordinary skill in the art “would” make these changes. *Belden Inc. v. Berk-Tek LLC*, 805 F.3d 1064, 1073 (Fed. Cir. 2015) (“obviousness concerns whether a

skilled artisan not only *could have made* but *would have been motivated to make* the combinations or modifications of prior art to arrive at the claimed invention”); *see ActiveVideo Networks, Inc. v. Verizon Commc’ns, Inc.*, 694 F.3d 1312, 1327 (Fed. Cir. 2012) (“[T]he expert’s testimony on obviousness was essentially a conclusory statement that a person of ordinary skill in the art would have known, based on the ‘modular’ nature of the claimed components, how to combine any of a number of references to achieve the claimed inventions. This is not sufficient and is fraught with hindsight bias.”). As discussed by Patent Owner, Dr. Subramanian conceded that there were multiple ways a person of ordinary skill in the art could add self-testing to Ellsberry. *See* PO Resp. 57 (citing Ex. 2004 ¶¶ 46, 47, 128, 129, 146; Ex. 2005, 121–125). For example, a person of ordinary skill in the art could have incorporated all of Jeddeloh’s self-test logic into Ellsberry’s Control ASIC or duplicated Jeddeloh’s self-test logic into Ellsberry’s Switch ASICs. *See* Ex. 2004 ¶ 129; Ex. 2005, 121–125, 184–185. Dr. Brogioli credibly testifies that these alternative designs would have been more similar to what was seen in the prior art. Ex. 2004 ¶¶ 46, 47, 128, 129, 146. Accordingly, the record supports that Petitioner’s specific modifications were driven by hindsight view of the roadmap provided by the claim limitations, e.g., the testing data and address/control logic is separated between the data module and the control module (limitations 1[e], 1[g][i], 1[g][ii], 1[g][iii]).

Petitioner also asserts that a person of ordinary skill in the art would have understood that Jeddeloh has a central hub 140 which functions like an AMB in an FBDIMM to buffer both data and address/command signals for the memory devices. Reply 20. Petitioner further asserts that Jeddeloh

teaches implementing the BIST self-test functionality in the central hub, “i.e., an AMB that buffers data and address/command signals.” *Id.* at 21 (citing Ex. 1061 ¶ 672; Ex. 1006, 5:47–49, 8:26–29, 9:44–60). Petitioner asserts that the ’523 patent “admits that self-test functionality could be implemented in an AMB.” *Id.* (citing Ex. 1001, 4:41–44). Dr. Subramanian testifies that the ’523 patent “express[es] concerns about concentration of self-test circuitry in a combined address/control and data buffer, as [Jeddeloh] does, and these known concerns with prior art implementations would have further motivated” a person of ordinary skill in the art to divide Jeddeloh’s circuitry functionality between Ellsberry’s Control ASIC and Switch ASICs to avoid “these very issues.” Ex. 1003 ¶ 156 (citing Ex. 1001, 4:41–56); *see* Reply 21.

We agree with Patent Owner (Sur-reply 12–13) that whether self-test can be implemented in an AMB or a centralized architecture such as Jeddeloh does not inform whether the combination of Ellsberry and Jeddeloh is obvious. *See Innogenetics N.V. v. Abbott Laboratories*, 512 F.3d 1363, 1373 (Fed. Cir. 2008) (“knowledge of a problem and motivation to solve it are entirely different from motivation to combine particular references to reach the particular claimed method”). Instead, motivation must be demonstrated to combine the particular references, i.e., Ellsberry and Jeddeloh. *See id.* at 1373–1374; *see also W.L. Gore & Assoc., Inc. v. Garlock, Inc.*, 721 F.2d 1540, 1553 (Fed. Cir. 1983) (“To imbue one of ordinary skill in the art with knowledge of the invention in suit, when no prior art reference or references of record convey or suggest that knowledge,

is to fall victim to the insidious effect of a hindsight syndrome wherein that which only the inventor taught is used against its teacher.”)

For the third rationale for distributing Jeddeloh’s functionality into Ellsberry, Petitioner relies upon several references—Tsern, Schaffer, and Averbuj—to support its contentions that a person of ordinary skill in the art “would have been motivated to make such a design because other references teach [a person of ordinary skill in the art] to place such self-test functionality in distributed data buffers similar to [Ellsberry’s] Switch ASICs.” Pet. 36 (citing Ex. 1019 ¶¶ 77, 97, Figs. 5, 18; Ex. 1020, 15:57–65, 18:63–19:23, 35:10–37:1, Figs. 5, 18, 36; Ex. 1007 ¶¶ 49–50, 53, 55–57, Figs. 1, 6–7; Ex. 1025, 30). In the Reply, Petitioner expands on this, arguing that a person of ordinary skill in the art “would be motivated to distribute the self-test functionalities in a way that matches how those functionalities are distributed in the memory module.” Reply 36–37 (citing Ex. 1061 ¶ 786 (citing Ex. 1016 at 3, 25; Ex. 1058 at 3, 26; Ex. 1017 at 2–12; Ex. 1018 at 2:38–67; Ex. 1019, Figs. 1, 8)).

Dr. Subramanian testifies that the prior art teaches the use of the same interfaces and signal traces that are used during normal operation. Ex. 1061 ¶ 783. For example, he testifies that Jeddeloh “teaches to put all the self-test functionality blocks in the central hub (self-test module 290) and use the existing data/address/command interfaces (270) and connections from the central hub to the memory devices (240) which are also used during normal operation.” *Id.* (citing Ex. 1006, Fig. 2). Dr. Subramanian further testifies that “Tsern also teaches to follow the existing structure of its module where each of the distributed buffers (100a-d) has existing address/command lines (103) and data lines (102) to the memory devices (101), and put a separate

self-test module (1883) in each of those buffers.” *Id.* (citing Ex. 1019, Figs. 1, 18). To that end, Dr. Subramanian testifies that “if the memory module as a central hub . . . which uses address/command and data signals to interface with the memory devices, then the corresponding self-test functionalities of generating test address/command and data signals would also be implemented in that central hub.” Ex. 1061 ¶ 786 (citing Ex. 1016 at 3, 26; Ex. 1017 at 2-12; Ex. 1018 at 2:38–67; Ex. 1058 at 3, 26). Dr. Subramanian further testifies that “if the memory module has distributed buffers,” like in Tsern, “where each buffer handles both address/command and data signals, the corresponding self-test functionalities of generating test address/command and data signals are distributed the same way, replicating the self-test functionalities in each buffer.” *Id.* (citing Ex. 1019, Figs. 1, 18). Dr. Subramanian compares Tsern’s distributed data buffers to Ellsberry’s Switch ASICs. Ex. 1003 ¶ 159; *see also* Pet. 36.

We do not see how these references support Petitioner’s proposed combination, or provide a motivation to a person of ordinary skill in the art to divide Jeddelloh’s circuitry and functionality and incorporate it into Ellsberry’s Switch ASIC and Control ASIC. Rather, Petitioner’s arguments here are similar to its arguments relying on Ellsberry’s distributed architecture as motivation to divide Jeddelloh’s circuitry and functionality, as discussed above.

Although each of Tsern, Schaffer, and Averbuj disclose a distributed buffer, akin to the Switch ASICs in Ellsberry, none of the references cited by Petitioner teaches separating testing data and address logic, as in the proposed combination. *See* Sur-reply 14. As Dr. Brogioli testifies, each of these references “kept data and address/control generation elements

consolidated on the PCBs of their respective memory modules, unlike how the '523 Patent teaches.” Ex. 2004 ¶ 149; *see* PO Resp. 58, Sur-reply 17. This testimony is supported by the disclosure in these references. For example, Tsern discloses a single redundancy and repair unit 1883 in buffer 100a that sends address/control signals over path 1005 and data signals over path 1006. PO Resp. 58; *see* Ex. 1019, Figs. 5, 18. Dr. Subramanian even testified that each of the repeating components in Tsern’s buffers (i.e., data slice) includes all of the same circuitry and functionality. Ex. 2005, 159:11–160:22; *see* Ex. 1019, Fig. 5. Schaffer provides a similar disclosure to Tsern. *See* Ex. 1020, Figs. 5, 18, 36. Averbuj discloses both address generation unit 42 and data generation unit 44 in memory interfaces 41. Ex. 1007, ¶¶ 48–51, Fig. 6. Moreover, Dr. Subramanian testified during his deposition that “none of these references teach breaking up the self-test functional blocks the way the '523 Patent does.” Ex. 2004 ¶ 148; Ex. 2004, 181:19–182:4, 182:9–19, 183:10–15.

Although these references may support placing self-test functionality into distributed data buffers, i.e., Ellsberry’s Switch ASICs, they do not fully support the combination proposed by Petitioner, that is, dividing the data logic in one location (i.e., Ellsberry’s Switch ASIC) and the address/control in a different location (i.e., Ellsberry’s Control ASIC) in the context of a self-test environment. In other words, all Petitioner has established is that the self-test functionality in the cited references was in the same location, e.g., a central hub or repeated in distributed buffers, but not that it was split, as in the proposed combination, and as is required by the claims. We find that the disclosure in these references does not support the proposed combination; rather, it further reinforces that the proposed combination is

driven by hindsight. We agree with Patent Owner that “Petitioner cites no example anywhere attempting disaggregated data and address handling for testing, much less doing so into distributed memory architecture like Ellsberry.” PO Resp. 38 (citing Ex. 2004 ¶¶ 123–125, 143).

For the fourth rationale for distributing Jeddeloh’s functionality into Ellsberry, Petitioner relies upon the Lepejian reference, asserting that it “teaches [a person of ordinary skill in the art] that, in order to reduce busing area, pattern generators should be distributed to each of the Switch ASICs.” Pet. 36 (citing Ex. 1021, 3:45–53; Ex. 1003 ¶ 162). However, similar to Petitioner’s reliance on the other references discussed above, this merely establishes that Lepejian discloses redundantly distributing functionality across Switch ASICs, but does not provide motivation for the proposed combination, i.e., distributing the self-test functionality across the Switch ASIC and Control ASIC. *See* PO Resp. 58–59.

In view of the number and extent of modifications necessary to achieve the proposed combination of Ellsberry and Jeddeloh, this indicates that the series of proposed modifications are the result of hindsight. *See Metalcraft of Mayville, Inc. v. The Toro Co.*, 848 F.3d 1358, 1367 (Fed. Cir. 2017) (“[W]e cannot allow hindsight bias to be the thread that stitches together prior art patches into something that is the claimed invention.”); *InTouch Techs., Inc. v. VGO Commc’ns, Inc.*, 751 F.3d 1327, 1351 (Fed. Cir. 2014) (criticizing use of the challenged patent as a “roadmap” for putting what the expert referred to as “pieces of a ‘jigsaw puzzle’” together). We, therefore, find that the evidence does not show that a person of ordinary skill in the art would have been motivated to make the alleged modifications to Jeddeloh and Ellsberry without the benefit and knowledge of the ’523

patent. See *In re NTP, Inc.*, 654 F.3d 1279, 1299 (Fed. Cir. 2011) (internal quotations omitted) (an obviousness analysis must “avoid hindsight reconstruction by using the patent in suit as a guide through the maze of prior art references, combining the right references in the right way so as to achieve the result of the claims in suit.”). Specifically, we determine that Petitioner has not provided sufficient reasoning establishing why a person of ordinary skill in the art would have been motivated to divide the self-test functionality of Jeddeloh’s Figure 3 between the Control ASIC and Switch ASICs of Ellsberry and that the skilled artisan would have had a reasonable expectation of success in doing so.

Accordingly, for the foregoing reasons, we determine that Petitioner has not established that claims 1–34 are unpatentable over Ellsberry and Jeddeloh.

E. Grounds 2–4; Obviousness Over Ellsberry, Jeddeloh, and Averbuj/Lee/Averbuj and Lee

Petitioner relies on the same motivation to combine Ellsberry and Jeddeloh in the remaining grounds. See Pet. 104–113. Therefore, for the same reasons as set forth above for Ground 1, we determine that Petitioner has not established that claims 1–34 are unpatentable.

IV. MOTIONS TO EXCLUDE

A. Petitioner’s Motion to Exclude

Petitioner contends that in its Sur-reply, Patent Owner improperly relies upon testimony from Petitioner’s expert reading from certain documents in order to “backdoor” certain exhibits into evidence. Pet. Mot. Excl. 4. Petitioner identifies these as Exhibits 2010, 2011, 2012, and 2014, but states that Patent Owner did not submit these exhibits with its Sur-reply.

Id. This testimony is purportedly related to when the labels “LRDIMM” and “FBDIMM” were first introduced. *Id.* at 3–4. Petitioner, therefore “requests that the Board exclude any attempt by Patent Owner to use any testimony related to Exhibits 2010, 2011, 2012, or 2014 concerning when the labels ‘LRDIMM’ and ‘FBDIMM’ were first introduced.” *Id.* at 5.

Patent Owner argues that the deposition excerpts at Exhibit 2016, 57:17–58:15 and 79:8–80:1 show permissible questioning regarding information within Dr. Subramanian’s personal knowledge. PO Opp. Mot. Excl. 1–4. In addition, Patent Owner contends that Petitioner’s motion to exclude is improper in that it (1) seeks to exclude evidence that is not cited or relied upon; (2) cites to no authority for the Board to strike testimony based on speculation as to what might occur in the future; and (3) is overbroad in that it is unclear what is encompassed by the request. *Id.* at 5.

In Reply, Petitioner argues that the cited testimony shows that Dr. Subramanian lacked personal knowledge, and, is therefore inadmissible. Pet. Mot. Excl. Reply 2–4.

Although Petitioner cites to testimony as part of its arguments, Petitioner does not clearly identify what testimony it seeks to exclude. Petitioner cites to Exhibit 2016 at 48:20–49:11, 49:17–50:6, 50:25–51:15, 52:3–7, 57:17–58:15, 71:21–72:16, 74:2–4, 76:1–24, 77:1–78:2, 79:8–80:1. *See* Pet. Mot. Excl. 3–5; Pet. Mot. Exclude Reply 2–3. In the Reply, Petitioner also cites to Exhibit 2016, 80:2–81:1. Pet. Mot. Excl. Reply 4.

We do not rely on any of the foregoing testimony in this Decision. *See supra*, Section III.D.3, including fn. 9. Therefore, Petitioner’s Motion to Exclude is dismissed as moot.

B. Patent Owner's Motion to Exclude

Patent Owner seeks exclusion, in whole or in part, of Exhibits 1012, 1014–1017, 1022, 1024, 1028, 1029, and 1031, as well as any related testimony that is relied upon by Petitioner. PO Mot. Excl. 1. Patent Owner contends that Petitioner (1) has not properly authenticated these exhibits and (2) that they are inadmissible hearsay. PO Mot. Excl. 5–8; PO Mot. Excl. Reply 1–4. In response, Petitioner contends that (1) for various reasons, each of the exhibits have been properly authenticated, and (2) the exhibits are not inadmissible hearsay because they are merely used to “show the state of the art and the knowledge of one of ordinary skill in the art.” Pet. Opp. Mot. Excl. 1–14.

We do not rely on any of these exhibits in rendering our Decision. *See supra* Section III.D.3. Therefore, Patent Owner's Motion to Exclude is dismissed as moot.

V. CONCLUSION

For the foregoing reasons, we are not persuaded that Petitioner has established by a preponderance of the evidence that claims 1–34 of the '523 patent are unpatentable.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–34	103(a)	Ellsberry, Jeddeloh		1–34
1–34	103(a)	Ellsberry, Jeddeloh, Averbuj		1–34
14, 17– 34	103(a)	Ellsberry, Jeddeloh, Lee		14, 17–34
14, 17– 34	103(a)	Ellsberry, Jeddeloh, Averbuj,		14, 17–34

		Lee		
Overall Outcome				1–34

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–34 of the '523 patent have not been shown to be unpatentable under 35 U.S.C. § 103(a); and

FURTHER ORDERED that Petitioner's Motion to Exclude (Paper 44) is *dismissed as moot*; and

FURTHER ORDERED that Patent Owner's Motion to Exclude (Paper 43) is *dismissed as moot*; and

FURTHER ORDERED that, because this is a final written decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2022-00063
Patent 10,217,523 B1

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(12) **INTER PARTES REVIEW CERTIFICATE** (4073rd)

United States Patent
Lee et al.

(10) **Number:** **US 10,217,523 K1**
(45) **Certificate Issued:** **Jul. 25, 2025**

(54) **MULTI-MODE MEMORY MODULE WITH
DATA HANDLERS**

(71) **Applicants:** **Hyun Lee; Jayesh R. Bhakta;
Soonju Choi**

(72) **Inventors:** **Hyun Lee; Jayesh R. Bhakta;
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Trial Number:

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Patent No.: 10,217,523
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The results of IPR2022-00063 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 10,217,523 K1
Trial No. IPR2022-00063
Certificate Issued Jul. 25, 2025

1

2

AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claims 1-34 are found patentable.

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